

# Estimation and Inference in Boundary Discontinuity Designs\*

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## Abstract

Boundary Discontinuity Designs are used to learn about treatment effects along a continuous boundary that splits units into control and treatment groups according to a bivariate score variable. These research designs are also called Multi-Score Regression Discontinuity Designs, a leading special case being Geographic Regression Discontinuity Designs. We study the statistical properties of commonly used local polynomial treatment effects estimators along the continuous treatment assignment boundary. We consider two distinct approaches: one based explicitly on the bivariate score variable for each unit, and the other based on their univariate distance to the boundary. For each approach, we present pointwise and uniform estimation and inference methods for the treatment effect function over the assignment boundary. Notably, we show that methods based on univariate distance to the boundary exhibit an irreducible large misspecification bias when the assignment boundary has kinks or other irregularities, making the distance-based approach unsuitable for empirical work in those settings. In contrast, methods based on the bivariate score variable do not suffer from that drawback. We illustrate our methods with an empirical application. Companion general-purpose software is provided.

*Keywords:* regression discontinuity, uniform estimation and inference, causal inference.

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# 1 Introduction

We study estimation and inference in Boundary Discontinuity Designs (Black, 1999; Dell, 2010; Jardim et al., 2024), where the goal is to learn about causal treatment effects along a continuous boundary that splits units into control and treatment groups according to the value of a bivariate location score variable. This setup is also known as a Multi-Score Regression Discontinuity (RD) Design (Papay et al., 2011; Reardon and Robinson, 2012), a leading special case being the Geographic RD Design and variations thereof (Keele and Titiunik, 2015, 2016; Keele et al., 2017; Galiani et al., 2017; Diaz and Zubizarreta, 2023). See Cattaneo and Titiunik (2022, Section 2.3) for an overview of the literature on Multi-Dimensional RD Designs, and Cattaneo et al. (2024, Section 5) for a practical introduction.

To describe the setup formally, suppose that  $(Y_i(0), Y_i(1), \mathbf{X}_i^\top)^\top$ ,  $i = 1, 2, \dots, n$ , is a random sample, where  $Y_i(0)$  and  $Y_i(1)$  denote the scalar potential outcomes for unit  $i$  under control and treatment assignment, respectively, and the score  $\mathbf{X}_i = (X_{1i}, X_{2i})^\top$  is a continuous bivariate vector with support  $\mathcal{X} \subseteq \mathbb{R}^2$ . Units are assigned to either the control group or treatment group according to their location  $\mathbf{X}_i$  relative to a known one-dimensional boundary curve  $\mathcal{B}$  splitting the support  $\mathcal{X}$  in two disjoint regions:  $\mathcal{X} = \mathcal{A}_0 \cup \mathcal{A}_1$  with  $\mathcal{A}_0$  and  $\mathcal{A}_1$  the control and treatment disjoint (connected) regions, respectively, and  $\mathcal{B} = \text{bd}(\mathcal{A}_0) \cap \text{bd}(\mathcal{A}_1)$ , where  $\text{bd}(\mathcal{A}_t)$  denotes the boundary of the set  $\mathcal{A}_t$ . The observed response variable is  $Y_i = \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_0) \cdot Y_i(0) + \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_1) \cdot Y_i(1)$ . Without loss of generality, we assume that the boundary belongs to the treatment group, that is,  $\text{bd}(\mathcal{A}_1) \subset \mathcal{A}_1$  and  $\mathcal{B} \cap \mathcal{A}_0 = \emptyset$ .

Boundary discontinuity designs are commonly used in quantitative social, behavioral, and biomedical sciences. For example, consider the substantive application in Londoño-Vélez, Rodríguez and Sánchez (2020). The authors studied the effects of a governmental subsidy for post-secondary education in Colombia, called *Ser Pilo Paga* (SPP), an anti-poverty policy providing tuition support for four-year or five-year undergraduate college students in any government-certified higher education institution with high-quality status. Eligibility to the program SSP was based on both merit and economic need: in order to qualify for the program, students had to obtain a high grade in Colombia’s national standardized high school exit exam, SABER 11, and they also had to come from economically disadvantaged families, as measured by the survey-based wealth index

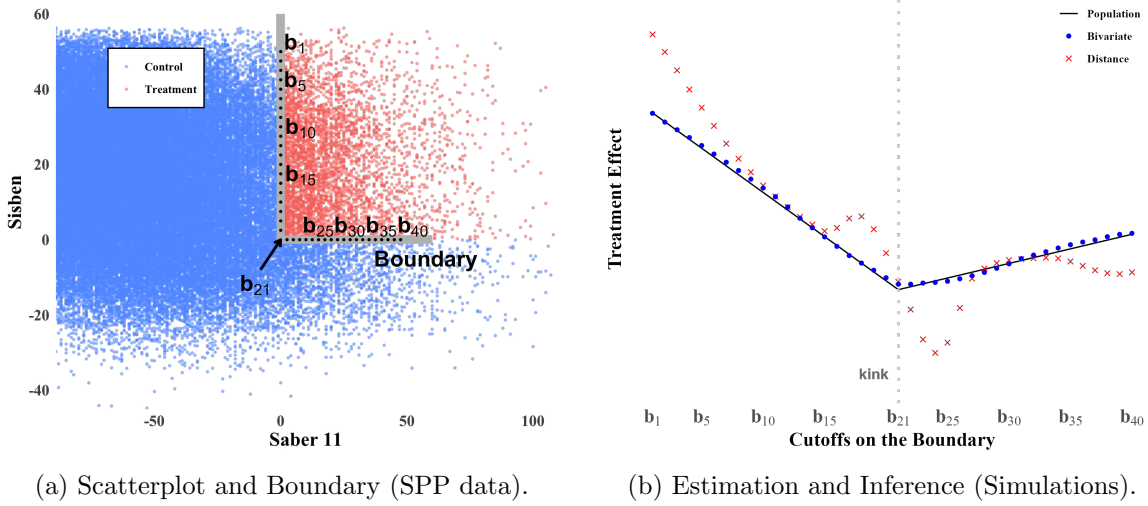


Figure 1: Scatterplot, Treatment Boundary, and Estimation and Inference Methods.

Note: Panel (a) presents a scatterplot of the bivariate score  $\mathbf{X}_i$  using the SPP data, and also plots the treatment boundary  $\mathcal{B}$  with 40 marked grid points. Panel (b) presents estimation and inference results over the 40 boundary grid points depicted in Panel (a) based on one simulated dataset calibrated using the real SPP data, where the population treatment effect curve  $\tau(\mathbf{x})$  is assumed linear to remove smoothing bias. Point estimates are implemented using a MSE-optimal bandwidth, and averaged over 1,000 Monte Carlo simulations. The companion R software package `rd2d` is used for implementation, and further details are available in the replication files and [Cattaneo et al. \(2025\)](#).

SISBEN. Eligibility followed a deterministic rule with a fixed bivariate cutoff: students had to obtain a SABER 11 score in the top 9 percent of scores or better, and they had to come from a household with SISBEN index below a region-specific threshold. Formally, each student was assigned a bivariate score  $\mathbf{X}_i = (\text{SABER11}_i, \text{SISBEN}_i)^\top$ , where  $X_{1i} = \text{SABER11}_i$  recorded the SABER11 score and  $X_{2i} = \text{SISBEN}_i$  recorded the SISBEN wealth score. After recentering each variable at its corresponding threshold, the treatment assignment boundary becomes  $\mathcal{B} = \{(\text{SABER11}, \text{SISBEN}) : (\text{SABER11}, \text{SISBEN}) \in \{\text{SABER11} \geq 0 \text{ and } \text{SISBEN} = 0\} \cup \{\text{SABER11} = 0 \text{ and } \text{SISBEN} \geq 0\}\}$ . Figure 1a presents a scatterplot of the bivariate score of the data of students in the 2014 cohort ( $n = 363,096$  observations), and also plots the bivariate assignment boundary  $\mathcal{B}$  together with 40 evenly-spaced cutoff points along the boundary.

Section 2 presents the core assumption underlying the causal inference framework, which generalizes the standard univariate RD design to boundary discontinuity designs. Given the standard (continuity and finite-moments) conditions in Assumption 1 (below), the goal is to conduct esti-

mation and inference for the *average treatment effect curve along the boundary*:

$$\tau(\mathbf{x}) = \mathbb{E}[Y_i(1) - Y_i(0) | \mathbf{X}_i = \mathbf{x}], \quad \mathbf{x} \in \mathcal{B},$$

both pointwise for each  $\mathbf{x} \in \mathcal{B}$ , and uniformly over  $\mathcal{B}$ . For example, in the SPP application, the outcome variable is  $Y_i = 1$  if student  $i$  attended college or  $Y_i = 0$  otherwise, and thus the causal parameter  $\tau(\mathbf{x})$  captures the treatment effect of SPP on the probability of college education for students at the margin of program eligibility, as determined by their bivariate score  $\mathbf{X}_i = (\text{SABER11}_i, \text{SISBEN}_i)^\top \in \mathcal{B}$ . The parameter  $\tau(\mathbf{x})$  captures policy-relevant heterogeneous treatment effects along the boundary  $\mathcal{B}$ : for example, in Figure 1a,  $\tau(\mathbf{b}_1)$  corresponds to a (local) average treatment effect for students with high SISBEN score (wealth) and low SABER11 score (academic), while  $\tau(\mathbf{b}_{40})$  corresponds to a (local) average treatment effect for students with low SISBEN score and higher SABER11 score. Identification of these boundary treatment effects parallels standard continuity-based univariate RD design arguments (Hahn et al., 2001): treatment assignment changes abruptly along the boundary  $\mathcal{B}$ , which implies that conditional expectations on each side of the assignment boundary can be used to identify  $\tau(\mathbf{x})$  whenever there is no systematic “sorting” of units across the boundary, that is, whenever  $\mathbb{E}[Y_i(0) | \mathbf{X}_i = \mathbf{x}]$  and  $\mathbb{E}[Y_i(1) | \mathbf{X}_i = \mathbf{x}]$  are continuous for all  $\mathbf{x} \in \mathcal{B}$  (Assumption 1 below). For more discussion, see Papay et al. (2011), Reardon and Robinson (2012), Keele and Titiunik (2015), and references therein.

Motivated by the local-to-the-assignment-boundary identifiability of  $\tau(\mathbf{x})$ , researchers employ flexible regression methods using only observations with their score near the boundary  $\mathcal{B}$ . Local polynomial methods are the preferred choice for estimation and inference because they are simple (weighted) linear regression estimates that intuitively incorporate localization to the assignment boundary, while also retaining most of the familiar features of least squares regression methods. Two distinct implementations can be considered in this setting:

- (i) regression analysis based on the *univariate distance* to the boundary  $\mathcal{B}$ , or
- (ii) regression analysis based on the *bivariate location* relative to the boundary  $\mathcal{B}$ .

The first approach is more commonly used in practice because it is perceived as simpler, while the second approach is less often encountered in applications. Despite their widespread use, however, there is no foundational understanding of the statistical properties and relative merits of these

alternative implementation methods. Our paper fills this gap in the literature by providing comprehensive large sample statistical results for each of the two approaches, which we then use to offer specific practical recommendations for the analysis and interpretation of boundary discontinuity designs. In particular, we provide pointwise and uniform (over  $\mathcal{B}$ ) estimation and inference methods for both local polynomial regression approaches, and demonstrate theoretical and practical advantages of the approach based on bivariate location over the approach based on distance (albeit both are shown to be valid under appropriate assumptions).

Using simulated data calibrated with the real SPP data, Figure 1b illustrates graphically the two local polynomial approaches for estimation of  $\tau(\mathbf{x})$ . The solid black line corresponds to the population treatment effect curve  $\tau(\mathbf{x})$ , taken to be linear in order to eliminate smoothing bias issues. The red crosses and blue dots correspond to local polynomial estimators based on, respectively, univariate distance to the boundary and bivariate location relative to the boundary, for the 40 grid points depicted on the assignment boundary  $\mathcal{B}$  in Figure 1a. These treatment effect estimators are implemented using a MSE-optimal bandwidth choice (developed below), and averaged over 1,000 Monte Carlo simulations. The estimator based on distance exhibits a noticeable higher bias near (but not at) the boundary kink point  $\mathbf{x} = \mathbf{b}_{21}$ , relative to the bivariate local polynomial estimator based on the bivariate score. A contribution of our paper is to provide a theory-based explanation of this empirical phenomena, which turns out to be a general limitation of distance-based methods.

## 1.1 Paper Organization and Main Contributions

After introducing the formal causal inference framework in Section 2, Section 3 presents pointwise and uniform estimation and inference methods for local polynomial regression methods based on the univariate distance to the boundary  $\mathcal{B}$ . We begin by providing interpretable sufficient conditions (Assumption 2 below) for identification of  $\tau(\mathbf{x})$ , which restrict the distance function in conjunction with the univariate kernel function used and the shape of the boundary  $\mathcal{B}$ . We then present an important negative result: near kinks of the boundary  $\mathcal{B}$ , a  $p$ th order distance-based local polynomial estimator exhibits an irreducible bias of order  $h$ , the bandwidth used for implementation, no matter the polynomial order  $p$  used. This drawback is due to the fact that the underlying population regression function is at most Lipschitz continuous near kinks of the boundary  $\mathcal{B}$ . Figure 1b illustrated the phenomenon numerically, and our paper provides a theoretical explanation for the

large bias of the distance-based treatment effect estimator. In contrast, when the boundary  $\mathcal{B}$  is smooth, we show that the  $p$ th order distance-based local polynomial estimator exhibits the usual bias of order  $h^{p+1}$ . Thus, our results show that the standard distance-based treatment effect estimator, while consistent for  $\tau(\mathbf{x})$ , both pointwise and uniformly over  $\mathcal{B}$ , it can exhibit a large bias affecting bandwidth selection, point estimation, and statistical inference in applications, whenever  $\mathcal{B}$  has kinks or other irregularities. For uncertainty quantification, Section 3 presents pointwise and uniform (over  $\mathcal{B}$ ) large sample distribution theory, which is used to propose both confidence intervals for  $\tau(\mathbf{x})$  and confidence bands for the entire treatment effect curve  $(\tau(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$ . The last part of Section 3 discusses implementation, and also compares our results with related methods for standard univariate RD designs.

The analysis based on univariate distance to the boundary is not as straight forward as it may seem, and can in fact lead to invalid empirical findings in applications. More precisely, our results show that standard methods in the literature for univariate RD estimation and inference will not be valid when deployed directly to a bivariate RD design, upon constructing distance to the boundary, because the specific features of the assignment boundary can lead to a large misspecification bias near kinks or other irregularities of  $\mathcal{B}$ . The large misspecification bias we highlight when the assignment boundary is non-smooth is particularly worrisome in Geographic RD designs (Keele and Titiunik, 2015), and similar settings, where the boundary naturally exhibits many kinks or other irregularities.

An alternative is to employ the bivariate location score directly. Section 4 studies this approach, and offers pointwise and uniform over  $\mathcal{B}$  estimation and inference methods. The pointwise results follow directly from the literature, provided an additional regularity condition on the bivariate kernel function and boundary  $\mathcal{B}$  hold (Assumption 3 below). On the other hand, the uniform inference results require some additional technical care, and are new to the literature. The main potential issue is that a uniform distributional approximation is established over the lower-dimensional manifold  $\mathcal{B}$ , and thus its shape can affect the validity of the results. Section 4 also discusses new bandwidth selection methods based on mean square error (MSE) expansions, robust bias-corrected inference, and related implementation details. Our results provide natural generalizations of well-established results for univariate RD designs; see Calonico et al. (2020) for bandwidth selection, and Calonico et al. (2014, 2018, 2022) for robust bias correction.

Section 5 deploys our theoretical and methodological results to the SPP data, revising the main results reported in [Londoño-Vélez, Rodríguez and Sánchez \(2020\)](#). In addition to providing further empirical evidence in favor of their empirical findings, we also find some evidence of treatment effect heterogeneity along the assignment boundary  $\mathcal{B}$ . All empirical results are obtained using the companion R package `rd2d`, and we provide complete replication files.

Section 6 concludes with specific recommendations for practice. The supplemental appendix presents generalizations of our theoretical results, reports their proofs, and gives other theoretical results that may be of independent interest. In particular, it presents a new strong approximation result for empirical process with bounded moments, extending recent work by [Cattaneo and Yu \(2025\)](#). Our companion software article ([Cattaneo et al., 2025](#)) discusses the general-purpose software R package `rd2d` (<https://rdpackages.github.io/rd2d>) implementing the methods developed in this paper.

## 2 Setup

We impose the following basic conditions on the underlying data generating process.

**Assumption 1** (Data Generating Process). *Let  $t \in \{0, 1\}$ .*

- (i)  $(Y_1(t), \mathbf{X}_1)^\top, \dots, (Y_n(t), \mathbf{X}_n)^\top$  are independent and identically distributed random vectors.
- (ii) The distribution of  $\mathbf{X}_i$  has a Lebesgue density  $f_X(\mathbf{x})$  that is continuous and bounded away from zero on its support  $\mathcal{X} = [a_1, b_1] \times [a_2, b_2]$ , for  $-\infty < a_l < b_l < \infty$  with  $l \in \{1, 2\}$ .
- (iii)  $\mu_t(\mathbf{x}) = \mathbb{E}[Y_i(t) | \mathbf{X}_i = \mathbf{x}]$  is  $(p+1)$ -times continuously differentiable on  $\mathcal{X}$ .
- (iv)  $\sigma_t^2(\mathbf{x}) = \mathbb{V}[Y_i(t) | \mathbf{X}_i = \mathbf{x}]$  is bounded away from zero and continuous on  $\mathcal{X}$ .
- (v)  $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[|Y_i(t)|^{2+v} | \mathbf{X}_i = \mathbf{x}] < \infty$  for some  $v \geq 2$ .

This assumption is on par with the usual assumptions encountered in the classical RD literature with an univariate score. In particular, part (ii) goes beyond the usual compact support restriction and further assumes a tensor product structure on  $\mathcal{X}$  to avoid technicalities underlying our strong approximation results used for uniform distribution theory: the assumption is not practically restrictive because all methods considered in this paper localize to the boundary. Part (iii) imposes standard smoothness conditions on the bivariate conditional expectations of interest, which will

play an important role in misspecification (smoothing bias) reduction in our upcoming results. Identification of  $\tau(\mathbf{x})$  follows directly from Assumption 1: see Papay et al. (2011), Reardon and Robinson (2012), Keele and Titiunik (2015), and references therein.

## 2.1 Notation

We employ standard concepts and notations from empirical process theory (van der Vaart and Wellner, 1996; Giné and Nickl, 2016) and geometric measure theory (Simon et al., 1984; Federer, 2014). For a random variable  $\mathbf{V}_i$ , we write  $\mathbb{E}_n[g(\mathbf{V}_i)] = n^{-1} \sum_{i=1}^n g(\mathbf{V}_i)$ . For a vector  $\mathbf{v} \in \mathbb{R}^k$ , the Euclidean norm is  $\|\mathbf{v}\| = (\sum_{i=1}^k \mathbf{v}_i^2)^{1/2}$ . For a matrix  $A \in \mathbb{R}^{m \times n}$ , the operator norm is  $\|A\| = \sup_{\|\mathbf{x}\|=1} \|A\mathbf{x}\|$ .  $C^k(\mathcal{X}, \mathcal{Y})$  denotes the class of  $k$ -times continuously differentiable functions from  $\mathcal{X}$  to  $\mathcal{Y}$ , and  $C^k(\mathcal{X})$  is a shorthand for  $C^k(\mathcal{X}, \mathbb{R})$ . For a Borel set  $\mathcal{S} \subseteq \mathcal{X}$ , the De Giorgi perimeter of  $\mathcal{S}$  is  $\text{perim}(\mathcal{S}) = \sup_{g \in \mathcal{D}_2(\mathcal{X})} \int_{\mathbb{R}^2} \mathbf{1}(\mathbf{x} \in \mathcal{S}) \text{div } g(\mathbf{x}) d\mathbf{x} / \|g\|_\infty$ , where  $\text{div}$  is the divergence operator, and  $\mathcal{D}_2(\mathcal{X})$  denotes the space of  $C^\infty$  functions with values in  $\mathbb{R}^2$  and with compact support included in  $\mathcal{X}$ . When  $\mathcal{S}$  is connected, and the boundary  $\text{bd}(\mathcal{S})$  is a smooth simple closed curve,  $\mathcal{S}$  simplifies to the curve length of  $\text{bd}(\mathcal{S})$ . A curve  $\mathcal{B} \subseteq \mathbb{R}^2$  is a *rectifiable curve* if there exists a Lipschitz continuous function  $\gamma : [0, 1] \mapsto \mathbb{R}^2$  such that  $\mathcal{B} = \gamma([0, 1])$ . For a function  $f : \mathbb{R}^2 \mapsto \mathbb{R}$ ,  $\text{Supp}(f)$  denotes closure of the set  $\{\mathbf{x} \in \mathbb{R}^2 : f(\mathbf{x}) \neq 0\}$ . For reals sequences  $a_n = o(b_n)$  if  $\limsup_{n \rightarrow \infty} \frac{|a_n|}{|b_n|} = 0$ ,  $|a_n| \lesssim |b_n|$  if there exists some constant  $C$  and  $N > 0$  such that  $n > N$  implies  $|a_n| \leq C|b_n|$ . For sequences of random variables  $a_n = o_{\mathbb{P}}(b_n)$  if  $\text{plim}_{n \rightarrow \infty} \frac{|a_n|}{|b_n|} = 0$ ,  $|a_n| \lesssim_{\mathbb{P}} |b_n|$  if  $\limsup_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}[|\frac{a_n}{b_n}| \geq M] = 0$ . Finally,  $\Phi(x)$  denotes the standard Gaussian cumulative distribution function.

## 3 Analysis based on Univariate Distance

For each unit  $i = 1, \dots, n$ , define their scalar distance-based score  $D_i(\mathbf{x}) = \mathcal{d}(\mathbf{X}_i, \mathbf{x})(\mathbf{1}(\mathbf{X}_i \in \mathcal{A}_1) - \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_0))$  to the point  $\mathbf{x} \in \mathcal{B}$ , where  $\mathcal{d}(\cdot, \cdot)$  denotes a distance function. It is customary to use the Euclidean distance  $\mathcal{d}(\mathbf{X}_i, \mathbf{x}) = \|\mathbf{X}_i - \mathbf{x}\| = \sqrt{(X_{1i} - x_1)^2 + (X_{2i} - x_2)^2}$  for  $\mathbf{x} = (x_1, x_2)^\top \in \mathcal{B}$  in applications, but other choices are sometimes encountered. For each  $\mathbf{x} \in \mathcal{B}$ , the setup thus reduces to a standard univariate RD design with  $D_i(\mathbf{x}) \in \mathbb{R}$  the score variable and  $c = 0$  the cutoff, where  $D_i(\mathbf{x}) \geq 0$  if unit  $i$  is assigned to treatment status and  $D_i(\mathbf{x}) < 0$  if unit  $i$  is assigned to

control status. The local polynomial treatment effect curve estimator based on distance is

$$\hat{\tau}_{\text{dis}}(\mathbf{x}) = \mathbf{e}_1^\top \hat{\gamma}_1(\mathbf{x}) - \mathbf{e}_1^\top \hat{\gamma}_0(\mathbf{x}), \quad \mathbf{x} \in \mathcal{B},$$

where, for  $t \in \{0, 1\}$ ,

$$\hat{\gamma}_t(\mathbf{x}) = \arg \min_{\gamma \in \mathbb{R}^{p+1}} \mathbb{E}_n \left[ (Y_i - \mathbf{r}_p(D_i(\mathbf{x}))^\top \gamma)^2 k_h(D_i(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right],$$

with  $\mathbf{r}_p(u) = (1, u, u^2, \dots, u^p)^\top$  the usual univariate polynomial basis,  $k_h(u) = k(u/h)/h^2$  for univariate kernel function  $k(\cdot)$  and bandwidth parameter  $h$ , and  $\mathcal{J}_0 = (-\infty, 0)$  and  $\mathcal{J}_1 = [0, \infty)$ . See [Cattaneo and Titiunik \(2022\)](#) for a literature review of RD designs, and [Cattaneo et al. \(2020, 2024\)](#) for practical introductions. The univariate kernel function typically down-weights observations as the distance to  $\mathbf{x} \in \mathcal{B}$  increases, while the bandwidth determines the level of localization to each point on the boundary  $\mathcal{B}$ . In this case, observations contribute relative to their distance  $\mathcal{d}(\mathbf{X}_i, \mathbf{x})$  to the point  $\mathbf{x} \in \mathcal{B}$ .

We impose the following conditions on the underlying features of the distance-based local polynomial estimator  $\hat{\tau}_{\text{dis}}(\mathbf{x})$ .

**Assumption 2** (Univariate Distance-Based Kernel). *Let  $t \in \{0, 1\}$ .*

- (i)  $\mathcal{d} : \mathbb{R}^2 \mapsto [0, \infty)$  satisfies  $\|\mathbf{x}_1 - \mathbf{x}_2\| \lesssim \mathcal{d}(\mathbf{x}_1, \mathbf{x}_2) \lesssim \|\mathbf{x}_1 - \mathbf{x}_2\|$  for all  $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}$ .
- (ii)  $k : \mathbb{R} \rightarrow [0, \infty)$  is compact supported and Lipschitz continuous, or  $k(u) = \mathbf{1}(u \in [-1, 1])$ .
- (iii)  $\liminf_{h \downarrow 0} \inf_{\mathbf{x} \in \mathcal{B}} \int_{\mathcal{A}_t} k_h(\mathcal{d}(\mathbf{u}, \mathbf{x})) d\mathbf{u} \gtrsim 1$ .

Part (i) of this assumption requires the distance function be equivalent (up to constants) to the Euclidean distance, while Assumption 2(ii) imposes standard conditions on the (univariate) kernel function. The last part of this assumption is novel to the literature: it implicitly restricts the geometry of the boundary  $\mathcal{B}$  relative to the kernel and distance functions. More precisely, it rules out settings where highly irregular boundary shapes will lead to regions with too “few” data: a necessary condition for the estimator  $\hat{\gamma}_t(\mathbf{x})$  to be well-defined in large samples is that  $\mathbb{E}[k_h(D_i(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t)] = \int_{\mathcal{A}_t} k_h(\mathcal{d}(\mathbf{u}, \mathbf{x})) f_X(\mathbf{u}) d\mathbf{u} > 0$ . Our theoretical results show that Assumption 2(iii) is a simple sufficient condition, due to Assumption 1(ii). Commonly encountered treatment assignment boundaries and distance functions typically satisfy Assumption 2; a potential

exception being highly irregular geographic boundaries with low density of observations.

### 3.1 Identification and Interpretation

For each treatment group  $t \in \{0, 1\}$ , boundary point  $\mathbf{x} \in \mathcal{B}$ , and distance score  $D_i(\mathbf{x})$ , the univariate distance-based local polynomial estimator  $\hat{\gamma}_t(\mathbf{x})$  is the sample analog of the coefficients associated with the best (weighted, local) mean square approximation of the conditional expectation  $\mathbb{E}[Y_i|D_i(\mathbf{x})]$  based on  $\mathbf{r}_p(D_i(\mathbf{x}))$ :

$$\gamma_t^*(\mathbf{x}) = \arg \min_{\gamma \in \mathbb{R}^{p+1}} \mathbb{E} \left[ (Y_i - \mathbf{r}_p(D_i(\mathbf{x}))^\top \gamma)^2 k_h(D_i(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right].$$

Therefore, letting  $\hat{\theta}_{t,\mathbf{x}}(0) = \mathbf{e}_1^\top \hat{\gamma}_t(\mathbf{x})$ ,  $\theta_{t,\mathbf{x}}^*(0) = \mathbf{e}_1^\top \gamma_t^*(\mathbf{x})$ , and  $\theta_{t,\mathbf{x}}(r) = \mathbb{E}[Y_i|D_i(\mathbf{x}) = r, D_i(\mathbf{x}) \in \mathcal{J}_t]$  for  $r \in \mathbb{R}$ , we have the standard least squares (best linear prediction) decomposition

$$\hat{\theta}_{t,\mathbf{x}}(0) - \theta_{t,\mathbf{x}}(0) = \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} + [\theta_{t,\mathbf{x}}^*(0) - \theta_{t,\mathbf{x}}(0)] + [\mathbf{e}_1^\top (\hat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1}) \mathbf{O}_{t,\mathbf{x}}] \quad (1)$$

for each group  $t \in \{0, 1\}$ , and where

$$\begin{aligned} \Psi_{t,\mathbf{x}} &= \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right)^\top k_h(D_i(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right], \\ \hat{\Psi}_{t,\mathbf{x}} &= \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right)^\top k_h(D_i(\mathbf{x})) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right], \quad \text{and} \\ \mathbf{O}_{t,\mathbf{x}} &= \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) k_h(D_i(\mathbf{x})) (Y_i - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))) \mathbf{1}(D_i(\mathbf{x}) \in \mathcal{J}_t) \right]. \end{aligned}$$

In the decomposition (1), the first term is the stochastic linear representation of the centered estimator,  $\hat{\theta}_{t,\mathbf{x}}(0) - \theta_{t,\mathbf{x}}(0)$ , since it is an average of (unconditional) mean-zero random variables, the second term is the mean square approximation bias, and the third term is a non-linearity error arising from the convergence of the Gram matrix associated with the (weighted) least squares estimator.

Since  $\hat{\tau}_{\text{dis}}(\mathbf{x}) = \hat{\theta}_{1,\mathbf{x}}(0) - \hat{\theta}_{0,\mathbf{x}}(0)$ , the following lemma characterizes the target estimand of the distance-based local polynomial estimator.

**Lemma 1** (Identification). *Suppose Assumptions 1(i)–(iii) and 2(i) hold. Then,  $\tau(\mathbf{x}) = \lim_{r \downarrow 0} \theta_{1,\mathbf{x}}(r) - \lim_{r \uparrow 0} \theta_{0,\mathbf{x}}(r)$  for all  $\mathbf{x} \in \mathcal{B}$ .*

This lemma is established by noting that, for each group  $t \in \{0, 1\}$  and  $r \in \mathbb{R}$ ,

$$\begin{aligned}\theta_{t,\mathbf{x}}(r) &= \mathbb{E}[Y_i | D_i(\mathbf{x}) = r, D_i(\mathbf{x}) \in \mathcal{J}_t] \\ &= \mathbb{E}[Y_i | \mathcal{d}(\mathbf{X}_i, \mathbf{x}) = |r|, \mathbf{X}_i \in \mathcal{A}_t] = \mathbb{E}[Y_i(t) | \mathcal{d}(\mathbf{X}_i, \mathbf{x}) = |r|],\end{aligned}$$

and then verifying that  $\lim_{u \downarrow 0} \mathbb{E}[Y_i(t) | \mathcal{d}(\mathbf{X}_i, \mathbf{x}) = u] = \mu_t(\mathbf{x})$  under the conditions imposed. Without restricting the data generating process, as well as the assignment boundary and distance function,  $\theta_{t,\mathbf{x}}(0)$  and  $\mu_t(\mathbf{x})$  need not agree. Employing Lemma 1 and the decomposition (1), we obtain:

$$\widehat{\tau}_{\text{dis}}(\mathbf{x}) - \tau(\mathbf{x}) = \mathfrak{L}_n(\mathbf{x}) + \mathfrak{B}_n(\mathbf{x}) + \mathfrak{Q}_n(\mathbf{x}), \quad \mathbf{x} \in \mathcal{B},$$

where  $\mathfrak{L}_n(\mathbf{x}) = \mathbf{e}_1^\top \Psi_{1,\mathbf{x}}^{-1} \mathbf{O}_{1,\mathbf{x}} - \mathbf{e}_1^\top \Psi_{0,\mathbf{x}}^{-1} \mathbf{O}_{0,\mathbf{x}}$  is a mean-zero linear statistic,  $\mathfrak{B}_n(\mathbf{x}) = \theta_{1,\mathbf{x}}^*(0) - \theta_{0,\mathbf{x}}^*(0) - \tau(\mathbf{x})$  is the bias of the estimator, and  $\mathfrak{Q}_n(\mathbf{x}) = \mathbf{e}_1^\top (\widehat{\Psi}_{1,\mathbf{x}}^{-1} - \Psi_{1,\mathbf{x}}^{-1}) \mathbf{O}_{1,\mathbf{x}} - \mathbf{e}_1^\top (\widehat{\Psi}_{0,\mathbf{x}}^{-1} - \Psi_{0,\mathbf{x}}^{-1}) \mathbf{O}_{0,\mathbf{x}}$  is the higher-order linearization error.

In standard local polynomial regression settings,  $\mathfrak{L}_n(\mathbf{x})$  is approximately Gaussian,  $\mathfrak{B}_n(\mathbf{x})$  is of order  $h^{p+1}$ , and  $\mathfrak{Q}_n(\mathbf{x})$  is neglected. However, because estimation is conducted along the boundary  $\mathcal{B}$ , not all of those standard results remain valid in the context of boundary discontinuity designs.

### 3.2 Bias Along the Boundary

Unlike the case of standard univariate local polynomial estimation, in boundary discontinuity designs the smoothness of  $\theta_{t,\mathbf{x}}(r) = \mathbb{E}[Y_i | D_i(\mathbf{x}) = r, D_i(\mathbf{x}) \in \mathcal{J}_t]$  depends on the smoothness of the boundary  $\mathcal{B}$ . More specifically, the bias of the distance-based local polynomial estimator can be affected by the shape of the boundary  $\mathcal{B}$ , regardless of the polynomial order  $p$  used when constructing the estimator. Figure 2 demonstrates the problem graphically: for a point  $\mathbf{x} \in \mathcal{B}$  that is close enough to a kink point  $(0, 0)$  on the boundary, the conditional expectation  $\theta_{1,\mathbf{x}}(r)$  is not differentiable for all  $r \geq 0$ . This problem arises because, given the distance function  $\mathcal{d}(\cdot, \cdot)$ , for any point  $\mathbf{x} \in \mathcal{B}$  near a kink, a “small”  $r$  gives a complete arc  $\{\mathbf{x} \in \mathcal{A}_1 : \mathcal{d}(\mathbf{X}_i, \mathbf{x}) = r\}$ , while for a “large”  $r$  the arc is truncated by the boundary. As a result, for the example in Figure 2,  $\theta_{1,\mathbf{x}}(r)$  is smooth for all  $r \leq r_3$  and  $r > r_3$ , but the function is not differentiable at  $r = r_3$ . Furthermore, at  $r = r_3$ , the left derivative is constant, but the right derivative is equal to infinity. The supplemental

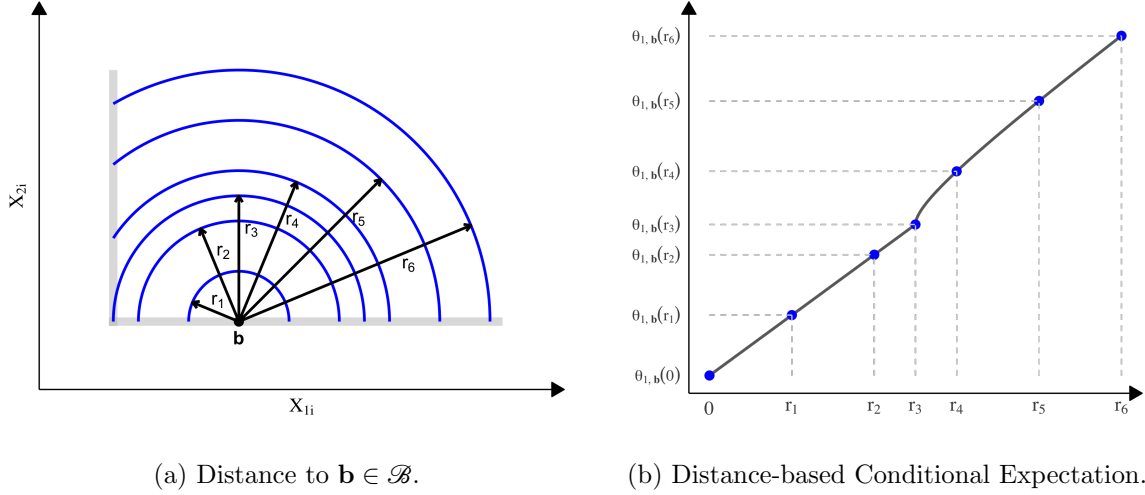


Figure 2: Lack of Smoothness of Distance-Based Conditional Expectation near a Kink.  
Note: Analytic example of  $\theta_{1,\mathbf{b}}(r) = \mathbb{E}[Y(1)|D_i(\mathbf{b}) = r]$ ,  $r \geq 0$ , for distance transformation  $D_i(\mathbf{b}) = \mathcal{d}(\mathbf{X}_i, \mathbf{b}) = \|\mathbf{X}_i - \mathbf{b}\|$  to point  $\mathbf{b} \in \mathcal{B}$  near a kink point on the boundary, based on location  $\mathbf{X}_i = (X_{1i}, X_{2i})^\top$ . The induced univariate conditional expectation  $r \mapsto \theta_{1,\mathbf{b}}(r)$  is continuous but not differentiable at  $r = r_3$ .

appendix gives details on this analytic example.

Although smoothness of the boundary  $\mathcal{B}$  can affect the smoothness of  $\theta_{t,\mathbf{x}}(r)$ , the fact that the distance-based estimator is “local” means that the *approximation error* will be no greater than that of a local constant estimator, regardless of the choice of polynomial order  $p$ . The following lemma formalizes this result, and also shows that this bias order cannot be improved by increasing  $p \geq 0$ .

**Lemma 2** (Approximation Error: Minimal Guarantee). *For some  $L > 0$ , let  $\mathcal{P}$  be the class of data generating processes satisfying Assumptions 1(i)-(iii) with  $\mathcal{X} \subseteq [-L, L]^2$ , Assumption 2, and the following three additional conditions:*

- (i)  $L^{-1} \leq \inf_{\mathbf{x} \in \mathcal{X}} f_X(\mathbf{x}) \leq \sup_{\mathbf{x} \in \mathcal{X}} f_X(\mathbf{x}) \leq L$ ,
- (ii)  $\max_{0 \leq |\nu| \leq p} \sup_{\mathbf{x} \in \mathcal{X}} |\partial^\nu \mu(\mathbf{x})| + \max_{0 \leq |\nu| \leq p} \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \frac{|\partial^\nu \mu(\mathbf{x}) - \partial^\nu \mu(\mathbf{y})|}{\|\mathbf{x} - \mathbf{y}\|} \leq L$ , and
- (iii)  $\liminf_{h \downarrow 0} \inf_{\mathbf{x} \in \mathcal{B}} \int_{\mathcal{A}_t} k_h(\mathcal{d}(\mathbf{u}, \mathbf{x})) d\mathbf{u} \geq L^{-1}$ .

For any  $p \geq 1$ , if  $nh^2 \rightarrow \infty$ , then

$$1 \lesssim \liminf_{n \rightarrow \infty} \sup_{\mathbb{P} \in \mathcal{P}} \sup_{\mathbf{x} \in \mathcal{B}} \frac{\mathfrak{B}_n(\mathbf{x})}{h} \leq \limsup_{n \rightarrow \infty} \sup_{\mathbb{P} \in \mathcal{P}} \sup_{\mathbf{x} \in \mathcal{B}} \frac{\mathfrak{B}_n(\mathbf{x})}{h} \lesssim 1.$$

This lemma characterizes precisely the uniform over  $\mathcal{B}$  (and data generating processes) bias of

the distance-based local polynomial estimator  $\hat{\tau}_{\text{dis}}(\mathbf{x})$ . The upper bound in Lemma 2 is established uniformly over the class of data generating processes because we can show that  $|\theta_t(0) - \theta_t(r)| \lesssim r$  for  $t \in \{0, 1\}$  in general. The lower bound is shown using the following example. Suppose  $\mathbf{X}_i \sim \text{Uniform}([-2, 2]^2)$ ,  $\mu_0(x_1, x_2) = 0$ ,  $\mu_1(x_1, x_2) = x_2$  for all  $(x_1, x_2) \in [-2, 2]^2$ , and  $Y_i(0)|\mathbf{X}_i \sim \text{Normal}(\mu_0(\mathbf{X}_i), 1)$  and  $Y_i(1)|\mathbf{X}_i \sim \text{Normal}(\mu_1(\mathbf{X}_i), 1)$ . Let  $\mathcal{d}(\cdot, \cdot)$  be the Euclidean distance, and suppose that the control and treatment regions are  $\mathcal{A}_1 = \{(x, y) \in \mathbb{R}^2 : x \leq 0, y \geq 0\}$  and  $\mathcal{A}_0 = \mathbb{R}^2 / \mathcal{A}_1$ , respectively, and hence  $\mathcal{B} = \{(x, y) \in \mathbb{R}^2 : -2 \leq x \leq 0, y = 0 \text{ or } x = 0, 0 \leq y \leq 2\}$ . This boundary is as in Figure 2a, having a  $90^\circ$  kink at  $\mathbf{x} = (0, 0)$ . It follows that the conditions of Lemma 2 hold, and hence this is an allowed data generating process. The supplemental appendix establishes the lower bound by careful analysis of the resulting approximation bias.

As a point of contrast, note that  $\mathfrak{B}_n(\mathbf{x}) \lesssim_{\mathbb{P}} h^{p+1}$  pointwise in  $\mathbf{x} \in \mathcal{B}$ , for small enough  $h$ , provided that the kinks on the boundary  $\mathcal{B}$  are sufficiently far apart of each other relative to the bandwidth. However, Lemma 2 demonstrates that, no matter how large the sample size is (i.e., how small the bandwidth is), there will always be a region near a kink of the boundary  $\mathcal{B}$  where the misspecification bias of the distance-based local polynomial estimator  $\hat{\tau}_{\text{dis}}(\mathbf{x})$  is at most of order  $h$  (i.e., not of order  $h^{p+1}$  as expected), regardless of the polynomial order  $p$  employed. The problem arises when the boundary  $\mathcal{B}$  changes non-smoothly, leading to a non-differentiable induced conditional expectation  $\theta_{t,\mathbf{x}}(r) = \mathbb{E}[Y_i | D_i(\mathbf{x}) = r, D_i(\mathbf{x}) \in \mathcal{J}_t]$ ,  $t \in \{0, 1\}$ , as illustrated in Figure 2b. On the other hand, if the boundary  $\mathcal{B}$  is smooth enough, then a better smoothing bias can be established.

**Lemma 3** (Approximation Error: Smooth Boundary). *Suppose Assumptions 1(i)-(iii) and 2 hold, with  $\mathcal{d}(\cdot, \cdot)$  the Euclidean distance.*

- (i) *For  $\mathbf{x} \in \mathcal{B}$ , and for some  $\delta, \varepsilon > 0$ , suppose that  $\mathcal{B} \cap \{\mathbf{y} : \|\mathbf{y} - \mathbf{x}\| \leq \varepsilon\} = \gamma([- \delta, \delta])$ , where  $\gamma : \mathbb{R} \rightarrow \mathbb{R}^2$  is a one-to-one function in  $C^{\kappa+2}([- \delta, \delta], \mathbb{R}^2)$ . Then,  $\theta_{0,\mathbf{x}}(\cdot)$  and  $\theta_{1,\mathbf{x}}(\cdot)$  are  $(\kappa \wedge (p+1))$ -times continuously differentiable on  $[0, \varepsilon]$ . Therefore, there exists a positive constant  $C$  such that  $|\mathfrak{B}_n(\mathbf{x})| \leq Ch^{\kappa \wedge (p+1)}$  for all  $h \in [0, \varepsilon]$  and  $\mathbf{x} \in \mathcal{B}$ .*
- (ii) *Suppose  $\mathcal{B} = \gamma([0, L])$  where  $\gamma$  is a one-to-one function in  $C^{\iota+2}([0, L], \mathbb{R}^2)$  for some  $L > 0$ . Suppose there exists  $\delta, \varepsilon > 0$  such that for all  $\mathbf{x} \in \mathcal{B}^\circ = \gamma([\delta, L - \delta])$ ,  $r \in [0, \varepsilon]$ , and  $t = 0, 1$ , the set  $\{\mathbf{u} \in \mathbb{R}^2 : \mathcal{d}(\mathbf{u}, \mathbf{x}) = r\}$  intersects with  $\text{bd}(\mathcal{A}_t)$  at only two points in  $\mathcal{B}$ . Then, for all*

$\mathbf{x} \in \gamma([\delta, L - \delta])$ ,  $\theta_{0,\mathbf{x}}(\cdot)$  and  $\theta_{1,\mathbf{x}}(\cdot)$  are  $(\iota \wedge (p+1))$ -times continuously differentiable on  $[0, \varepsilon]$ , and  $\lim_{r \downarrow 0} \frac{d^v}{dr^v} \theta_{t,\mathbf{x}}(r)$  exists and is finite for all  $0 \leq v \leq p+1$  and  $t \in \{0, 1\}$ . Therefore, there exists a positive constant  $C$  such that  $\sup_{\mathbf{x} \in \mathcal{B}^o} |\mathfrak{B}_n(\mathbf{x})| \leq Ch^{\iota \wedge (p+1)}$  for all  $h \in [0, r]$ .

This lemma gives sufficient conditions in terms of smoothness of the boundary  $\mathcal{B}$  to achieve a smaller misspecification bias of  $\hat{\tau}_{\text{dis}}(\mathbf{x})$ , when compared to the minimal guarantee given by Lemma 2. In particular, for a uniform bound on misspecification bias, the sufficient condition in Lemma 3 requires the boundary  $\mathcal{B}$  be *uniformly smooth* in that it can be parameterized by *one* smooth function. The sufficient condition ensuring a smooth boundary everywhere is crucial because, as shown in Lemma 2, even a piecewise smooth  $\mathcal{B}$  with only one kink will have detrimental effects on the convergence rate of the bias (Figure 2).

In Appendix A, we also show that if we consider the class of rectifiable boundaries, then no estimator based on univariate distance, not just local polynomial estimators, can achieve a uniform convergence rate better than  $n^{-1/4}$ , that is, the non-parametric mean square minimax rate for estimating bivariate Lipschitz functions (Stone, 1982). Since we show below that the local polynomial estimator based on univariate distance has a uniform convergence no larger than  $(n/\log n)^{-1/4}$ , it is *nearly* minimax optimal (up to the  $\log^{1/4} n$  term). This minimax result means that no matter which univariate distance-based estimator is used, it is always possible to find a rectifiable boundary and a data generating process such that the estimation error is no better than  $n^{-1/4}$ .

### 3.3 Treatment Effect Estimation and Inference

Using technical results established in the supplemental appendix, we obtain the following convergence rates for the univariate distance-based local polynomial treatment effect estimator.

**Theorem 1** (Rates of Convergence). *Suppose Assumptions 1 and 2 hold. If  $nh^2/\log(1/h) \rightarrow \infty$ , then*

- (i)  $|\hat{\tau}_{\text{dis}}(\mathbf{x}) - \tau(\mathbf{x})| \lesssim_{\mathbb{P}} \frac{1}{\sqrt{nh^2}} + \frac{1}{n^{\frac{1+v}{2+v}} h^2} + |\mathfrak{B}_n(\mathbf{x})|$  for  $\mathbf{x} \in \mathcal{B}$ , and
- (ii)  $\sup_{\mathbf{x} \in \mathcal{B}} |\hat{\tau}_{\text{dis}}(\mathbf{x}) - \tau(\mathbf{x})| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^2}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^2} + \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_n(\mathbf{x})|.$

This theorem establishes the pointwise and uniform (over  $\mathcal{B}$ ) convergence rates for the distance-based treatment effect estimator. By Lemma 2,  $\hat{\tau}_{\text{dis}}(\mathbf{x}) \rightarrow_{\mathbb{P}} \tau(\mathbf{x})$ , both pointwise and uniformly,

if  $h \rightarrow 0$ . However,  $\hat{\tau}_{\text{dis}}(\mathbf{x})$  has a variance convergence rate of order  $n^{-1}h^{-2}$  despite being a “univariate” local polynomial estimator, which would have naïvely suggested a variance convergence of order  $n^{-1}h^{-1}$  instead. In addition to exhibiting a bivariate curse of dimensionality, the convergence rate of the treatment effect estimator  $\hat{\tau}_{\text{dis}}(\mathbf{x})$  along the boundary  $\mathcal{B}$  is determined by the smoothness of the  $\mathcal{B}$ , as discussed in the previous subsection. Finally, it is not difficult to establish valid pointwise, or integrated over  $\mathcal{B}$ , MSE convergence rates matching those in Theorem 1(i). See the supplemental appendix for omitted details. Section 3.4 discusses the implications of these results for implementation, leveraging standard methods from univariate RD designs.

To develop companion pointwise and uniform inference procedures along the treatment assignment boundary  $\mathcal{B}$ , we consider the feasible t-statistic at each boundary point  $\mathbf{x} \in \mathcal{B}$  (for given a bandwidth choice):

$$\hat{T}_{\text{dis}}(\mathbf{x}) = \frac{\hat{\tau}_{\text{dis}}(\mathbf{x}) - \tau(\mathbf{x})}{\sqrt{\hat{\Xi}_{\mathbf{x},\mathbf{x}}}},$$

where, using standard least squares algebra, for all  $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}$  and  $t \in \{0, 1\}$ , we define

$$\hat{\Xi}_{\mathbf{x}_1, \mathbf{x}_2} = \hat{\Xi}_{0, \mathbf{x}_1, \mathbf{x}_2} + \hat{\Xi}_{1, \mathbf{x}_1, \mathbf{x}_2}, \quad \hat{\Xi}_{t, \mathbf{x}_1, \mathbf{x}_2} = \frac{1}{nh^2} \mathbf{e}_1^\top \hat{\Psi}_{t, \mathbf{x}_1}^{-1} \hat{\Upsilon}_{t, \mathbf{x}_1, \mathbf{x}_2} \hat{\Psi}_{t, \mathbf{x}_2}^{-1} \mathbf{e}_1,$$

and

$$\begin{aligned} \hat{\Upsilon}_{t, \mathbf{x}_1, \mathbf{x}_2} = h^2 \mathbb{E}_n \Big[ & \mathbf{r}_p \left( \frac{D_i(\mathbf{x}_1)}{h} \right) k_h(D_i(\mathbf{x}_1)) (Y_i - \mathbf{r}_p(D_i(\mathbf{x}_1))^\top \hat{\gamma}_t(\mathbf{x}_1)) \mathbf{1}(D_i(\mathbf{x}_1) \in \mathcal{J}_t) \\ & \times \mathbf{r}_p \left( \frac{D_i(\mathbf{x}_2)}{h} \right)^\top k_h(D_i(\mathbf{x}_2)) (Y_i - \mathbf{r}_p(D_i(\mathbf{x}_2))^\top \hat{\gamma}_t(\mathbf{x}_2)) \mathbf{1}(D_i(\mathbf{x}_2) \in \mathcal{J}_t) \Big]. \end{aligned}$$

Thus, feasible confidence intervals and confidence bands over  $\mathcal{B}$  take the form:

$$\hat{I}_{\text{dis}}(\mathbf{x}; \alpha) = \left[ \hat{\tau}_{\text{dis}}(\mathbf{x}) - \varphi_\alpha \sqrt{\hat{\Xi}_{\mathbf{x}, \mathbf{x}}}, \hat{\tau}_{\text{dis}}(\mathbf{x}) + \varphi_\alpha \sqrt{\hat{\Xi}_{\mathbf{x}, \mathbf{x}}} \right], \quad \mathbf{x} \in \mathcal{B},$$

for any  $\alpha \in [0, 1]$ , and where  $\varphi_\alpha$  denotes the appropriate quantile depending on the desired inference procedure.

For pointwise inference, it is straightforward to show that  $\sup_{t \in \mathbb{R}} |\mathbb{P}[\hat{T}_{\text{dis}}(\mathbf{x}) \leq t] - \Phi(t)| \rightarrow 0$  for each  $\mathbf{x} \in \mathcal{B}$ , under standard regularity conditions, and provided that  $\sqrt{nh^2} |\mathfrak{B}_n(\mathbf{x})| \rightarrow 0$ . Thus,

in this case,  $q_\alpha = \Phi^{-1}(1 - \alpha/2)$  is an asymptotically valid choice. For uniform inference, we first establish a novel strong approximation for the entire stochastic process  $(\widehat{T}_{\text{dis}}(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$ , assuming that  $\sqrt{nh^2} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_n(\mathbf{x})| \rightarrow 0$ . With some technical work, we can then choose the appropriate  $q_\alpha$  for uniform inference because

$$\mathbb{P}[\tau(\mathbf{x}) \in \widehat{I}_{\text{dis}}(\mathbf{x}; \alpha), \text{ for all } \mathbf{x} \in \mathcal{B}] = \mathbb{P}\left[\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{T}_{\text{dis}}(\mathbf{x})| \leq q_\alpha\right],$$

and the distribution of  $\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{T}_{\text{dis}}(\mathbf{x})|$  can be deduced from the strong approximation of the stochastic process  $(\widehat{T}_{\text{dis}}(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$ .

**Theorem 2** (Statistical Inference). *Suppose Assumptions 1 and 2 hold. Let  $\mathcal{U}_n$  be the  $\sigma$ -algebra generated by  $((Y_i, (D_i(\mathbf{x}) : \mathbf{x} \in \mathcal{B})) : 1 \leq i \leq n)$ .*

(i) *For all  $\mathbf{x} \in \mathcal{B}$ , if  $n^{\frac{v}{2+v}} h^2 \rightarrow \infty$  and  $nh^2 \mathfrak{B}_n^2(\mathbf{x}) \rightarrow 0$ , then*

$$\mathbb{P}[\tau(\mathbf{x}) \in \widehat{I}_{\text{dis}}(\mathbf{x}; \alpha)] \rightarrow 1 - \alpha,$$

*for  $q_\alpha = \Phi^{-1}(1 - \alpha/2)$ .*

(ii) *If  $n^{\frac{v}{2+v}} h^2 / \log n \rightarrow \infty$ ,  $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$ ,  $nh^2 \sup_{\mathbf{x} \in \mathcal{B}} \mathfrak{B}_n^2(\mathbf{x}) \rightarrow 0$ , and  $\text{perim}(\{\mathbf{y} \in \mathcal{A}_t : \mathcal{d}(\mathbf{y}, \mathbf{x})/h \in \text{Supp}(k)\}) \lesssim h$  for all  $\mathbf{x} \in \mathcal{B}$  and  $t \in \{0, 1\}$ , then*

$$\mathbb{P}[\tau(\mathbf{x}) \in \widehat{I}_{\text{dis}}(\mathbf{x}; \alpha), \text{ for all } \mathbf{x} \in \mathcal{B}] \rightarrow 1 - \alpha,$$

*for  $q_\alpha = \inf\{c > 0 : \mathbb{P}[\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}_n(\mathbf{x})| \geq c | \mathcal{U}_n] \leq \alpha\}$ , where  $(\widehat{Z}_n : \mathbf{x} \in \mathcal{B})$  is a Gaussian process conditional on  $\mathcal{U}_n$ , with  $\mathbb{E}[\widehat{Z}_n(\mathbf{x}_1) | \mathcal{U}_n] = 0$  and  $\mathbb{E}[\widehat{Z}_n(\mathbf{x}_1) \widehat{Z}_n(\mathbf{x}_2) | \mathcal{U}_n] = \widehat{\Xi}_{\mathbf{x}_1, \mathbf{x}_2} / \sqrt{\widehat{\Xi}_{\mathbf{x}_1, \mathbf{x}_1} \widehat{\Xi}_{\mathbf{x}_2, \mathbf{x}_2}}$ , for all  $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}$ .*

This theorem establishes asymptotically valid inference procedures using the distance-based local polynomial treatment effect estimator  $\widehat{\tau}_{\text{dis}}(\mathbf{x})$ . For uniform inference, an additional restriction on the assignment boundary  $\mathcal{B}$  is imposed: the De-Giorgi perimeter condition can be verified when the boundary of  $\{\mathbf{y} \in \mathcal{A}_t : \mathcal{d}(\mathbf{y}, \mathbf{x})/h \in \text{Supp}(k)\}$  is a curve of length no greater than  $h$  up to a constant; since the set is contained in a  $h$ -ball centered at  $\mathbf{x}$ , the curve length condition holds as long as the curve  $\mathcal{B} \cap \{\mathbf{y} \in \mathbb{R}^2 : \mathcal{d}(\mathbf{y}, \mathbf{x}) \leq h\}$  is not “too wiggly”.

### 3.4 Discussion and Implementation

Although the distance-based estimator  $\hat{\tau}_{\text{dis}}(\mathbf{x})$  looks like an univariate local polynomial estimation procedure, based on the scalar score variable  $D_i(\mathbf{x})$ , Theorem 1 shows that its pointwise and uniform variance convergence rates are equal to that of a bivariate nonparametric estimator (which are unimprovable). Theorem 2 shows that inference results derived using univariate local polynomial regression methods can be deployed directly in distance-based settings, provided the side conditions are satisfied. This result follows from the fact that  $\hat{T}_{\text{dis}}(\mathbf{x})$  is constructed as a self-normalizing statistic, and therefore it is *adaptive* to the fact that the univariate covariate  $D_i(\mathbf{x})$  is actually based on the bivariate covariate  $\mathbf{X}_i$ . This finding documents another advantage of employing pre-asymptotic variance estimators and self-normalizing statistics for distributional approximation and inference (Calonico et al., 2018). It follows that standard estimation and inference methods from the univariate RD design literature could be deployed to boundary discontinuity designs employing univariate distance to the boundary, provided an appropriate bandwidth is chosen to ensure that the bias due to the shape of the assignment boundary  $\mathcal{B}$  (documented in Section 3.2) is small.

For implementation, consider first the case that  $\mathfrak{B}_n(\mathbf{x}) \lesssim h^{p+1}$ , that is, the assignment boundary  $\mathcal{B}$  is smooth (in the sense of Lemma 3). Establishing a precise MSE expansion for  $\hat{\tau}_{\text{dis}}(\mathbf{x})$  is cumbersome due to the added complexity introduced by the distance transformation, but the convergence rates can be deduced from Theorem 1. The *incorrect* univariate MSE-optimal bandwidth is  $h_{1d} \asymp n^{-1/(3+2p)}$ , while the *correct* MSE-optimal bandwidth is  $h_{\text{dis}} \asymp n^{-1/(4+2p)}$ , implying that taking the distance variable as the univariate covariate, and thus ignoring its intrinsic bivariate dimension, implies that the resulting bandwidth choice will be smaller  $n^{-1/(3+2p)} < n^{-1/(4+2p)}$ , thereby undersmoothing the point estimator (relative to the *correct* optimal MSE bandwidth choice). As a consequence, the point estimator will not be MSE-optimal but rather exhibit more variance and less bias, and the associated inference procedures will be more conservative. An obvious solution is to rescale the *incorrect* univariate bandwidth,  $\frac{n^{1/(3+2p)}}{n^{1/(4+2p)}} h_{1d}$ , but this may not be necessary if the empirical implementation of  $h_{1d}$  employs a pre-asymptotic variance estimator, as in the software package `rdrobust` (<https://rdpackages.github.io/rdrobust/>). See Calonico et al. (2020), and references therein, for details on bandwidth selection for standard univariate RD designs.

When the assignment boundary  $\mathcal{B}$  exhibits kinks (or other irregularities), no matter how large

the sample size or the polynomial order, there will always be a region near each kink where the bias will be large, that is, of irreducible order  $h$ , as shown by Lemma 2. In this case, near each kink the *correct* MSE-optimal bandwidth is  $h_{\text{dis}} \asymp n^{-1/4}$ . Away from each kink, the MSE-optimal bandwidths are as discussed in the preceeding paragraph. This phenomenon is generated by the lack of smoothness of  $\mathcal{B}$ , and leads to different MSE-optimal bandwidths for each  $\mathbf{x} \in \mathcal{B}$ , thus making automatic implementation more difficult. A simple solution to this problem is to set  $h_{\text{dis}} \asymp n^{-1/4}$  for all  $\mathbf{x} \in \mathcal{B}$ , which is generically suboptimal but always no larger than the pointwise MSE-optimal bandwidth, thus delivering a more variable (than optimal) point estimator and more conservative (than possible) associated inference procedure.

Given a choice of bandwidth  $h$ , valid statistical inference can be developed by controlling the remaining misspecification bias. When the boundary  $\mathcal{B}$  is smooth, robust bias correction for standard univariate RD designs continues to be valid in the context of distance-based estimation (Calonico et al., 2014, 2018, 2022). On the other hand, when  $\mathcal{B}$  exhibits kinks, undersmoothing relative to the MSE-optimal bandwidth for  $p = 0$  is needed due to the fact that increasing  $p$  does not necessarily imply a reduction in misspecification bias, and thus bias correction techniques are ineffective (uniformly over  $\mathbf{x} \in \mathcal{B}$ ).

It remains to explain how uniform inference is implemented based on Theorem 5(ii). In practice, we discretize along the boundary with point of evaluations  $\mathbf{b}_1, \dots, \mathbf{b}_M \in \mathcal{B}$ , as in Figure 1a, and hence the feasible (conditional) Gaussian process  $(\hat{Z}_n(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$  is reduced to the  $M$ -dimensional (conditional) Gaussian random vector  $\hat{\mathbf{Z}}_n = (\hat{Z}_n(\mathbf{b}_1), \dots, \hat{Z}_n(\mathbf{b}_M))$  with covariance matrix having a typical element  $\mathbb{E}[\hat{Z}_n(\mathbf{b}_1)\hat{Z}_n(\mathbf{b}_2)|\mathcal{U}]$ . Finding  $q_\alpha$  reduces to finding the  $\alpha$ -quantile of the distribution of  $\max_{1 \leq l \leq M} |\hat{Z}_n(\mathbf{b}_l)|$ , which can be easily simulated.

In the companion software package `rd2d`, we employ a simple rule-of-thumb bandwidth selector targeting  $h_{\text{dis}} \asymp n^{-1/4}$  to implement the univariate distance-based estimator  $\hat{\tau}_{\text{dis}}(\mathbf{x})$  as default for all  $\mathbf{x} \in \mathcal{B}$ , which is valid for estimation whether  $\mathcal{B}$  is smooth or not. For inference, following Calonico et al. (2018), the package employs the undersmoothed choice of order  $n^{-1/3}$ . If  $\mathcal{B}$  is known to be smooth, then the package implements a rule-of-thumb bandwidth selector targeting  $h_{\text{dis}} \asymp n^{-1/(4+2p)}$  for point estimation, and then employs robust bias corrected inference based on that bandwidth choice. Section 5 illustrates the performance of these methods with the SPP empirical application, and also considers the software package `rdrobust` for bandwidth selection

with  $p = 0$  as another simple rule-of-thumb bandwidth implementation. Further implementation details and simulation evidence are given in [Cattaneo et al. \(2025\)](#).

## 4 Analysis based on Bivariate Location

The previous section demonstrated the potential detrimental aspects of employing distance-based local polynomial regression to analyze boundary discontinuity designs. When the boundary has kinks, as it is common in many applications based on sharp or geographic assignment rules, univariate methods based on distance to the boundary can lead to large biases near kinks or other irregularities on the assignment boundary. This section shows that an easy way to avoid the distance-based bias is to employ bivariate local polynomial regression methods.

The location-based treatment effect curve estimator of  $\tau(\mathbf{x})$  is

$$\hat{\tau}(\mathbf{x}) = \mathbf{e}_1^\top \hat{\boldsymbol{\beta}}_1(\mathbf{x}) - \mathbf{e}_1^\top \hat{\boldsymbol{\beta}}_0(\mathbf{x}), \quad \mathbf{x} \in \mathcal{B},$$

where, for  $t \in \{0, 1\}$ ,

$$\hat{\boldsymbol{\beta}}_t(\mathbf{x}) = \arg \min_{\boldsymbol{\beta} \in \mathbb{R}^{p_p+1}} \mathbb{E}_n \left[ (Y_i - \mathbf{R}_p(\mathbf{X}_i - \mathbf{x})^\top \boldsymbol{\beta})^2 K_h(\mathbf{X}_i - \mathbf{x}) \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_t) \right],$$

with  $p_p = (2 + p)(1 + p)/2 - 1$ ,  $\mathbf{R}_p(\mathbf{u}) = (1, u_1, u_2, u_1^2, u_2^2, u_1 u_2, \dots, u_1^p, u_2^p)^\top$  denotes the  $p$ th order polynomial expansion of the bivariate vector  $\mathbf{u} = (u_1, u_2)^\top$ ,  $K_h(\mathbf{u}) = K(u_1/h, u_2/h)/h^2$  for a bivariate kernel function  $K(\cdot)$  and a bandwidth parameter  $h$ . We employ the same bandwidth for both dimensions of  $\mathbf{X}_i$  only for simplicity, and because it is common practice to first standardize each dimension of the bivariate score.

We impose the following assumption on the bivariate kernel function and assignment boundary.

**Assumption 3** (Kernel Function). *Let  $t \in \{0, 1\}$ .*

- (i)  $K : \mathbb{R}^2 \rightarrow [0, \infty)$  is compact supported and Lipschitz continuous, or  $K(\mathbf{u}) = \mathbf{1}(\mathbf{u} \in [-1, 1]^2)$ .
- (ii)  $\liminf_{h \downarrow 0} \inf_{\mathbf{x} \in \mathcal{B}} \int_{\mathcal{A}_t} K_h(\mathbf{u} - \mathbf{x}) d\mathbf{u} \gtrsim 1$ .

The first condition is standard in the literature, while the second condition is new. Similar to the case of distance-based estimation (Assumption 2(iii)), the goal of Assumption 3(ii) is to ensure

enough data is available in large samples for each point  $\mathbf{x} \in \mathcal{B}$  because  $\mathbb{E}[K_h(\mathbf{X}_i - \mathbf{x})\mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t)] \gtrsim \int_{\mathcal{A}_t} K_h(\mathbf{u} - \mathbf{x})d\mathbf{u}$  under Assumption 1. Given these conditions, pointwise and uniform estimation results, as well as valid MSE expansions, follow from standard local polynomial calculations and empirical process theory. On the other hand, uniform distribution theory needs to be established with some extra technical care: the supplemental appendix establishes a key new strong approximation result that takes into account the specific features of the estimator and manifold  $\mathcal{B}$ .

#### 4.1 Treatment Effect Estimation

Using standard concentration techniques from empirical process theory, we obtain the pointwise and uniform convergence rate of  $\hat{\tau}(\mathbf{x})$ .

**Theorem 3** (Rate of Convergence). *Suppose Assumptions 1 and 3 hold. If  $nh^2/\log(n) \rightarrow \infty$ , then*

- (i)  $|\hat{\tau}(\mathbf{x}) - \tau(\mathbf{x})| \lesssim_{\mathbb{P}} \frac{1}{\sqrt{nh^2}} + \frac{1}{n^{\frac{1+v}{2+v}}h^2} + h^{p+1}$  for  $\mathbf{x} \in \mathcal{B}$ , and
- (ii)  $\sup_{\mathbf{x} \in \mathcal{B}} |\hat{\tau}(\mathbf{x}) - \tau(\mathbf{x})| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^2}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}}h^2} + h^{p+1}$ .

This theorem immediately establishes consistency of the treatment effect estimator based on the bivariate location score, provided that  $h \rightarrow 0$ . More importantly, the theorem shows that the bias is of order  $h^{p+1}$  regardless of whether there are kinks or other irregularities in  $\mathcal{B}$ . Comparing Theorems 1 and 3, it follows that both estimators can achieve the same (optimal) convergence rate when the boundary  $\mathcal{B}$  is smooth (Lemma 3), but otherwise  $\hat{\tau}(\mathbf{x})$  can achieve a faster (indeed, optimal) convergence rate, while  $\hat{\tau}_{\text{dis}}(\mathbf{x})$  cannot because of the large order- $h$  bias when  $\mathcal{B}$  is not smooth (Lemma 2).

Furthermore, it is interesting to note that the radial (univariate) kernel  $K(\mathbf{u}) = k(\|\mathbf{u}\|)$  satisfies Assumption 3(b). Therefore, the large (order- $h$ ) bias of the distance-based estimator  $\hat{\tau}_{\text{dis}}(\mathbf{x})$  due to a non-smooth assignment boundary, as demonstrated in Lemma 2, is not necessarily due to the choice of distance-based kernel smoothing, but rather due to the choice of distance-based (univariate) local polynomial approximation: e.g.,  $\mathbf{r}_p(\|\mathbf{u}\|)$  (in  $\hat{\tau}_{\text{dis}}(\mathbf{x})$ ) vs.  $\mathbf{R}_p(\mathbf{u})$  (in  $\hat{\tau}(\mathbf{x})$ ). In other words, Theorem 3 demonstrates that a bivariate local polynomial approximation does not suffer from the bias documented in Lemma 2 even when the radial kernel is used for localization.

Given its more standard structure, it is possible to establish pointwise and integrated (conditional) MSE expansions for the estimator  $\hat{\tau}(\mathbf{x})$ . Using standard multi-index notation, define the

leading conditional bias  $B_{\mathbf{x}} = B_{1,\mathbf{x}} - B_{0,\mathbf{x}}$  with

$$B_{t,\mathbf{x}} = \mathbf{e}_1^\top \widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1} \sum_{|\mathbf{k}|=p+1} \frac{\mu_t^{(\mathbf{k})}(\mathbf{x})}{\mathbf{k}!} \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right)^{\mathbf{k}} K_h(\mathbf{X}_i - \mathbf{x}) \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_t) \right]$$

and  $\widehat{\mathbf{\Gamma}}_{t,\mathbf{x}} = \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \mathbf{r}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right)^\top K_h(\mathbf{X}_i - \mathbf{x}) \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_t) \right]$ , for  $t \in \{0, 1\}$ . Similarly, the leading conditional variance is  $V_{\mathbf{x}} = V_{1,\mathbf{x}} + V_{0,\mathbf{x}}$  with  $V_{t,\mathbf{x}} = \mathbf{e}_1^\top \widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1} \mathbf{\Sigma}_{t,\mathbf{x},\mathbf{x}} \widehat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1} \mathbf{e}_1$  and

$$\mathbf{\Sigma}_{t,\mathbf{x},\mathbf{x}} = h^2 \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \mathbf{r}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right)^\top K_h(\mathbf{X}_i - \mathbf{x})^2 \sigma_t^2(\mathbf{X}_i) \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_t) \right],$$

for  $t \in \{0, 1\}$ . The following theorem gives the MSE expansions.

**Theorem 4** (MSE Expansions). *Suppose Assumptions 1 and 3 hold, and let  $w(\mathbf{x})$  be a non-negative continuous function on  $\mathcal{B}$  such that  $\int_{\mathcal{B}} w(\mathbf{x}) d\mathbf{x} < \infty$ . If  $nh^2/\log(n) \rightarrow \infty$ , then*

- (i)  $\mathbb{E}[(\widehat{\tau}(\mathbf{x}) - \tau(\mathbf{x}))^2 | \mathbf{X}] = h^{2(p+1)} B_{\mathbf{x}}^2 + \frac{1}{nh^2} V_{\mathbf{x}} + o_{\mathbb{P}}(r_n)$ , and
- (ii)  $\int_{\mathcal{B}} \mathbb{E}[(\widehat{\tau}(\mathbf{x}) - \tau(\mathbf{x}))^2 | \mathbf{X}] w(\mathbf{x}) d\mathbf{x} = h^{2(p+1)} \int_{\mathcal{B}} B_{\mathbf{x}}^2 dw(\mathbf{x}) + \frac{1}{nh^2} \int_{\mathcal{B}} V_{\mathbf{x}} w(\mathbf{x}) d\mathbf{x} + o_{\mathbb{P}}(r_n)$ ,

with  $r_n = h^{2p+2} + n^{-1}h^{-2} + n^{-\frac{2(1+v)}{2+v}} h^{-4}$ .

Ignoring the asymptotically constant and higher-order terms, approximate MSE-optimal and IMSE-optimal bandwidth choices are

$$h_{\text{MSE},\mathbf{x}} = \left( \frac{2V_{\mathbf{x}}}{(2p+2)B_{\mathbf{x}}^2} \frac{1}{n} \right)^{1/(2p+4)} \quad \text{and} \quad h_{\text{IMSE}} = \left( \frac{2 \int_{\mathcal{B}} V_{\mathbf{x}} w(\mathbf{x}) d\mathbf{x}}{(2p+2) \int_{\mathcal{B}} B_{\mathbf{x}}^2 w(\mathbf{x}) d\mathbf{x}} \frac{1}{n} \right)^{1/(2p+4)},$$

provided that  $B_{\mathbf{x}} \neq 0$  and  $\int_{\mathcal{B}} B_{\mathbf{x}}^2 w(\mathbf{x}) d\mathbf{x} \neq 0$ , respectively. These choices are infeasible because a preliminary bandwidth, as well as estimates of the conditional variances and higher-order derivatives of the conditional mean, are needed. We discuss implementation below and in our companion software article (Cattaneo et al., 2025).

## 4.2 Uncertainty Quantification

Given a bandwidth choice, the feasible t-statistic at each boundary point  $\mathbf{x} \in \mathcal{B}$  is

$$\widehat{\mathbf{T}}(\mathbf{x}) = \frac{\widehat{\tau}(\mathbf{x}) - \tau(\mathbf{x})}{\sqrt{\widehat{\Omega}_{\mathbf{x},\mathbf{x}}}},$$

where, using standard least squares algebra, for all  $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}$  and  $t \in \{0, 1\}$ ,

$$\widehat{\Omega}_{\mathbf{x}_1, \mathbf{x}_2} = \widehat{\Omega}_{0, \mathbf{x}_1, \mathbf{x}_2} + \widehat{\Omega}_{1, \mathbf{x}_1, \mathbf{x}_2}, \quad \widehat{\Omega}_{t, \mathbf{x}_1, \mathbf{x}_2} = \frac{1}{nh^2} \mathbf{e}_1^\top \widehat{\Gamma}_{t, \mathbf{x}_1}^{-1} \widehat{\Sigma}_{t, \mathbf{x}_1, \mathbf{x}_2} \widehat{\Gamma}_{t, \mathbf{x}_2}^{-1} \mathbf{e}_1$$

with

$$\widehat{\Sigma}_{t, \mathbf{x}_1, \mathbf{x}_2} = h^2 \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{\mathbf{X}_i - \mathbf{x}_1}{h} \right) \mathbf{r}_p \left( \frac{\mathbf{X}_i - \mathbf{x}_2}{h} \right)^\top K_h(\mathbf{X}_i - \mathbf{x}_1) K_h(\mathbf{X}_i - \mathbf{x}_2) \varepsilon_i(\mathbf{x}_1) \varepsilon_i(\mathbf{x}_2) \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_t) \right]$$

and  $\varepsilon_i(\mathbf{x}) = Y_i - \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_0) \mathbf{R}_p(\mathbf{X}_i - \mathbf{x})^\top \widehat{\beta}_0(\mathbf{x}) - \mathbf{1}(\mathbf{X}_i \in \mathcal{A}_1) \mathbf{R}_p(\mathbf{X}_i - \mathbf{x})^\top \widehat{\beta}_1(\mathbf{x})$ .

Feasible confidence intervals and confidence bands over the treatment boundary  $\mathcal{B}$  take the form:

$$\widehat{\mathbf{I}}(\mathbf{x}; \alpha) = \left[ \widehat{\tau}(\mathbf{x}) - \varphi_\alpha \sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}}}, \widehat{\tau}(\mathbf{x}) + \varphi_\alpha \sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}}} \right], \quad \mathbf{x} \in \mathcal{B},$$

for any  $\alpha \in [0, 1]$ , and where  $\varphi_\alpha$  denotes the appropriate quantile depending on the targeted inference procedure. For pointwise inference, as in the previous section, it is a textbook exercise to show that  $\sup_{t \in \mathbb{R}} |\mathbb{P}[\widehat{\mathbf{T}}(\mathbf{x}) \leq t] - \Phi(t)| \rightarrow 0$  for each  $\mathbf{x} \in \mathcal{B}$ , under standard regularity conditions, and provided that  $nh^{2p+4} \rightarrow 0$ . For uniform inference, as in the previous section, we approximate the distribution of the entire stochastic process  $(\widehat{\mathbf{T}}(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$  first, and then deduce an approximation for  $\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{\mathbf{T}}(\mathbf{x})|$ . This approach enable us to construct asymptotically valid confidence bands because, as noted previously,

$$\mathbb{P}[\tau(\mathbf{x}) \in \widehat{\mathbf{I}}(\mathbf{x}; \alpha), \text{ for all } \mathbf{x} \in \mathcal{B}] = \mathbb{P}\left[\sup_{\mathbf{x} \in \mathcal{B}} |\widehat{\mathbf{T}}(\mathbf{x})| \leq \varphi_\alpha\right].$$

See the supplemental appendix for omitted technical details.

**Theorem 5** (Inferences). *Suppose Assumptions 1 and 3 hold. Let  $\mathbf{W}_n = ((Y_1, \mathbf{X}_1^\top)^\top, \dots, (Y_n, \mathbf{X}_n^\top)^\top)$ .*

(i) *For all  $\mathbf{x} \in \mathcal{B}$ , if  $n^{\frac{v}{2+v}} h^2 \rightarrow \infty$  and  $nh^{2p+4} \rightarrow 0$ , then*

$$\mathbb{P}[\tau(\mathbf{x}) \in \widehat{\mathbf{I}}(\mathbf{x}; \alpha)] \rightarrow 1 - \alpha,$$

*for  $\varphi_\alpha = \Phi^{-1}(1 - \alpha/2)$ .*

(ii) *If  $n^{\frac{v}{2+v}} h^2 / \log n \rightarrow \infty$ ,  $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$ ,  $nh^{2p+4} \rightarrow 0$ , and  $\text{perim}(\{\mathbf{y} \in \mathcal{A}_t : (\mathbf{y} - \mathbf{x})/h \in$*

$\text{Supp}(K)\} \lesssim h$  for all  $\mathbf{x} \in \mathcal{B}$  and  $t \in \{0, 1\}$ , then

$$\mathbb{P}[\tau(\mathbf{x}) \in \hat{\mathbf{I}}(\mathbf{x}; \alpha), \text{ for all } \mathbf{x} \in \mathcal{B}] \rightarrow 1 - \alpha,$$

for  $q_\alpha = \inf\{c > 0 : \mathbb{P}[\sup_{\mathbf{x} \in \mathcal{B}} |\hat{Z}_n(\mathbf{x})| \geq c | \mathbf{W}_n] \leq \alpha\}$ , where  $(\hat{Z}_n : \mathbf{x} \in \mathcal{B})$  is a Gaussian process conditional on  $\mathbf{W}_n$  with  $\mathbb{E}[\hat{Z}_n(\mathbf{x}_1) | \mathbf{W}_n] = 0$  and  $\mathbb{E}[\hat{Z}_n(\mathbf{x}_1)\hat{Z}_n(\mathbf{x}_2) | \mathbf{W}_n] = \hat{\Omega}_{\mathbf{x}_1, \mathbf{x}_2} / \sqrt{\hat{\Omega}_{\mathbf{x}_1, \mathbf{x}_1} \hat{\Omega}_{\mathbf{x}_2, \mathbf{x}_2}}$ , for all  $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}$ .

This theorem gives valid pointwise and inference procedures for the boundary treatment effect  $\tau(\mathbf{x})$  based on the bivariate local polynomial estimator. Unlike the case of distance-based estimation, the bias condition is the same for  $\mathbf{x} \in \mathcal{B}$ , making implementation substantially easier. However, as in the case of Theorem 2, and additional technical restriction on  $\mathcal{B}$  is needed in order to avoid overly “wiggly” boundary designs that would lead to invalid statistical inference.

### 4.3 Implementation

The bivariate local polynomial estimator  $\hat{\tau}(\mathbf{x})$  and associated t-statistic  $\hat{\mathbf{T}}(\mathbf{x})$  is fully adaptive to kinks and other irregularities of the boundary  $\mathcal{B}$ . Therefore, it is straightforward to implement local and global bandwidth selectors based on Theorem 4 and the discussion given above. In particular, replacing the (asymptotic) bias and variance constants,  $B_{\mathbf{x}}$  and  $V_{\mathbf{x}}$ , with preliminary estimators, we obtain the feasible plug-in bandwidth selectors

$$\hat{h}_{\text{MSE}} = \left( \frac{2\hat{V}_{\mathbf{x}}}{(2p+2)\hat{B}_{\mathbf{x}}^2} \frac{1}{n} \right)^{1/(2p+4)} \quad \text{and} \quad \hat{h}_{\text{IMSE}} = \left( \frac{2 \int_{\mathcal{B}} \hat{V}_{\mathbf{x}} w(\mathbf{x}) d\mathbf{x}}{(2p+2) \int_{\mathcal{B}} \hat{B}_{\mathbf{x}}^2 w(\mathbf{x}) d\mathbf{x}} \frac{1}{n} \right)^{1/(2p+4)},$$

where  $\hat{B}_{\mathbf{x}}$  is an appropriate estimator of  $B_{\mathbf{x}}$ , and  $\hat{V}_{\mathbf{x}} = nh^2 \hat{\Omega}_{\mathbf{x}, \mathbf{x}}$ . Omitted details are given in the supplemental appendix. These bandwidth choices can now be used to implement (I)MSE-optimal  $\hat{\tau}(\mathbf{x})$  point treatment effect estimators pointwise and uniformly along the boundary  $\mathcal{B}$ . Furthermore, leveraging the results in Theorem 5, a simple application of robust bias-corrected inference proceeds by employing the same (I)MSE-optimal bandwidth (for  $p$ th order point estimation), but then constructing the t-statistic  $\hat{\mathbf{T}}(\mathbf{x})$  with a choice of  $p+1$  (instead of  $p$ ). This inference approach has several theoretical advantages (Calonico et al., 2014, 2018, 2022), and has been validated empirically (Hyytinen et al., 2018; De Magalhães et al., 2025). Our companion software, `rd2d`, implements

the procedures described above. See [Cattaneo et al. \(2025\)](#) for more details.

## 5 Empirical Applications

We illustrate our proposed methodology with the SPP application. We consider both univariate distance-based and bivariate location-based estimation and inference. The data set has  $n = 363,096$  complete observations for the first cohort of the program (2014). Each observation corresponds to one student, and the bivariate score  $\mathbf{X}_i = (X_{1i}, X_{2i})^\top = (\text{SABER11}_i, \text{SISBEN}_i)^\top$  is composed by the student’s SABER11 test score (ranging from  $-310$  to  $172$ ) and SISBEN wealth index (ranging from  $-103.41$  to  $127.21$ ). As discussed in the introduction, and without loss of generality, the score is recentered at their corresponding cutoff for program eligibility, so that the treatment assignment boundary is as shown in Figure [1a](#). Furthermore, also without loss of generality, each dimension of  $\mathbf{X}_i$  is standardized in order to more naturally accommodate a common bandwidth  $h$ . Recall that  $Y_i$  is an indicator variable equal to one if the student enrolled in a higher education institution after receiving the SPP subsidy, or equal to zero otherwise. All the results in this section are implemented using our companion R software package `rd2d`, and omitted details are given in the replication files.

Figure [3](#) presents the empirical results using our proposed methods, both pointwise across the 40 boundary points depicted in [1a](#) and uniformly over  $\mathcal{B}$ . Figure [3a](#) compares the two point estimation approaches, and demonstrates empirically the presence of some bias associated with the distance-based approach: the blue dots correspond to  $\hat{\tau}(\mathbf{x})$  implemented using the data-driven MSE-optimal bandwidth; the red crosses correspond to  $\hat{\tau}_{\text{dis}}$  implemented a rule-of-thumb bandwidth ignoring the presence of the kink at  $\mathbf{x} = (0, 0) \in \mathcal{B}$ ; and the gray circles correspond to  $\hat{\tau}(\mathbf{x})$  implemented using a rule-of-thumb bandwidth of order  $n^{-1/4}$ . Figure [3b](#) demonstrates empirically the pointwise and uniform inference results from Section [4](#) based on  $\hat{\tau}(\mathbf{x})$  implemented using the data-driven MSE-optimal bandwidth. The treatment effects are highly statistically significant (different from zero) along the assignment boundary, indicating mostly homogeneous treatment effects along poverty ( $\mathbf{b}_1$ – $\mathbf{b}_{21}$ ) and some degree of heterogeneity in treatment effects along academic performance ( $\mathbf{b}_{21}$ – $\mathbf{b}_{40}$ ). Specifically, college attendance remains constant as students become wealthier ( $\hat{\tau}(\mathbf{x}) \approx 0.3$  for  $\mathbf{x} \in \{\mathbf{b}_1, \dots, \mathbf{b}_{21}\}$ ), but the treatment effects appear to decrease as students exhibit higher

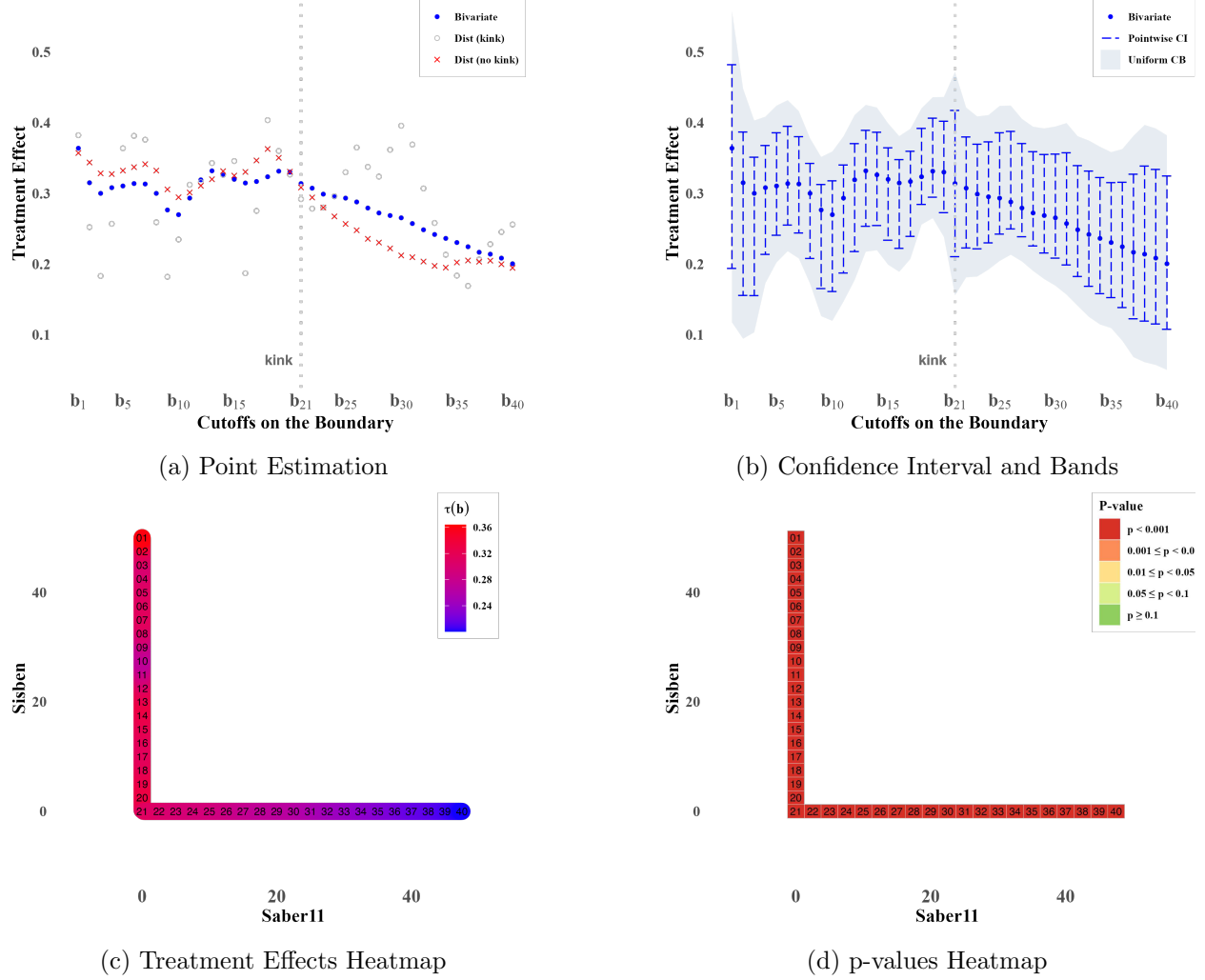
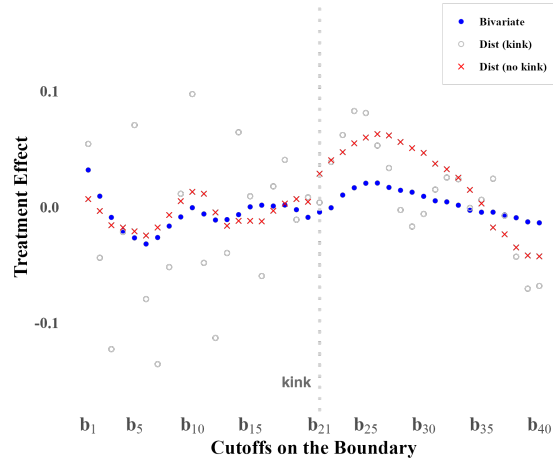


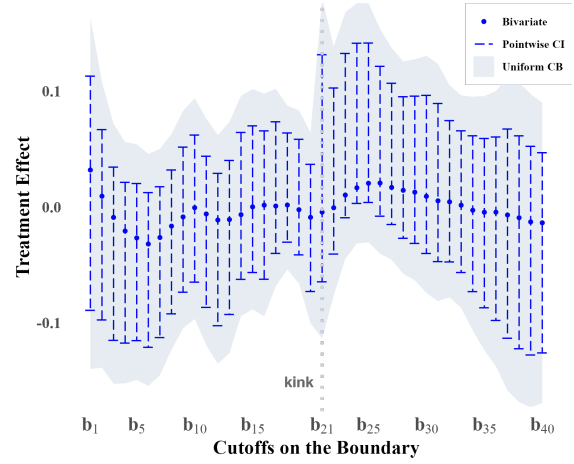
Figure 3: Boundary Treatment Effect Estimation and Inference (SPP application). Outcome: College enrollment.

academic performance (from  $\hat{\tau}(\mathbf{b}_{21}) \approx 0.3$  to  $\hat{\tau}(\mathbf{b}_{40}) \approx 0.2$ ). Finally, Figures 3c and 3d offer “heat maps” for the point estimates and associated RBC p-values along the assignment boundary  $\mathcal{B}$ .

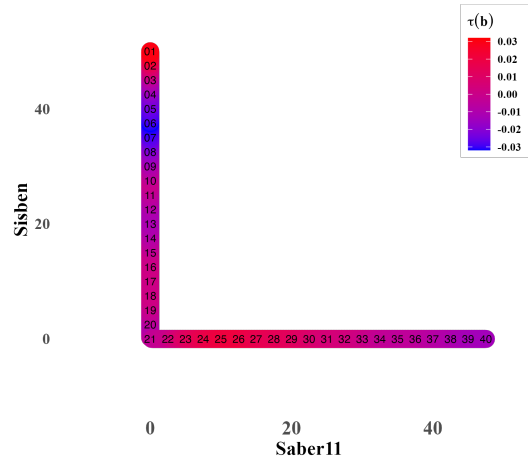
To demonstrate the credibility of the boundary discontinuity design, we repeat the empirical analysis using a pre-intervention covariate: mother’s education of the student. This corresponds to a standard “placebo” analysis, where the treatment effect is expected to be statistically indistinguishable from zero. Figure 4 shows that this is indeed the case.



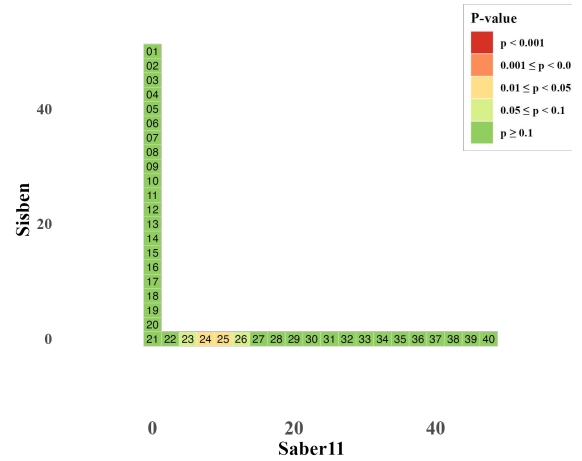
(a) Point Estimation



(b) Confidence Interval and Bands



(c) Heatmap of Treatment Effect



(d) Heatmap of P-value

Figure 4: Boundary Treatment Effect Estimation and Inference (SPP application). Outcome: Mother's education (placebo).

## 6 Conclusion

We studied the pointwise and uniform statistical properties of the two most popular local polynomial methods for treatment effect estimation and inference in boundary discontinuity designs. Our theoretical and numerical results demonstrated that methods based on the univariate distance to the assignment boundary can exhibit a large bias in the presence of kinks or other irregularities of  $\mathcal{B}$ . In contrast, methods based on the bivariate score do not suffer from those problems, and thus can perform better in applications. We thus recommend employing bivariate local polynomial analysis whenever possible in boundary discontinuity designs. Our companion software package `rd2d` (<https://rdpackages.github.io/rd2d>) offers general-purpose implementations of all the estimation and inference methods developed.

Building on the framework and results presented, there are (at least) two directions for future research. First, it would be of interest to incorporate predetermined covariates for either efficiency improvements in estimation and inference, or for treatment effect heterogeneity analysis. Second, it would be of interest to consider functionals of the average treatment effect curve along the boundary,  $\tau(\mathbf{x})$ , which can help summarize empirical findings. For example, interesting scalar parameters include the *weighted average treatment effect along the boundary*  $\tau_p = \int_{\mathcal{R}} \tau(\mathbf{x})w(\mathbf{x})d\mathbf{x}$ , and the *largest average treatment effect along the boundary*  $\tau_\infty = \sup_{\mathbf{x} \in \mathcal{R}} \tau(\mathbf{x})$ , for some region  $\mathcal{R} \subseteq \mathcal{B}$ . In research underway we are investigating how our estimation and inference results can be extended to study the properties of covariate-adjusted estimators of  $\tau(\mathbf{x})$ , and of the plug-in estimators  $\hat{\tau}_p = \int_{\mathcal{R}} \hat{\tau}(\mathbf{x})w(\mathbf{x})d\mathbf{x}$  and  $\hat{\tau}_\infty = \sup_{\mathbf{x} \in \mathcal{R}} \hat{\tau}(\mathbf{x})$ . (For the latter estimator, the results in the supplemental appendix already apply directly.)

## A Minimax Convergence Rate

The following theorem presents a minimax uniform convergence rate result for estimation of bivariate compact supported nonparametric functions employing a class of estimators based on distance to the evaluation point.

**Theorem 6** (Distance-based Minimax Convergence Rate). *For a constants  $q \geq 1$  and  $L > 0$ , let  $\mathcal{P}_{\text{NP}} = \mathcal{P}_{\text{NP}}(L, q)$  be the class of (joint) probability laws  $\mathbb{P}$  of  $(Y_1, \mathbf{X}_1), \dots, (Y_n, \mathbf{X}_n)$  satisfying the*

following:

- (i)  $((Y_i, \mathbf{X}_i) : 1 \leq i \leq n)$  are i.i.d. taking values in  $\mathbb{R} \times \mathbb{R}^2$ .
- (ii)  $\mathbf{X}_i$  admits a Lebesgue density  $f$  that is continuous on its compact support  $\mathcal{X} \subseteq [-L, L]^2$ , with  $L^{-1} \leq \inf_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) \leq \sup_{\mathbf{x} \in \mathcal{X}} f(\mathbf{x}) \leq L$ , and  $\mathcal{B} = \text{bd}(\mathcal{X})$  is a rectifiable curve.
- (iii)  $\mu(\mathbf{x}) = \mathbb{E}[Y_i | \mathbf{X}_i = \mathbf{x}]$  is  $q$ -times continuously differentiable on  $\mathcal{X}$  with

$$\max_{0 \leq |\nu| \leq \lfloor q \rfloor} \sup_{\mathbf{x} \in \mathcal{X}} |\partial_\nu \mu(\mathbf{x})| + \max_{|\nu| = q} \sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{X}} \frac{|\partial_\nu \mu(\mathbf{x}_1) - \partial_\nu \mu(\mathbf{x}_2)|}{\|\mathbf{x}_1 - \mathbf{x}_2\|^{q - \lfloor q \rfloor}} \leq L.$$

- (iv)  $\sigma^2(\mathbf{x}) = \mathbb{V}[Y_i | \mathbf{X}_i = \mathbf{x}]$  is continuous on  $\mathcal{X}$  with  $L^{-1} \leq \inf_{\mathbf{x} \in \mathcal{X}} \sigma^2(\mathbf{x}) \leq \sup_{\mathbf{x} \in \mathcal{X}} \sigma^2(\mathbf{x}) \leq L$ .

In addition, let  $\mathcal{T}$  be the class of all distance-based estimators  $T_n(\mathbf{U}_n(\mathbf{x}))$  with  $\mathbf{U}_n(\mathbf{x}) = ((Y_i, \|\mathbf{X}_i - \mathbf{x}\|)^T : 1 \leq i \leq n)$  for each  $\mathbf{x} \in \mathcal{X}$ . Then,

$$\liminf_{n \rightarrow \infty} n^{1/4} \inf_{T_n \in \mathcal{T}} \sup_{\mathbb{P} \in \mathcal{P}_{\text{NP}}} \mathbb{E}_{\mathbb{P}} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |T_n(\mathbf{U}_n(\mathbf{x})) - \mu(\mathbf{x})| \right] \gtrsim 1,$$

where  $\mathbb{E}_{\mathbb{P}}[\cdot]$  denotes an expectation taken under the data generating process  $\mathbb{P}$ .

In sharp contrast to Theorem 6, under the assumptions imposed, a classical result of Stone (1982) shows that the uniform minimax convergence rate is

$$\liminf_{n \rightarrow \infty} \left( \frac{n}{\log n} \right)^{\frac{q}{2q+2}} \inf_{S_n \in \mathcal{S}} \sup_{\mathbb{P} \in \mathcal{P}_{\text{NP}}} \mathbb{E}_{\mathbb{P}} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |S_n(\mathbf{x}; \mathbf{W}_n) - \mu(\mathbf{x})| \right] \gtrsim 1,$$

where  $\mathcal{S}$  is the (unrestricted class) of all estimators based on  $(\mathbf{W}_n = (Y_i, \mathbf{X}_i^\top)^\top : 1 \leq i \leq n)$ . Therefore, Theorem 6 shows that if we restrict the class of estimators to only using the distance  $\|\mathbf{X}_i - \mathbf{x}\|$  to the evaluation point  $\mathbf{x}$  (as opposed to using  $\mathbf{X}_i$  directly) for estimation of  $\mu(\mathbf{x})$ , then for any such estimator there is a data generating process such that the largest estimation error uniformly along the boundary is at least of order  $n^{-1/4}$ , when the boundary can have possibly countably many kinks. Notably, increasing smoothness of the underlying bivariate regression function  $\mu(\mathbf{x})$  does not improve the uniform estimation accuracy of estimators in  $\mathcal{T}$  due to the possible lack of smoothness introduced by a non-smooth boundary  $\mathcal{B}$  of the support  $\mathcal{X}$ .

Finally, by Theorem 1 and Lemma 2, the univariate distance-based  $p$ -th order local polynomial

nonparametric regression estimator

$$\hat{\mu}_{\text{dis}}(\mathbf{x}) = \mathbf{e}_1^\top \hat{\boldsymbol{\gamma}}(\mathbf{x}), \quad \hat{\boldsymbol{\gamma}}(\mathbf{x}) = \arg \min_{\boldsymbol{\gamma} \in \mathbb{R}^{p+1}} \mathbb{E}_n \left[ (Y_i - \mathbf{r}_p(D_i(\mathbf{x}))^\top \boldsymbol{\gamma})^2 k_h(D_i(\mathbf{x})) \right], \quad \mathbf{x} \in \mathcal{B},$$

satisfies

$$\limsup_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P} \left[ \left( \frac{n}{\log n} \right)^{1/4} \sup_{\mathbf{x} \in \mathcal{B}} |\hat{\mu}_{\text{dis}}(\mathbf{x}) - \mu(\mathbf{x})| \geq M \right] = 0.$$

Since  $\hat{\mu}_{\text{dis}}(\mathbf{x}) \in \mathcal{T}$  in Theorem 6, it follows that the distance-based local polynomial regression estimator is minimax optimal in the sense of Theorem 6, up to the  $\log^{1/4} n$  factor, which we conjecture is unimprovable (i.e., the lower bound in Theorem 6 is only loose by the factor  $\log^{1/4} n$ ).

## References

- Black, S. E. (1999), “Do Better Schools Matter? Parental Valuation of Elementary Education,” *Quarterly Journal of Economics*, 114, 577–599.
- Calonico, S., Cattaneo, M. D., and Farrell, M. H. (2018), “On the Effect of Bias Estimation on Coverage Accuracy in Nonparametric Inference,” *Journal of the American Statistical Association*, 113, 767–779.
- (2020), “Optimal Bandwidth Choice for Robust Bias Corrected Inference in Regression Discontinuity Designs,” *Econometrics Journal*, 23, 192–210.
- (2022), “Coverage Error Optimal Confidence Intervals for Local Polynomial Regression,” *Bernoulli*, 28, 2998–3022.
- Calonico, S., Cattaneo, M. D., and Titiunik, R. (2014), “Robust Nonparametric Confidence Intervals for Regression-Discontinuity Designs,” *Econometrica*, 82, 2295–2326.
- Cattaneo, M. D., Idrobo, N., and Titiunik, R. (2020), *A Practical Introduction to Regression Discontinuity Designs: Foundations*, Cambridge University Press.
- (2024), *A Practical Introduction to Regression Discontinuity Designs: Extensions*, Cambridge University Press.

- Cattaneo, M. D., and Titiunik, R. (2022), “Regression Discontinuity Designs,” *Annual Review of Economics*, 14, 821–851.
- Cattaneo, M. D., Titiunik, R., and Yu, R. R. (2025), “rd2d: Causal Inference in Boundary Discontinuity Designs,” *arXiv preprint arXiv:2505.xxxx*.
- Cattaneo, M. D., and Yu, R. R. (2025), “Strong Approximations for Empirical Processes Indexed by Lipschitz Functions,” *Annals of Statistics*.
- De Magalhães, L., Hangartner, D., Hirvonen, S., Meriläinen, J., Ruiz, N. A., and Tukiainen, J. (2025), “When Can We Trust Regression Discontinuity Design Estimates from Close Elections? Evidence from Experimental Benchmarks,” *Political Analysis*.
- Dell, M. (2010), “The Persistent Effects of Peru’s Mining Mita,” *Econometrica*, 78, 1863–1903.
- Diaz, J. D., and Zubizarreta, J. R. (2023), “Complex Discontinuity Designs Using Covariates for Policy Impact Evaluation,” *Annals of Applied Statistics*, 17, 67–88.
- Federer, H. (2014), *Geometric measure theory*, Springer.
- Galiani, S., McEwan, P. J., and Quistorff, B. (2017), “External and Internal Validity of a Geographic Quasi-Experiment Embedded in a Cluster-Randomized Experiment,” in *Regression Discontinuity Designs: Theory and Applications (Advances in Econometrics, volume 38)*, eds. M. D. Cattaneo and J. C. Escanciano, Emerald Group Publishing, pp. 195–236.
- Giné, E., and Nickl, R. (2016), *Mathematical Foundations of Infinite-dimensional Statistical Models*, New York: Cambridge University Press.
- Hahn, J., Todd, P., and van der Klaauw, W. (2001), “Identification and Estimation of Treatment Effects with a Regression-Discontinuity Design,” *Econometrica*, 69, 201–209.
- Hyttinen, A., Meriläinen, J., Saarimaa, T., Toivanen, O., and Tukiainen, J. (2018), “When does regression discontinuity design work? Evidence from random election outcomes,” *Quantitative Economics*, 9, 1019–1051.

- Jardim, E., Long, M. C., Plotnick, R., Vigdor, J., and Wiles, E. (2024), “Local minimum wage laws, boundary discontinuity methods, and policy spillovers,” *Journal of Public Economics*, 234, 105131.
- Keele, L., and Titiunik, R. (2016), “Natural experiments based on geography,” *Political Science Research and Methods*, 4, 65–95.
- Keele, L. J., Lorch, S., Passarella, M., Small, D., and Titiunik, R. (2017), “An Overview of Geographically Discontinuous Treatment Assignments with an Application to Children’s Health Insurance,” in *Regression Discontinuity Designs: Theory and Applications (Advances in Econometrics, volume 38)*, eds. M. D. Cattaneo and J. C. Escanciano, Emerald Group Publishing, pp. 147–194.
- Keele, L. J., and Titiunik, R. (2015), “Geographic Boundaries as Regression Discontinuities,” *Political Analysis*, 23, 127–155.
- Londoño-Vélez, J., Rodríguez, C., and Sánchez, F. (2020), “Upstream and downstream impacts of college merit-based financial aid for low-income students: Ser Pilo Paga in Colombia,” *American Economic Journal: Economic Policy*, 12, 193–227.
- Papay, J. P., Willett, J. B., and Murnane, R. J. (2011), “Extending the regression-discontinuity approach to multiple assignment variables,” *Journal of Econometrics*, 161, 203–207.
- Reardon, S. F., and Robinson, J. P. (2012), “Regression Discontinuity Designs with Multiple Rating-Score Variables,” *Journal of Research on Educational Effectiveness*, 5, 83–104.
- Simon, L. et al. (1984), *Lectures on geometric measure theory*, Centre for Mathematical Analysis, Australian National University Canberra.
- Stone, C. J. (1982), “Optimal Global Rates of Convergence for Nonparametric Regression,” *Annals of Statistics*, 10, 1040–1053.
- van der Vaart, A. W., and Wellner, J. A. (1996), *Weak Convergence and Empirical Processes*, Springer.

# Estimation and Inference in Boundary Discontinuity Designs

## Supplemental Appendix\*

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### Abstract

This supplemental appendix presents more general theoretical results encompassing those discussed in the main paper, and their proofs.

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## SA-1 Setup

This supplemental appendix collects all the technical work underlying the results presented in the main paper. It considers a generalized version of the problems studied in the main paper: the location variable  $\mathbf{X}_i$  is  $d$ -dimensional with  $d \geq 1$  (its support is  $\mathcal{X} \subseteq \mathbb{R}^d$ ), and the boundary region  $\mathcal{B}$  is a low dimensional manifold with “effective dimension”  $d - 1$ . The special case considered in the main paper is  $d = 2$ , that is,  $\mathbf{X}_i$  is bivariate and  $\mathcal{B}$  is a one-dimensional (boundary) curve.

Assumption 1 from the main paper is generalized to the following:

### Assumption SA–1 (Data Generating Process)

Let  $t \in \{0, 1\}$ .

- (i)  $(Y_1(t), \mathbf{X}_1^\top)^\top, \dots, (Y_n(t), \mathbf{X}_n^\top)^\top$  are independent and identically distributed random vectors with  $\mathcal{X} = \prod_{l=1}^d [a_l, b_l]$  for  $-\infty < a_l < b_l < \infty$  for  $l = 1, \dots, d$ .
- (ii) The distribution of  $\mathbf{X}_i$  has a Lebesgue density  $f_X(\mathbf{x})$  that is continuous and bounded away from zero on  $\mathcal{X}$ .
- (iii)  $\mu_t(\mathbf{x}) = \mathbb{E}[Y_i(t)|\mathbf{X}_i = \mathbf{x}]$  is  $(p + 1)$ -times continuously differentiable on  $\mathcal{X}$ .
- (iv)  $\sigma_t^2(\mathbf{x}) = \mathbb{V}[Y_i(t)|\mathbf{X}_i = \mathbf{x}]$  is bounded away from zero and continuous on  $\mathcal{X}$ .
- (v)  $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[|Y_i(t)|^{2+v}|\mathbf{X}_i = \mathbf{x}] < \infty$  for some  $v \geq 2$ .

We partition  $\mathcal{X}$  into two areas,  $\mathcal{A}_t \subset \mathbb{R}^d$  with  $t \in \{0, 1\}$ , which represent the control and treatment regions, respectively. That is,  $\mathcal{X} = \mathcal{A}_0 \cup \mathcal{A}_1$ , where  $\mathcal{A}_0$  and  $\mathcal{A}_1$  are two disjoint regions in  $\mathbb{R}^d$ , and  $\text{cl}(\mathcal{A}_t)$  denotes the closure of  $\mathcal{A}_t$ ,  $t \in \{0, 1\}$ . The observed outcome is  $Y_i = \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_0)Y_i(0) + \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1)Y_i(1)$ .  $\mathcal{B} = \text{bd}(\mathcal{A}_0) \cap \text{bd}(\mathcal{A}_1)$  denotes the boundary determined by the assignment regions  $\mathcal{A}_t$ ,  $t \in \{0, 1\}$ , where  $\text{bd}(\mathcal{A}_t)$  denotes the topological boundary of  $\mathcal{A}_t$ . The treatment effect curve along the boundary is

$$\tau(\mathbf{x}) = \mathbb{E}[Y_i(1) - Y_i(0)|\mathbf{X}_i = \mathbf{x}], \quad \mathbf{x} \in \mathcal{B}.$$

### SA-1.1 Notation and Definitions

For textbook references on empirical process, see [van der Vaart and Wellner \(1996\)](#), [Dudley \(2014\)](#), and [Giné and Nickl \(2016\)](#). For textbook reference on geometric measure theory, see [Simon et al. \(1984\)](#), [Federer \(2014\)](#), and [Folland \(2002\)](#).

- (i) *Multi-index Notations.* For a multi-index  $\mathbf{u} = (u_1, \dots, u_d) \in \mathbb{N}^d$ , denote  $|\mathbf{u}| = \sum_{i=1}^d u_i$ ,  $\mathbf{u}! = \prod_{i=1}^d u_i!$ . Denote  $\mathbf{R}_p(\mathbf{u}) = (1, u_1, \dots, u_d, u_1^2, \dots, u_d^2, \dots, u_1^p, \dots, u_d^p)$ , that is, all monomials  $u_1^{\alpha_1} \dots u_d^{\alpha_d}$  such that  $\alpha_i \in \mathbb{N}$  and  $\sum_{i=1}^d \alpha_i \leq p$ . Define  $\mathbf{e}_{1+\nu}$  to be the  $p_d = \frac{(d+p)!}{d!p!}$ -dimensional vector such that  $\mathbf{e}_{1+\nu}^\top \mathbf{R}_p(\mathbf{u}) = \mathbf{u}^\nu$  for all  $\mathbf{u} \in \mathbb{R}^d$ .
- (ii) *Norms.* For a vector  $\mathbf{v} \in \mathbb{R}^k$ ,  $\|\mathbf{v}\| = (\sum_{i=1}^k \mathbf{v}_i^2)^{1/2}$ ,  $\|\mathbf{v}\|_\infty = \max_{1 \leq i \leq k} |\mathbf{v}_i|$ . For a matrix  $A \in \mathbb{R}^{m \times n}$ ,  $\|A\|_p = \sup_{\|\mathbf{x}\|_p=1} \|A\mathbf{x}\|_p$ ,  $p \in \mathbb{N} \cup \{\infty\}$ . For a function  $f$  on a metric space  $(S, d)$ ,  $\|f\|_\infty = \sup_{x \in \mathcal{X}} |f|$ ,  $\|f\|_{\text{Lip}, \infty} = \sup_{\mathbf{x}, \mathbf{x}' \in S} \frac{|f(\mathbf{x}) - f(\mathbf{x}')|}{d(\mathbf{x}, \mathbf{x}')}$ . For a probability measure  $Q$  on  $(\mathcal{S}, \mathcal{S})$  and  $p \geq 1$ , define  $\|f\|_{Q, p} = (\int_{\mathcal{S}} |f|^p dQ)^{1/p}$ . For a set  $E \subseteq \mathbb{R}^d$ , denote by  $\mathbf{m}(E)$  the Lebesgue measure of  $E$ .
- (iii) *Empirical Process.* We use standard empirical process notations:  $\mathbb{E}_n[g(\mathbf{v}_i)] = \frac{1}{n} \sum_{i=1}^n g(\mathbf{v}_i)$  and  $\mathbb{G}_n[g(\mathbf{v}_i)] = \frac{1}{\sqrt{n}} \sum_{i=1}^n (g(\mathbf{v}_i) - \mathbb{E}[g(\mathbf{v}_i)])$ . Let  $(\mathcal{S}, d)$  be a semi-metric space. The covering number  $N(\mathcal{S}, d, \varepsilon)$  is the minimal number of balls  $B_s(\varepsilon) = \{t : d(t, s) < \varepsilon\}$  needed to cover  $\mathcal{S}$ . A

$\mathbb{P}$ -Brownian bridge is a mean-zero Gaussian random function  $W_n(f), f \in L_2(\mathcal{X}, \mathbb{P})$  with the covariance  $\mathbb{E}[W_{\mathbb{P}}(f)W_{\mathbb{P}}(g)] = \mathbb{P}(fg) - \mathbb{P}(f)\mathbb{P}(g)$ , for  $f, g \in L_2(\mathcal{X}, \mathbb{P})$ . A class  $\mathcal{F} \subseteq L_2(\mathcal{X}, \mathbb{P})$  is  $\mathbb{P}$ -pregaussian if there is a version of  $\mathbb{P}$ -Brownian bridge  $W_{\mathbb{P}}$  such that  $W_{\mathbb{P}} \in C(\mathcal{F}; \rho_{\mathbb{P}})$  almost surely, where  $\rho_{\mathbb{P}}$  is the semi-metric on  $L_2(\mathcal{X}, \mathbb{P})$  is defined by  $\rho_{\mathbb{P}}(f, g) = (\|f - g\|_{\mathbb{P}, 2}^2 - (\int f d\mathbb{P} - \int g d\mathbb{P})^2)^{1/2}$ , for  $f, g \in L_2(\mathcal{X}, \mathbb{P})$ .

- (iv) *Geometric Measure Theory.* For a set  $E \subseteq \mathcal{X}$ , the *De Giorgi perimeter* of  $E$  related to  $\mathcal{X}$  is  $\mathcal{L}(E) = \text{TV}_{\{\mathbf{1}_E\}, \mathcal{X}}$ .  $B$  is a *rectifiable curve* if there exists a Lipschitz continuous function  $\gamma : [0, 1] \rightarrow \mathbb{R}$  such that  $B = \gamma([0, 1])$ . We define the curve length function of  $B$  to be  $\mathfrak{L}(B) = \sup_{\pi \in \Pi} s(\pi, \gamma)$ , where  $\Pi = \{(t_0, t_1, \dots, t_N) : N \in \mathbb{N}, 0 \leq t_0 < t_1 < \dots \leq t_N \leq 1\}$  and  $s(\pi, \gamma) = \sum_{i=0}^N \|\gamma(t_i) - \gamma(t_{i+1})\|_2$  for  $\pi = (t_0, t_1, \dots, t_N)$ .
- (v) *Bounds and Asymptotics.* For reals sequences  $|a_n| = o(|b_n|)$  if  $\limsup \frac{a_n}{b_n} = 0$ ,  $|a_n| \lesssim |b_n|$  if there exists some constant  $C$  and  $N > 0$  such that  $n > N$  implies  $|a_n| \leq C|b_n|$ . For sequences of random variables  $a_n = o_{\mathbb{P}}(b_n)$  if  $\text{plim}_{n \rightarrow \infty} \frac{a_n}{b_n} = 0$ ,  $|a_n| \lesssim_{\mathbb{P}} |b_n|$  if  $\limsup_{M \rightarrow \infty} \limsup_{n \rightarrow \infty} \mathbb{P}[|\frac{a_n}{b_n}| \geq M] = 0$ .
- (vi) *Distributions and Statistical Distances.* For  $\boldsymbol{\mu} \in \mathbb{R}^k$  and  $\boldsymbol{\Sigma}$  a  $k \times k$  positive definite matrix,  $\mathbf{N}(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  denotes the Gaussian distribution with mean  $\boldsymbol{\mu}$  and covariance  $\boldsymbol{\Sigma}$ . For  $-\infty < a < b < \infty$ ,  $\text{Unif}([a, b])$  denotes the uniform distribution on  $[a, b]$ .  $\text{Bern}(p)$  denotes the Bernoulli distribution with success probability  $p$ .  $\Phi(\cdot)$  denotes the standard Gaussian cumulative distribution function. For two distributions  $P$  and  $Q$ ,  $d_{\text{KL}}(P, Q)$  denotes the KL-distance between  $P$  and  $Q$ , and  $d_{\chi^2}(P, Q)$  denotes the  $\chi^2$  distance between  $P$  and  $Q$ .

## SA-1.2 Mapping between Main Paper and Supplement

The results in the main paper are special cases of the results in this supplemental appendix as follows.

- Lemma 1 in the paper corresponds to Lemma SA-3.1 with  $d = 2$ .
- Lemma 2 in is proven in Section SA-7.1.
- Lemma 3 in is proven in Section SA-7.2.
- Theorem 1(i) in the paper corresponds to Theorem SA-3.1 with  $d = 2$ .
- Theorem 1(ii) in the paper corresponds to Theorem SA-3.3 with  $d = 2$ .
- Theorem 2(i) in the paper corresponds to Theorem SA-3.2 with  $d = 2$ .
- Theorem 2(ii) in the paper corresponds to Theorem SA-3.6 with  $d = 2$ .
- Theorem 3(i) in the paper corresponds to Theorem SA-2.1 with  $d = 2$ .
- Theorem 3(ii) in the paper corresponds to Theorem SA-2.5 with  $d = 2$ .
- Theorem 4 in the paper corresponds to Theorem SA-2.2 with  $d = 2$ .
- Theorem 5(i) in the paper corresponds to Theorem SA-2.4 with  $d = 2$ .
- Theorem 5(ii) in the paper corresponds to Theorem SA-2.9 with  $d = 2$ .
- Theorem 6 is proven in Section SA-7.3.

## SA-2 Location-Based Methods

We consider a more general setting compared to the main paper, where the parameter of interest is

$$\tau^{(\nu)}(\mathbf{x}) = \mu_1^{(\nu)}(\mathbf{x}) - \mu_0^{(\nu)}(\mathbf{x}), \quad \mathbf{x} \in \mathcal{B},$$

where  $\nu$  is a multi-index with  $|\nu| \leq p$ . Thus, the treatment effect curve estimator is  $(\hat{\tau}^{(\nu)}(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$ , where

$$\hat{\tau}^{(\nu)}(\mathbf{x}) = \hat{\mu}_1^{(\nu)}(\mathbf{x}) - \hat{\mu}_0^{(\nu)}(\mathbf{x}), \quad \mathbf{x} \in \mathcal{B},$$

where, for  $t \in \{0, 1\}$ ,  $\hat{\mu}_t^{(\nu)}(\mathbf{x}) = \mathbf{e}_{1+\nu}^\top \hat{\beta}_t(\mathbf{x})$  with

$$\hat{\beta}_t(\mathbf{x}) = \underset{\beta \in \mathbb{R}^{p+1}}{\operatorname{argmin}} \mathbb{E}_n \left[ (Y_i - \mathbf{R}_p(\mathbf{X}_i - \mathbf{x})^\top \beta)^2 K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right], \quad \mathbf{x} \in \mathcal{B},$$

with  $\mathbf{p}_p = \frac{(d+p)!}{d!p!}$ ,  $\mathbf{R}_p(\mathbf{u}) = (1, u_1, u_2, \dots, u_d, u_1^2, u_1 u_2, u_1 u_2, \dots, u_d^2, \dots, u_1^p, u_1^{p-1} u_2, \dots, u_2^p)^\top$  denotes the  $p$ th order polynomial expansion of the  $d$ -variate vector  $\mathbf{u} = (u_1, \dots, u_d)^\top$ ,  $K_h(\mathbf{u}) = K(u_1/h, \dots, u_d/h)/h^d$  for a  $d$ -variate kernel function  $K(\cdot)$  and a bandwidth parameter  $h$ .

We impose the following assumption on  $d$ -variate kernel function and assignment boundary.

### Assumption SA-2 (Kernel Function and Bandwidth)

Let  $t \in \{0, 1\}$ .

- (i)  $K : \mathbb{R}^d \rightarrow [0, \infty)$  is compact supported and Lipschitz continuous, or  $K(\mathbf{u}) = \mathbb{1}(\mathbf{u} \in [-1, 1]^d)$ .
- (ii)  $\liminf_{h \downarrow 0} \inf_{\mathbf{x} \in \mathcal{B}} \int_{\mathcal{A}_t} K_h(\mathbf{u} - \mathbf{x}) d\mathbf{u} \gtrsim 1$ .

Under the assumptions imposed, for  $t \in \{0, 1\}$ , we have

$$\hat{\beta}_t(\mathbf{x}) = \mathbf{H}^{-1} \hat{\Gamma}_{t,\mathbf{x}}^{-1} \mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) Y_i \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right],$$

where  $\mathbf{H} = \operatorname{diag}((h^{|\mathbf{v}|})_{0 \leq |\mathbf{v}| \leq p})$  with  $\mathbf{v}$  running through all  $\frac{d+p}{d!p!}$  multi-indices such that  $|\mathbf{v}| \leq p$ , and

$$\hat{\Gamma}_{t,\mathbf{x}} = \mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right)^\top K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right].$$

In particular,  $\|\mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1}\|_2 = \|\mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1}\|_\infty = h^{-|\nu|}$ .

For  $\mathbf{x}, \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}$  and  $t \in \{0, 1\}$ , we introduce the following quantities:

$$\begin{aligned} \Gamma_{t,\mathbf{x}} &= \mathbb{E} \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right)^\top K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right], \\ \hat{\Sigma}_{t,\mathbf{x}_1,\mathbf{x}_2} &= h^d \mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}_1}{h} \right) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}_2}{h} \right)^\top K_h(\mathbf{X}_i - \mathbf{x}_1) K_h(\mathbf{X}_i - \mathbf{x}_2) \varepsilon_i^2 \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right], \\ \Sigma_{t,\mathbf{x}_1,\mathbf{x}_2} &= h^d \mathbb{E} \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}_1}{h} \right) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}_2}{h} \right)^\top K_h(\mathbf{X}_i - \mathbf{x}_1) K_h(\mathbf{X}_i - \mathbf{x}_2) \sigma_i^2(\mathbf{X}_i) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right], \\ \hat{\Omega}_{t,\mathbf{x}_1,\mathbf{x}_2}^{(\nu)} &= \frac{1}{nh^{d+2|\nu|}} \mathbf{e}_{1+\nu}^\top \hat{\Gamma}_{t,\mathbf{x}_1}^{-1} \hat{\Sigma}_{t,\mathbf{x}_1,\mathbf{x}_2} \hat{\Gamma}_{t,\mathbf{x}_2}^{-1} \mathbf{e}_{1+\nu}, \quad \hat{\Omega}_{\mathbf{x}_1,\mathbf{x}_2}^{(\nu)} = \hat{\Omega}_{0,\mathbf{x}_1,\mathbf{x}_2}^{(\nu)} + \hat{\Omega}_{1,\mathbf{x}_1,\mathbf{x}_2}^{(\nu)}, \\ \Omega_{t,\mathbf{x}_1,\mathbf{x}_2}^{(\nu)} &= \frac{1}{nh^{d+2|\nu|}} \mathbf{e}_{1+\nu}^\top \Gamma_{t,\mathbf{x}_1}^{-1} \Sigma_{t,\mathbf{x}_1,\mathbf{x}_2} \Gamma_{t,\mathbf{x}_2}^{-1} \mathbf{e}_{1+\nu}, \quad \Omega_{\mathbf{x}_1,\mathbf{x}_2}^{(\nu)} = \Omega_{0,\mathbf{x}_1,\mathbf{x}_2}^{(\nu)} + \Omega_{1,\mathbf{x}_1,\mathbf{x}_2}^{(\nu)}, \end{aligned}$$

where  $\varepsilon_i = Y_i - \sum_{t \in \{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \hat{\beta}_t(\mathbf{x})^\top \mathbf{R}_p(\mathbf{X}_i - \mathbf{x})$  and  $\sigma_t^2(\mathbf{x}) = \mathbb{V}[Y_i(t) | \mathbf{X}_i = \mathbf{x}]$ . Denote

$$\begin{aligned} \hat{B}_{t,\mathbf{x}}^{(\nu)} &= \mathbf{e}_{1+\nu}^\top \hat{\Gamma}_{t,\mathbf{x}}^{-1} \sum_{|\omega|=p+1} \frac{\mu_t^{(\omega)}(\mathbf{x})}{\omega!} \mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right)^\omega K_h(\mathbf{X}_i - \mathbf{x}) \right], & \hat{B}_{\mathbf{x}}^{(\nu)} &= \hat{B}_{1,\mathbf{x}}^{(\nu)} - \hat{B}_{0,\mathbf{x}}^{(\nu)}, \\ B_{t,\mathbf{x}}^{(\nu)} &= \mathbf{e}_{1+\nu}^\top \Gamma_{t,\mathbf{x}}^{-1} \sum_{|\omega|=p+1} \frac{\mu_t^{(\omega)}(\mathbf{x})}{\omega!} \mathbb{E} \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right)^\omega K_h(\mathbf{X}_i - \mathbf{x}) \right], & B_{\mathbf{x}}^{(\nu)} &= B_{1,\mathbf{x}}^{(\nu)} - B_{0,\mathbf{x}}^{(\nu)}, \\ \mathbf{Q}_{t,\mathbf{x}} &= \mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \varepsilon_i \right], \\ \hat{V}_{t,\mathbf{x}}^{(\nu)} &= \mathbf{e}_{1+\nu}^\top \hat{\Gamma}_{t,\mathbf{x}}^{-1} \hat{\Sigma}_{t,\mathbf{x}} \hat{\Gamma}_{t,\mathbf{x}}^{-1} \mathbf{e}_{1+\nu}, & \hat{V}_{\mathbf{x}}^{(\nu)} &= \hat{V}_{0,\mathbf{x}}^{(\nu)} + \hat{V}_{1,\mathbf{x}}^{(\nu)}, \\ V_{t,\mathbf{x}}^{(\nu)} &= \mathbf{e}_{1+\nu}^\top \Gamma_{t,\mathbf{x}}^{-1} \Sigma_{t,\mathbf{x}} \Gamma_{t,\mathbf{x}}^{-1} \mathbf{e}_{1+\nu}, & V_{\mathbf{x}}^{(\nu)} &= V_{0,\mathbf{x}}^{(\nu)} + V_{1,\mathbf{x}}^{(\nu)}, \end{aligned}$$

where  $u_i = Y_i - \sum_{t \in \{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \mu_t(\mathbf{X}_i)$ .

### SA-2.1 Preliminary Lemmas

In what follows, we denote  $\mathbf{X} = (\mathbf{X}_1^\top, \dots, \mathbf{X}_n^\top)$  and  $\mathbf{W}_n = ((\mathbf{X}_1^\top, Y_1), \dots, (\mathbf{X}_n^\top, Y_n))^\top$ .

#### Lemma SA-2.1 (Gram)

Suppose Assumption SA-1(i)(ii) and Assumption SA-2 hold. If  $\frac{\log(1/h)}{nh^d} = o(1)$ , then

$$\begin{aligned} \sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\Gamma}_{t,\mathbf{x}} - \Gamma_{t,\mathbf{x}}\| &\lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}}, & 1 &\lesssim \mathbb{P} \inf_{\mathbf{x} \in \mathcal{B}} \|\hat{\Gamma}_{t,\mathbf{x}}\| \lesssim \sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\Gamma}_{t,\mathbf{x}}\| \lesssim \mathbb{P} 1, \\ \sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\Gamma}_{t,\mathbf{x}}^{-1} - \Gamma_{t,\mathbf{x}}^{-1}\| &\lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}}, \end{aligned}$$

for  $t \in \{0,1\}$ .

#### Lemma SA-2.2 (Bias)

Suppose Assumption SA-1(i)(ii)(iii) and Assumption SA-2 hold. If  $\frac{\log(1/h)}{nh^d} = o(1)$ , then

$$\sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbb{E}[\hat{\mu}_t^{(\nu)}(\mathbf{x}) | \mathbf{X}] - \mu_t^{(\nu)}(\mathbf{x}) \right| \lesssim \mathbb{P} h^{p+1-|\nu|},$$

for  $t \in \{0,1\}$ . If, in addition,  $h = o(1)$ , then

$$\sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbb{E}[\hat{\mu}_t^{(\nu)}(\mathbf{x}) | \mathbf{X}] - \mu_t^{(\nu)}(\mathbf{x}) - h^{p+1-|\nu|} \hat{B}_{t,\mathbf{x}}^{(\nu)} \right| = o_{\mathbb{P}}(h^{p+1-|\nu|}),$$

for  $t \in \{0,1\}$ . Moreover,  $\sup_{\mathbf{x} \in \mathcal{B}} |\hat{B}_{t,\mathbf{x}}^{(\nu)} - B_{t,\mathbf{x}}^{(\nu)}| \lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}}$ , which implies  $\sup_{\mathbf{x} \in \mathcal{B}} |\hat{B}_{t,\mathbf{x}}^{(\nu)}| \lesssim \mathbb{P} 1$  for  $t \in \{0,1\}$ .

#### Lemma SA-2.3 (Stochastic Linear Approximation)

Suppose Assumption SA-1(i)(ii)(iv)(v) and Assumption SA-2 hold. Suppose  $\frac{\log(1/h)}{nh^d} = o(1)$ , then

$$\begin{aligned} \sup_{\mathbf{x} \in \mathcal{B}} \|\mathbf{Q}_{t,\mathbf{x}}\| &\lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d}, \\ \sup_{\mathbf{x} \in \mathcal{B}} \left| \hat{\mu}_t^{(\nu)}(\mathbf{x}) - \mathbb{E}[\hat{\mu}_t^{(\nu)}(\mathbf{x}) | \mathbf{X}] - \mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1} \Gamma_{t,\mathbf{x}}^{-1} \mathbf{Q}_{t,\mathbf{x}} \right| &\lesssim \mathbb{P} h^{-|\nu|} \sqrt{\frac{\log(1/h)}{nh^d}} \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right), \end{aligned}$$

for  $t \in \{0, 1\}$ .

**Lemma SA-2.4 (Covariance)**

Suppose Assumptions SA-1 and SA-2 hold. If  $\frac{\log(1/h)}{nh^d} = o(1)$ , then

$$\begin{aligned} \sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}} \|\widehat{\Sigma}_{t, \mathbf{x}_1, \mathbf{x}_2} - \Sigma_{t, \mathbf{x}_1, \mathbf{x}_2}\| &\lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} + h^{p+1}, \\ \sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}} \left| \widehat{\Omega}_{\mathbf{x}_1, \mathbf{x}_2}^{(\nu)} - \Omega_{\mathbf{x}_1, \mathbf{x}_2}^{(\nu)} \right| &\lesssim \mathbb{P} (nh^{d+2|\nu|})^{-1} \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} + h^{p+1} \right), \\ \sup_{\mathbf{x} \in \mathcal{B}} \left| (\widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)})^{-\frac{1}{2}} - (\Omega_{\mathbf{x}, \mathbf{x}}^{(\nu)})^{-\frac{1}{2}} \right| &\lesssim \mathbb{P} \sqrt{nh^{d+2|\nu|}} \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} + h^{p+1} \right), \end{aligned}$$

for  $t \in \{0, 1\}$ .

**SA-2.2 Point Estimation**

**Theorem SA-2.1 (Pointwise Convergence Rate)**

Suppose Assumptions SA-1 and SA-2 hold. If  $nh^d \rightarrow \infty$ , then

$$\sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{\tau}^{(\nu)}(\mathbf{x}) - \tau^{(\nu)}(\mathbf{x}) \right| \lesssim \mathbb{P} h^{-|\nu|} \left( h^{p+1} + \frac{1}{\sqrt{nh^d}} + \frac{1}{n^{\frac{1+v}{2+v}} h^d} \right).$$

The conditional mean-squared error (MSE) is

$$\text{MSE}_\nu(\mathbf{x}) = \mathbb{E} \left[ (\widehat{\tau}^{(\nu)}(\mathbf{x}) - \tau^{(\nu)}(\mathbf{x}))^2 \middle| \mathbf{X} \right], \quad \mathbf{x} \in \mathcal{B},$$

and, for some non-negative weighting function  $\omega$  satisfying  $\int_{\mathcal{B}} \omega(\mathbf{x}) d\mathbf{x} < \infty$ , the conditional integrated mean-squared error (IMSE) is defined to be

$$\text{IMSE}_\nu = \int_{\mathcal{B}} \text{MSE}_\nu(\mathbf{x}) \omega(\mathbf{x}) dH^{d-1}(\mathbf{x}),$$

where  $H^{d-1}$  is the  $(d-1)$  dimensional Hausdorff measure, also known as “area” element on  $\mathcal{B}$  (Folland, 2002; Federer, 2014).

**Theorem SA-2.2 (MSE Expansions)**

Suppose Assumptions SA-1 and SA-2 hold. If  $\frac{\log(1/h)}{nh^d} = o(1)$  and  $h = o(1)$ , then

$$\begin{aligned} \text{MSE}_\nu(\mathbf{x}) &= (h^{p+1-|\nu|} B_{\mathbf{x}}^{(\nu)})^2 + n^{-1} h^{-d-2|\nu|} V_{\mathbf{x}}^{(\nu)} + o_{\mathbb{P}}(h^{2p+2-2|\nu|} + n^{-1} h^{-d-2|\nu|}), \quad \mathbf{x} \in \mathcal{B}, \\ \text{IMSE}_\nu &= \int_{\mathcal{B}} \left[ (h^{p+1-|\nu|} B_{\mathbf{x}}^{(\nu)})^2 + n^{-1} h^{-d-2|\nu|} V_{\mathbf{x}}^{(\nu)} \right] \omega(\mathbf{x}) dH^{d-1}(\mathbf{x}) + o_{\mathbb{P}}(h^{2p+2-2|\nu|} + n^{-1} h^{-d-2|\nu|}). \end{aligned}$$

With the estimated  $\widehat{B}_{\mathbf{x}}^{(\nu)}$  and  $\widehat{V}_{\mathbf{x}}^{(\nu)}$ , suppose  $\frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} = o(1)$  and  $h = o(1)$ , then

$$\begin{aligned} \text{MSE}_\nu(\mathbf{x}) &= (h^{p+1-|\nu|} \widehat{B}_{\mathbf{x}}^{(\nu)})^2 + n^{-1} h^{-d-2|\nu|} \widehat{V}_{\mathbf{x}}^{(\nu)} + o_{\mathbb{P}}(h^{2p+2-2|\nu|} + n^{-1} h^{-d-2|\nu|}), \quad \mathbf{x} \in \mathcal{B}, \\ \text{IMSE}_\nu &= \int_{\mathcal{B}} \left[ (h^{p+1-|\nu|} \widehat{B}_{\mathbf{x}}^{(\nu)})^2 + n^{-1} h^{-d-2|\nu|} \widehat{V}_{\mathbf{x}}^{(\nu)} \right] \omega(\mathbf{x}) dH^{d-1}(\mathbf{x}) + o_{\mathbb{P}}(h^{2p+2-2|\nu|} + n^{-1} h^{-d-2|\nu|}). \end{aligned}$$

If  $\widehat{B}_{\mathbf{x}}^{(\nu)} \neq 0$ , the asymptotic MSE-optimal bandwidth is

$$h_{\text{MSE}, \nu, p}(\mathbf{x}) = \left( \frac{(d + 2|\nu|) \widehat{V}_{\mathbf{x}}^{(\nu)}}{(2p + 2 - 2|\nu|)n(\widehat{B}_{\mathbf{x}}^{(\nu)})^2} \right)^{\frac{1}{2p+d+2}}, \quad \mathbf{x} \in \mathcal{B}.$$

If  $\int_{\mathcal{B}} (B_{\mathbf{x}}^{(\nu)})^2 \omega(\mathbf{x}) dH^{d-1}(\mathbf{x}) \neq 0$ , the asymptotic IMSE-optimal bandwidth is

$$h_{\text{IMSE}, \nu, p} = \left( \frac{(d + 2|\nu|) \int_{\mathcal{B}} \widehat{V}_{\mathbf{x}}^{(\nu)} \omega(\mathbf{x}) dH^{d-1}(\mathbf{x})}{(2p + 2 - 2|\nu|)n \int_{\mathcal{B}} (\widehat{B}_{\mathbf{x}}^{(\nu)})^2 \omega(\mathbf{x}) dH^{d-1}(\mathbf{x})} \right)^{\frac{1}{2p+d+2}}.$$

### SA-2.3 Pointwise Inference

For  $|\nu| \leq p$ , define the feasible  $t$ -statistics:

$$\widehat{T}^{(\nu)}(\mathbf{x}) = \frac{\widehat{\tau}^{(\nu)}(\mathbf{x}) - \tau^{(\nu)}(\mathbf{x})}{\sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)}}}, \quad \mathbf{x} \in \mathcal{B}.$$

#### Theorem SA-2.3 (Asymptotic Normality)

Suppose Assumptions SA-1 and SA-2 hold. If  $nh^d \rightarrow \infty$  and  $nh^d h^{2(p+1)} \rightarrow 0$ , then

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P}(\widehat{T}^{(\nu)}(\mathbf{x}) \leq u) - \Phi(u) \right| = o(1), \quad \mathbf{x} \in \mathcal{B}.$$

For any  $0 < \alpha < 1$ , define the confidence interval:

$$\widehat{\mathbf{I}}_{\alpha}^{(\nu)}(\mathbf{x}) = \left[ \widehat{\tau}^{(\nu)}(\mathbf{x}) - \mathbf{c}_{\alpha} \sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)}}, \widehat{\tau}^{(\nu)}(\mathbf{x}) + \mathbf{c}_{\alpha} \sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)}} \right],$$

where  $\mathbf{c}_{\alpha} = \inf\{c > 0 : \mathbb{P}(|\widehat{Z}| \geq c | \mathbf{W}_n) \leq \alpha\}$  with  $\widehat{Z} | \mathbf{X} \sim \text{Normal}(0, \widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)})$ , for each  $\mathbf{x} \in \mathcal{B}$ .

#### Theorem SA-2.4 (Confidence Intervals)

Suppose Assumptions SA-1 and SA-2 hold. If  $nh^d \rightarrow \infty$  and  $nh^d h^{2(p+1)} \rightarrow 0$ , then

$$\mathbb{P}[\mu^{(\nu)}(\mathbf{x}) \in \widehat{\mathbf{I}}_{\alpha}^{(\nu)}(\mathbf{x})] = 1 - \alpha + o(1), \quad \mathbf{x} \in \mathcal{B}.$$

### SA-2.4 Uniform Inference

#### Theorem SA-2.5 (Uniform Convergence Rate)

Suppose Assumptions SA-1 and SA-2 hold. If  $\frac{\log(1/h)}{nh^d} = o(1)$ , then

$$\sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{\tau}^{(\nu)}(\mathbf{x}) - \tau^{(\nu)}(\mathbf{x}) \right| \lesssim_{\mathbb{P}} h^{-|\nu|} \left( h^{p+1} + \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right).$$

$\widehat{T}^{(\nu)}$  is not directly a sum of i.i.d terms. For  $\mathbf{x} \in \mathcal{B}$ , we define the *stochastic linearization* of  $\widehat{T}^{(\nu)}(\mathbf{x})$  to be

$$\overline{T}^{(\nu)}(\mathbf{x}) = \mathbb{E}_n \left[ \mathbf{e}_{1+\nu}^{\top} \mathbf{H}^{-1} (\mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \Gamma_{1, \mathbf{x}}^{-1} - \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_0) \Gamma_{0, \mathbf{x}}^{-1}) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) u_i(\Omega_{\mathbf{x}, \mathbf{x}}^{(\nu)})^{-1/2} \right],$$

with  $u_i = Y_i - \sum_{t \in \{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \mu_t(\mathbf{X}_i)$ .

**Theorem SA-2.6 (Stochastic Linearization)**

Suppose Assumptions SA-1 and SA-2 hold. If  $\frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} = o(1)$ , then

$$\sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \overline{\mathbf{T}}^{(\nu)}(\mathbf{x}) \right| \lesssim_{\mathbb{P}} h^{p+1} \sqrt{nh^d} + \sqrt{\log(1/h)} \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right).$$

Next, we exploit a structure of  $(\overline{\mathbf{T}}^{(\nu)}(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$ . Define the following function indexed by  $\mathbf{x} \in \mathcal{B}$ .

$$\begin{aligned} g_{\mathbf{x}}(\mathbf{u}) &= \mathbb{1}(\mathbf{u} \in \mathcal{A}_1) \mathcal{K}_1^{(\nu)}(\mathbf{u}; \mathbf{x}) - \mathbb{1}(\mathbf{u} \in \mathcal{A}_0) \mathcal{K}_0^{(\nu)}(\mathbf{u}; \mathbf{x}), & \mathbf{u} \in \mathcal{X}, \\ \mathcal{K}_t^{(\nu)}(\mathbf{u}; \mathbf{x}) &= n^{-1/2} (\Omega_{\mathbf{x}, \mathbf{x}}^{(\nu)})^{-1/2} \mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1} \Gamma_{t, \mathbf{x}}^{-1} \mathbf{R}_p \left( \frac{\mathbf{u} - \mathbf{x}}{h} \right) K_h(\mathbf{u} - \mathbf{x}), & \mathbf{u} \in \mathcal{X}, t \in \{0, 1\}, \end{aligned}$$

and define the class of functions  $\mathcal{G} = \{g_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$  and  $\mathcal{R} = \{\text{Id}\}$ , where  $\text{Id}(x) = x$ , for all  $x \in \mathbb{R}$ . Define the residual-based empirical process by

$$R_n(g, r) = n^{-1/2} \sum_{i=1}^n \left[ g(\mathbf{X}_i) r(Y_i) - g(\mathbf{X}_i) \mathbb{E}[r(Y_i) | \mathbf{X}_i] \right], \quad g \in \mathcal{G}, r \in \mathcal{R}.$$

Then,

$$\overline{\mathbf{T}}^{(\nu)}(\mathbf{x}) = R_n(g_{\mathbf{x}}, \text{Id}), \quad \mathbf{x} \in \mathcal{B}.$$

In Lemma SA-4.1, we provide a generic bound on the rate of Gaussian strong approximation for residual-based empirical process. This lemma generalizes Cattaneo and Yu (2025, Theorem 3) to allow for polynomial moment bound on the conditional distribution of  $Y_i$  given  $\mathbf{X}_i$ .

**Theorem SA-2.7 (Strong Approximation of  $\overline{\mathbf{T}}^{(\nu)}$ )**

Suppose Assumptions SA-1 and SA-2 hold. Suppose there exists a constant  $C > 0$  such that for  $t \in \{0, 1\}$  and for any  $\mathbf{x} \in \mathcal{B}$ , the De Giorgi perimeter of the set  $E_{t, \mathbf{x}} = \{\mathbf{y} \in \mathcal{A}_t : (\mathbf{y} - \mathbf{x})/h \in \text{Supp}(K)\}$  satisfies  $\mathcal{L}(E_{t, \mathbf{x}}) \leq Ch^{d-1}$ . Suppose  $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$  and  $nh^d \rightarrow \infty$  as  $n \rightarrow \infty$ . Then, on a possibly enlarged probability space, there exists a mean-zero Gaussian process  $Z^{(\nu)}$  indexed by  $\mathcal{B}$  with almost surely continuous sample path such that

$$\mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} \left| \overline{\mathbf{T}}^{(\nu)}(\mathbf{x}) - Z^{(\nu)}(\mathbf{x}) \right| \right] \lesssim (\log n)^{\frac{3}{2}} \left( \frac{1}{nh^d} \right)^{\frac{1}{2d+2} \cdot \frac{v}{v+2}} + \log(n) \left( \frac{1}{n^{\frac{v}{2+v}} h^d} \right)^{\frac{1}{2}},$$

where  $\lesssim$  is up to a universal constant, and  $Z^{(\nu)}$  has the same covariance structure as  $\overline{\mathbf{T}}^{(\nu)}$ ; that is,  $\text{Cov}[\overline{\mathbf{T}}^{(\nu)}(\mathbf{x}_1), \overline{\mathbf{T}}^{(\nu)}(\mathbf{x}_2)] = \text{Cov}[Z^{(\nu)}(\mathbf{x}_1), Z^{(\nu)}(\mathbf{x}_2)]$  for all  $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}$ .

For confidence bands, let  $\widehat{Z}^{(\nu)}(\mathbf{x})$ ,  $\mathbf{x} \in \mathcal{B}$ , be a mean-zero Gaussian process with feasible (conditional) covariance function given by

$$\text{Cov} \left[ \widehat{Z}^{(\nu)}(\mathbf{x}_1), \widehat{Z}^{(\nu)}(\mathbf{x}_2) \middle| \mathbf{W}_n \right] = \frac{\widehat{\Omega}_{\mathbf{x}_1, \mathbf{x}_2}^{(\nu)}}{\sqrt{\widehat{\Omega}_{\mathbf{x}_1, \mathbf{x}_1}^{(\nu)} \widehat{\Omega}_{\mathbf{x}_2, \mathbf{x}_2}^{(\nu)}}}, \quad \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}.$$

### Theorem SA-2.8 (Distributional Approximation for Suprema)

Suppose Assumptions SA-1 and SA-2 hold. Suppose  $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$ ,  $h^{p+1} \sqrt{nh^d} \rightarrow 0$  and  $\frac{n^{\frac{v}{2+v}} h^d}{(\log n)^3} \rightarrow \infty$ , then

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{\mathbf{T}}^{(\nu)}(\mathbf{x}) \right| \leq u \right) - \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{Z}^{(\nu)}(\mathbf{x}) \right| \leq u \mid \mathbf{W}_n \right) \right| = o_{\mathbb{P}}(1),$$

where  $\mathbf{W}_n = ((\mathbf{X}_1^\top, Y_1), \dots, (\mathbf{X}_n^\top, Y_n))^\top$ .

For any  $0 < \alpha < 1$ , define the confidence bands by

$$\widehat{\mathbf{I}}_\alpha^{(\nu)}(\mathbf{x}) = \left[ \widehat{\tau}^{(\nu)}(\mathbf{x}) - \mathbf{c}_\alpha \sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)}}, \widehat{\tau}^{(\nu)}(\mathbf{x}) + \mathbf{c}_\alpha \sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)}} \right], \quad \mathbf{x} \in \mathcal{B},$$

where  $\mathbf{c}_\alpha = \inf \left\{ c > 0 : \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{Z}^{(\nu)}(\mathbf{x}) \right| \geq c \mid \mathbf{W}_n \right) \leq \alpha \right\}$ .

### Theorem SA-2.9 (Confidence bands)

Suppose Assumptions SA-1 and SA-2 hold. Suppose  $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$ ,  $h^{p+1} \sqrt{nh^d} \rightarrow 0$  and  $\frac{n^{\frac{v}{2+v}} h^d}{(\log n)^3} \rightarrow \infty$ , then

$$\mathbb{P}[\mu^{(\nu)}(\mathbf{x}) \in \widehat{\mathbf{I}}_\alpha^{(\nu)}(\mathbf{x}), \forall \mathbf{x} \in \mathcal{B}] = 1 - \alpha - o(1).$$

## SA-3 Distance-Based Methods

The treatment effect curve estimator for  $(\tau(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$  is

$$\widehat{\tau}_{\text{dis}}(\mathbf{x}) = \widehat{\theta}_{1, \mathbf{x}}(0) - \widehat{\theta}_{0, \mathbf{x}}(0), \quad \mathbf{x} \in \mathcal{B},$$

where, for  $t \in \{0, 1\}$ ,  $\widehat{\theta}_{t, \mathbf{x}}(0) = \mathbf{e}_1^\top \widehat{\gamma}_t(\mathbf{x})$  with

$$\widehat{\gamma}_t(\mathbf{x}) = \underset{\gamma \in \mathbb{R}^{p+1}}{\text{argmin}} \mathbb{E}_n \left[ (Y_i - \mathbf{r}_p(D_i(\mathbf{x}))^\top \gamma)^2 k_h(D_i(\mathbf{x})) \mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x})) \right],$$

where the univariate distance score is

$$D_i(\mathbf{x}) = \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \mathcal{d}(\mathbf{X}_i, \mathbf{x}) - \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_0) \mathcal{d}(\mathbf{X}_i, \mathbf{x}), \quad \mathbf{x} \in \mathcal{B},$$

$\mathbf{r}_p(u) = (1, u, \dots, u^p)^\top$ ,  $k_h(u) = k(u/h)/h^2$  for a univariate kernel  $k(\cdot)$  and a bandwidth parameter  $h$ , and  $\mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x})) = \mathbb{1}(D_i(\mathbf{x}) \in \mathcal{J}_t)$  with  $\mathcal{J}_0 = (-\infty, 0)$  and  $\mathcal{J}_1 = [0, \infty)$ . More generally,

$$\widehat{\theta}_{t, \mathbf{x}}(D_i(\mathbf{x})) = \mathbf{r}_p(D_i(\mathbf{x}))^\top \widehat{\gamma}_t(\mathbf{x}), \quad t \in \{0, 1\}, \quad \mathbf{x} \in \mathcal{B}.$$

We impose the following assumptions on the distance function, kernel function, and assignment boundary.

### Assumption SA-3 (Regularity Conditions for Distance)

$\mathcal{d} : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$  is a metric on  $\mathbb{R}^d$  equivalent to the Euclidean distance, that is, there exists positive constants  $C_u$  and  $C_l$  such that  $C_l \|\mathbf{x} - \mathbf{x}'\| \leq \mathcal{d}(\mathbf{x}, \mathbf{x}') \leq C_u \|\mathbf{x} - \mathbf{x}'\|$  for all  $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$ .

### Assumption SA-4 (Kernel Function)

Let  $t \in \{0, 1\}$ .

- (i)  $k : \mathbb{R} \rightarrow [0, \infty)$  is compact supported and Lipschitz continuous, or  $k(u) = \mathbb{1}(u \in [-1, 1])$ .
- (ii)  $\liminf_{h \downarrow 0} \inf_{\mathbf{x} \in \mathcal{B}} \int_{\mathcal{A}_t} k_h(\mathcal{D}(\mathbf{u}, \mathbf{x})) d\mathbf{u} \gtrsim 1$ .

For each  $t \in \{0, 1\}$ , the induced conditional expectation based on univariate distance is

$$\theta_{t,\mathbf{x}}(r) = \mathbb{E}[Y_i | D_i(\mathbf{x}) = r] = \mathbb{E}[Y_i | \mathcal{D}(\mathbf{X}_i, \mathbf{x}) = |r|, \mathbf{X}_i \in \mathcal{A}_t], \quad r \in \mathcal{J}_t, \quad \mathbf{x} \in \mathcal{B}.$$

More rigorously, for each  $t \in \{0, 1\}$ , let  $S_{t,\mathbf{x}}(r) = \{\mathbf{v} \in \mathcal{X} : \mathcal{D}(\mathbf{v}, \mathbf{x}) = r, \mathbf{v} \in \mathcal{A}_t\}$  for  $r \geq 0$  and  $\mathbf{x} \in \mathcal{B}$ . Letting  $H_{d-1}$  denote the  $(d-1)$ -dimensional Hausdorff measure, then our definition means

$$\theta_{t,\mathbf{x}}(r) = \mathbb{E}[Y_i | \mathcal{D}(\mathbf{X}_i, \mathbf{x}) = |r|, \mathbf{X}_i \in \mathcal{A}_t] = \frac{\int_{S_{t,\mathbf{x}}(|r|)} \mu_t(\mathbf{v}) f_X(\mathbf{v}) H_{d-1}(d\mathbf{v})}{\int_{S_{t,\mathbf{x}}(|r|)} f_X(\mathbf{v}) H_{d-1}(d\mathbf{v})},$$

for  $|r| > 0, \mathbf{x} \in \mathcal{B}, t \in \{0, 1\}$ . For  $r = 0, \mathbf{x} \in \mathcal{B}, t \in \{0, 1\}$ , then

$$\theta_{t,\mathbf{x}}(0) = \lim_{r \rightarrow 0} \mathbb{E}[Y_i | \mathcal{D}(\mathbf{X}_i, \mathbf{x}) = |r|, \mathbf{X}_i \in \mathcal{A}_t] = \lim_{r \rightarrow 0} \frac{\int_{S_{t,\mathbf{x}}(|r|)} \mu_t(\mathbf{v}) f_X(\mathbf{v}) H_{d-1}(d\mathbf{v})}{\int_{S_{t,\mathbf{x}}(|r|)} f_X(\mathbf{v}) H_{d-1}(d\mathbf{v})}.$$

Under our assumptions, the above limit exists, and thus we obtain the following identification result.

**Lemma SA-3.1 (Distance-Based Identification)**

Suppose Assumption SA-1 (i)-(iii), and Assumption SA-3 hold. Then,  $\theta_{t,\mathbf{x}}(0) = \mu_t(\mathbf{x})$ , for all  $t \in \{0, 1\}$  and  $\mathbf{x} \in \mathcal{B}$ .

For  $t \in \{0, 1\}$ , define the best mean square approximation

$$\theta_{t,\mathbf{x}}^*(D_i(\mathbf{x})) = \mathbf{r}_p(D_i(\mathbf{x}))^\top \boldsymbol{\gamma}_t^*(\mathbf{x}),$$

where

$$\boldsymbol{\gamma}_t^*(\mathbf{x}) = \underset{\boldsymbol{\gamma} \in \mathbb{R}^{p+1}}{\operatorname{argmin}} \mathbb{E} \left[ (Y_i - \mathbf{r}_p(D_i(\mathbf{x}))^\top \boldsymbol{\gamma})^2 k_h(D_i(\mathbf{x})) \mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x})) \right].$$

The estimation error decomposes into *linear error*, *approximation error*, and *non-linear error*:

$$\begin{aligned} \hat{\theta}_{t,\mathbf{x}}(0) - \theta_{t,\mathbf{x}}(0) &= \mathbf{e}_1^\top \hat{\boldsymbol{\Psi}}_{t,\mathbf{x}}^{-1} \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) k_h(D_i(\mathbf{x})) Y_i \right] - \theta_{t,\mathbf{x}}(0) \\ &= \mathbf{e}_1^\top \hat{\boldsymbol{\Psi}}_{t,\mathbf{x}}^{-1} \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) k_h(D_i(\mathbf{x})) (Y_i - \theta_{t,\mathbf{x}}^*(D_i)) \right] + \theta_{t,\mathbf{x}}^*(0) - \theta_{t,\mathbf{x}}(0) \\ &= \underbrace{\mathbf{e}_1^\top \boldsymbol{\Psi}_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}}}_{\text{linear error}} + \underbrace{\theta_{t,\mathbf{x}}^*(0) - \theta_{t,\mathbf{x}}(0)}_{\text{approximation error}} + \underbrace{\mathbf{e}_1^\top (\hat{\boldsymbol{\Psi}}_{t,\mathbf{x}}^{-1} - \boldsymbol{\Psi}_{t,\mathbf{x}}^{-1}) \mathbf{O}_{t,\mathbf{x}}}_{\text{non-linear error}}, \end{aligned} \tag{SA-3.1}$$

for all  $t \in \{0, 1\}$  and  $\mathbf{x} \in \mathcal{B}$ , where

$$\begin{aligned}\widehat{\Psi}_{t,\mathbf{x}} &= \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right)^\top k_h(D_i(\mathbf{x})) \mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x})) \right], \\ \Psi_{t,\mathbf{x}} &= \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right)^\top k_h(D_i(\mathbf{x})) \mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x})) \right], \\ \mathbf{O}_{t,\mathbf{x}} &= \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) k_h(D_i(\mathbf{x})) (Y_i - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))) \mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x})) \right],\end{aligned}$$

and the misspecification bias is

$$\mathfrak{B}_{n,t}(\mathbf{x}) = \theta_{t,\mathbf{x}}^*(0) - \theta_{t,\mathbf{x}}(0). \quad (\text{SA-3.2})$$

In the main text,  $\mathfrak{B}_n(\mathbf{x}) = \mathfrak{B}_{n,1}(\mathbf{x}) - \mathfrak{B}_{n,0}(\mathbf{x})$ . Define the following for variance analysis: For  $t \in \{0, 1\}$ ,  $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}$ ,

$$\begin{aligned}\widehat{\Upsilon}_{t,\mathbf{x}_1,\mathbf{x}_2} &= h^d \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x}_1)}{h} \right) \mathbf{r}_p \left( \frac{D_i(\mathbf{x}_2)}{h} \right)^\top k_h(D_i(\mathbf{x}_1)) k_h(D_i(\mathbf{x}_2)) (Y_i - \widehat{\theta}_{t,\mathbf{x}_1}(D_i(\mathbf{x}_1))) \right. \\ &\quad \left. (Y_i - \widehat{\theta}_{t,\mathbf{x}_2}(D_i(\mathbf{x}_2))) \mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x}_1)) \right], \\ \Upsilon_{t,\mathbf{x}_1,\mathbf{x}_2} &= h^d \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x}_1)}{h} \right) \mathbf{r}_p \left( \frac{D_i(\mathbf{x}_2)}{h} \right)^\top k_h(D_i(\mathbf{x}_1)) k_h(D_i(\mathbf{x}_2)) (Y_i - \theta_{t,\mathbf{x}_1}^*(D_i(\mathbf{x}_1))) \right. \\ &\quad \left. (Y_i - \theta_{t,\mathbf{x}_2}^*(D_i(\mathbf{x}_2))) \mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x}_1)) \right], \\ \widehat{\Xi}_{t,\mathbf{x}_1,\mathbf{x}_2} &= \frac{1}{nh^d} \mathbf{e}_1^\top \widehat{\Psi}_{t,\mathbf{x}_1}^{-1} \widehat{\Upsilon}_{t,\mathbf{x}_1,\mathbf{x}_2} \widehat{\Psi}_{t,\mathbf{x}_2}^{-1} \mathbf{e}_1, & \widehat{\Xi}_{\mathbf{x}_1,\mathbf{x}_2} &= \widehat{\Xi}_{0,\mathbf{x}_1,\mathbf{x}_2} + \widehat{\Xi}_{1,\mathbf{x}_1,\mathbf{x}_2}, \\ \Xi_{t,\mathbf{x}_1,\mathbf{x}_2} &= \frac{1}{nh^d} \mathbf{e}_1^\top \Psi_{t,\mathbf{x}_1}^{-1} \Upsilon_{t,\mathbf{x}_1,\mathbf{x}_2} \Psi_{t,\mathbf{x}_2}^{-1} \mathbf{e}_1, & \Xi_{\mathbf{x}_1,\mathbf{x}_2} &= \Xi_{0,\mathbf{x}_1,\mathbf{x}_2} + \Xi_{1,\mathbf{x}_1,\mathbf{x}_2}.\end{aligned}$$

### SA-3.1 Preliminary Lemmas

#### Lemma SA-3.2 (Gram)

Suppose Assumption SA-1 (i)(ii), Assumption SA-3 and Assumption SA-4 hold. If  $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ , then

$$\begin{aligned}\sup_{x \in \mathcal{B}} \|\widehat{\Psi}_{t,\mathbf{x}} - \Psi_{t,\mathbf{x}}\| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}, \\ 1 &\lesssim_{\mathbb{P}} \inf_{\mathbf{x} \in \mathcal{B}} \|\widehat{\Psi}_{t,\mathbf{x}}\| \lesssim \sup_{\mathbf{x} \in \mathcal{B}} \|\widehat{\Psi}_{t,\mathbf{x}}\| \lesssim_{\mathbb{P}} 1, \\ \sup_{\mathbf{x} \in \mathcal{B}} \|\widehat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1}\| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}},\end{aligned}$$

for  $t \in \{0, 1\}$ .

#### Lemma SA-3.3 (Stochastic Linear Approximation)

Suppose Assumption SA-1 (i)(ii)(iii)(v), Assumption SA-3 and Assumption SA-4 hold. If  $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ ,

then

$$\begin{aligned} \sup_{\mathbf{x} \in \mathcal{B}} \|\mathbf{O}_{t,\mathbf{x}}\| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d}, \\ \sup_{\mathbf{x} \in \mathcal{B}} |\mathbf{e}_1^\top \boldsymbol{\Psi}_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}}| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d}, \\ \sup_{\mathbf{x} \in \mathcal{B}} |\mathbf{e}_1^\top (\hat{\boldsymbol{\Psi}}_{t,\mathbf{x}}^{-1} - \boldsymbol{\Psi}_{t,\mathbf{x}}^{-1}) \mathbf{O}_{t,\mathbf{x}}| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right), \end{aligned}$$

for  $t \in \{0, 1\}$ .

**Lemma SA-3.4 (Approximation Error: Minimal Guarantee)**

Suppose Assumption SA-1 (i)(ii)(iii), Assumption SA-3 and Assumption SA-4 hold. Then,

$$\sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_n(\mathbf{x})| \lesssim h.$$

**Lemma SA-3.5 (Covariance)**

Suppose Assumptions SA-1, SA-3 and SA-4 hold. If  $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ , then

$$\begin{aligned} \max_{t \in \{0,1\}} \sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}} \|\hat{\boldsymbol{\Upsilon}}_{t,\mathbf{x}_1,\mathbf{x}_2} - \boldsymbol{\Upsilon}_{t,\mathbf{x}_1,\mathbf{x}_2}\| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}, \\ \max_{t \in \{0,1\}} \sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}} nh^d |\hat{\Xi}_{t,\mathbf{x}_1,\mathbf{x}_2} - \Xi_{t,\mathbf{x}_1,\mathbf{x}_2}| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}. \end{aligned}$$

If, in addition,  $\frac{n^{\frac{v}{2+v}} h^d}{\log(1/h)} \rightarrow \infty$ , then

$$\begin{aligned} \inf_{\mathbf{x} \in \mathcal{B}} \lambda_{\min}(\hat{\boldsymbol{\Upsilon}}_{t,\mathbf{x},\mathbf{x}}) &\gtrsim_{\mathbb{P}} 1, \\ \inf_{\mathbf{x} \in \mathcal{B}} \hat{\Xi}_{t,\mathbf{x},\mathbf{x}} &\gtrsim_{\mathbb{P}} (nh^d)^{-1}, \\ \sup_{\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}} \left| \frac{\hat{\Xi}_{t,\mathbf{x}_1,\mathbf{x}_2}}{\sqrt{\hat{\Xi}_{t,\mathbf{x}_1,\mathbf{x}_2} \hat{\Xi}_{t,\mathbf{x}_2,\mathbf{x}_2}}} - \frac{\Xi_{t,\mathbf{x}_1,\mathbf{x}_2}}{\sqrt{\Xi_{t,\mathbf{x}_2,\mathbf{x}_2} \Xi_{t,\mathbf{x}_2,\mathbf{x}_2}}} \right| &\lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}. \end{aligned}$$

Since we consider a covariance estimator based on the best linear approximation, instead of the population conditional mean functions, no bias condition appears in the estimates above.

## SA-3.2 Pointwise Inference

**Theorem SA-3.1 (Convergence Rate)**

Suppose Assumptions SA-1, SA-3 and SA-4 hold. If  $nh^d \rightarrow \infty$ , then

$$|\hat{\tau}_{\text{dis}}(\mathbf{x}) - \tau(\mathbf{x})| \lesssim_{\mathbb{P}} \frac{1}{\sqrt{nh^d}} + \frac{1}{n^{\frac{1+v}{2+v}} h^d} + |\mathfrak{B}_n(\mathbf{x})|,$$

for all  $\mathbf{x} \in \mathcal{B}$ .

Define the feasible t-statistics by

$$\hat{T}_{\text{dis}}(\mathbf{x}) = \frac{\hat{\tau}_{\text{dis}}(\mathbf{x}) - \tau(\mathbf{x})}{\sqrt{\hat{\Xi}_{\mathbf{x},\mathbf{x}}}}, \quad \mathbf{x} \in \mathcal{B}.$$

**Theorem SA-3.2 (Asymptotic Normality)**

Suppose Assumptions SA-1, SA-3 and SA-4 hold. If  $n^{\frac{v}{2+v}} h^d \rightarrow \infty$  and  $\sqrt{nh^d} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_n(\mathbf{x})| \rightarrow 0$ , then

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left( \hat{T}_{\text{dis}}(\mathbf{x}) \leq u \right) - \Phi(u) \right| = o(1), \quad \forall \mathbf{x} \in \mathcal{B}.$$

For any  $0 < \alpha < 1$ , take  $\mathbf{c}_\alpha = \inf \{c > 0 : \mathbb{P}(|Z| \geq c) \leq \alpha\}$  where  $Z \sim N(0, 1)$ , and define  $\hat{I}_{\text{dis}}(\mathbf{x}, \alpha) = \left( \hat{\tau}_{\text{dis}}(\mathbf{x}) - \mathbf{c}_\alpha \sqrt{\hat{\Xi}_{\mathbf{x},\mathbf{x}}}, \hat{\tau}_{\text{dis}}(\mathbf{x}) + \mathbf{c}_\alpha \sqrt{\hat{\Xi}_{\mathbf{x},\mathbf{x}}} \right)$ . Then,

$$\mathbb{P} \left( \tau(\mathbf{x}) \in \hat{I}_{\text{dis}}(\mathbf{x}, \alpha) \right) \rightarrow 1 - \alpha, \quad \mathbf{x} \in \mathcal{B}.$$

**SA-3.3 Uniform Inference**

**Theorem SA-3.3 (Uniform Convergence Rate)**

Suppose Assumptions SA-1, SA-3 and SA-4 hold. If  $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ , then

$$\sup_{\mathbf{x} \in \mathcal{B}} |\hat{\tau}_{\text{dis}}(\mathbf{x}) - \tau(\mathbf{x})| \lesssim \mathbb{P} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} + \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_n(\mathbf{x})|.$$

Define  $\bar{T}_{\text{dis}}(\mathbf{x})$  to be the stochastic linearization of  $\hat{T}_{\text{dis}}(\mathbf{x})$ , that is, we define

$$\bar{T}_{\text{dis}}(\mathbf{x}) = \Xi_{\mathbf{x},\mathbf{x}}^{-1/2} (\mathbf{e}_1^\top \Psi_{1,\mathbf{x}}^{-1} \mathbf{O}_{1,\mathbf{x}} - \mathbf{e}_1^\top \Psi_{0,\mathbf{x}}^{-1} \mathbf{O}_{0,\mathbf{x}}), \quad \mathbf{x} \in \mathcal{B}.$$

**Theorem SA-3.4 (Stochastic Linearization)**

Suppose Assumptions SA-1, SA-3 and SA-4 hold. Suppose  $\frac{nh^d}{\log(1/h)} \rightarrow \infty$ . Then,

$$\sup_{\mathbf{x} \in \mathcal{B}} \left| \hat{T}_{\text{dis}}(\mathbf{x}) - \bar{T}_{\text{dis}}(\mathbf{x}) \right| \lesssim \mathbb{P} \sqrt{\log(1/h)} \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right) + \sqrt{nh^d} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_n(\mathbf{x})|.$$

To establish a Gaussian strong approximation for  $\bar{T}_{\text{dis}}(\mathbf{x})$ , consider the class of functions  $\mathcal{G} = \{g_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$  and  $\mathcal{H} = \{h_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$ , where

$$\begin{aligned} g_{\mathbf{x}}(\mathbf{u}) &= \mathbb{1}_{\mathcal{A}_1}(\mathbf{u}) \mathfrak{K}_1(\mathbf{u}; \mathbf{x}) - \mathbb{1}_{\mathcal{A}_0}(\mathbf{u}) \mathfrak{K}_0(\mathbf{u}; \mathbf{x}), & \mathbf{u} \in \mathcal{X}, \\ \mathfrak{K}_t(\mathbf{u}; \mathbf{x}) &= \frac{1}{\sqrt{n \Xi_{\mathbf{x},\mathbf{x}}}} \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{r}_p \left( \frac{\mathcal{d}(\mathbf{u}, \mathbf{x})}{h} \right) k_h(\mathcal{d}(\mathbf{u}, \mathbf{x})), & \mathbf{u} \in \mathcal{X}, \mathbf{x} \in \mathcal{B}, t \in \{0, 1\}, \\ h_{\mathbf{x}}(\mathbf{u}) &= -\mathbb{1}_{\mathcal{A}_1}(\mathbf{u}) \mathfrak{K}_1(\mathbf{u}; \mathbf{x}) \theta_{1,\mathbf{x}}^*(\mathcal{d}(\mathbf{u}, \mathbf{x})) + \mathbb{1}_{\mathcal{A}_0}(\mathbf{u}) \mathfrak{K}_0(\mathbf{u}; \mathbf{x}) \theta_{0,\mathbf{x}}^*(\mathcal{d}(\mathbf{u}, \mathbf{x})), & \mathbf{u} \in \mathcal{X}, \mathbf{x} \in \mathcal{B}, \end{aligned} \quad (\text{SA-3.3})$$

and  $\mathcal{R}$  is the singleton of identity function  $\text{Id} : \mathbb{R} \mapsto \mathbb{R}$ ,  $\text{Id}(x) = x$ . For classes of functions  $\mathcal{G}, \mathcal{H}$  from  $\mathbb{R}^d$  to

$\mathbb{R}$  and  $\mathcal{R}$  from  $\mathbb{R}$  to  $\mathbb{R}$ . Then, for  $\mathbf{x} \in \mathcal{B}$ ,  $\bar{T}_{\text{dis}}(\mathbf{x})$  can be represented by

$$\bar{T}_{\text{dis}}(\mathbf{x}) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[ g_{\mathbf{x}}(\mathbf{X}_i) \text{Id}(y_i) + h_{\mathbf{x}}(\mathbf{X}_i) - \mathbb{E}[g_{\mathbf{x}}(\mathbf{X}_i) \text{Id}(y_i) + h_{\mathbf{x}}(\mathbf{X}_i)] \right].$$

Define the multiplicative separable empirical processes by

$$M_n(g, r) = \frac{1}{\sqrt{n}} \sum_{i=1}^n [g(\mathbf{x}_i)r(y_i) - \mathbb{E}[g(\mathbf{x}_i)r(y_i)]], \quad g \in \mathcal{G}, r \in \mathcal{R}.$$

Then,  $\bar{T}_{\text{dis}}(\mathbf{x})$  has the representation

$$\bar{T}_{\text{dis}}(\mathbf{x}) = M_n(g_{\mathbf{x}}, \text{Id}) + M_n(h_{\mathbf{x}}, 1), \quad \mathbf{x} \in \mathcal{B}.$$

In Lemma SA-4.2, we give upper bounds for Gaussian strong approximation of *additive empirical process* of the form  $(M_n(g, r) + M_n(h, s) : g \in \mathcal{G}, r \in \mathcal{R}, h \in \mathcal{H}, s \in \mathcal{S})$ . Since upper bounds for empirical processes of the form  $(M_n(g, r) : g \in \mathcal{G}, r \in \mathcal{R})$  has already been studied in (Cattaneo and Yu, 2025, Theorem SA.1), Lemma SA-4.2 is given as its simple extension, considering the worse case between  $\mathcal{G}$  and  $\mathcal{H}$ , and between  $\mathcal{R}$  and  $\mathcal{S}$ . Applying Lemma SA-4.2, we get the following theorem on Gaussian strong approximation of  $(\bar{T}_{\text{dis}}(\mathbf{x}) : \mathbf{x} \in \mathcal{B})$ .

**Theorem SA-3.5 (Strong Approximation of t-statistics)**

Suppose Assumption SA-1, SA-3 and SA-4 hold. Suppose there exists a constant  $C > 0$  such that for  $t \in \{0, 1\}$  and for any  $\mathbf{x} \in \mathcal{B}$ , the De Giorgi perimeter of the set  $E_{t, \mathbf{x}} = \{\mathbf{y} \in \mathcal{A}_t : (\mathbf{y} - \mathbf{x})/h \in \text{Supp}(K)\}$  satisfies  $\mathcal{L}(E_{t, \mathbf{x}}) \leq Ch^{d-1}$ . Suppose  $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$  and  $nh^d \rightarrow \infty$  as  $n \rightarrow \infty$ . Then, on a possibly enlarged probability space there exists a mean-zero Gaussian process  $z$  indexed by  $\mathcal{B}$  with almost surely continuous sample path such that

$$\mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}_{\text{dis}}(\mathbf{x}) - z(\mathbf{x})| \right] \lesssim (\log n)^{\frac{3}{2}} \left( \frac{1}{nh^d} \right)^{\frac{1}{2d+2} \cdot \frac{v}{v+2}} + \log(n) \left( \frac{1}{n^{\frac{v}{2+v}} h^d} \right)^{\frac{1}{2}},$$

where  $\lesssim$  is up to a universal constant. Moreover,  $z$  has the same covariance structure as  $\bar{T}_{\text{dis}}$ , that is,

$$\text{Cov} [\bar{T}_{\text{dis}}(\mathbf{x}_1), \bar{T}_{\text{dis}}(\mathbf{x}_2)] = \text{Cov} [z(\mathbf{x}_1), z(\mathbf{x}_2)], \quad \forall \mathbf{x}, \mathbf{y} \in \mathcal{B}.$$

**Theorem SA-3.6 (Confidence Bands)**

Suppose Assumption SA-1, SA-3 and SA-4 hold. Suppose  $\liminf_{n \rightarrow \infty} \frac{\log h}{\log n} > -\infty$ ,  $\frac{n^{\frac{v}{2+v}} h^d}{(\log n)^3} \rightarrow \infty$ , and  $\sqrt{nh^d} \sup_{\mathbf{x} \in \mathcal{B}} \sum_{t \in \{0, 1\}} |\mathfrak{B}_{n,t}(\mathbf{x})| \rightarrow 0$ . Suppose  $\hat{z}$  is a mean-zero Gaussian process indexed by  $\mathcal{B}$  s.t.

$$\text{Cov} [\hat{z}(\mathbf{x}_1), \hat{z}(\mathbf{x}_2)] = \frac{\hat{\Xi}_{\mathbf{x}_1, \mathbf{x}_2}}{\sqrt{\hat{\Xi}_{\mathbf{x}_1, \mathbf{x}_1} \hat{\Xi}_{\mathbf{x}_2, \mathbf{x}_2}}}, \quad \mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}.$$

Let  $\mathcal{U}_n$  be the  $\sigma$ -algebra generated by  $((Y_i, (D_i(\mathbf{x}) : \mathbf{x} \in \mathcal{B})) : 1 \leq i \leq n)$ . Then

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\hat{T}_{\text{dis}}(\mathbf{x})| \leq u \right) - \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\hat{z}(\mathbf{x})| \leq u \middle| \mathcal{U}_n \right) \right| = o_{\mathbb{P}}(1).$$

For any  $0 < \alpha < 1$ , if we define  $\mathbf{c}_\alpha = \inf \{c > 0 : \mathbb{P}(\sup_{\mathbf{x} \in \mathcal{B}} |\hat{z}(\mathbf{x})| \geq c|\mathcal{U}_n) \leq \alpha\}$  and define  $\hat{\mathbf{I}}_\alpha(\mathbf{x}) = \left( \hat{\tau}_{\text{dis}}(\mathbf{x}) - \mathbf{c}_\alpha \sqrt{\hat{\Xi}_{\mathbf{x}, \mathbf{x}}}, \hat{\tau}_{\text{dis}}(\mathbf{x}) + \mathbf{c}_\alpha \sqrt{\hat{\Xi}_{\mathbf{x}, \mathbf{x}}} \right)$  for all  $\mathbf{x} \in \mathcal{B}$ , then

$$\mathbb{P} \left( \tau(\mathbf{x}) \in \hat{\mathbf{I}}_\alpha(\mathbf{x}), \forall \mathbf{x} \in \mathcal{B} \right) = 1 - \alpha - o(1).$$

## SA-4 Gaussian Strong Approximation Lemmas

We present two Gaussian strong approximation lemmas that are the key technical tools behind Theorem SA-2.7 and Theorem SA-3.5, building on and generalizing the results in Cattaneo and Yu (2025). Consider the *residual-based empirical process* given by

$$R_n[g, r] = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[ g(\mathbf{x}_i) r(y_i) - \mathbb{E}[g(\mathbf{x}_i) r(y_i) | \mathbf{x}_i] \right], \quad g \in \mathcal{G}, r \in \mathcal{R},$$

where  $\mathcal{G}$  and  $\mathcal{R}$  are classes of functions satisfying certain regularity conditions. In addition, consider the *multiplicative-separable empirical process* given by

$$M_n[g, r] = \frac{1}{\sqrt{n}} \sum_{i=1}^n \left[ g(\mathbf{x}_i) r(y_i) - \mathbb{E}[g(\mathbf{x}_i) r(y_i)] \right], \quad g \in \mathcal{G}, r \in \mathcal{R}.$$

### SA-4.1 Definitions for Function Spaces

Let  $\mathcal{F}$  be a class of measurable functions from a probability space  $(\mathbb{R}^q, \mathcal{B}(\mathbb{R}^q), \mathbb{P})$  to  $\mathbb{R}$ . We introduce several definitions that capture properties of  $\mathcal{F}$ .

- (i)  $\mathcal{F}$  is pointwise measurable if it contains a countable subset  $\mathcal{G}$  such that for any  $f \in \mathcal{F}$ , there exists a sequence  $(g_m : m \geq 1) \subseteq \mathcal{G}$  such that  $\lim_{m \rightarrow \infty} g_m(\mathbf{u}) = f(\mathbf{u})$  for all  $\mathbf{u} \in \mathbb{R}^q$ .
- (ii) Let  $\text{Supp}(\mathcal{F}) = \cup_{f \in \mathcal{F}} \text{Supp}(f)$ . A probability measure  $\mathbb{Q}_{\mathcal{F}}$  on  $(\mathbb{R}^q, \mathcal{B}(\mathbb{R}^q))$  is a surrogate measure for  $\mathbb{P}$  with respect to  $\mathcal{F}$  if

(i)  $\mathbb{Q}_{\mathcal{F}}$  agrees with  $\mathbb{P}$  on  $\text{Supp}(\mathbb{P}) \cap \text{Supp}(\mathcal{F})$ .

(ii)  $\mathbb{Q}_{\mathcal{F}}(\text{Supp}(\mathcal{F}) \setminus \text{Supp}(\mathbb{P})) = 0$ .

Let  $\mathcal{Q}_{\mathcal{F}} = \text{Supp}(\mathbb{Q}_{\mathcal{F}})$ .

- (iii) For  $q = 1$  and an interval  $\mathcal{J} \subseteq \mathbb{R}$ , the pointwise total variation of  $\mathcal{F}$  over  $\mathcal{J}$  is

$$\text{pTV}_{\mathcal{F}, \mathcal{J}} = \sup_{f \in \mathcal{F}} \sup_{P \geq 1} \sup_{\mathcal{P}_P \in \mathcal{J}} \sum_{i=1}^{P-1} |f(a_{i+1}) - f(a_i)|,$$

where  $\mathcal{P}_P = \{(a_1, \dots, a_P) : a_1 \leq \dots \leq a_P\}$  denotes the collection of all partitions of  $\mathcal{J}$ .

- (iv) For a non-empty  $\mathcal{C} \subseteq \mathbb{R}^q$ , the total variation of  $\mathcal{F}$  over  $\mathcal{C}$  is

$$\text{TV}_{\mathcal{F}, \mathcal{C}} = \inf_{\mathcal{U} \in \mathcal{O}(\mathcal{C})} \sup_{f \in \mathcal{F}} \sup_{\phi \in \mathcal{D}_q(\mathcal{U})} \int_{\mathbb{R}^q} f(\mathbf{u}) \text{div}(\phi)(\mathbf{u}) d\mathbf{u} / \|\phi\|_2, \infty,$$

where  $\mathcal{O}(\mathcal{C})$  denotes the collection of all open sets that contains  $\mathcal{C}$ , and  $\mathcal{D}_q(\mathcal{U})$  denotes the space of infinitely differentiable functions from  $\mathbb{R}^q$  to  $\mathbb{R}^q$  with compact support contained in  $\mathcal{U}$ .

- (v) For a non-empty  $\mathcal{C} \subseteq \mathbb{R}^q$ , the local total variation constant of  $\mathcal{F}$  over  $\mathcal{C}$ , is a positive number  $K_{\mathcal{F},\mathcal{C}}$  such that for any cube  $\mathcal{D} \subseteq \mathbb{R}^q$  with edges of length  $\ell$  parallel to the coordinate axes,

$$\text{TV}_{\mathcal{F},\mathcal{D} \cap \mathcal{C}} \leq K_{\mathcal{F},\mathcal{C}} \ell^{d-1}.$$

- (vi) For a non-empty  $\mathcal{C} \subseteq \mathbb{R}^q$ , the envelopes of  $\mathcal{F}$  over  $\mathcal{C}$  are

$$M_{\mathcal{F},\mathcal{C}} = \sup_{\mathbf{u} \in \mathcal{C}} M_{\mathcal{F},\mathcal{C}}(\mathbf{u}), \quad M_{\mathcal{F},\mathcal{C}}(\mathbf{u}) = \sup_{f \in \mathcal{F}} |f(\mathbf{u})|, \quad \mathbf{u} \in \mathcal{C}.$$

- (vii) For a non-empty  $\mathcal{C} \subseteq \mathbb{R}^q$ , the Lipschitz constant of  $\mathcal{F}$  over  $\mathcal{C}$  is

$$L_{\mathcal{F},\mathcal{C}} = \sup_{f \in \mathcal{F}} \sup_{\mathbf{u}_1, \mathbf{u}_2 \in \mathcal{C}} \frac{|f(\mathbf{u}_1) - f(\mathbf{u}_2)|}{\|\mathbf{u}_1 - \mathbf{u}_2\|_\infty}.$$

- (viii) For a non-empty  $\mathcal{C} \subseteq \mathbb{R}^q$ , the  $L_1$  bound of  $\mathcal{F}$  over  $\mathcal{C}$  is

$$E_{\mathcal{F},\mathcal{C}} = \sup_{f \in \mathcal{F}} \int_{\mathcal{C}} |f| d\mathbb{P}.$$

- (ix) For a non-empty  $\mathcal{C} \subseteq \mathbb{R}^q$ , the uniform covering number of  $\mathcal{F}$  with envelope  $M_{\mathcal{F},\mathcal{C}}$  over  $\mathcal{C}$  is

$$N_{\mathcal{F},\mathcal{C}}(\delta, M_{\mathcal{F},\mathcal{C}}) = \sup_{\mu} N(\mathcal{F}, \|\cdot\|_{\mu,2}, \delta \|M_{\mathcal{F},\mathcal{C}}\|_{\mu,2}), \quad \delta \in (0, \infty),$$

where the supremum is taken over all finite discrete measures on  $(\mathcal{C}, \mathcal{B}(\mathcal{C}))$ . We assume that  $M_{\mathcal{F},\mathcal{C}}(\mathbf{u})$  is finite for every  $\mathbf{u} \in \mathcal{C}$ .

- (x) For a non-empty  $\mathcal{C} \subseteq \mathbb{R}^q$ , the uniform entropy integral of  $\mathcal{F}$  with envelope  $M_{\mathcal{F},\mathcal{C}}$  over  $\mathcal{C}$  is

$$J_{\mathcal{C}}(\delta, \mathcal{F}, M_{\mathcal{F},\mathcal{C}}) = \int_0^\delta \sqrt{1 + \log N_{\mathcal{F},\mathcal{C}}(\varepsilon, M_{\mathcal{F},\mathcal{C}})} d\varepsilon,$$

where it is assumed that  $M_{\mathcal{F},\mathcal{C}}(\mathbf{u})$  is finite for every  $\mathbf{u} \in \mathcal{C}$ .

- (xi) For a non-empty  $\mathcal{C} \subseteq \mathbb{R}^q$ ,  $\mathcal{F}$  is a VC-type class with envelope  $M_{\mathcal{F},\mathcal{C}}$  over  $\mathcal{C}$  if (i)  $M_{\mathcal{F},\mathcal{C}}$  is measurable and  $M_{\mathcal{F},\mathcal{C}}(\mathbf{u})$  is finite for every  $\mathbf{u} \in \mathcal{C}$ , and (ii) there exist  $c_{\mathcal{F},\mathcal{C}} > 0$  and  $d_{\mathcal{F},\mathcal{C}} > 0$  such that

$$N_{\mathcal{F},\mathcal{C}}(\varepsilon, M_{\mathcal{F},\mathcal{C}}) \leq c_{\mathcal{F},\mathcal{C}} \varepsilon^{-d_{\mathcal{F},\mathcal{C}}}, \quad \varepsilon \in (0, 1).$$

If a surrogate measure  $\mathbb{Q}_{\mathcal{F}}$  for  $\mathbb{P}$  with respect to  $\mathcal{F}$  has been assumed, and it is clear from the context, we drop the dependence on  $\mathcal{C} = \mathcal{Q}_{\mathcal{F}}$  for all quantities in the previous definitions. That is, to save notation, we set  $\text{TV}_{\mathcal{F}} = \text{TV}_{\mathcal{F},\mathcal{Q}_{\mathcal{F}}}$ ,  $K_{\mathcal{F}} = K_{\mathcal{F},\mathcal{Q}_{\mathcal{F}}}$ ,  $M_{\mathcal{F}} = M_{\mathcal{F},\mathcal{Q}_{\mathcal{F}}}$ ,  $M_{\mathcal{F}}(\mathbf{u}) = M_{\mathcal{F},\mathcal{Q}_{\mathcal{F}}}(\mathbf{u})$ ,  $L_{\mathcal{F}} = L_{\mathcal{F},\mathcal{Q}_{\mathcal{F}}}$ , and so on, whenever there is no confusion.

## SA-4.2 Residual-based Empirical Process

The following Lemma SA-4.1 generalizes Cattaneo and Yu (2025, Theorem 2) by allowing  $y_i$  to have bounded moments conditional on  $\mathbf{x}_i$ .

### Lemma SA-4.1 (Strong Approximation for Residual-based Empirical Processes)

Suppose  $(\mathbf{z}_i = (\mathbf{x}_i, y_i) : 1 \leq i \leq n)$  are i.i.d. random vectors taking values in  $(\mathbb{R}^{d+1}, \mathcal{B}(\mathbb{R}^{d+1}))$  with common law  $\mathbb{P}_Z$ , where  $\mathbf{x}_i$  has distribution  $\mathbb{P}_X$  supported on  $\mathcal{X} \subseteq \mathbb{R}^d$ ,  $y_i$  has distribution  $\mathbb{P}_Y$  supported on  $\mathcal{Y} \subseteq \mathbb{R}$ ,  $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[|y_i|^{2+v} | \mathbf{x}_i = \mathbf{x}] \leq 2$  for some  $v > 0$ , and the following conditions hold.

- (i)  $\mathcal{G}$  is a real-valued pointwise measurable class of functions on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mathbb{P}_X)$ .
- (ii) There exists a surrogate measure  $\mathbb{Q}_{\mathcal{G}}$  for  $\mathbb{P}_X$  with respect to  $\mathcal{G}$  such that  $\mathbb{Q}_{\mathcal{G}} = \mathbf{m} \circ \phi_{\mathcal{G}}$ , where the normalizing transformation  $\phi_{\mathcal{G}} : \mathbb{Q}_{\mathcal{G}} \mapsto [0, 1]^d$  is a diffeomorphism.
- (iii)  $\mathcal{G}$  is a VC-type class with envelope  $\mathbb{M}_{\mathcal{G}}$  over  $\mathbb{Q}_{\mathcal{G}}$  with  $\mathbf{c}_{\mathcal{G}} \geq e$  and  $\mathbf{d}_{\mathcal{G}} \geq 1$ .
- (iv)  $\mathcal{R}$  is a real-valued pointwise measurable class of functions on  $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_Y)$ .
- (v)  $\mathcal{R}$  is a VC-type class with envelope  $M_{\mathcal{R}, \mathcal{Y}}$  over  $\mathcal{Y}$  with  $\mathbf{c}_{\mathcal{R}, \mathcal{Y}} \geq e$  and  $\mathbf{d}_{\mathcal{R}, \mathcal{Y}} \geq 1$ , where  $M_{\mathcal{R}, \mathcal{Y}}(y) + \mathbf{p} \text{TV}_{\mathcal{R}, (-|y|, |y|)} \leq \mathbf{v}(1 + |y|)$  for all  $y \in \mathcal{Y}$ , for some  $\mathbf{v} > 0$ .
- (vi) There exists a constant  $\mathbf{k}$  such that  $|\log_2 \mathbf{E}_{\mathcal{G}}| + |\log_2 \text{TV}| + |\log_2 \mathbb{M}_{\mathcal{G}}| \leq \mathbf{k} \log_2 n$ , where  $\text{TV} = \max\{\text{TV}_{\mathcal{G}}, \text{TV}_{\mathcal{G} \times \mathcal{V}_{\mathcal{R}}, \mathbb{Q}_{\mathcal{G}}}\}$  with  $\mathcal{V}_{\mathcal{R}} = \{\theta(\cdot, r) : r \in \mathcal{R}\}$ , and  $\theta(\mathbf{x}, r) = \mathbb{E}[r(y_i) | \mathbf{x}_i = \mathbf{x}]$ .

Define the residual based empirical process

$$R_n(g, r) = \frac{1}{\sqrt{n}} \sum_{i=1}^n g(\mathbf{x}_i)(r(y_i) - \mathbb{E}[r(y_i) | \mathbf{x}_i]), \quad g \in \mathcal{G}, r \in \mathcal{R}.$$

Then, on a possibly enlarged probability space, there exists a sequence of mean-zero Gaussian processes  $(Z_n^R(g, r) : g \in \mathcal{G}, r \in \mathcal{R})$  with almost sure continuous trajectories such that:

- $\mathbb{E}[R_n(g_1, r_1)R_n(g_2, r_2)] = \mathbb{E}[Z_n^R(g_1, r_1)Z_n^R(g_2, r_2)]$  for all  $(g_1, r_1), (g_2, r_2) \in \mathcal{G} \times \mathcal{R}$ , and
- $\mathbb{E}[\|R_n - Z_n^R\|_{\mathcal{G} \times \mathcal{R}}] \leq C\mathbf{v}((\mathbf{d} \log(\mathbf{c}n))^{\frac{3}{2}} \mathbf{r}_n^{\frac{\mathbf{v}}{\mathbf{v}+2}} (\sqrt{\mathbb{M}_{\mathcal{G}} \mathbf{E}_{\mathcal{G}}})^{\frac{2}{\mathbf{v}+2}} + \mathbf{d} \log(\mathbf{c}n) \mathbb{M}_{\mathcal{G}} n^{-\frac{\mathbf{v}/2}{2+\mathbf{v}}} + \mathbf{d} \log(\mathbf{c}n) \mathbb{M}_{\mathcal{G}} n^{-\frac{1}{2}} \left( \frac{\sqrt{\mathbb{M}_{\mathcal{G}} \mathbf{E}_{\mathcal{G}}}}{\mathbf{r}_n} \right)^{\frac{2}{\mathbf{v}+2}}),$

where  $C$  is a universal constant,  $\mathbf{c} = \mathbf{c}_{\mathcal{G}} + \mathbf{c}_{\mathcal{R}, \mathcal{Y}} + \mathbf{k}$ ,  $\mathbf{d} = \mathbf{d}_{\mathcal{G}} \mathbf{d}_{\mathcal{R}, \mathcal{Y}} \mathbf{k}$ , and

$$\mathbf{r}_n = \min \left\{ \frac{(\mathbf{c}_1^d \mathbb{M}_{\mathcal{G}}^{d+1} \text{TV}^d \mathbf{E}_{\mathcal{G}})^{1/(2d+2)}}{n^{1/(2d+2)}}, \frac{(\mathbf{c}_1^{d/2} \mathbf{c}_2^{d/2} \mathbb{M}_{\mathcal{G}} \text{TV}^{d/2} \mathbf{E}_{\mathcal{G}} \mathbf{L}^{d/2})^{1/(d+2)}}{n^{1/(d+2)}} \right\},$$

$$\mathbf{c}_1 = d \sup_{\mathbf{x} \in \mathcal{Q}_{\mathcal{G}}} \prod_{j=1}^{d-1} \sigma_j(\nabla \phi_{\mathcal{G}}(\mathbf{x})), \quad \mathbf{c}_2 = \sup_{\mathbf{x} \in \mathcal{Q}_{\mathcal{G}}} \frac{1}{\sigma_d(\nabla \phi_{\mathcal{G}}(\mathbf{x}))}.$$

## SA-4.3 Multiplicative-Separable Empirical Process

The following Lemma SA-4.2 generalizes Cattaneo and Yu (2025, Theorem SA.1) by allowing  $y_i$  to have bounded moments conditional on  $\mathbf{x}_i$ .

**Lemma SA-4.2 (Strong Approximation for  $(M_n(g, r) + M_n(h, s) : g \in \mathcal{G}, r \in \mathcal{R}, h \in \mathcal{H}, s \in \mathcal{S})$ )**

Suppose  $(\mathbf{z}_i = (\mathbf{x}_i, y_i) : 1 \leq i \leq n)$  are i.i.d. random vectors taking values in  $(\mathbb{R}^{d+1}, \mathcal{B}(\mathbb{R}^{d+1}))$  with common law  $\mathbb{P}_Z$ , where  $\mathbf{x}_i$  has distribution  $\mathbb{P}_X$  supported on  $\mathcal{X} \subseteq \mathbb{R}^d$ ,  $y_i$  has distribution  $\mathbb{P}_Y$  supported on  $\mathcal{Y} \subseteq \mathbb{R}$ ,  $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[|y_i|^{2+v} | \mathbf{x}_i = \mathbf{x}] \leq 2$  for some  $v > 0$ , and the following conditions hold.

- (i)  $\mathcal{G}$  and  $\mathcal{H}$  are real-valued pointwise measurable classes of functions on  $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d), \mathbb{P}_X)$ .
- (ii) There exists a surrogate measure  $\mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}$  for  $\mathbb{P}_X$  with respect to  $\mathcal{G} \cup \mathcal{H}$  such that  $\mathbb{Q}_{\mathcal{G} \cup \mathcal{H}} = \mathbf{m} \circ \phi_{\mathcal{G} \cup \mathcal{H}}$ , where the normalizing transformation  $\phi_{\mathcal{G} \cup \mathcal{H}} : \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}} \mapsto [0, 1]^d$  is a diffeomorphism.
- (iii)  $\mathcal{G}$  is a VC-type class with envelope  $\mathbf{M}_{\mathcal{G}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}}$  over  $\mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}$  with  $\mathbf{c}_{\mathcal{G}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}} \geq e$  and  $\mathbf{d}_{\mathcal{G}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}} \geq 1$ .  $\mathcal{H}$  is a VC-type class with envelope  $\mathbf{M}_{\mathcal{H}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}}$  over  $\mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}$  with  $\mathbf{c}_{\mathcal{H}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}} \geq e$  and  $\mathbf{d}_{\mathcal{H}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}} \geq 1$ .
- (iv)  $\mathcal{R}$  and  $\mathcal{S}$  are real-valued pointwise measurable classes of functions on  $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mathbb{P}_Y)$ .
- (v)  $\mathcal{R}$  is a VC-type class with envelope  $M_{\mathcal{R}, \mathcal{Y}}$  over  $\mathcal{Y}$  with  $\mathbf{c}_{\mathcal{R}, \mathcal{Y}} \geq e$  and  $\mathbf{d}_{\mathcal{R}, \mathcal{Y}} \geq 1$ , where  $M_{\mathcal{R}, \mathcal{Y}}(y) + \mathbf{pTV}_{\mathcal{R}, (-|y|, |y|)} \leq \mathbf{v}(1 + |y|)$  for all  $y \in \mathcal{Y}$ , for some  $\mathbf{v} > 0$ .  $\mathcal{S}$  is a VC-type class with envelope  $M_{\mathcal{S}, \mathcal{Y}}$  over  $\mathcal{Y}$  with  $\mathbf{c}_{\mathcal{S}, \mathcal{Y}} \geq e$  and  $\mathbf{d}_{\mathcal{S}, \mathcal{Y}} \geq 1$ , where  $M_{\mathcal{S}, \mathcal{Y}}(y) + \mathbf{pTV}_{\mathcal{S}, (-|y|, |y|)} \leq \mathbf{v}(1 + |y|)$  for all  $y \in \mathcal{Y}$ , for some  $\mathbf{v} > 0$ .
- (vi) There exists a constant  $\mathbf{k}$  such that  $|\log_2 \mathbf{E}| + |\log_2 \mathbf{TV}| + |\log_2 \mathbf{M}| \leq \mathbf{k} \log_2(n)$ , where  $\mathbf{E} = \max\{\mathbf{E}_{\mathcal{G}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}}, \mathbf{E}_{\mathcal{H}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}}\}$ ,  $\mathbf{TV} = \max\{\mathbf{TV}_{\mathcal{G}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}}, \mathbf{TV}_{\mathcal{H}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}}\}$  and  $\mathbf{M} = \max\{\mathbf{M}_{\mathcal{G}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}}, \mathbf{M}_{\mathcal{H}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}}\}$ .

Consider the empirical process

$$A_n(g, h, r, s) = M_n(g, r) + M_n(h, s), \quad g \in \mathcal{G}, r \in \mathcal{R}, h \in \mathcal{H}, s \in \mathcal{S}.$$

Then, on a possibly enlarged probability space, there exists a sequence of mean-zero Gaussian processes  $(Z_n^A(g, h, r, s) : g \in \mathcal{G}, h \in \mathcal{H}, r \in \mathcal{R}, s \in \mathcal{S})$  with almost sure continuous trajectories such that:

- $\mathbb{E}[A_n(g_1, h_1, r_1, s_1)A_n(g_2, h_2, r_2, s_2)] = \mathbb{E}[Z_n^A(g_1, h_1, r_1, s_1)Z_n^A(g_2, h_2, r_2, s_2)]$  holds for all  $(g_1, h_1, r_1, s_1), (g_2, h_2, r_2, s_2) \in \mathcal{G} \times \mathcal{H} \times \mathcal{R} \times \mathcal{S}$ , and
- $\mathbb{E}[\|A_n - Z_n^A\|_{\mathcal{G} \times \mathcal{H} \times \mathcal{R} \times \mathcal{S}}] \leq C \mathbf{v} ((\mathbf{d} \log(\mathbf{c}n))^{\frac{3}{2}} \mathbf{r}_n^{\frac{\mathbf{v}}{\mathbf{v}+2}} (\sqrt{\mathbf{ME}})^{\frac{2}{\mathbf{v}+2}} + \mathbf{d} \log(\mathbf{c}n) \mathbf{M} n^{-\frac{\mathbf{v}}{2+\mathbf{v}}} + \mathbf{d} \log(\mathbf{c}n) \mathbf{M} n^{-\frac{1}{2}} \left(\frac{\sqrt{\mathbf{ME}}}{\mathbf{r}_n}\right)^{\frac{2}{\mathbf{v}+2}}),$

where  $C$  is a universal constant,  $\mathbf{c} = \mathbf{c}_{\mathcal{G}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}} + \mathbf{c}_{\mathcal{H}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}} + \mathbf{c}_{\mathcal{R}, \mathcal{Y}} + \mathbf{c}_{\mathcal{S}, \mathcal{Y}} + \mathbf{k}$ ,  $\mathbf{d} = \mathbf{d}_{\mathcal{G}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}} \mathbf{d}_{\mathcal{H}, \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}} \mathbf{d}_{\mathcal{R}, \mathcal{Y}} \mathbf{d}_{\mathcal{S}, \mathcal{Y}} \mathbf{k}$ ,

$$\mathbf{r}_n = \min \left\{ \frac{(\mathbf{c}_1^d \mathbf{M}^{d+1} \mathbf{TV}^d \mathbf{E})^{1/(2d+2)}}{n^{1/(2d+2)}}, \frac{(\mathbf{c}_1^{\frac{d}{2}} \mathbf{c}_2^{\frac{d}{2}} \mathbf{MTV}^{\frac{d}{2}} \mathbf{EL}^{\frac{d}{2}})^{1/(d+2)}}{n^{1/(d+2)}} \right\},$$

$$\mathbf{c}_1 = d \sup_{\mathbf{x} \in \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}} \prod_{j=1}^{d-1} \sigma_j(\nabla \phi_{\mathcal{G} \cup \mathcal{H}}(\mathbf{x})), \quad \mathbf{c}_2 = \sup_{\mathbf{x} \in \mathbb{Q}_{\mathcal{G} \cup \mathcal{H}}} \frac{1}{\sigma_d(\nabla \phi_{\mathcal{G} \cup \mathcal{H}}(\mathbf{x}))}.$$

## SA-5 Proofs for Section SA-2

### SA-5.1 Proof of Lemma SA-2.1

Since  $\hat{\Gamma}_{t, \mathbf{x}}$  is a finite dimensional matrix, it suffices to show the stated rate of convergence for each entry. Let  $\mathbf{v}$  be a multi-index such that  $|\mathbf{v}| \leq |2p|$ . Define

$$g_n(\xi, \mathbf{x}) = \left( \frac{\xi - \mathbf{x}}{h} \right)^{\mathbf{v}} \frac{1}{h^d} K \left( \frac{\xi - \mathbf{x}}{h} \right) \mathbb{1}(\xi \in \mathcal{A}_t), \quad \xi \in \mathcal{X}, \mathbf{x} \in \mathcal{B}.$$

Define  $\mathcal{F} = \{g_n(\cdot, \mathbf{x}) : \mathbf{x} \in \mathcal{B}\}$ . We will show  $\mathcal{F}$  is a VC-type of class. In order to do this, we study the following quantities.

**Constant Envelope Function:** We assume  $K$  is continuous and has compact support, or  $K = \mathbb{1}(\cdot \in [-1, 1]^d)$ . Hence there exists a constant  $C_1$  such that for all  $l \in \mathcal{F}$ , for all  $\mathbf{x} \in \mathcal{B}$ ,

$$|l(\mathbf{x})| \leq C_1 h^{-d} = F.$$

**Diameter of  $\mathcal{F}$  in  $L_2$ :**  $\sup_{l \in \mathcal{F}} \|l\|_{\mathbb{P}, 2} = \sup_{\mathbf{x} \in \mathcal{B}} \left( \int_{\frac{\mathcal{A}_t - \mathbf{x}}{h}} \frac{1}{h^d} \mathbf{y}^{2\mathbf{v}} K(\mathbf{y})^2 f_X(\mathbf{x} + h\mathbf{y}) d\mathbf{y} \right)^{1/2} \leq C_2 h^{-d/2}$  for some constant  $C_2$ . We can take  $C_1$  large enough so that

$$\sigma = C_2 h^{-d/2} \leq F = C_1 h^{-d}.$$

**Ratio:** For some constant  $C_3$ ,

$$\delta = \frac{\sigma}{F} = C_3 \sqrt{h^d}.$$

**Covering Numbers:** *Case 1: When  $K$  is Lipschitz.* Let  $\mathbf{x}, \mathbf{x}' \in \mathcal{B}$ . Then,

$$\begin{aligned} \sup_{\xi \in \mathcal{X}} |g_n(\xi, \mathbf{x}) - g_n(\xi, \mathbf{x}')| &\leq \left| \left( \frac{\xi_1 - \mathbf{x}_1}{h} \right)^{v_1} \cdots \left( \frac{\xi - \mathbf{x}_d}{h} \right)^{v_d} - \left( \frac{\xi_1 - \mathbf{x}'_1}{h} \right)^{v_1} \cdots \left( \frac{\xi - \mathbf{x}'_d}{h} \right)^{v_d} \right| K_h(\xi - \mathbf{x}) \\ &\quad + \left( \frac{\xi_1 - \mathbf{x}'_1}{h} \right)^{v_1} \cdots \left( \frac{\xi - \mathbf{x}'_d}{h} \right)^{v_d} |K_h(\xi - \mathbf{x}) - K_h(\xi - \mathbf{x}')| \\ &\lesssim h_n^{-d-1} \|\mathbf{x} - \mathbf{x}'\|_\infty, \end{aligned}$$

since we have assumed that  $K$  has compact support and is Lipschitz continuous. Hence for any  $\varepsilon \in (0, 1]$  and for any finitely supported measure  $Q$  and metric  $\|\cdot\|_{Q, 2}$  based on  $L_2(Q)$ ,

$$N(\mathcal{F}, \|\cdot\|_{Q, 2}, \varepsilon \|F\|_{Q, 2}) \leq N(\mathcal{X}, \|\cdot\|_\infty, \varepsilon \|F\|_{Q, 2} h^{d+1}) \stackrel{(1)}{\lesssim} \left( \frac{\text{diam}(\mathcal{X})}{\varepsilon \|F\|_{Q, 2} h^{d+1}} \right)^d \lesssim \left( \frac{\text{diam}(\mathcal{X})}{\varepsilon h} \right)^d,$$

where in (1) we used the fact that  $\varepsilon \|F\|_{Q, 2} h^{d+1} \lesssim \varepsilon h \lesssim 1$ . Hence  $\mathcal{F}$  forms a VC-type class, and taking  $A_1 = \text{diam}(\mathcal{X})/h$  and  $A_2 = d$ ,

$$\sup_Q N(\mathcal{F}, \|\cdot\|_{Q, 2}, \varepsilon \|F\|_{Q, 2}) \lesssim (A_1/\varepsilon)^{A_2}, \quad \varepsilon \in (0, 1],$$

where the supremum is over all finite discrete measure.

*Case 2: When  $K = \mathbb{1}(\cdot \in [-1, 1]^d)$ .* Consider

$$h_n(\xi, \mathbf{x}) = \left( \frac{\xi - \mathbf{x}}{h} \right)^{\mathbf{v}} \frac{1}{h^d} \mathbb{1}(\xi \in \mathcal{A}_t), \quad \xi, \mathbf{x} \in \mathcal{X},$$

$\mathcal{H} = \{h_n(\cdot, \mathbf{x}) : \mathbf{x} \in \mathcal{B}\}$  and the constant envelope function  $H = C_4 h^{-|\mathbf{v}|-d}$ , for some constant  $C_4$  only depending on diameter of  $\mathcal{X}$ . The same argument as before shows that for any discrete measure  $Q$ , we have

$$N(\mathcal{H}, \|\cdot\|_{Q, 2}, \varepsilon \|H\|_{Q, 2}) \leq N(\mathcal{X}, \|\cdot\|_\infty, \varepsilon \|H\|_{Q, 2} h^{d+|\mathbf{v}|+1}) \lesssim \left( \frac{\text{diam}(\mathcal{X})}{\varepsilon \|H\|_{Q, 2} h^{d+|\mathbf{v}|+1}} \right)^d \lesssim \left( \frac{\text{diam}(\mathcal{X})}{\varepsilon h} \right)^d.$$

The class  $\mathcal{G} = \{\mathbb{1}(\cdot - \mathbf{x} \in [-1, 1]^d) : \mathbf{x} \in \mathcal{B}\}$  has VC dimension no greater than  $2d$  (van der Vaart and Wellner, 1996, Example 2.6.1), and by van der Vaart and Wellner (1996, Theorem 2.6.4), for any discrete measure  $Q$ , we have

$$N(\mathcal{G}, \|\cdot\|_{Q,2}, \varepsilon) \leq 2d(4e)^{2d}\varepsilon^{-4d}, \quad 0 < \varepsilon \leq 1.$$

It then follows that for any discrete measure  $Q$ ,

$$N(\mathcal{F}, \|\cdot\|_{Q,2}, \varepsilon\|H\|_{Q,2}) \lesssim N(\mathcal{H}, \|\cdot\|_{Q,2}, \varepsilon/2\|H\|_{Q,2}) + N(\mathcal{G}, \|\cdot\|_{Q,2}, \varepsilon/2) \lesssim 2^d h^{-d} \varepsilon^{-d} + 2d(32e)^d \varepsilon^{-4d}.$$

Hence taking  $A_1 = (2^d h^{-d} + 2d(32e)^d)h^{-|\mathbf{v}|}$  and  $A_2 = 4d$ ,

$$\sup_Q N(\mathcal{F}, \|\cdot\|_{Q,2}, \varepsilon\|F\|_{Q,2}) \lesssim (A_1/\varepsilon)^{A_2}, \quad \varepsilon \in (0, 1],$$

the supremum is over all finite discrete measure.

**Maximal Inequality:** Using Corollary 5.1 in Chernozhukov et al. (2014b) for the empirical process on class  $\mathcal{F}$ ,

$$\begin{aligned} \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |\mathbb{E}_n [g_n(\mathbf{X}_i, \mathbf{x})] - \mathbb{E}[g_n(\mathbf{X}_i, \mathbf{x})]| \right] &\lesssim \frac{\sigma}{\sqrt{n}} \sqrt{A_2 \log(A_1/\delta)} + \frac{\|F\|_{\mathbb{P},2} A_2 \log(A_1/\delta)}{n} \\ &\lesssim \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{nh^d}, \end{aligned}$$

where  $A_1, A_2, \sigma, F, \delta$  are all given previously. Assuming  $\frac{\log(h^{-1})}{nh^d} \rightarrow 0$  as  $n \rightarrow \infty$ , we conclude that  $\sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\mathbf{\Gamma}}_{t,\mathbf{x}} - \mathbf{\Gamma}_{t,\mathbf{x}}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}$ . Hence,  $1 \lesssim_{\mathbb{P}} \inf_{\mathbf{x} \in \mathcal{B}} \|\hat{\mathbf{\Gamma}}_{t,\mathbf{x}}\| \lesssim_{\mathbb{P}} \sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\mathbf{\Gamma}}_{t,\mathbf{x}}\| \lesssim_{\mathbb{P}} 1$ . By Weyl's Theorem,  $\sup_{\mathbf{x} \in \mathcal{B}} |\lambda_{\min}(\hat{\mathbf{\Gamma}}_{t,\mathbf{x}}) - \lambda_{\min}(\mathbf{\Gamma}_{t,\mathbf{x}})| \leq \sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\mathbf{\Gamma}}_{t,\mathbf{x}} - \mathbf{\Gamma}_{t,\mathbf{x}}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}$ . Therefore, we can lower bound the minimum eigenvalue by

$$\inf_{\mathbf{x} \in \mathcal{B}} \lambda_{\min}(\hat{\mathbf{\Gamma}}_{t,\mathbf{x}}) \geq \inf_{\mathbf{x} \in \mathcal{B}} \lambda_{\min}(\mathbf{\Gamma}_{t,\mathbf{x}}) - \sup_{\mathbf{x} \in \mathcal{B}} |\lambda_{\min}(\hat{\mathbf{\Gamma}}_{t,\mathbf{x}}) - \lambda_{\min}(\mathbf{\Gamma}_{t,\mathbf{x}})| \gtrsim_{\mathbb{P}} 1.$$

It follows that  $\sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1}\| \lesssim_{\mathbb{P}} 1$  and hence

$$\sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1} - \mathbf{\Gamma}_{t,\mathbf{x}}^{-1}\| \leq \sup_{\mathbf{x} \in \mathcal{B}} \|\mathbf{\Gamma}_{t,\mathbf{x}}^{-1}\| \|\mathbf{\Gamma}_{t,\mathbf{x}} - \hat{\mathbf{\Gamma}}_{t,\mathbf{x}}\| \|\hat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}.$$

This completes the proof. ■

## SA-5.2 Proof of Lemma SA-2.2

We introduce the following notation for an approximation error and an empirical average:

$$\begin{aligned} \mathbf{r}_t(\xi; \mathbf{x}) &= \mu_t(\xi) - \sum_{0 \leq |\boldsymbol{\omega}| \leq p} \frac{\mu_t^{(\boldsymbol{\omega})}(\mathbf{x})}{\boldsymbol{\omega}!} (\xi - \mathbf{x})^{\boldsymbol{\omega}}, \\ \boldsymbol{\chi}_{t,\mathbf{x}} &= \mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \mathbf{r}_t(\mathbf{X}_i; \mathbf{x}) \right]. \end{aligned}$$

Since we have assumed  $\mu_t$  is  $(p+1)$  times continuously differentiable, there exists  $\alpha_{\mathbf{x}, \mathbf{x}_i, t} \in \mathbb{R}^{p+1}$  such that

$$\begin{aligned}\|\chi_{t, \mathbf{x}}\|^2 &= \left\| \frac{1}{n} \sum_{i=1}^n \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right)^\top (\mathbf{0}^\top, \alpha_{\mathbf{x}, \mathbf{x}_i, t}^\top)^\top \right\|^2 h^{2(p+1)} \\ &\leq \left( \frac{1}{n} \sum_{i=1}^n \left\| \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right)^\top \right\|^2 \right) \left( \frac{1}{n} \sum_{i=1}^n \|\alpha_{\mathbf{x}, \mathbf{x}_i, t}\|^2 \right) h^{2(p+1)},\end{aligned}$$

where  $\sup_{\mathbf{x} \in \mathcal{B}} \max_{t \in \{0,1\}} \max_{1 \leq i \leq n} \|\alpha_{\mathbf{x}, \mathbf{x}_i, t}\| \lesssim 1$ . Assume  $\frac{\log(1/h)}{nh^d} = o(1)$ , the same argument as the proof of Lemma SA-2.1 shows

$$\frac{1}{n} \sum_{i=1}^n \left\| \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right)^\top \right\|^2 \lesssim_{\mathbb{P}} 1.$$

It then follows from Lemma SA-2.1 that

$$\sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbb{E}[\hat{\mu}_t^{(\nu)}(\mathbf{x}) | \mathbf{X}] - \mu_t^{(\nu)}(\mathbf{x}) \right| = \sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1} \hat{\Gamma}_{t, \mathbf{x}}^{-1} \chi_{t, \mathbf{x}} \right| \lesssim_{\mathbb{P}} h^{p+1-|\nu|}.$$

Now assume further that  $h = o(1)$ . Since  $\gamma_{\mathbf{v}}(\xi; \mathbf{x}) = \frac{|\mathbf{v}|}{\mathbf{v}!} \int_0^1 (1-t)^{|\mathbf{v}|-1} \partial_{\mathbf{v}} \mu_t(\mathbf{x} + t(\xi - \mathbf{x})) dt$ , then for all  $\mathbf{x} \in \mathcal{B}$ ,  $\xi \in \mathcal{X}$ ,

$$\mathbb{1}(K_h(\xi - \mathbf{x}) \neq 0) \left| \gamma_{\mathbf{v}}(\xi; \mathbf{x}) - \frac{|\mathbf{v}|}{\mathbf{v}!} \partial_{\mathbf{v}} \mu_t(\mathbf{x}) \right| \leq \frac{|\mathbf{v}|}{\mathbf{v}!} \sup_{\|\mathbf{u} - \mathbf{u}'\| \leq h} |\partial_{\mathbf{v}} \mu_t(\mathbf{u}) - \partial_{\mathbf{v}} \mu_t(\mathbf{u}')| = M_n.$$

By Assumption SA-1(iii),  $\partial_{\mathbf{v}} \mu_t$  is uniformly continuous on the compact set  $\mathcal{X}$ . This implies that when  $h = o(1)$ ,  $M_n = o(1)$ . Denote

$$\tilde{\chi}_{t, \mathbf{x}} = \mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h_n} \right) K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \left( \sum_{|\mathbf{v}|=p+1} \frac{|\mathbf{v}|}{\mathbf{v}!} \partial_{\mathbf{v}} \mu_t(\mathbf{x}) (\mathbf{X}_i - \mathbf{x})^{\mathbf{v}} \right) \right],$$

then

$$\begin{aligned}\sup_{\mathbf{x} \in \mathcal{B}} \|\chi_{t, \mathbf{x}} - \tilde{\chi}_{t, \mathbf{x}}\| &\lesssim M_n \sup_{\mathbf{x} \in \mathcal{B}} \left\| \mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \left( \sum_{|\mathbf{v}|=p+1} \frac{|\mathbf{v}|}{\mathbf{v}!} |\mathbf{X}_i - \mathbf{x}|^{\mathbf{v}} \right) \right] \right\| \\ &= o_{\mathbb{P}}(h^{p+1}),\end{aligned}$$

where in the last equality, we have used the same as in the proof of Lemma SA-2.1 maximal inequality to bound the deviation of the term on the left hand side from its expectation. Hence

$$\sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbb{E}[\hat{\mu}_t^{(\nu)}(\mathbf{x}) | \mathbf{X}] - \mu_t^{(\nu)}(\mathbf{x}) - h^{p+1-|\nu|} \hat{B}_{t, \mathbf{x}}^{(\nu)} \right| = \sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1} \hat{\Gamma}_{t, \mathbf{x}}^{-1} \chi_{t, \mathbf{x}} - \mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1} \hat{\Gamma}_{t, \mathbf{x}}^{-1} \tilde{\chi}_{t, \mathbf{x}} \right| = o_{\mathbb{P}}(h^{p+1-|\nu|}).$$

Using Lemma SA-2.1 and maximal inequality as in the proof of Lemma SA-2.1, we can show

$$\max_{t \in \{0,1\}} \sup_{\mathbf{x} \in \mathcal{B}} |\hat{B}_{t, \mathbf{x}}^{(\nu)} - B_{t, \mathbf{x}}^{(\nu)}| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}.$$

Since  $\max_{t \in \{0,1\}} \sup_{\mathbf{x} \in \mathcal{B}} |B_{t,\mathbf{x}}^{(\nu)}| \lesssim 1$ , the inequality above implies

$$\max_{t \in \{0,1\}} \sup_{\mathbf{x} \in \mathcal{B}} |\widehat{B}_{t,\mathbf{x}}^{(\nu)}| \lesssim_{\mathbb{P}} 1.$$

This completes the proof. ■

### SA-5.3 Proof of Lemma SA-2.3

The proof will be similar to the proof of Lemma SA-2.1. Let  $\mathbf{v}$  be a multi-index such that  $0 \leq |\mathbf{v}| \leq \mathbf{p}$ . Denote

$$g_n(\xi, \mathbf{x}) = \left( \frac{\xi - \mathbf{x}}{h} \right)^{\mathbf{v}} K_h \left( \frac{\xi - \mathbf{x}}{h} \right) \mathbb{1}(\xi \in \mathcal{A}_t), \quad \xi, \mathbf{x} \in \mathcal{X}.$$

Define the class of functions  $\mathcal{F} = \{(\xi, u) \in \mathcal{X} \times \mathbb{R} \mapsto g_n(\xi, \mathbf{x}) : \mathbf{x} \in \mathcal{B}\}$ . Consider the following quantities.

**Envelope Function:** Since  $K$  is continuous on its compact support, there exists a constant  $C_1 > 0$  such that  $|g_n(\xi, \mathbf{x})u| \leq C_1 \frac{|u|}{h^d} \forall \xi, \mathbf{x} \in \mathcal{X}, u \in \mathbb{R}$ . We define the envelope function  $F(\xi, u) = C_1 h^{-d} |u|, \xi \in \mathcal{X}, u \in \mathbb{R}$ . Moreover, by (v) in Assumption SA-1, denote  $M = \max_{1 \leq i \leq n} F(\mathbf{X}_i, u_i)$ , then

$$\mathbb{E}[M^2]^{1/2} \lesssim h^{-d} \mathbb{E} \left[ \max_{1 \leq i \leq n} |u_i|^2 \right]^{1/2} \lesssim h^{-d} \mathbb{E} \left[ \max_{1 \leq i \leq n} |u_i|^{2+v} \right]^{1/(2+v)} \lesssim n^{1/(2+v)} h^{-d}.$$

**Diameter of  $\mathcal{F}$  in  $L_2$ :** By (v) in Assumption SA-1, recall we denote  $u_i = Y_i - \mathbb{E}[Y_i | \mathbf{X}_i]$ ,

$$\sup_{l \in \mathcal{F}} \mathbb{E}[l(\mathbf{X}_i, u_i)^2]^{1/2} \leq \sup_{\xi \in \mathcal{X}} \mathbb{E}[u_i^2 | \mathbf{X}_i = \xi]^{1/2} \sup_{\xi \in \mathcal{X}} \mathbb{E}[g_n(\mathbf{X}_i, \xi)^2]^{1/2} \leq C_3 h^{-d/2} = \sigma.$$

**Ratio:**  $\delta = \frac{\sigma}{\|F\|_{\mathbb{R},2}} \lesssim h^{d/2}$ .

**Covering Numbers:** *Case 1:  $K$  is Lipschitz.* Let  $\mathbb{Q}$  be a finite distribution on  $(\mathcal{X} \times \mathbb{R}, \mathcal{B}(\mathcal{X}) \otimes \mathcal{B}(\mathbb{R}))$ . Let  $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$ . In the proof of Lemma SA-2.1, we have shown  $\sup_{\xi \in \mathcal{X}} \sup_{\mathbf{x}, \mathbf{x}' \in \mathcal{X}} \frac{|g_n(\xi, \mathbf{x}) - g_n(\xi, \mathbf{x}')|}{\|\mathbf{x} - \mathbf{x}'\|_{\infty}} \lesssim h^{-d-1}$ . Hence

$$\|g_n(\mathbf{X}_i, \mathbf{x})u_i - g_n(\mathbf{X}_i, \mathbf{x}')u_i\|_{\mathbb{Q},2} \leq \|g_n(\cdot, \mathbf{x}) - g_n(\cdot, \mathbf{x}')\|_{\infty} \|u_i\|_{\mathbb{Q},2} \lesssim h^{-1} \|F\|_{\mathbb{Q},2} \|\mathbf{x} - \mathbf{x}'\|_{\infty}.$$

It follows that  $\sup_{\mathbb{Q}} N(\mathcal{F}, \|\cdot\|_{\mathbb{Q},2}, \epsilon \|F\|_{\mathbb{Q},2}) \lesssim \left( \frac{\text{diam}(\mathcal{X})}{\epsilon h} \right)^d$ , where sup is over all finite probability distribution on  $(\mathcal{X} \times \mathbb{R}, \mathcal{B}(\mathcal{X}) \otimes \mathcal{B}(\mathbb{R}))$ . Denote  $A_1 = \frac{\text{diam}(\mathcal{X})}{h}, A_2 = d$ . We have

$$\sup_{\mathbb{Q}} N(\mathcal{F}, \|\cdot\|_{\mathbb{Q},2}, \epsilon \|F\|_{\mathbb{Q},2}) \lesssim (A_1/\epsilon)^{A_2}, \quad \epsilon \in (0, 1].$$

*Case 2:  $K$  is the uniform kernel.* Consider

$$h_n(\xi, \mathbf{x}) = \left( \frac{\xi - \mathbf{x}}{h} \right)^{\mathbf{v}} \frac{1}{h^d} \mathbb{1}(\xi \in \mathcal{A}_t), \quad \xi, \mathbf{x} \in \mathcal{X},$$

and  $\mathcal{H} = \{(\xi, u) \in \mathcal{X} \times \mathbb{R} \mapsto h_n(\xi, \mathbf{x})u : \mathbf{x} \in \mathcal{B}\}$ . By similar arguments as *Case 1* and the proof of Lemma SA-2.1, we can show

$$\sup_{\mathbb{Q}} N(\mathcal{H}, \|\cdot\|_{\mathbb{Q},2}, \epsilon \|H\|_{\mathbb{Q},2}) \lesssim \left( \frac{\text{diam}(\mathcal{X})}{\epsilon h} \right)^d,$$

where the supremum is taken over all finite discrete measures. Taking  $\mathcal{G} = \{\mathbb{1}(\cdot - \mathbf{x} \in [-1, 1]^d) : \mathbf{x} \in \mathcal{B}\}$ , the proof of Lemma SA-2.1 shows

$$\sup_Q N(\mathcal{G}, \|\cdot\|_{Q,2}, \varepsilon) \leq 2d(4e)^{2d}\varepsilon^{-4d}, \quad 0 < \varepsilon \leq 1,$$

where the supremum is taken over all finite discrete measures. Taking  $A_1 = (2^d h^{-d} + 2d(32e)^d)h^{-|\mathbf{v}|}$  and  $A_2 = 4d$ , we have

$$\sup_Q N(\mathcal{F}, \|\cdot\|_{Q,2}, \varepsilon \|F\|_{Q,2}) \lesssim (A_1/\varepsilon)^{A_2}, \quad \varepsilon \in (0, 1],$$

the supremum is over all finite discrete measure.

**Maximal Inequality:** By Corollary 5.1 in Chernozhukov et al. (2014b),

$$\begin{aligned} \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{X}} \left| \frac{1}{n} \sum_{i=1}^n g_n(\mathbf{X}_i, \mathbf{x}) u_i \right| \right] &\lesssim \frac{\sigma}{\sqrt{n}} \sqrt{A_2 \log(A_1/\delta)} + \frac{\|M\|_{\mathbb{P},2} A_2 \log(A_1/\delta)}{n} \\ &\lesssim \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d}. \end{aligned}$$

Since  $\mathbf{Q}_{t,\mathbf{x}}$  is finite-dimensional, entrywise-convergence implies convergence in norm in the same rate. Hence  $\sup_{\mathbf{x} \in \mathcal{X}} \|\mathbf{Q}_{t,\mathbf{x}}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d}$ . By Lemma SA-2.1,

$$\begin{aligned} \sup_{\mathbf{x} \in \mathcal{X}} \left| \hat{\mu}_t^{(\nu)}(\mathbf{x}) - \mathbb{E} \left[ \hat{\mu}_t^{(\nu)}(\mathbf{x}) | \mathbf{X} \right] - \mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1} \mathbf{\Gamma}_{t,\mathbf{x}}^{-1} \mathbf{Q}_{t,\mathbf{x}} \right| &= \sup_{\mathbf{x} \in \mathcal{X}} \left| \mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1} \left( \hat{\mathbf{\Gamma}}_{t,\mathbf{x}}^{-1} - \mathbf{\Gamma}_{t,\mathbf{x}}^{-1} \right) \mathbf{Q}_{t,\mathbf{x}} \right| \\ &\lesssim_{\mathbb{P}} h^{-|\nu|} \sqrt{\frac{\log(1/h)}{nh^d}} \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right), \\ \sup_{\mathbf{x} \in \mathcal{X}} \left| \hat{\mu}_t^{(\nu)}(\mathbf{x}) - \mathbb{E} \left[ \hat{\mu}_t^{(\nu)}(\mathbf{x}) | \mathbf{X} \right] \right| &\lesssim_{\mathbb{P}} h^{-|\nu|} \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right). \end{aligned}$$

This completes the proof. ■

#### SA-5.4 Proof of Lemma SA-2.4

Recall we denote  $\varepsilon_i = Y_i - \sum_{t \in \{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \hat{\beta}_t(\mathbf{x})^\top \mathbf{R}_p(\mathbf{X}_i - \mathbf{x})$ , and  $u_i = Y_i - \sum_{t \in \{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \mu_t(\mathbf{X}_i)$ . Denote  $\eta_i = \sum_{t \in \{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) (\mu_t(\mathbf{X}_i) - \hat{\beta}_t(\mathbf{x})^\top \mathbf{R}_p(\mathbf{X}_i - \mathbf{x}))$ . Then, for all  $\mathbf{x}, \mathbf{y} \in \mathcal{B}$ , the difference between estimated and true sandwich matrix can be decomposed by

$$\hat{\Sigma}_{t,\mathbf{x},\mathbf{y}} - \Sigma_{t,\mathbf{x},\mathbf{y}} = \mathbf{M}_{1,\mathbf{x},\mathbf{y}} + \mathbf{M}_{2,\mathbf{x},\mathbf{y}} + \mathbf{M}_{3,\mathbf{x},\mathbf{y}} + \mathbf{M}_{4,\mathbf{x},\mathbf{y}}$$

where

$$\begin{aligned}
\mathbf{M}_{1,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right)^\top \frac{1}{h^d} K \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right) \eta_i^2 \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right], \\
\mathbf{M}_{2,\mathbf{x},\mathbf{y}} &= 2\mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right)^\top \frac{1}{h^d} K \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right) \eta_i u_i \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right], \\
\mathbf{M}_{3,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right)^\top \frac{1}{h^d} K \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right) (u_i^2 - \sigma_t(\mathbf{X}_i)^2) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right], \\
\mathbf{M}_{4,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right)^\top \frac{1}{h^d} K \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right) \sigma_t(\mathbf{X}_i)^2 \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right] \\
&\quad - \mathbb{E} \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right)^\top \frac{1}{h^d} K \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right) \sigma_t(\mathbf{X}_i)^2 \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right].
\end{aligned}$$

Let  $\mathbf{u}, \mathbf{v}$  be multi-indices. Denote  $g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y}) = \frac{1}{h^d} \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right)^\mathbf{u} \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right)^\mathbf{v} K \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t)$ . For notational simplicity, denote in what follows

$$\alpha_n = \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+\mathbf{v}}{2+\mathbf{v}}} h^d}.$$

First, we present a bound on  $\max_{1 \leq i \leq n} |\eta_i| \mathbb{1}((\mathbf{X}_i - \mathbf{x})/h \in \text{Supp}(K))$ . By Lemma SA-2.2 and Lemma SA-2.3, and multi-index  $\boldsymbol{\nu}$  such that  $|\boldsymbol{\nu}| \leq p$ ,

$$\sup_{\mathbf{x} \in \mathcal{B}} |\mathbf{e}_{1+\boldsymbol{\nu}}^\top \hat{\mu}_t(\mathbf{x}) - \mathbf{e}_{1+\boldsymbol{\nu}}^\top \mu_t(\mathbf{x})| \lesssim_{\mathbb{P}} h^{-|\boldsymbol{\nu}|} (h^{p+1} + \alpha_n).$$

Since  $K$  is compactly supported, we have

$$\max_{1 \leq i \leq n} \left| \sum_{t \in \{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) (\hat{\beta}_t(\mathbf{x}) - \beta_t(\mathbf{x}))^\top \mathbf{R}_p(\mathbf{X}_i - \mathbf{x}) \mathbb{1}((\mathbf{X}_i - \mathbf{x})/h \in \text{Supp}(K)) \right| \lesssim_{\mathbb{P}} h^{p+1} + \alpha_n.$$

Since  $\mu_t$  is  $p+1$  times continuously differentiable,

$$\max_{1 \leq i \leq n} \left| \sum_{t \in \{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) (\mu_t(\mathbf{X}_i) - \beta_t(\mathbf{x}))^\top \mathbf{R}_p(\mathbf{X}_i - \mathbf{x}) \mathbb{1}((\mathbf{X}_i - \mathbf{x})/h \in \text{Supp}(K)) \right| \lesssim h^{p+1}.$$

It follows that

$$\max_{1 \leq i \leq n} |\eta_i| \mathbb{1}((\mathbf{X}_i - \mathbf{x})/h \in \text{Supp}(K)) \lesssim_{\mathbb{P}} h^{p+1} + \alpha_n.$$

**Term  $\mathbf{M}_{1,\mathbf{x},\mathbf{y}}$ .** From the proof for Lemma SA-2.1,  $\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} |\mathbb{E}_n[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})] - \mathbb{E}[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})]| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}$ . Moreover,  $\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} |\mathbb{E}[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})]| \lesssim_{\mathbb{P}} 1$ . Hence  $\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} |\mathbb{E}_n[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})]| \lesssim_{\mathbb{P}} 1$ . So

$$\begin{aligned}
\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} |\mathbb{E}_n[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y}) \eta_i^2]| &\leq \max_{1 \leq i \leq n} |\eta_i| \mathbb{1}((\mathbf{X}_i - \mathbf{x})/h \in \text{Supp}(K)) \cdot \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} |\mathbb{E}_n[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})]| \\
&\lesssim_{\mathbb{P}} (h^{p+1} + \alpha_n)^2,
\end{aligned}$$

where we have use Theorem SA-2.5, which does not depend on this lemma, for  $\sup_{\mathbf{x} \in \mathcal{B}} |\hat{\mu}_t(\mathbf{x}) - \mu_t(\mathbf{x})| \lesssim_{\mathbb{P}}$

$h^{p+1} + \alpha_n$ . Finite dimensionality of  $\mathbf{M}_{1,\mathbf{x},\mathbf{y}}$  then implies

$$\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \|\mathbf{M}_{1,\mathbf{x},\mathbf{y}}\| \lesssim_{\mathbb{P}} (h^{p+1} + \alpha_n)^2.$$

**Term  $\mathbf{M}_{2,\mathbf{x},\mathbf{y}}$ .** From the proof of Lemma SA-2.3,  $\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} |\mathbb{E}_n[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})u_i] - \mathbb{E}[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})u_i]| \lesssim_{\mathbb{P}} \alpha_n$ . Moreover,  $\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} |\mathbb{E}[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})u_i]| \lesssim_{\mathbb{P}} 1$ . Hence  $\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} |\mathbb{E}_n[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})u_i]| \lesssim_{\mathbb{P}} 1$ .

$$\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} |\mathbb{E}_n[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})\eta_i u_i]| \leq \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} |\hat{\mu}_t(\mathbf{x}) - \mu_t(\mathbf{x})| \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \mathbb{E}_n[|g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})u_i|] \lesssim_{\mathbb{P}} h^{p+1} + \alpha_n,$$

implying

$$\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \|\mathbf{M}_{2,\mathbf{x},\mathbf{y}}\| \lesssim_{\mathbb{P}} h^{p+1} + \alpha_n.$$

**Term  $\mathbf{M}_{3,\mathbf{x},\mathbf{y}}$ .** Define  $l_n(\cdot, \cdot; \mathbf{x}, \mathbf{y}) : \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R}$  by

$$l_n(\xi, \varepsilon; \mathbf{x}, \mathbf{y}) = \frac{1}{h^d} \left( \frac{\xi - \mathbf{x}}{h} \right)^{\mathbf{u}} \left( \frac{\xi - \mathbf{y}}{h} \right)^{\mathbf{v}} K \left( \frac{\xi - \mathbf{x}}{h} \right) K \left( \frac{\xi - \mathbf{y}}{h} \right) \mathbb{1}(\xi \in \mathcal{A}_t)(\varepsilon^2 - \sigma_t(\xi)^2),$$

and consider  $\mathcal{L} = \{l_n(\cdot, \cdot; \mathbf{x}, \mathbf{y}) : \mathbf{x}, \mathbf{y} \in \mathcal{X}\}$ . Define  $L : \mathcal{X} \times \mathbb{R} \rightarrow \mathbb{R}$  by  $L(\xi, \varepsilon) = \frac{c}{h^d} |\varepsilon^2 - \sigma_t(\xi)^2|$  where  $c = \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \left| \left( \frac{\xi - \mathbf{x}}{h} \right)^{\mathbf{u}} \left( \frac{\xi - \mathbf{y}}{h} \right)^{\mathbf{v}} K \left( \frac{\xi - \mathbf{x}}{h} \right) K \left( \frac{\xi - \mathbf{y}}{h} \right) \right|$ . By similar argument as in the proof for Lemma SA-2.3, we can show  $\mathcal{L}$  is a VC-type class such that  $\mathbb{E}[l_n(\mathbf{X}_i, u_i; \mathbf{x}, \mathbf{y})] = 0, \forall \mathbf{x}, \mathbf{y} \in \mathcal{X}$  and

$$\begin{aligned} \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \mathbb{E}[l_n(\mathbf{X}_i, \varepsilon; \mathbf{x}, \mathbf{y})^2]^{\frac{1}{2}} &\lesssim \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \mathbb{E}[g_n(\mathbf{X}_i, u_i; \mathbf{x}, \mathbf{y})^2]^{\frac{1}{2}} \sup_{\xi \in \mathcal{X}} \mathbb{V}[u_i^2 | \mathbf{X}_i = \xi] \lesssim h^{-d/2}, \\ \mathbb{E} \left[ \max_{1 \leq i \leq n} L(\mathbf{X}_i, u_i)^2 \right]^{\frac{1}{2}} &\lesssim h^{-d} \mathbb{E} \left[ \max_{1 \leq i \leq n} u_i^4 \right]^{1/2} \lesssim h^{-d} \mathbb{E} \left[ \max_{1 \leq i \leq n} u_i^{2+v} \right]^{\frac{2}{2+v}} \lesssim h^{-d} n^{\frac{2}{2+v}}. \end{aligned}$$

Apply Corollary 5.1 in Chernozhukov et al. (2014b), we get

$$\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} |\mathbb{E}_n[l_n(\mathbf{X}_i, u_i; \mathbf{x}, \mathbf{y})]| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}, \quad \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \|\mathbf{M}_{3,\mathbf{x},\mathbf{y}}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}.$$

**Term  $\mathbf{M}_{4,\mathbf{x},\mathbf{y}}$ .** Notice that  $\{g_n(\cdot; \mathbf{x}, \mathbf{y})\sigma_t^2(\cdot) : \mathbf{x}, \mathbf{y} \in \mathcal{B}\}$  is a VC-type of class with constant envelope function  $Ch^{-d}$  for some suitable  $C$  and

$$\begin{aligned} \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \sup_{\xi \in \mathcal{X}} |g_n(\xi; \mathbf{x}, \mathbf{y})\sigma_t^2(\xi)| &\lesssim h^{-d}, \\ \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \mathbb{E}[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})^2 \sigma_t(\mathbf{X}_i)^2]^{\frac{1}{2}} &\lesssim h^{-d/2}. \end{aligned}$$

Then, similar to the proof of  $\mathbf{M}_{1,\mathbf{x},\mathbf{y}}$  we can get

$$\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} |\mathbb{E}_n[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})] - \mathbb{E}[g_n(\mathbf{X}_i; \mathbf{x}, \mathbf{y})]| \lesssim \sqrt{\frac{\log(1/h)}{nh^d}}, \quad \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \|\mathbf{M}_{4,\mathbf{x},\mathbf{y}}\| \lesssim \sqrt{\frac{\log(1/h)}{nh^d}}.$$

**Putting Together.** Combining the the upper bounds of the four terms, we get

$$\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \|\widehat{\Sigma}_{1, \mathbf{x}, \mathbf{y}} - \Sigma_{1, \mathbf{x}, \mathbf{y}}\| \lesssim_{\mathbb{P}} h^{p+1} + \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d},$$

which implies  $\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \|\widehat{\Sigma}_{1, \mathbf{x}, \mathbf{y}}\| \lesssim_{\mathbb{P}} 1$ . It follows that

$$\begin{aligned} & \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} |\widehat{\Omega}_{1, \mathbf{x}, \mathbf{y}}^{(\nu)} - \Omega_{1, \mathbf{x}, \mathbf{y}}^{(\nu)}| \\ & \leq \frac{1}{nh^{d+2|\nu|}} \left( \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \|\widehat{\Gamma}_{1, \mathbf{x}}^{-1} - \Gamma_{1, \mathbf{x}}^{-1}\| \|\widehat{\Sigma}_{1, \mathbf{x}, \mathbf{y}}\| \|\widehat{\Gamma}_{1, \mathbf{y}}^{-1}\| + \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \|\Gamma_{1, \mathbf{x}}^{-1}\| \|\widehat{\Sigma}_{1, \mathbf{x}, \mathbf{y}} - \Sigma_{1, \mathbf{x}, \mathbf{y}}\| \|\widehat{\Gamma}_{1, \mathbf{y}}^{-1}\| \right. \\ & \quad \left. + \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{B}} \|\Gamma_{1, \mathbf{x}}^{-1}\| \|\Sigma_{1, \mathbf{x}, \mathbf{y}}\| \|\widehat{\Gamma}_{1, \mathbf{y}}^{-1} - \Gamma_{1, \mathbf{y}}^{-1}\| \right) \\ & \leq \frac{1}{nh^{d+2|\nu|}} \left( h^{p+1} + \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right). \end{aligned}$$

By Assumption SA-1(iv) and Assumption SA-2(ii),  $\inf_{\mathbf{x} \in \mathcal{B}} \Omega_{\mathbf{x}, \mathbf{x}}^{(\nu)} \gtrsim_{\mathbb{P}} (nh^{d+2|\nu|})^{-1}$ . Hence  $\inf_{\mathbf{x} \in \mathcal{B}} \widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)} \gtrsim (nh^{d+2|\nu|})^{-1}$ .

$$\begin{aligned} & \sup_{\mathbf{x} \in \mathcal{B}} \left| \sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)}} - \sqrt{\Omega_{\mathbf{x}, \mathbf{x}}^{(\nu)}} \right| \lesssim_{\mathbb{P}} \sup_{\mathbf{x} \in \mathcal{B}} \sqrt{nh^{d+2|\nu|}} \left| \widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)} - \Omega_{\mathbf{x}, \mathbf{x}}^{(\nu)} \right| \lesssim_{\mathbb{P}} \frac{1}{\sqrt{nh^{d+2|\nu|}}} \left( h^{p+1} + \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right), \\ & \sup_{\mathbf{x} \in \mathcal{B}} \left| \frac{h^{-|\nu|}}{\sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)}}} - \frac{h^{-|\nu|}}{\sqrt{\Omega_{\mathbf{x}, \mathbf{x}}^{(\nu)}}} \right| = h^{-|\nu|} \sup_{\mathbf{x} \in \mathcal{B}} \left| \frac{\sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)}} - \sqrt{\Omega_{\mathbf{x}, \mathbf{x}}^{(\nu)}}}{\sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}}^{(\nu)}} \sqrt{\Omega_{\mathbf{x}, \mathbf{x}}^{(\nu)}}} \right| \lesssim_{\mathbb{P}} \sqrt{nh^d} \left( h^{p+1} + \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right). \end{aligned}$$

This completes the proof. ■

### SA-5.5 Proof of Theorem SA-2.1

We can use the same argument as in the proof for Lemma SA-2.1, SA-2.2 and SA-2.3, with  $\mathcal{B} = \{\mathbf{x}\}$ , to get that under the conditions specified we have

$$|\mathbb{E}[\widehat{\mu}_t^{(\nu)}(\mathbf{x})|\mathbf{X}] - \mu_t^{(\nu)}(\mathbf{x})| \lesssim_{\mathbb{P}} h^{p+1-|\nu|},$$

and

$$|\widehat{\mu}_t^{(\nu)}(\mathbf{x}) - \mathbb{E}[\widehat{\mu}_t^{(\nu)}(\mathbf{x})|\mathbf{X}]| \lesssim_{\mathbb{P}} h^{-|\nu|} \left( \frac{1}{\sqrt{nh^d}} + \frac{1}{n^{\frac{1+v}{2+v}} h^d} \right).$$

In particular, when applying concentration inequalities, we always apply the singleton class of functions that correspond to the point of evaluation  $\mathbf{x}$ . Putting together, we get the claimed result. ■

For the proof of Theorem SA-2.2, we define the following matrices: For  $\mathbf{x}, \mathbf{y} \in \mathcal{B}$ ,

$$\begin{aligned}\bar{\Sigma}_{t,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n \left[ \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{y}}{h} \right)^\top h^d K_h(\mathbf{X}_i - \mathbf{x}_1) K_h(\mathbf{X}_i - \mathbf{y}) \sigma_t^2(\mathbf{X}_i) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right], \\ \bar{\Omega}_{t,\mathbf{x},\mathbf{y}}^{(\nu)} &= \frac{1}{nh^{d+2|\nu|}} \mathbf{e}_{1+\nu}^\top \mathbf{\Gamma}_{t,\mathbf{x}}^{-1} \bar{\Sigma}_{t,\mathbf{x},\mathbf{y}} \mathbf{\Gamma}_{t,\mathbf{y}}^{-1} \mathbf{e}_{1+\nu}, \quad \bar{\Omega}_{\mathbf{x},\mathbf{y}}^{(\nu)} = \bar{\Omega}_{0,\mathbf{x},\mathbf{y}}^{(\nu)} + \bar{\Omega}_{1,\mathbf{x},\mathbf{y}}^{(\nu)}, \\ \bar{V}_{t,\mathbf{x}}^{(\nu)} &= \mathbf{e}_{1+\nu}^\top \mathbf{\Gamma}_{t,\mathbf{x}}^{-1} \bar{\Sigma}_{t,\mathbf{x},\mathbf{x}} \mathbf{\Gamma}_{t,\mathbf{x}}^{-1} \mathbf{e}_{1+\nu}, \quad \bar{V}_{\mathbf{x}}^{(\nu)} = \bar{V}_{0,\mathbf{x}}^{(\nu)} + \bar{V}_{1,\mathbf{x}}^{(\nu)}.\end{aligned}$$

The following lemma is used for the convergence of  $\bar{\Omega}_{t,\mathbf{x},\mathbf{y}}^{(\nu)}$ .

**Lemma SA-5.1 (Conditional Variance)**

Suppose Assumption SA-1 (i), (ii), (iv) and Assumption SA-2 hold. If  $\frac{\log(1/h)}{nh^d} = o(1)$ , then

$$\sup_{\mathbf{x} \in \mathcal{B}} \|\Sigma_{t,\mathbf{x},\mathbf{x}} - \bar{\Sigma}_{t,\mathbf{x},\mathbf{x}}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}, \quad t \in \{0, 1\}, \quad \text{and} \quad \sup_{\mathbf{x} \in \mathcal{B}} |V_{\mathbf{x}}^{(\nu)} - \bar{V}_{\mathbf{x}}^{(\nu)}| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}.$$

**Proof of Lemma SA-5.1.** The proof will be similar to the proof of Lemma SA-2.1. Let  $\mathbf{u}, \mathbf{v}$  be multi-indices such that  $|\mathbf{u}| \leq p$  and  $|\mathbf{v}| \leq p$ . Fix  $t \in \{0, 1\}$ . For  $\xi \in \mathcal{X}$  and  $\mathbf{x} \in \mathcal{B}$ , define

$$g_n(\xi, \mathbf{x}) = \left( \frac{\xi - \mathbf{x}}{h} \right)^{\mathbf{u}+\mathbf{v}} \frac{1}{h^d} K^2 \left( \frac{\xi - \mathbf{x}}{h} \right) \sigma_t^2(\xi) \mathbb{1}(\xi \in \mathcal{A}_t).$$

Consider the class of functions  $\mathcal{F} = \{g_n(\cdot, \mathbf{x}) : \mathbf{x} \in \mathcal{B}\}$ . Then, by the same maximal inequality argument as in the proof of Lemma SA-2.1,

$$\mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |\mathbb{E}_n [g_n(\mathbf{X}_i, \mathbf{x})] - \mathbb{E}[g_n(\mathbf{X}_i, \mathbf{x})]| \right] \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}.$$

Since  $\Sigma_{t,\mathbf{x},\mathbf{x}}$  is finite dimensional,  $\sup_{\mathbf{x} \in \mathcal{B}} \|\Sigma_{t,\mathbf{x},\mathbf{x}} - \bar{\Sigma}_{t,\mathbf{x},\mathbf{x}}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}$  and hence  $\sup_{\mathbf{x} \in \mathcal{B}} \|V_{\mathbf{x}} - \bar{V}_{\mathbf{x}}\| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}$ . ■

## SA-5.6 Proof of Theorem SA-2.2

For conditional bias, by Lemma SA-2.2,

$$\begin{aligned}& \sup_{\mathbf{x} \in \mathcal{B}} |\mathbb{E}[\hat{\tau}^{(\nu)}(\mathbf{x}) - \tau^{(\nu)}(\mathbf{x}) | \mathbf{X}]^2 - (h^{p+1-|\nu|} \hat{B}_{\mathbf{x}}^{(\nu)})^2| \\ & \leq \sup_{\mathbf{x} \in \mathcal{B}} |\mathbb{E}[\hat{\tau}^{(\nu)}(\mathbf{x}) - \tau^{(\nu)}(\mathbf{x}) | \mathbf{X}] - h^{p+1-|\nu|} \hat{B}_{\mathbf{x}}^{(\nu)}| \cdot \sup_{\mathbf{x} \in \mathcal{B}} |\mathbb{E}[\hat{\tau}^{(\nu)}(\mathbf{x}) - \tau^{(\nu)}(\mathbf{x}) | \mathbf{X}] + h^{p+1-|\nu|} \hat{B}_{\mathbf{x}}^{(\nu)}| \\ & = o_{\mathbb{P}}(h^{p+1-|\nu|}).\end{aligned}$$

Since we know  $\sup_{\mathbf{x} \in \mathcal{B}} |\hat{B}_{t,\mathbf{x}}^{(\nu)} - B_{t,\mathbf{x}}^{(\nu)}| \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}}$  from Lemma SA-2.2,

$$\sup_{\mathbf{x} \in \mathcal{B}} |\mathbb{E}[\hat{\tau}^{(\nu)}(\mathbf{x}) - \tau^{(\nu)}(\mathbf{x}) | \mathbf{X}]^2 - (h^{p+1-|\nu|} B_{\mathbf{x}}^{(\nu)})^2| = o_{\mathbb{P}}(h^{p+1-|\nu|}).$$

For conditional variance, by Lemma SA-5.1,

$$\sup_{\mathbf{x} \in \mathcal{B}} |\mathbb{V}[\widehat{\tau}^{(\nu)}(\mathbf{x})|\mathbf{X}] - (nh^{d+2|\nu|})^{-1}V_{\mathbf{x}}^{(\nu)}| = \sup_{\mathbf{x} \in \mathcal{B}} |(nh^{d+2|\nu|})^{-1}\overline{V}_{\mathbf{x}}^{(\nu)} - (nh^{d+2|\nu|})^{-1}V_{\mathbf{x}}^{(\nu)}| = o_{\mathbb{P}}((nh^{d+2|\nu|})^{-1}).$$

Since  $(nh^{d+2|\nu|})^{-1} \sup_{\mathbf{x} \in \mathcal{B}} |V_{\mathbf{x}}^{(\nu)} - \widehat{V}_{\mathbf{x}}^{(\nu)}| = \sup_{\mathbf{x} \in \mathcal{B}} |\Omega_{\mathbf{x},\mathbf{x}}^{(\nu)} - \widehat{\Omega}_{\mathbf{x},\mathbf{x}}^{(\nu)}| = o_{\mathbb{P}}((nh^{d+2|\nu|})^{-1})$  from Lemma SA-2.4,

$$\sup_{\mathbf{x} \in \mathcal{B}} |\mathbb{V}[\widehat{\tau}^{(\nu)}(\mathbf{x})|\mathbf{X}] - (nh^{d+2|\nu|})^{-1}\widehat{V}_{\mathbf{x}}^{(\nu)}| = \sup_{\mathbf{x} \in \mathcal{B}} |(nh^{d+2|\nu|})^{-1}\overline{V}_{\mathbf{x}}^{(\nu)} - (nh^{d+2|\nu|})^{-1}\widehat{V}_{\mathbf{x}}^{(\nu)}| = o_{\mathbb{P}}((nh^{d+2|\nu|})^{-1}).$$

Putting together we get the two MSE results. And

$$\begin{aligned} & |\text{IMSE}_{\nu} - \int_{\mathcal{B}} [(h^{p+1-|\nu|}B_{\mathbf{x}}^{(\nu)})^2 + (nh^{d+2|\nu|})^{-1}V_{\mathbf{x}}^{(\nu)}]\omega(\mathbf{x})dH^{d-1}(\mathbf{x})| \\ & \leq \int_{\mathcal{B}} \omega(\mathbf{x})dH^{d-1}(\mathbf{x}) \cdot \sup_{\mathbf{x} \in \mathcal{B}} |\text{MSE}_{\nu}(\mathbf{x}) - (h^{p+1-|\nu|}B_{\mathbf{x}}^{(\nu)})^2 - (nh^{d+2|\nu|})^{-1}V_{\mathbf{x}}^{(\nu)}| \\ & = o_{\mathbb{P}}(h^{2p+2-2|\nu|} + (nh^{d+2|\nu|})^{-1}). \end{aligned}$$

Similarly, we can get

$$|\text{IMSE}_{\nu} - \int_{\mathcal{B}} [(h^{p+1-|\nu|}\widehat{B}_{\mathbf{x}}^{(\nu)})^2 + (nh^{d+2|\nu|})^{-1}\widehat{V}_{\mathbf{x}}^{(\nu)}]\omega(\mathbf{x})dH^{d-1}(\mathbf{x})| = o_{\mathbb{P}}(h^{2p+2-2|\nu|} + (nh^{d+2|\nu|})^{-1}).$$

■

## SA-5.7 Proof of Theorem SA-2.3

Consider  $\overline{\mathbf{T}}^{(\nu)}(\mathbf{x}) = (\Omega_{\mathbf{x},\mathbf{x}}^{(\nu)})^{-1/2} \mathbf{e}_{1+\nu}^{\top} \mathbf{H}^{-1} \Gamma_{t,\mathbf{x}}^{-1} \mathbf{Q}_{t,\mathbf{x}}$  and  $u_i = Y_i - \sum_{t \in \{0,1\}} \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \mu_t(\mathbf{X}_i)$ . Define

$$Z_i = \sum_{t \in \{0,1\}} n^{-1} (\Omega_{\mathbf{x},\mathbf{x}}^{(\nu)})^{-1/2} \mathbf{e}_{1+\nu}^{\top} \mathbf{H}^{-1} \Gamma_{t,\mathbf{x}}^{-1} \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) u_i.$$

Then,  $\overline{\mathbf{T}}^{(\nu)}(\mathbf{x}) = \sum_{i=1}^n Z_i$  and  $\mathbb{E}[Z_i] = 0$  and  $\mathbb{V}[Z_i] = n^{-1}$ . By Berry-Essen Theorem,

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left( \overline{\mathbf{T}}^{(\nu)}(\mathbf{x}) \leq u \right) - \Phi(u) \right| \lesssim B_n^{-1} \sum_{i=1}^n \mathbb{E}[|Z_i|^3],$$

where  $B_n = \sum_{i=1}^n \mathbb{V}[Z_i] = 1$ . Moreover,

$$\begin{aligned} \sum_{i=1}^n \mathbb{E}[|Z_i|^3] &= n^{-3} (\Omega_{\mathbf{x},\mathbf{x}}^{(\nu)})^{-3/2} \sum_{i=1}^n \mathbb{E} \left[ \left| \sum_{t \in \{0,1\}} \mathbf{e}_{1+\nu}^{\top} \mathbf{H}^{-1} \Gamma_{t,\mathbf{x}}^{-1} \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) u_i \right|^3 \right] \\ &\lesssim n^{-3} (\Omega_{\mathbf{x},\mathbf{x}}^{(\nu)})^{-3/2} \sum_{i=1}^n \mathbb{E} \left[ \left| \sum_{t \in \{0,1\}} \mathbf{e}_{1+\nu}^{\top} \mathbf{H}^{-1} \Gamma_{t,\mathbf{x}}^{-1} \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right|^3 \right] \\ &\lesssim n^{-2} h^{-|\nu|-d} (\Omega_{\mathbf{x},\mathbf{x}}^{(\nu)})^{-3/2} \mathbb{E} \left[ \left| \sum_{t \in \{0,1\}} \mathbf{e}_{1+\nu}^{\top} \mathbf{H}^{-1} \Gamma_{t,\mathbf{x}}^{-1} \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right|^2 \right] \\ &\lesssim n^{-1} h^{-|\nu|-d} (\Omega_{\mathbf{x},\mathbf{x}}^{(\nu)})^{-1/2} \\ &\lesssim (nh^d)^{-1/2}, \end{aligned}$$

where in the second line we used Assumption SA-1(v), in the third line we used

$$\left| \sum_{t \in \{0,1\}} \mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1} \mathbf{\Gamma}_{t,\mathbf{x}}^{-1} \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h} \right) K_h(\mathbf{X}_i - \mathbf{x}) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right| \lesssim h^{-|\nu|-d},$$

in the fourth line we used the definition of  $\Omega_{\mathbf{x},\mathbf{x}}^{(\nu)}$ . Hence the Berry-Esseen inequality gives

$$\mathbf{s}_n = \sup_{u \in \mathbb{R}} \left| \mathbb{P} \left( \bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) \leq u \right) - \Phi(u) \right| = o(1). \quad (\text{SA-5.1})$$

Although Lemma SA-2.1 to Lemma SA-2.4 provides convergence results uniformly in  $\mathbf{x}$ , for pointwise result with fix  $\mathbf{x} \in \mathcal{B}$ , we can replace the class of functions in the proof by one *singleton* corresponding to  $\mathbf{x}$ , and get: If  $h^{p+1} \sqrt{nh^d} \rightarrow 0$  and  $n^{\frac{\nu}{2+\nu}} h^d \rightarrow 0$ , then

$$\left| \hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) \right| \lesssim_{\mathbb{P}} \mathbf{r}_n, \quad (\text{SA-5.2})$$

where  $\mathbf{r}_n = h^{p+1} \sqrt{nh^d} + 1/\sqrt{nh^d} + 1/(n^{\frac{\nu}{2+\nu}} h^d)$ . Take  $Z$  to be a standard Gaussian random variable and using anti-concentration arguments, for any  $t \in \mathbb{R}$ ,

$$\begin{aligned} \mathbb{P}(\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) \leq t) &= \mathbb{P}(\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) \leq t, |\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \leq \mathbf{r}_n) + \mathbb{P}(\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) \leq t, |\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \geq \mathbf{r}_n) \\ &\leq \mathbb{P}(\bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) \leq t + \mathbf{r}_n) + \mathbb{P}(|\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \geq \mathbf{r}_n) \\ &\leq \mathbb{P}(Z \leq t + \mathbf{r}_n) + \mathbb{P}(|\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \geq \mathbf{r}_n) + \mathbf{s}_n \\ &= \Phi(t) + \sup_{t \in \mathbb{R}} |\mathbb{P}(t \leq Z \leq t + \mathbf{r}_n)| + \mathbb{P}(|\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \geq \mathbf{r}_n) + \mathbf{s}_n, \end{aligned}$$

where in the third line we used Equation (SA-5.1), in the fourth line we used Equation (SA-5.2) and  $\mathbb{P}(t \leq Z \leq t + \mathbf{r}_n) = o(1)$ . Similarly, for any  $t \in \mathbb{R}$ ,

$$\begin{aligned} \mathbb{P}(\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) \leq t) &= \mathbb{P}(\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) \leq t, |\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \leq \mathbf{r}_n) + \mathbb{P}(\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) \leq t, |\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \geq \mathbf{r}_n) \\ &\geq \mathbb{P}(\bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) \leq t - \mathbf{r}_n) - \mathbb{P}(|\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \geq \mathbf{r}_n) \\ &\geq \mathbb{P}(Z \leq t - \mathbf{r}_n) - \mathbb{P}(|\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \geq \mathbf{r}_n) - \mathbf{s}_n \\ &= \Phi(t) - \sup_{t \in \mathbb{R}} |\mathbb{P}(t - \mathbf{r}_n \leq Z \leq t)| - \mathbb{P}(|\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \geq \mathbf{r}_n) - \mathbf{s}_n. \end{aligned}$$

It follows that

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P}(\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) \leq t) - \Phi(t) \right| \leq \sup_{t \in \mathbb{R}} |\mathbb{P}(t - \mathbf{r}_n \leq Z \leq t + \mathbf{r}_n)| + \mathbb{P}(|\hat{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \geq \mathbf{r}_n) + \mathbf{s}_n = o(1). \quad \blacksquare$$

## SA-5.8 Proof of Theorem SA-2.4

The proof follows directly from Theorem SA-2.3. \blacksquare

### SA-5.9 Proof of Theorem SA-2.5

The result follows from Lemma SA-2.2 and Lemma SA-2.3. ■

### SA-5.10 Proof of Theorem SA-2.6

The feasible  $t$ -statistic can be decomposed as follows:

$$\hat{T}^{(\nu)}(\mathbf{x}) = \frac{\hat{\tau}^{(\nu)}(\mathbf{x}) - \tau^{(\nu)}(\mathbf{x})}{\sqrt{\hat{\Omega}_{\mathbf{x},\mathbf{x}}^{(\nu)}}} = \bar{T}^{(\nu)}(\mathbf{x}) + G_1^{(\nu)}(\mathbf{x}) + G_2^{(\nu)}(\mathbf{x}), \quad \mathbf{x} \in \mathcal{B},$$

where

$$\begin{aligned} G_1^{(\nu)}(\mathbf{x}) &= \left( \mathbb{E}[\hat{\tau}^{(\nu)}(\mathbf{x}) | \mathbf{X}] - \tau^{(\nu)}(\mathbf{x}) \right) (\hat{\Omega}_{\mathbf{x},\mathbf{x}}^{(\nu)})^{-1/2}, \\ G_2^{(\nu)}(\mathbf{x}) &= \mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1} \left[ \left( \hat{\Gamma}_{1,\mathbf{x}}^{-1} \mathbf{Q}_{1,\mathbf{x}} - \hat{\Gamma}_{0,\mathbf{x}}^{-1} \mathbf{Q}_{0,\mathbf{x}} \right) (\hat{\Omega}_{\mathbf{x},\mathbf{x}}^{(\nu)})^{-\frac{1}{2}} - \left( \Gamma_{1,\mathbf{x}}^{-1} \mathbf{Q}_{1,\mathbf{x}} - \Gamma_{0,\mathbf{x}}^{-1} \mathbf{Q}_{0,\mathbf{x}} \right) (\Omega_{\mathbf{x},\mathbf{x}}^{(\nu)})^{-\frac{1}{2}} \right]. \end{aligned}$$

By Lemma SA-2.2 and Lemma SA-2.4,

$$\sup_{\mathbf{x} \in \mathcal{B}} |G_1^{(\nu)}(\mathbf{x})| \lesssim_{\mathbb{P}} h^{p+1-|\nu|} (nh^{d+2|\nu|})^{1/2} \lesssim h^{p+1} \sqrt{nh^d}.$$

By Lemma SA-2.1, Lemma SA-2.3 and Lemma SA-2.4, for  $t \in \{0, 1\}$  we have

$$\sup_{\mathbf{x} \in \mathcal{B}} |e_{1+\nu}^\top \mathbf{H}^{-1} [\hat{\Gamma}_{t,\mathbf{x}}^{-1} - \Gamma_{t,\mathbf{x}}^{-1}] \mathbf{Q}_{t,\mathbf{x}} (\hat{\Omega}_{\mathbf{x},\mathbf{x}}^{(\nu)})^{-1/2}| \lesssim \sqrt{\log n} \left( \sqrt{\frac{\log n}{nh^d}} + \frac{\log n}{n^{\frac{1+\nu}{2+\nu}} h^d} \right)$$

By Lemma SA-2.1, Lemma SA-2.3 and Lemma SA-2.4, for  $t \in \{0, 1\}$  we have

$$\begin{aligned} & \sup_{\mathbf{x} \in \mathcal{B}} \left| e_{1+\nu}^\top \mathbf{H}^{-1} \Gamma_{t,\mathbf{x}}^{-1} \mathbf{Q}_{t,\mathbf{x}} \left[ (\hat{\Omega}_{\mathbf{x},\mathbf{x}}^{(\nu)})^{-1/2} - (\Omega_{\mathbf{x},\mathbf{x}}^{(\nu)})^{-1/2} \right] \right| \\ & \lesssim_{\mathbb{P}} h^{-|\nu|} \cdot \left( \sqrt{\frac{\log n}{nh^d}} + \frac{\log n}{n^{\frac{1+\nu}{2+\nu}} h^d} \right) \cdot \sqrt{nh^{d+2\nu}} \left( \sqrt{\frac{\log n}{nh^d}} + \frac{\log n}{n^{\frac{\nu}{2+\nu}} h^d} + h^{p+1} \right) \\ & \lesssim \frac{\log n}{\sqrt{nh^d}} + \frac{(\log n)^{3/2}}{n^{\frac{\nu}{2+\nu}} h^d}. \end{aligned}$$

Combining the previous two displays, we get

$$\sup_{\mathbf{x} \in \mathcal{B}} |G_2^{(\nu)}(\mathbf{x})| \lesssim_{\mathbb{P}} \sqrt{\log n} \left( \sqrt{\frac{\log n}{nh^d}} + \frac{\log n}{n^{\frac{\nu}{2+\nu}} h^d} \right).$$

It follows from the decomposition of  $\hat{T}^{(\nu)}(\mathbf{x}) - \bar{T}^{(\nu)}(\mathbf{x})$  that

$$\sup_{\mathbf{x} \in \mathcal{B}} |\hat{T}^{(\nu)}(\mathbf{x}) - \bar{T}^{(\nu)}(\mathbf{x})| \lesssim_{\mathbb{P}} h^{p+1} \sqrt{nh^d} + \sqrt{\log n} \left( \sqrt{\frac{\log n}{nh^d}} + \frac{\log n}{n^{\frac{\nu}{2+\nu}} h^d} \right). ■$$

The following lemma is used in the proof of Theorem SA-3.5.

**Lemma SA-5.2 (VC Class to VC2 Class)**

Assume  $\mathcal{F}$  is a VC class on a measure space  $(\mathcal{X}, \mathcal{B})$  in the sense that there exists an envelope function  $F$  and positive constants  $c(\mathcal{F}), d(\mathcal{F})$  such that for all  $0 < \varepsilon < 1$ ,

$$\sup_{\mathbb{Q} \in \mathcal{A}(\mathcal{X})} N(\mathcal{F}, \|\cdot\|_{\mathbb{Q},1}, \varepsilon \|F\|_{\mathbb{Q},1}) \leq c(\mathcal{F}) \varepsilon^{-d(\mathcal{F})}.$$

Then,  $\mathcal{F}$  is also VC2 in the sense that for all  $0 < \varepsilon < 1$ ,

$$\sup_{\mathbb{Q} \in \mathcal{A}(\mathcal{X})} N(\mathcal{F}, \|\cdot\|_{\mathbb{Q},2}, \varepsilon \|F\|_{\mathbb{Q},2}) \leq c(\mathcal{F}) (\varepsilon^2/2)^{-d(\mathcal{F})}.$$

**Proof of Lemma SA-5.2.** Let  $\mathbb{Q}$  be a finite discrete probability measure. Let  $f, g \in \mathcal{F}$ . Then

$$\int |f - g|^2 d\mathbb{Q} \leq 2 \int |f - g| F d\mathbb{Q}.$$

Suppose  $\mathbb{Q}$  is supported on  $\{c_1, \dots, c_p\}$ . Define another probability measure  $\tilde{\mathbb{Q}}(c_k) = F(c_k) \mathbb{Q}(c_k) / \|F\|_{\mathbb{Q},1}$ . Then,

$$\begin{aligned} \int |f - g|^2 d\mathbb{Q} &\leq 2 \|F\|_{\mathbb{Q},1} \int |f - g| d\tilde{\mathbb{Q}} \\ &\leq 2 \|F\|_{\mathbb{Q},1} \|f - g\|_{\tilde{\mathbb{Q}},1}. \end{aligned}$$

Hence if we take an  $\varepsilon^2/2$ -net in  $(\mathcal{F}, \|\cdot\|_{\tilde{\mathbb{Q}},1})$  with cardinality no greater than  $c(\mathcal{F}) \varepsilon^{-d(\mathcal{F})}$ , then for any  $f \in \mathcal{F}$ , there exists a  $g \in \mathcal{F}$  such that  $\|f - g\|_{\tilde{\mathbb{Q}},1} \leq \varepsilon^2/2 \|F\|_{\tilde{\mathbb{Q}},1}$ , and hence

$$\|f - g\|_{\mathbb{Q},2}^2 \leq 2 \varepsilon^2/2 \|F\|_{\mathbb{Q},1} \|F\|_{\tilde{\mathbb{Q}},1} \leq \varepsilon^2 \|F\|_{\mathbb{Q},2}^2.$$

Hence  $\sup_{\mathbb{Q} \in \mathcal{A}(\mathcal{X})} N(\mathcal{F}, \|\cdot\|_{\mathbb{Q},2}, \varepsilon \|F\|_{\mathbb{Q},2}) \leq c(\mathcal{F}) (\varepsilon^2/2)^{-d(\mathcal{F})}$ . ■

### SA-5.11 Proof of Theorem SA-2.7

First, we consider the class of functions  $\mathcal{F}_t = \{\mathcal{K}_t^{(\nu)}(\cdot; \mathbf{x}) : \mathbf{x} \in \mathcal{B}\}$ ,  $t \in \{0, 1\}$ . W.l.o.g., we can assume  $\mathcal{X} = [0, 1]^d$ , and  $\mathbb{Q}_{\mathcal{F}_t} = \mathbb{P}_X$  is a valid surrogate measure for  $\mathbb{P}_X$  with respect to  $\mathcal{F}_t$ , and  $\phi_{\mathcal{F}_t} = \text{Id}$  is a valid normalizing transformation (as in Lemma SA-4.1). This implies the constants  $c_1$  and  $c_2$  from Lemma SA-4.1 are all 1.

#### I. Properties of $\mathcal{F}_t$

**Envelope Function:** By Lemma SA-2.1 and Lemma SA-2.4 and the fact that  $\text{Supp}(K)$  is compact,

$$\sup_{\mathbf{x} \in \mathcal{B}} \sup_{\xi \in \mathcal{X}} |\mathcal{K}_t^{(\nu)}(\xi; \mathbf{x})| \lesssim \frac{1}{\sqrt{n} h^{d+|\nu|}} \sup_{\mathbf{x} \in \mathcal{B}} (\|\Gamma_{1,\mathbf{x}}^{-1}\| + \|\Gamma_{0,\mathbf{x}}^{-1}\|) \sup_{\mathbf{x} \in \mathcal{B}} \left| \left( \Omega_{\mathbf{x},\mathbf{x}}^{(\nu)} \right)^{-\frac{1}{2}} \right| \lesssim h^{-d/2}.$$

Hence there exists a constant  $C_1 > 0$  such that  $M_{\mathcal{F}_t} = C_1 h^{-d/2}$  is a constant envelope function of  $\mathcal{F}$ .

**$L_1$  Bound:**

$$E_{\mathcal{F}_t} = \sup_{\mathbf{x} \in \mathcal{B}} \mathbb{E} \left[ |\mathcal{K}_t^{(\nu)}(\mathbf{X}_i; \mathbf{x})| \right] \lesssim h^{d/2}.$$

**Uniform Variation:** *Case 1: Suppose  $K$  is Lipschitz.* By (iv) in Assumption SA-1 and Assumption SA-2,

$$\mathbf{L}_{\mathcal{F}_t} = \sup_{\mathbf{x} \in \mathcal{B}} \sup_{\xi, \xi' \in \mathcal{X}} \frac{|\mathcal{K}_t^{(\nu)}(\xi; \mathbf{x}) - \mathcal{K}_t^{(\nu)}(\xi'; \mathbf{x})|}{\|\xi - \xi'\|_\infty} \lesssim h^{-d/2-1}.$$

Each entry of  $\mathbf{\Gamma}_{t, \mathbf{x}}$  and  $\mathbf{\Sigma}_{t, \mathbf{x}}$  are of the form  $\int \left(\frac{\xi - \mathbf{x}}{h}\right)^{\mathbf{u} + \mathbf{v}} K_h(\xi - \mathbf{x}) \mathbb{1}(\xi \in \mathcal{A}_t) f(\xi) d\xi$  and  $\int \left(\frac{\xi - \mathbf{x}}{h}\right)^{\mathbf{u} + \mathbf{v}} K_h(\xi - \mathbf{x}) \sigma_t(\xi)^2 \mathbb{1}(\xi \in \mathcal{A}_t) d\xi$  for some multi-index  $\mathbf{u}$  and  $\mathbf{v}$ , respectively. Hence by Assumption SA-2, each entry of  $\mathbf{\Gamma}_{t, \mathbf{x}}$  and  $\mathbf{\Sigma}_{t, \mathbf{x}}$  are  $h^{-1}$ -Lipschitz in  $\mathbf{x}$ . Hence there exists a constant  $C_2$  such that for all  $\mathbf{x}, \mathbf{x}' \in \mathcal{B}$ ,

$$\|\mathbf{\Gamma}_{t, \mathbf{x}}^{-1} - \mathbf{\Gamma}_{t, \mathbf{x}'}^{-1}\| \leq \|\mathbf{\Gamma}_{t, \mathbf{x}}^{-1}\| \|\mathbf{\Gamma}_{t, \mathbf{x}} - \mathbf{\Gamma}_{t, \mathbf{x}'}\| \|\mathbf{\Gamma}_{t, \mathbf{x}'}^{-1}\| \leq C_2 h^{-1} \|\mathbf{x} - \mathbf{x}'\|.$$

Also by definition of  $\Omega_{t, \mathbf{x}}$  and (iv) in Assumption SA-2, there exists  $C_3$  such that for all  $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$ ,

$$\begin{aligned} \left| \Omega_{t, \mathbf{x}}^{(\nu)} - \Omega_{t, \mathbf{x}'}^{(\nu)} \right| &\leq C_3 (nh^{d+2|\nu|+1})^{-1} \|\mathbf{x} - \mathbf{x}'\|_\infty, \\ \left| \left( \Omega_{t, \mathbf{x}}^{(\nu)} \right)^{-1/2} - \left( \Omega_{t, \mathbf{x}'}^{(\nu)} \right)^{-1/2} \right| &\leq \frac{1}{2} \inf_{\mathbf{z} \in \mathcal{X}} \left( \Omega_{t, \mathbf{z}}^{(\nu)} \right)^{-3/2} \left| \Omega_{t, \mathbf{x}}^{(\nu)} - \Omega_{t, \mathbf{x}'}^{(\nu)} \right| \leq \frac{1}{2} C_3 h^{-1} (nh^{d+2|\nu|})^{1/2} \|\mathbf{x} - \mathbf{x}'\|_\infty. \end{aligned}$$

It then follows that we have a uniform Lipschitz property with respect to the point of evaluation:

$$\mathbf{L}_{\mathcal{F}_t} = \sup_{\xi \in \mathcal{X}} \sup_{\mathbf{x}, \mathbf{x}' \in \mathcal{B}} \frac{|\mathcal{K}_t^{(\nu)}(\xi; \mathbf{x}) - \mathcal{K}_t^{(\nu)}(\xi; \mathbf{x}')|}{\|\mathbf{x} - \mathbf{x}'\|_\infty} \lesssim h^{-d/2-1}.$$

Let  $\mathbf{x} \in \mathcal{B}$ . Then,  $\mathcal{K}_t^{(\nu)}(\cdot; \mathbf{x})$  is supported on  $\mathbf{x} + \mathbf{c}[-h, h]^d$ . Then,

$$\mathbf{TV}_{\mathcal{F}_t} \lesssim \mathbf{m}(\mathbf{c}[-h, h]^d) \mathbf{L}_{\mathcal{F}_t} \lesssim h^{d/2-1}$$

*Case 2: Suppose  $K = \mathbb{1}(\cdot \in [-1, 1]^d)$ .* Consider

$$\tilde{\mathcal{K}}_t^{(\nu)}(\mathbf{u}; \mathbf{x}) = n^{-1/2} (\Omega_{\mathbf{x}, \mathbf{x}}^{(\nu)})^{-1/2} \mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1} \mathbf{\Gamma}_{t, \mathbf{x}}^{-1} \mathbf{R}_p \left( \frac{\mathbf{u} - \mathbf{x}}{h} \right) h^{-d}, \quad \mathbf{u} \in \mathcal{X}, t \in \{0, 1\}.$$

Then,  $\mathcal{K}^{(\nu)}(\mathbf{u}; \mathbf{x}) = \tilde{\mathcal{K}}^{(\nu)}(\mathbf{u}; \mathbf{x}) \mathbb{1}(\mathbf{u} - \mathbf{x} \in [-1, 1]^d)$  for all  $\mathbf{u} \in \mathcal{X}, \mathbf{x} \in \mathcal{B}$ . Consider  $\tilde{\mathcal{F}}_t = \{\tilde{\mathcal{K}}^{(\nu)}(\cdot; \mathbf{x}) : \mathbf{x} \in \mathcal{B}\}$ ,  $t \in \{0, 1\}$ . Then, the argument above implies

$$\mathbf{TV}_{\tilde{\mathcal{F}}_t} \lesssim \mathbf{m}(\mathbf{c}[-h, h]^d) \mathbf{L}_{\mathcal{F}_t} \lesssim h^{d/2-1}.$$

Consider  $\mathcal{L} = \{\mathbb{1}((\cdot - \mathbf{x})/h \in [-1, 1]^d) : \mathbf{x} \in \mathcal{B}\}$ . Then, using a product rule, we have

$$\mathbf{TV}_{\mathcal{F}_t} \leq \mathbf{TV}_{\tilde{\mathcal{F}}_t} \mathbf{M}_{\mathcal{L}} + \mathbf{M}_{\tilde{\mathcal{F}}_t} \mathbf{TV}_{\mathcal{L}} \lesssim h^{d/2-1} \cdot 1 + h^{-d/2} h^{d-1} \lesssim h^{d/2-1}.$$

**VC-type Class:** *Case 1: Suppose  $K$  is Lipschitz.* We will use Cattaneo et al. (2024, Lemma 7). To make the notation consistent, define

$$f_{\mathbf{x}}(\cdot) = \frac{1}{\sqrt{n\Omega_{\mathbf{x}, \mathbf{x}}^{(\nu)}}} \mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1} \mathbf{\Gamma}_t^{-1} \mathbf{R}_p(\cdot) K(\cdot), \mathbf{x} \in \mathcal{B},$$

and  $\mathcal{H} = \{g_{\mathbf{x}}(\frac{\cdot - \mathbf{x}}{h}) : \mathbf{x} \in \mathcal{B}\}$ . Notice that  $f_{\mathbf{x}}(\frac{\cdot - \mathbf{x}}{h}) = h^d \frac{1}{\sqrt{n\Omega_{\mathbf{x}, \mathbf{x}}^{(\nu)}}} \mathbf{e}_{1+\nu}^\top \mathbf{H}^{-1} \mathbf{\Gamma}^{-1} \mathbf{R}_p(\frac{\cdot - \mathbf{x}}{h}) K_h(\cdot - \mathbf{x})$ . Then, the following conditions for Lemma 7 in Cattaneo et al. (2024) hold:

$$\begin{aligned}
(i) \text{ boundedness} & \sup_{\mathbf{z}} \sup_{\mathbf{z}'} |f_{\mathbf{z}}(\mathbf{z}')| \leq \mathbf{c}, \\
(ii) \text{ compact support} & \text{supp}(f_{\mathbf{z}}(\cdot)) \subseteq [-\mathbf{c}, \mathbf{c}]^d, \forall \mathbf{z} \in \mathcal{X}, \\
(iii) \text{ Lipschitz continuity} & \sup_{\mathbf{z}} |f_{\mathbf{z}}(\mathbf{z}') - f_{\mathbf{z}}(\mathbf{z}'')| \leq \mathbf{c} |\mathbf{z}' - \mathbf{z}''| \\
& \sup_{\mathbf{z}} |f_{\mathbf{z}'}(\mathbf{z}) - f_{\mathbf{z}''}(\mathbf{z})| \leq \mathbf{c} h^{-1} |\mathbf{z}' - \mathbf{z}''|.
\end{aligned}$$

Then, by Cattaneo et al. (2024, Lemma 7), there exists a constant  $\mathbf{c}'$  only depending on  $\mathbf{c}$  and  $d$  that for any  $0 \leq \varepsilon \leq 1$ ,

$$\sup_{Q \in \mathcal{A}(\mathcal{X})} N(\mathcal{H}, \|\cdot\|_{Q,1}, (2c+1)^{d+1} \varepsilon) \leq \mathbf{c}' \varepsilon^{-d-1} + 1,$$

where  $\mathcal{A}(\mathcal{X})$  denotes the collections of all finite discrete measures on  $\mathcal{X} = [0, 1]^d$ . It then follows from Lemma SA-5.2 that with the constant envelope function  $M_{\mathcal{F}_t} = h^{-d/2}$ , for any  $0 \leq \varepsilon \leq 1$ ,

$$\sup_{Q \in \mathcal{A}(\mathcal{X})} N(\mathcal{F}_t, \|\cdot\|_{Q,2}, (2c+1)^{d+1} \varepsilon M_{\mathcal{F}_t}) \leq \mathbf{c}' \varepsilon^{-d-1} + 1.$$

*Case 2: Suppose  $K = \mathbb{1}(\cdot \in [-1, 1]^d)$ .* Recall  $\tilde{\mathcal{F}}_t$  and  $\mathcal{L}$  defined in the **Uniform Variation** section. The same argument as before shows

$$\sup_{Q \in \mathcal{A}(\mathcal{X})} N(\tilde{\mathcal{F}}_t, \|\cdot\|_{Q,2}, (2c+1)^{d+1} \varepsilon M_{\tilde{\mathcal{F}}_t}) \leq \mathbf{c}' \varepsilon^{-d-1} + 1, \quad \varepsilon \in (0, 1],$$

where  $\tilde{\mathcal{F}}_t = h^{-d/2}$ . By van der Vaart and Wellner (1996, Example 2.6.1), the class  $\mathcal{L} = \{\mathbb{1}((\cdot - \mathbf{x})/h \in [-1, 1]^d) : \mathbf{x} \in \mathcal{B}\}$  has VC dimension no greater than  $2d$ , and by van der Vaart and Wellner (1996, Theorem 2.6.4),

$$\sup_{Q \in \mathcal{A}(\mathcal{X})} N(\mathcal{L}, \|\cdot\|_{Q,2}, \varepsilon) \leq 2d(4e)^{2d} \varepsilon^{-4d}, \quad 0 < \varepsilon \leq 1.$$

Putting together, we have

$$\sup_{Q \in \mathcal{A}(\mathcal{X})} N(\mathcal{F}_t, \|\cdot\|_{Q,2}, \varepsilon C_1 M_{\tilde{\mathcal{F}}_t}) \leq C_2 \varepsilon^{-4d},$$

where  $C_1, C_2$  are constants only depending on  $d$ .

## II. Properties of $\mathcal{G}$

Recall for each  $\mathbf{x} \in \mathcal{B}$ ,

$$g_{\mathbf{x}}(\mathbf{u}) = \mathbb{1}_{\mathcal{A}_1}(\mathbf{u}) \mathcal{K}_t^{(\nu)}(\mathbf{u}; \mathbf{x}) - \mathbb{1}_{\mathcal{A}_0}(\mathbf{u}) \mathcal{K}_t^{(\nu)}(\mathbf{u}; \mathbf{x}), \mathbf{u} \in \mathcal{X},$$

and  $\mathcal{G} = \{g_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$ . Hence

$$\mathbf{M}_{\mathcal{G}} \lesssim h^{-d/2}, \quad \mathbf{E}_{\mathcal{G}} \lesssim h^{d/2}, \quad \sup_Q N(\mathcal{G}, \|\cdot\|_{Q,2}, \varepsilon(2c+1)^{d+1} \mathbf{M}_{\mathcal{G}}) \leq 2\mathbf{c}'\varepsilon^{-d-1} + 2.$$

**Total Variation:** Observe that  $\mathbb{1}_{\mathcal{A}_t}(\mathbf{u})\mathcal{K}_t^{(\nu)}(\mathbf{u}; \mathbf{x}) \neq 0$  implies  $E_{t,\mathbf{x}} = \mathbf{u} \in \{\mathbf{y} \in \mathcal{A}_t : (\mathbf{y} - \mathbf{x})/h \in \text{Supp}(K)\}$ , and

$$\mathbb{1}(\mathbf{u} \in \mathcal{A}_t)\mathcal{K}_t^{(\nu)}(\mathbf{u}; \mathbf{x}) = \mathbb{1}(\mathbf{u} \in E_{t,\mathbf{x}})\mathcal{K}_t^{(\nu)}(\mathbf{u}; \mathbf{x}), \quad \forall \mathbf{u} \in \mathcal{X}.$$

By the assumption that the De Giorgi perimeter of  $E_{t,\mathbf{x}}$  satisfies  $\mathcal{L}(E_{t,\mathbf{x}}) \leq Ch^{d-1}$  and using  $\text{TV}_{\{gh\}} \leq \mathbf{M}_{\{g\}}\text{TV}_{\{h\}} + \mathbf{M}_{\{h\}}\text{TV}_{\{g\}}$ , we have

$$\text{TV}_{\mathcal{G}} = \sup_{\mathbf{x} \in \mathcal{B}} \text{TV}_{\{g_{\mathbf{x}}\}} \leq \sup_{\mathbf{x} \in \mathcal{B}} \sum_{t \in \{0,1\}} \text{TV}_{\{\mathbb{1}_{\mathcal{A}_t}\mathcal{K}_t^{(\nu)}(\cdot; \mathbf{x})\}} \leq \sup_{\mathbf{x} \in \mathcal{B}} \sum_{t \in \{0,1\}} \text{TV}_{\{\mathcal{K}_t^{(\nu)}(\cdot; \mathbf{x})\}} + \mathbf{M}_{\mathcal{F}_t} \text{TV}_{\{\mathbb{1}_{E_{t,\mathbf{x}}}\}} \lesssim h^{d/2-1}.$$

Then, by Lemma SA-4.1, on a possibly enlarged probability space, there exists a mean-zero Gaussian process  $Z^{(\nu)}$  with the same covariance structure such that

$$\mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} \left| \bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) - Z^{(\nu)}(\mathbf{x}) \right| \right] \lesssim (\log n)^{\frac{3}{2}} \left( \frac{1}{nh^d} \right)^{\frac{1}{d+2} \cdot \frac{v}{v+2}} + \log(n) \left( \frac{1}{n^{\frac{v}{2+v}} h^d} \right)^{\frac{1}{2}}.$$

■

To build up the proof for confidence bands, we need the following lemmas.

### Lemma SA-5.3 (Distance Between Infeasible Gaussian and Bahadur Representation)

Suppose the conditions of Theorem SA-2.7 hold. Then, for any multi-index  $|\nu| \leq p$ , we have

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| \bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) \right| \leq u \right) - \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| Z^{(\nu)}(\mathbf{x}) \right| \leq u \right) \right| \lesssim \left[ (\log n)^{\frac{3}{2}} \left( \frac{1}{nh^d} \right)^{\frac{1}{d+2} \cdot \frac{v}{v+2}} + \log(n) \sqrt{\frac{1}{n^{\frac{v}{2+v}} h^d}} \right]^{1/2}.$$

**Proof of Lemma SA-5.3.** Denote  $R_n = (\log n)^{\frac{3}{2}} \left( \frac{1}{nh^d} \right)^{\frac{1}{d+2} \cdot \frac{v}{v+2}} + \log(n) \sqrt{\frac{1}{n^{\frac{v}{2+v}} h^d}}$ . Let  $\alpha_n$  to be determined. For any  $u > 0$ ,

$$\begin{aligned} & \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| \bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) \right| \leq u \right) \\ & \leq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| Z^{(\nu)}(\mathbf{x}) \right| \leq \sup_{\mathbf{x} \in \mathcal{B}} \left| \bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) - Z^{(\nu)}(\mathbf{x}) \right| + u \right) \\ & \leq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| Z^{(\nu)}(\mathbf{x}) \right| \leq u + \alpha_n \right) + \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| Z^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) \right| > \alpha_n \right) \\ & \leq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| Z^{(\nu)}(\mathbf{x}) \right| \leq u \right) + 4\alpha_n \left( \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} \left| Z^{(\nu)}(\mathbf{x}) \right| \right] + 1 \right) + \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| Z^{(\nu)}(\mathbf{x}) - \bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) \right| > \alpha_n \right) \\ & \leq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| Z^{(\nu)}(\mathbf{x}) \right| \leq u \right) + 4\alpha_n \left( \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} \left| Z^{(\nu)}(\mathbf{x}) \right| \right] + 1 \right) + \frac{CR_n}{\alpha_n}, \end{aligned}$$

where in the fourth line we have used the Gaussian Anti-concentration Inequality in (Chernozhukov et al., 2014a, Theorem 2.1), and in the last line we have used the tail bound in Theorem SA-2.7. Similarly, for any

$u > 0$ , we have the lower bound

$$\begin{aligned}
& \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}^{(\nu)}(\mathbf{x})| \leq u \right) \\
& \geq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \leq u - \sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}^{(\nu)}(\mathbf{x}) - Z^{(\nu)}(\mathbf{x})| \right) \\
& \geq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \leq u - \alpha_n \right) - \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x}) - \bar{T}^{(\nu)}(\mathbf{x})| > \alpha_n \right) \\
& \geq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \leq u \right) - 4\alpha_n \left( \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \right] + 1 \right) - \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x}) - \bar{T}^{(\nu)}(\mathbf{x})| > \alpha_n \right) \\
& \geq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \leq u \right) - 4\alpha_n \left( \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \right] + 1 \right) - \frac{CR_n}{\alpha_n}.
\end{aligned}$$

Notice that  $Z^{(\nu)}(\mathbf{x}), \mathbf{x} \in \mathcal{B}$  is a mean-zero Gaussian process such that

$$\begin{aligned}
d \left( Z^{(\nu)}(\mathbf{x}), Z^{(\nu)}(\mathbf{y}) \right) &= \mathbb{E} \left[ \left( Z^{(\nu)}(\mathbf{x}) - Z^{(\nu)}(\mathbf{y}) \right)^2 \right]^{\frac{1}{2}} = \mathbb{E} \left[ \left( \bar{T}^{(\nu)}(\mathbf{x}) - G_0^{(\nu)}(\mathbf{y}) \right)^2 \right]^{\frac{1}{2}} \\
&= \mathbb{E} \left[ \left( \mathcal{K}(\mathbf{X}_i, \mathbf{x}) - \mathcal{K}(\mathbf{X}_i, \mathbf{y}) \right)^2 \sigma^2(\mathbf{X}_i) \right]^{\frac{1}{2}} \leq C' l_{n,2} \|\mathbf{x} - \mathbf{y}\|_{\infty}, \\
\sup_{\mathbf{x} \in \mathcal{B}} d(Z^{(\nu)}(\mathbf{x}), Z^{(\nu)}(\mathbf{x})) &= \sup_{\mathbf{x} \in \mathcal{B}} \mathbb{E} [\mathcal{K}(\mathbf{X}_i, \mathbf{x})^2 \sigma^2(\mathbf{X}_i)] \lesssim 1.
\end{aligned}$$

where  $C'$  is a constant and  $l_{n,2} \asymp h_n^{-1}$ . Then, by Corollary 2.2.8 in [van der Vaart and Wellner \(1996\)](#), we have

$$\mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \right] \leq \mathbb{E} [|Z_n(\mathbf{x}_0)|] + \int_0^{2 \sup_{\mathbf{x} \in \mathcal{B}} d(Z^{(\nu)}(\mathbf{x}), Z^{(\nu)}(\mathbf{x}))} \sqrt{d \log \left( \frac{C'' l_{n,2}}{\varepsilon} \right)} \lesssim 1.$$

Hence by choosing  $\alpha_n^* \asymp \sqrt{R_n}$ , we have

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}^{(\nu)}(\mathbf{x})| \leq u \right) - \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \leq u \right) \right| \lesssim \sqrt{R_n}.$$

■

#### Lemma SA-5.4 (Distance Between Bahadur Representation and t-statistics)

Suppose the conditions in Theorem SA-2.7 hold. Then

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\hat{T}^{(\nu)}(\mathbf{x})| \leq u \right) - \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\bar{T}^{(\nu)}(\mathbf{x})| \leq u \right) \right| = o(1).$$

For notational simplicity, define  $r_n$  and  $\alpha_n$  to be sequences such that

$$\begin{aligned}
r_n &= \left[ (\log n)^{\frac{3}{2}} \left( \frac{1}{nh^d} \right)^{\frac{1}{d+2} \cdot \frac{v}{v+2}} + \sqrt{\frac{(\log n)^2}{n^{\frac{v}{v+2}} h^d}} \right]^{1/2}, \\
\alpha_n &\ll \sqrt{\log(1/h)} \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right) + h_n^{p+1} \sqrt{nh_n^d}.
\end{aligned}$$

Then,  $\sup_{\mathbf{x} \in \mathcal{B}} |\bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \hat{\mathbf{T}}(\mathbf{x})| = o_{\mathbb{P}}(\alpha_n)$ . Hence for any  $u > 0$ ,

$$\begin{aligned}
& \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\hat{\mathbf{T}}(\mathbf{x})| \leq u \right) \\
& \leq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \leq u + \alpha_n \right) + \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \hat{\mathbf{T}}(\mathbf{x})| \geq \alpha_n \right) \\
& \leq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \leq u + \alpha_n \right) + r_n + o(1) \\
& \leq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \leq u \right) + 4\alpha_n \left( \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \right] + 1 \right) + r_n + o(1) \\
& \leq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \leq u \right) + 4\alpha_n \left( \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \right] + 1 \right) + 2r_n + o(1),
\end{aligned}$$

where in the third line we have used Lemma SA-5.3 and  $\sup_{\mathbf{x} \in \mathcal{B}} |\bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \hat{\mathbf{T}}(\mathbf{x})| = o_{\mathbb{P}}(\alpha_n)$ , in the fourth line we use the (Chernozhukov et al., 2014a, Theorem 2.1), and in the last line we have used Lemma SA-5.3 again. Similarly,

$$\begin{aligned}
& \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\hat{\mathbf{T}}(\mathbf{x})| \leq u \right) \\
& \geq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \leq u - \alpha_n \right) - \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\bar{\mathbf{T}}^{(\nu)}(\mathbf{x}) - \hat{\mathbf{T}}(\mathbf{x})| \geq \alpha_n \right) \\
& \geq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \leq u - \alpha_n \right) - r_n + o(1) \\
& \geq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \leq u \right) - 4\alpha_n \left( \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \right] + 1 \right) - r_n + o(1) \\
& \geq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \leq u \right) - 4\alpha_n \left( \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \right] + 1 \right) - 2r_n + o(1).
\end{aligned}$$

From the proof of Lemma SA-5.3,  $\mathbb{E} [\sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})|] \lesssim 1$ . Hence under the rate restrictions in this lemma,

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\hat{\mathbf{T}}^{(\nu)}(\mathbf{x})| \leq u \right) - \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\bar{\mathbf{T}}^{(\nu)}(\mathbf{x})| \leq u \right) \right| = o(1).$$

■

#### Lemma SA-5.5 (Distance Between Feasible Gaussian and Infeasible Gaussian)

Suppose the conditions for Theorem SA-2.7 hold. Then, for any multi-index  $|\nu| \leq p$ ,

$$\sup_{\mathbf{u} \in \mathbb{R}} \left| \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \leq u \right) - \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\hat{Z}^{(\nu)}(\mathbf{x})| \leq u \mid \mathbf{W}_n \right) \right| \lesssim_{\mathbb{P}} \log n \left( \sqrt{\frac{\log n}{nh^d}} + \frac{\log n}{n^{\frac{v}{2+v}} h^d} + h^{p+1} \right)^{\frac{1}{2}}.$$

**Proof of Lemma SA-5.5.** First, using Lemma SA-2.4, we provide an upper bound between covariance

functions of the feasible Gaussian process and the infeasible Gaussian process.

$$\begin{aligned}
& \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \left| \mathbf{\Pi}_{\mathbf{x}, \mathbf{y}} - \widehat{\mathbf{\Pi}}_{\mathbf{x}, \mathbf{y}} \right| \\
&= \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \left| \Omega_{\mathbf{x}, \mathbf{y}} / \sqrt{\Omega_{\mathbf{x}, \mathbf{x}} \Omega_{\mathbf{y}, \mathbf{y}}} - \widehat{\Omega}_{\mathbf{x}_1, \mathbf{x}_2} / \sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}} \widehat{\Omega}_{\mathbf{y}, \mathbf{y}}} \right| \\
&= \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \left| \left( \Omega_{\mathbf{x}, \mathbf{y}} - \widehat{\Omega}_{\mathbf{x}, \mathbf{y}} \right) / \sqrt{\Omega_{\mathbf{x}, \mathbf{x}} \Omega_{\mathbf{y}, \mathbf{y}}} + \frac{\widehat{\Omega}_{\mathbf{x}, \mathbf{y}}}{\sqrt{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}} \widehat{\Omega}_{\mathbf{y}, \mathbf{y}}}} \left( \sqrt{\frac{\widehat{\Omega}_{\mathbf{x}, \mathbf{x}} \widehat{\Omega}_{\mathbf{y}, \mathbf{y}}}{\Omega_{\mathbf{x}, \mathbf{x}} \Omega_{\mathbf{y}, \mathbf{y}}}} - 1 \right) \right|
\end{aligned}$$

From Lemma SA-2.4 and the fact that  $|\sqrt{x} - \sqrt{y}| \leq (x \wedge y)^{-1/2} |x - y|/2$  for  $x, y > 0$ ,

$$\begin{aligned}
& \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \frac{\left| \left( \widehat{\Omega}_{\mathbf{x}, \mathbf{x}} \widehat{\Omega}_{\mathbf{y}, \mathbf{y}} \right)^{1/2} - \left( \Omega_{\mathbf{x}, \mathbf{x}} \Omega_{\mathbf{y}, \mathbf{y}} \right)^{1/2} \right|}{\left( \Omega_{\mathbf{x}, \mathbf{x}} \Omega_{\mathbf{y}, \mathbf{y}} \right)^{1/2}} \lesssim \frac{\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \left| \widehat{\Omega}_{\mathbf{x}, \mathbf{x}} \widehat{\Omega}_{\mathbf{y}, \mathbf{y}} - \Omega_{\mathbf{x}, \mathbf{x}} \Omega_{\mathbf{y}, \mathbf{y}} \right|}{\inf_{\mathbf{x}, \mathbf{y}} \widehat{\Omega}_{\mathbf{x}, \mathbf{x}} \widehat{\Omega}_{\mathbf{y}, \mathbf{y}} \wedge \inf_{\mathbf{x}, \mathbf{y}} \Omega_{\mathbf{x}, \mathbf{x}} \Omega_{\mathbf{y}, \mathbf{y}}} \lesssim_{\mathbb{P}} h^{p+1} + \sqrt{\frac{\log n}{nh^d}} + \frac{\log n}{n^{\frac{v}{2+v}} h^d}, \\
& \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \frac{\left| \Omega_{\mathbf{x}_1, \mathbf{x}_2} - \widehat{\Omega}_{\mathbf{x}_1, \mathbf{x}_2} \right|}{\sqrt{\Omega_{\mathbf{x}, \mathbf{x}} \Omega_{\mathbf{y}, \mathbf{y}}}} \lesssim_{\mathbb{P}} h^{p+1} + \sqrt{\frac{\log n}{nh^d}} + \frac{\log n}{n^{\frac{v}{2+v}} h^d}.
\end{aligned}$$

For simplicity, denote  $a_n = \sqrt{\frac{\log n}{nh^d}} + \frac{\log n}{n^{\frac{v}{2+v}} h^d}$ . Then, it follows that

$$\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \left| \mathbf{\Pi}_{\mathbf{x}, \mathbf{y}} - \widehat{\mathbf{\Pi}}_{\mathbf{x}, \mathbf{y}} \right| \lesssim_{\mathbb{P}} h^{p+1} + a_n.$$

Then, we bound the Kolmogorov-Smirnov distance between the maximum of  $Z_n$  and  $\widehat{Z}^{(\nu)}$  on a  $\delta_n$ -net of  $\mathcal{X}$ , denoted by  $\mathcal{X}_{\delta_n}$ , i.e. for all  $\mathbf{x} \in \mathcal{B}$ , there exists  $\mathbf{z} \in \mathcal{X}_{\delta_n}$  such that  $\|\mathbf{x} - \mathbf{z}\|_{\infty} \leq \delta_n$ . Since  $\mathcal{X}$  is compact, we can assume  $M := \text{Card}(\mathcal{X}_{\delta_n}) \lesssim \delta_n^{-d}$ . Denote  $\mathbf{Z}_n^{\delta_n}$  and  $\widehat{\mathbf{Z}}_n^{\delta_n}$  to the process  $Z_n$  and  $\widehat{Z}^{(\nu)}$  restricted on  $\mathcal{X}_{\delta_n}$ , respectively. Then, by the Gaussian Comparison Inequality Theorem 2.1 from Chernozhukov et al. (2022),

$$\sup_{\mathbf{y} \in \mathbb{R}^M} \left| \mathbb{P}(\mathbf{Z}_n^{\delta_n} \leq \mathbf{y}) - \mathbb{P}(\widehat{\mathbf{Z}}_n^{\delta_n} \leq \mathbf{y} | \mathbf{X}) \right| \lesssim \log M \left( \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \left| \mathbf{\Pi}_{\mathbf{x}, \mathbf{y}} - \widehat{\mathbf{\Pi}}_{\mathbf{x}, \mathbf{y}} \right| \right)^{\frac{1}{2}} \lesssim_{\mathbb{P}} \log M (a_n + h_n^{p+1})^{\frac{1}{2}}.$$

Consequently,

$$\begin{aligned}
& \sup_{x \in \mathbb{R}} \left| \mathbb{P}(\|\mathbf{Z}_n^{\delta_n}\|_{\infty} \leq x) - \mathbb{P}(\|\widehat{\mathbf{Z}}_n^{\delta_n}\|_{\infty} \leq x | \mathbf{X}) \right| \leq \sup_{x \in \mathbb{R}} \left| \mathbb{P}(-x\mathbf{1} \leq \mathbf{Z}_n^{\delta_n} \leq x\mathbf{1}) - \mathbb{P}(-x\mathbf{1} \leq \widehat{\mathbf{Z}}_n^{\delta_n} \leq x\mathbf{1} | \mathbf{X}) \right| \\
& \lesssim_{\mathbb{P}} \log M (a_n + h_n^{p+1})^{\frac{1}{2}} = R_M.
\end{aligned}$$

Then, we bound the Kolmogorov-Smirnov distance on the whole  $\mathcal{X}$  with the help of some  $\alpha_n > 0$  to be determined. For simplicity, denote

$$\begin{aligned}
\Phi_{\delta_n}(\alpha_n) &= \mathbb{P} \left( \sup_{\|\mathbf{x} - \mathbf{y}\|_{\infty} \leq \delta_n} \left| Z^{(\nu)}(\mathbf{x}) - Z^{(\nu)}(\mathbf{y}) \right| \geq \alpha_n \right), \\
\widehat{\Phi}_{\delta_n}(\alpha_n) &= \mathbb{P} \left( \sup_{\|\mathbf{x} - \mathbf{y}\|_{\infty} \leq \delta_n} \left| \widehat{Z}^{(\nu)}(\mathbf{x}) - \widehat{Z}^{(\nu)}(\mathbf{y}) \right| \geq \alpha_n | \mathbf{X} \right),
\end{aligned}$$

then for all  $t > 0$ ,

$$\begin{aligned}
& \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \leq t \right) \\
& \leq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}_{\delta_n}} |Z^{(\nu)}(\mathbf{x})| \leq t + \alpha_n \right) + \Phi_{\delta_n}(\alpha_n) \\
& \leq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}_{\delta_n}} |\widehat{Z}^{(\nu)}(\mathbf{x})| \leq t + \alpha_n \middle| \mathbf{X} \right) + \Phi_{\delta_n}(\alpha_n) + R_M \\
& \leq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}^{(\nu)}(\mathbf{x})| \leq t + \alpha_n \middle| \mathbf{X} \right) + \Phi_{\delta_n}(\alpha_n) + \widehat{\Phi}_{\delta_n}(\alpha_n) + R_M \\
& \leq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}^{(\nu)}(\mathbf{x})| \leq t \middle| \mathbf{X} \right) + 4\alpha_n \left( \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}^{(\nu)}(\mathbf{x})| \middle| \mathbf{X} \right] + 1 \right) + \Phi_{\delta_n}(\alpha_n) + \widehat{\Phi}_{\delta_n}(\alpha_n) + R_M.
\end{aligned}$$

Similary, we get for all  $t > 0$ ,

$$\begin{aligned}
& \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |Z^{(\nu)}(\mathbf{x})| \leq t \right) \\
& \geq \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}^{(\nu)}(\mathbf{x})| \leq t \middle| \mathbf{X} \right) - 4\alpha_n \left( \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |\widehat{Z}^{(\nu)}(\mathbf{x})| \middle| \mathbf{X} \right] + 1 \right) - \Phi_{\delta_n}(\alpha_n) - \widehat{\Phi}_{\delta_n}(\alpha_n) - R_M.
\end{aligned}$$

Heuristically,  $R_M$  depends on  $\delta_n$  through  $\log M \asymp \log(\delta_n^{-d})$ . By choosing  $\delta_n = n^{-s}$  for large enough  $s$ , the  $R_M$  term will dominates the terms  $\Phi_{\delta_n}(\alpha_n)$  and  $\widehat{\Phi}_{\delta_n}(\alpha_n)$ . Precisely, for any  $\delta$ ,

$$\begin{aligned}
& \sup_{\|\mathbf{x}-\mathbf{y}\|_\infty \leq \delta} \mathbb{E} \left[ \left( \widehat{Z}^{(\nu)}(\mathbf{x}) - \widehat{Z}^{(\nu)}(\mathbf{y}) \right)^2 \middle| \mathbf{X} \right] \\
& = \sup_{\|\mathbf{x}-\mathbf{y}\|_\infty \leq \delta} \left( \widehat{\Omega}_{\mathbf{x},\mathbf{x}} \widehat{\Omega}_{\mathbf{y}} \right)^{-\frac{1}{2}} \left( \frac{1}{nh_n^d} \right)^2 \sum_{i=1}^n \widehat{\varepsilon}_i^2 \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \cdot \\
& \quad \left( \mathbf{e}_1^T (\widehat{\Gamma}_{1,\mathbf{x}})^{-1} \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h_n} \right) K \left( \frac{\mathbf{X}_i - \mathbf{x}}{h_n} \right) - \mathbf{e}_1^T (\widehat{\Gamma}_{1,\mathbf{x}})^{-1} \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{y}}{h_n} \right) K \left( \frac{\mathbf{X}_i - \mathbf{y}}{h_n} \right) \right)^2 \\
& + \sup_{\|\mathbf{x}-\mathbf{y}\|_\infty \leq \delta} \left( \widehat{\Omega}_{\mathbf{x},\mathbf{x}} \widehat{\Omega}_{\mathbf{y}} \right)^{-\frac{1}{2}} \left( \frac{1}{nh_n^d} \right)^2 \sum_{i=1}^n \widehat{\varepsilon}_i^2 \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_0) \cdot \\
& \quad \left( \mathbf{e}_1^T (\widehat{\Gamma}_{0,\mathbf{x}})^{-1} \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{x}}{h_n} \right) K \left( \frac{\mathbf{X}_i - \mathbf{x}}{h_n} \right) - \mathbf{e}_1^T (\widehat{\Gamma}_{0,\mathbf{x}})^{-1} \mathbf{R}_p \left( \frac{\mathbf{X}_i - \mathbf{y}}{h_n} \right) K \left( \frac{\mathbf{X}_i - \mathbf{y}}{h_n} \right) \right)^2 \\
& \lesssim_{\mathbb{P}} h_n^{-d-2} \delta^2,
\end{aligned}$$

where in the last line we have used the scale of covariance matrices from Lemma SA-2.4, the scale of Gram matrices from Lemma SA-2.1, and the almost sure bound on the Lipschitz constant from the proof of Theorem SA-2.7 and  $C > 0$  is a constant. Similarly, for any  $\delta > 0$ ,

$$\begin{aligned}
& \sup_{\|\mathbf{x}-\mathbf{y}\|_\infty \leq \delta} \mathbb{E} \left[ \left( Z^{(\nu)}(\mathbf{x}) - Z^{(\nu)}(\mathbf{y}) \right)^2 \right] \\
& = \sup_{\|\mathbf{x}-\mathbf{y}\|_\infty \leq \delta} \mathbb{E} \left[ \left( \mathcal{K}(\mathbf{X}_i, \mathbf{x}) - \mathcal{K}(\mathbf{X}_i, \mathbf{y}) \right)^2 \varepsilon_i^2 \right] \leq C' h_n^{-2} \delta^2,
\end{aligned}$$

Then, by Corollary 2.2.5 from [van der Vaart and Wellner \(1996\)](#),

$$\begin{aligned}\mathbb{E} \left[ \sup_{\|\mathbf{x}-\mathbf{y}\|_\infty \leq \delta_n} \left| \widehat{Z}^{(\nu)}(\mathbf{x}) - \widehat{Z}^{(\nu)}(\mathbf{y}) \right| \middle| \mathbf{X} \right] &\lesssim_{\mathbb{P}} \int_0^{Ch_n^{-d/2-1}\delta_n} \sqrt{d \log \left( \frac{1}{\varepsilon h_n^{d/2+1}} \right)} d\varepsilon \lesssim \sqrt{\log n} h_n^{-d/2-1} \delta_n, \\ \mathbb{E} \left[ \sup_{\|\mathbf{x}-\mathbf{y}\|_\infty \leq \delta_n} \left| Z^{(\nu)}(\mathbf{x}) - Z^{(\nu)}(\mathbf{y}) \right| \right] &\lesssim \int_0^{Ch_n^{-1}\delta_n} \sqrt{d \log \left( \frac{1}{\varepsilon h_n} \right)} d\varepsilon \lesssim \sqrt{\log n} h_n^{-1} \delta_n.\end{aligned}$$

Also using the fact that  $\mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{Z}^{(\nu)}(\mathbf{x}) \right| \middle| \mathbf{X} \right] \lesssim 1$ , by choosing  $\alpha_n^* \asymp \left( \sqrt{\log n} h_n^{-d/2-1} \delta_n \right)^{\frac{1}{2}}$  and  $\delta_n \asymp n^{-s}$  for some large constant  $s > 0$ , we have

$$\begin{aligned}4\alpha_n \left( \mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{Z}^{(\nu)}(\mathbf{x}) \right| \middle| \mathbf{X} \right] + 1 \right) &+ \Phi_{\delta_n}(\alpha_n) + \widehat{\Phi}_{\delta_n}(\alpha_n) + R_M \\ &\lesssim_{\mathbb{P}} \left( \sqrt{\log n} h_n^{-d/2-1} \delta_n \right)^{\frac{1}{2}} + d \log(\delta_n^{-1}) (a_n + h_n^{p+1})^{\frac{1}{2}} \\ &\lesssim_{\mathbb{P}} d \log n (a_n + h_n^{p+1})^{\frac{1}{2}}.\end{aligned}$$

Putting together, we have

$$\sup_{\mathbf{u} \in \mathbb{R}} \left| \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| Z^{(\nu)}(\mathbf{x}) \right| \leq u \right) - \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{Z}^{(\nu)}(\mathbf{x}) \right| \leq u \middle| \mathbf{X} \right) \right| \lesssim \log n (a_n + h_n^{p+1})^{\frac{1}{2}}.$$

■

## SA-5.12 Proof of Theorem [SA-2.8](#)

The result follows from Lemma [SA-5.3](#), Lemma [SA-5.4](#) and Lemma [SA-5.5](#).

■

## SA-5.13 Proof of Theorem [SA-2.9](#)

Lemma [SA-2.8](#) and Dominated Convergence Theorem give

$$\sup_{u \in \mathbb{R}} \left| \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{T}^{(\nu)}(\mathbf{x}) \right| \leq u \right) - \mathbb{P} \left( \sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{Z}^{(\nu)}(\mathbf{x}) \right| \leq u \right) \right| = o(1).$$

Then, by definition of  $\widehat{B}_\alpha^{(\nu)}(\mathbf{x})$ ,

$$\begin{aligned}\mathbb{P}[\mu^{(\nu)}(\mathbf{x}) \in \widehat{B}_\alpha^{(\nu)}(\mathbf{x}), \forall \mathbf{x} \in \mathcal{B}] &= \mathbb{P} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |\widehat{T}^{(\nu)}(\mathbf{x})| \leq \mathfrak{c}_\alpha \right] \\ &= \mathbb{P} \left[ \sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{Z}^{(\nu)}(\mathbf{x}) \right| \leq \mathfrak{c}_\alpha \right] + o(1) \\ &= \mathbb{E} \left[ \mathbb{P} \left[ \sup_{\mathbf{x} \in \mathcal{B}} \left| \widehat{Z}^{(\nu)}(\mathbf{x}) \right| \leq \mathfrak{c}_\alpha \middle| \mathbf{W}_n \right] \right] + o(1) \\ &= 1 - \alpha + o(1).\end{aligned}$$

■

## SA-6 Proofs for Section SA-3

### SA-6.1 Proof of Lemma SA-3.1

By Assumption SA-1(iii) and Assumption SA-3, for any  $r \neq 0$ , for any  $\mathbf{x} \in \mathcal{B}$  and  $\mathbf{y} \in S_{t,\mathbf{x}}(r)$ ,

$$|\mu_t(\mathbf{y}) - \mu_t(\mathbf{x})| \lesssim |r|.$$

Hence for any  $r \neq 0$ , for any  $\mathbf{x} \in \mathcal{B}$ ,  $t \in \{0, 1\}$ ,

$$|\theta_{t,\mathbf{x}}(r) - \mu_t(\mathbf{x})| \leq \frac{\int_{S_{t,\mathbf{x}}(|r|)} |\mu_t(\mathbf{y}) - \mu_t(\mathbf{x})| f_X(\mathbf{y}) H_{d-1}(d\mathbf{y})}{\int_{S_{t,\mathbf{x}}(|r|)} f_X(\mathbf{y}) H_{d-1}(d\mathbf{y})} \lesssim r.$$

implying

$$|\theta_{t,\mathbf{x}}(0) - \mu_t(\mathbf{x})| \leq \lim_{r \rightarrow 0} |\theta_{t,\mathbf{x}}(r) - \mu_t(\mathbf{x})| = 0.$$

■

### SA-6.2 Proof of Lemma SA-3.2

The proof will be similar to the proof of Lemma SA-2.1. Let  $0 \leq v \leq p$ . Instead of  $g_n$ , we study the function  $k_n$  defined by

$$k_n(\xi, \mathbf{x}) = \left( \frac{d(\xi, \mathbf{x})}{h} \right)^v \frac{1}{h^d} K \left( \frac{d(\xi, \mathbf{x})}{h} \right), \xi, \mathbf{x} \in \mathcal{X}.$$

Define  $\mathcal{H} = \{k_n(\cdot, \mathbf{x}) \mathbb{1}(\cdot \in \mathcal{A}_t) : \mathbf{x} \in \mathcal{X}\}$ . We will show  $\mathcal{H}$  is a VC-type of class.

**Constant Envelope Function** We assume  $K$  is continuous and has compact support. Hence there exists a constant  $C_1$  such that  $\sup_{\mathbf{x} \in \mathcal{X}} \|k_n(\cdot, \mathbf{x})\|_\infty \leq C_1 h^{-d} = H$ .

**Diameter in  $L_2$**  For each  $\mathbf{x} \in \mathcal{X}$ ,  $k_n(\cdot, \mathbf{x})$  is supported in  $\{\xi : \mathcal{d}(\xi, \mathbf{x}) \leq h\}$ . By Assumption SA-1(ii) and Assumption SA-3(i),  $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{P}(\mathcal{d}(\mathbf{X}_i, \mathbf{x}) \leq h) \lesssim h^d$ . It follows that  $\sup_{\mathbf{x} \in \mathcal{X}} \|k_n(\cdot, \mathbf{x})\|_{\mathbb{P}, 2} \leq C_2 h^{-d/2}$  for some constant  $C_2$ . We can take  $C_1$  large enough so that  $\sigma = C_2 h^{-d/2} \leq F = C_1 h^{-d}$ .

**Ratio**  $\delta = \frac{\sigma}{F} = C_3 \sqrt{h^d}$ , for some constant  $C_3$ .

**Covering Numbers Case 1:  $k$  is Lipschitz.** Let  $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$ . By Assumption SA-3 and Assumption SA-2,

$$\begin{aligned} & \sup_{\xi \in \mathcal{X}} |k_n(\xi, \mathbf{x}) - k_n(\xi, \mathbf{x}')| \\ & \leq \sup_{\xi \in \mathcal{X}} \left[ \left( \frac{\mathcal{d}(\xi, \mathbf{x})}{h} \right)^v - \left( \frac{\mathcal{d}(\xi, \mathbf{x}')}{h} \right)^v \right] k_h(\mathcal{d}(\xi, \mathbf{x})) + \left( \frac{\mathcal{d}(\xi, \mathbf{x}')}{h} \right)^v [k_h(\mathcal{d}(\xi, \mathbf{x})) - k_h(\mathcal{d}(\xi, \mathbf{x}'))] \\ & \lesssim h_n^{-d-1} \|\mathbf{x} - \mathbf{x}'\|_\infty, \end{aligned}$$

By Lipschitz continuity property of  $\mathcal{F}$ , for any  $\varepsilon \in (0, 1]$  and for any finitely supported measure  $Q$  and metric  $\|\cdot\|_{Q,2}$  based on  $L_2(Q)$ ,

$$N(\{k_n(\cdot, \mathbf{x}) : \mathbf{x} \in \mathcal{X}\}, \|\cdot\|_{Q,2}, \varepsilon \|H\|_{Q,2}) \leq N(\mathcal{X}, \|\cdot\|_\infty, \varepsilon \|H\|_{Q,2} h^{d+1}) \stackrel{(1)}{\lesssim} \left( \frac{\text{diam}(\mathcal{X})}{\varepsilon \|H\|_{Q,2} h^{d+1}} \right)^d \lesssim \left( \frac{\text{diam}(\mathcal{X})}{\varepsilon h} \right)^d,$$

where in (1) we used the fact that  $\varepsilon \|H\|_{\mathbb{Q},2} h^{d+1} \lesssim \varepsilon h \lesssim 1$ . Hence  $\mathcal{H}$  forms a VC-type class in that  $\sup_Q N(\mathcal{H}, \|\cdot\|_{Q,2}, \varepsilon \|H\|_{\mathbb{Q},2}) \lesssim (C_1/\varepsilon)^{C_2}$  for all  $\varepsilon \in (0, 1]$  with  $C_1 = \frac{\text{diam}(\mathcal{X})}{h}$  and  $C_2 = d$ . Moreover, for any discrete measure  $Q$ , and for any  $\mathbf{x}, \mathbf{x}' \in \mathcal{X}$ ,  $\|k_n(\cdot, \mathbf{x})\mathbb{1}(\cdot \in A_t) - k_n(\cdot, \mathbf{x}')\mathbb{1}(\cdot \in A_t)\|_{Q,2} \leq \|k_n(\cdot, \mathbf{x}) - k_n(\cdot, \mathbf{x}')\|_{Q,2}$ . Hence

$$\sup_{Q \in \mathcal{A}(\mathcal{X})} N(\mathcal{H}, \|\cdot\|_{Q,2}, \varepsilon \|H\|_{\mathbb{Q},2}) \leq N(\{k_n(\cdot, \mathbf{x}) : \mathbf{x} \in \mathcal{X}\}, \|\cdot\|_{Q,2}, \varepsilon \|H\|_{\mathbb{Q},2}) \leq (C_1/\varepsilon)^{C_2}, \quad \varepsilon \in (0, 1],$$

where  $\mathcal{A}(\mathcal{X})$  denotes the collection of all finite discrete measures on  $\mathcal{X}$ .

*Case 2:*  $k = \mathbb{1}(\cdot \in [-1, 1])$ . The same argument as in the proof of Lemma SA-4.1 and the fact that  $\mathcal{L} = \{\mathbb{1}((\cdot - \mathbf{x})/h \in [-1, 1]^d) : \mathbf{x} \in \mathcal{B}\}$  has VC dimension no greater than  $2d$  implies again we have,

$$\sup_{Q \in \mathcal{A}(\mathcal{X})} N(\mathcal{H}, \|\cdot\|_{Q,2}, \varepsilon \|H\|_{\mathbb{Q},2}) \leq N(\{k_n(\cdot, \mathbf{x}) : \mathbf{x} \in \mathcal{X}\}, \|\cdot\|_{Q,2}, \varepsilon \|H\|_{\mathbb{Q},2}) \leq (C_1/\varepsilon)^{C_2}, \varepsilon \in (0, 1].$$

Hence, by Chernozhukov et al. (2014b, Corollary 5.1)

$$\begin{aligned} \mathbb{E} \left[ \sup_{l \in \mathcal{H}} |\mathbb{E}_n[l(\mathbf{X}_i)] - \mathbb{E}[l(\mathbf{X}_i)]| \right] &\lesssim \frac{\sigma}{\sqrt{n}} \sqrt{C_2 \log(C_1/\delta)} + \frac{\|M\|_{\mathbb{P},2} C_2 \log(C_1/\delta)}{n} \\ &\lesssim \frac{1}{\sqrt{nh^d}} \sqrt{d \log \left( \frac{\text{diam}(\mathcal{X})}{h^{1+d/2}} \right)} + \frac{1}{nh^d} d \log \left( \frac{\text{diam}(\mathcal{X})}{h^{1+d/2}} \right) \lesssim \sqrt{\frac{\log n}{nh^d}}. \end{aligned}$$

We conclude that  $\sup_{\mathbf{x} \in \mathcal{X}} \|\hat{\Psi}_{t,\mathbf{x}} - \Psi_{t,\mathbf{x}}\| \lesssim \mathbb{P} \sqrt{\frac{\log n}{nh^d}}$ . By Weyl's Theorem,  $\sup_{\mathbf{x} \in \mathcal{X}} |\lambda_{\min}(\hat{\Psi}_{t,\mathbf{x}}) - \lambda_{\min}(\Psi_{t,\mathbf{x}})| \leq \sup_{\mathbf{x} \in \mathcal{X}} \|\hat{\Psi}_{t,\mathbf{x}} - \Psi_{t,\mathbf{x}}\| \lesssim \mathbb{P} \sqrt{\frac{\log n}{nh^d}}$ . Therefore we can lower bound the minimum eigenvalue by  $\inf_{\mathbf{x} \in \mathcal{X}} \lambda_{\min}(\hat{\Psi}_{t,\mathbf{x}}) \geq \inf_{\mathbf{x} \in \mathcal{X}} \lambda_{\min}(\Psi_{t,\mathbf{x}}) - \sup_{\mathbf{x} \in \mathcal{X}} |\lambda_{\min}(\hat{\Psi}_{t,\mathbf{x}}) - \lambda_{\min}(\Psi_{t,\mathbf{x}})| \gtrsim \mathbb{P} 1$ . It follows that  $\sup_{\mathbf{x} \in \mathcal{X}} \|\hat{\Psi}_{t,\mathbf{x}}^{-1}\| \lesssim \mathbb{P} 1$  and hence

$$\sup_{\mathbf{x} \in \mathcal{X}} \|\hat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1}\| \leq \sup_{\mathbf{x} \in \mathcal{X}} \|\Psi_{t,\mathbf{x}}^{-1}\| \|\Psi_{t,\mathbf{x}} - \hat{\Psi}_{t,\mathbf{x}}\| \|\hat{\Psi}_{t,\mathbf{x}}^{-1}\| \lesssim \mathbb{P} \sqrt{\frac{\log n}{nh^d}}.$$

■

### SA-6.3 Proof of Lemma SA-3.3

Consider the class  $\mathcal{F} = \{(\mathbf{z}, u) \mapsto \mathbf{e}_\nu^\top g_\mathbf{x}(\mathbf{z})(u - h_\mathbf{x}(\mathbf{z})) : \mathbf{x} \in \mathcal{B}\}$ ,  $0 \leq \nu \leq p$ , where for  $\mathbf{z} \in \mathcal{X}$ ,

$$g_\mathbf{x}(\mathbf{z}) = \mathbf{r}_p \left( \frac{\mathcal{d}(\mathbf{z}, \mathbf{x})}{h} \right) k_h(\mathcal{d}(\mathbf{z}, \mathbf{x})), \quad h_\mathbf{x}(\mathbf{z}) = \gamma_t^*(\mathbf{x})^\top \mathbf{r}_p(\mathcal{d}(\mathbf{z}, \mathbf{x})).$$

By definition of  $\gamma_t^*(\mathbf{x})$ ,

$$\gamma_t^*(\mathbf{x}) = \mathbf{H}^{-1} \Psi_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}}, \quad \mathbf{S}_{t,\mathbf{x}} = \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) k_h(D_i(\mathbf{x})) Y_i \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right]. \quad (\text{SA-6.1})$$

Assumption SA-1 implies  $\mathbf{S}_{t,\mathbf{x}}$  is continuous in  $\mathbf{x}$ , hence  $\sup_{\mathbf{x} \in \mathcal{X}} \|\mathbf{S}_{t,\mathbf{x}}\| \lesssim 1$ . And by Assumption SA-2(ii),  $\inf_{\mathbf{x} \in \mathcal{X}} \lambda_{\min}(\Psi_{t,\mathbf{x}}) \gtrsim 1$ . Hence

$$\sup_{\mathbf{x} \in \mathcal{B}} \|\Psi_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}}\| \lesssim 1. \quad (\text{SA-6.2})$$

Now, consider properties of  $\mathcal{F}$ . Definition of  $\gamma_t^*(\mathbf{x})$  implies  $\mathbb{E}[f(\mathbf{X}_i, Y_i)] = 0$  for all  $f \in \mathcal{F}$ . Since  $K$  is compactly supported, there exists  $C_1, C_2 > 0$  such that  $F(\mathbf{z}, u) = C_1 h^{-d}(|u| + C_2)$  is an envelope function for  $\mathcal{F}$ . Denote  $M = \max_{1 \leq i \leq n} F(\mathbf{X}_i, Y_i)$ , then

$$\begin{aligned} \mathbb{E}[M^2]^{1/2} &\lesssim h^{-d} \mathbb{E} \left[ \max_{1 \leq i \leq n} |Y_i|^2 + 1 \right]^{1/2} \lesssim h^{-d} \mathbb{E} \left[ \max_{1 \leq i \leq n} |Y_i|^{2+v} \right]^{1/(2+v)} \\ &\lesssim h^{-d} \left[ \sum_{i=1}^n \mathbb{E}[\varepsilon_i + \sum_{t \in \{0,1\}} \mathbb{1}(\mathbf{x} \in \mathcal{A}_t) \mu_t(\mathbf{x})]^{2+v} \right]^{1/(2+v)} \lesssim h^{-d} n^{1/(2+v)}, \end{aligned}$$

where we have used  $\mathbf{X}$  is compact and  $\mu_t$  is continuous, hence  $\sup_{\mathbf{x} \in \mathcal{X}} |\sum_{t \in \{0,1\}} \mathbb{1}(\mathbf{x} \in \mathcal{A}_t) \mu_t(\mathbf{x})| \lesssim 1$ . Denote  $\sigma = \sup_{f \in \mathcal{F}} \mathbb{E}[f(\mathbf{X}_i, Y_i)^2]^{1/2}$ . Then,

$$\sigma^2 \lesssim \sup_{\mathbf{x} \in \mathcal{B}} \mathbb{E}[\|\mathbf{e}_\nu^\top g_{\mathbf{x}}\|_\infty^2 (|Y_i| + \|\mathbf{e}_\nu^\top h_{\mathbf{x}}\|_\infty)^2 \mathbb{1}(k_h(D_i(\mathbf{x})) \neq 0)] \lesssim h^{-d}.$$

To check for the covering number of  $\mathcal{F}$ , notice that compare to the proof of Lemma SA-2.1, we have one more term  $\mathbf{e}_\nu^\top g_{\mathbf{x}} h_{\mathbf{x}} = \mathbf{r}_p \left( \frac{\mathcal{d}(\mathbf{z}, \mathbf{x})}{h} \right) k_h(\mathcal{d}(\mathbf{z}, \mathbf{x})) \gamma_t^*(\mathbf{x})^\top \mathbf{r}_p(\mathcal{d}(\mathbf{z}, \mathbf{x}))$ . All terms except for  $\gamma_t^*(\mathbf{x})$  can be handled as in the proof of Lemma SA-2.1. Recall Equation (SA-6.1), and consider  $l_{t,\mathbf{x}} = \mathbf{e}_\nu^\top [\mathbf{R}(\mathcal{d}(\cdot, \mathbf{x})/h) k_h(\mathcal{d}(\cdot, \mathbf{x})) \mu_t] \mathbb{1}(\cdot \in \mathcal{A}_t)$  and  $\mathcal{L}_t = \{l_{t,\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$ ,  $\mathbf{v}$  is a any multi-index. Then, for any  $\mathbf{x}_1, \mathbf{x}_2 \in \mathcal{B}$ ,

$$|\mathbf{S}_{t,\mathbf{x}_1} - \mathbf{S}_{t,\mathbf{x}_2}| \leq \|l_{t,\mathbf{x}_1} - l_{t,\mathbf{x}_2}\|_{\mathbb{P}_X, 2},$$

and hence

$$N(\{\mathbf{e}_\nu^\top \mathbf{S}_{t,\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}, \|\cdot\|, \varepsilon h^{-d}) \leq N(\mathcal{L}_t, \|\cdot\|_{\mathbb{P}_X, 2}, \varepsilon h^{-d}) \leq \sup_Q N(\mathcal{L}_t, \|\cdot\|_{Q, 2}, \varepsilon h^{-d}),$$

Same argument as paragraph **Covering Numbers** in the proof of Lemma SA-3.2 then shows

$$\begin{aligned} \sup_Q N(\{g_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}, \|\cdot\|_{Q, 2}, \varepsilon C_1 h^{-d}) &\leq \left( \frac{\text{diam}(\mathcal{X})}{h\varepsilon} \right)^d, \quad 0 < \varepsilon \leq 1, \\ \sup_Q N(\{g_{\mathbf{x}} h_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}, \|\cdot\|_{Q, 2}, \varepsilon C_1 h^{-d}) &\leq \left( \frac{\text{diam}(\mathcal{X})}{h\varepsilon} \right)^d, \quad 0 < \varepsilon \leq 1, \end{aligned}$$

where sup is taken over all discrete measures on  $\mathcal{X}$ . Product  $\{g_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$  with the singleton of identity function  $\{u \mapsto u, u \in \mathbb{R}\}$ , and adding  $\{g_{\mathbf{x}} h_{\mathbf{x}} : \mathbf{x} \in \mathcal{B}\}$ ,

$$\sup_Q N(\mathcal{F}, \|\cdot\|_{Q, 2}, \varepsilon \|F\|_{Q, 2}) \leq 2 \left( \frac{2 \text{diam}(\mathcal{X})}{h\varepsilon} \right)^d, \quad 0 < \varepsilon \leq 1,$$

where sup is taken over all discrete measures on  $\mathcal{X} \times \mathbb{R}$ . Denote  $\mathbf{C}_1 = d$ ,  $\mathbf{C}_2 = \frac{2(2 \text{diam}(\mathcal{X}))^d}{h^d}$ . Hence, by

Chernozhukov et al. (2014b, Corollary 5.1)

$$\begin{aligned}
\mathbb{E} \left[ \sup_{\mathbf{x} \in \mathcal{B}} |\mathbf{e}_\nu^\top \mathbf{O}_{t,\mathbf{x}}| \right] &= \mathbb{E} \left[ \sup_{f \in \mathcal{F}} |\mathbb{E}_n [f(\mathbf{X}_i, Y_i)] - \mathbb{E}[f(\mathbf{X}_i, Y_i)]| \right] \\
&\lesssim \frac{\sigma}{\sqrt{n}} \sqrt{\mathbf{C}_2 \log(\mathbf{C}_1 \|M\|_{\mathbb{P},2}/\sigma)} + \frac{\|M\|_{\mathbb{P},2} \mathbf{C}_2 \log(\mathbf{C}_1 \|M\|_{\mathbb{P},2}/\sigma)}{n} \\
&\lesssim \frac{1}{\sqrt{nh^d}} \sqrt{d \log \left( \frac{\text{diam}(\mathcal{X})}{h^{1+d/2}} \right)} + \frac{1}{n^{\frac{1+v}{2+v}} h^d} d \log \left( \frac{\text{diam}(\mathcal{X})}{h^{1+d/2}} \right) \\
&\lesssim \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d}.
\end{aligned}$$

The rest follows from finite dimensionality of  $\mathbf{O}_{t,\mathbf{x}}$ , and Lemma SA-3.2. ■

#### SA-6.4 Proof of Lemma SA-3.4

By Lemma SA-3.1 and Equation (SA-6.1), we have

$$\begin{aligned}
\sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_{n,t}(\mathbf{x})| &= \sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbf{e}_1^\top \boldsymbol{\Psi}_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}} - \mu_t(\mathbf{x}) \right| \\
&= \sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbf{e}_1^\top \boldsymbol{\Psi}_{t,\mathbf{x}}^{-1} \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) k_h(D_i(\mathbf{x})) \mathbf{R}_p(D_i(\mathbf{x}))^\top (\mu_t(\mathbf{X}_i) - \mu_t(\mathbf{x}), 0, \dots, 0) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right] \right| \\
&\lesssim \sup_{\mathbf{x} \in \mathcal{B}} \sup_{\mathbf{z} \in \mathcal{X}} |\mu_t(\mathbf{x}) - \mu_t(\mathbf{z})| \mathbb{1}(k_h(\mathcal{A}(\mathbf{z}, \mathbf{x})) > 0) \\
&\lesssim h.
\end{aligned}$$
■

#### SA-6.5 Proof of Lemma SA-3.5

Denote  $\eta_{i,t,\mathbf{x}} = Y_i - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))$  and  $\xi_{i,t,\mathbf{x}} = \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x})) - \hat{\theta}_{t,\mathbf{x}}(D_i(\mathbf{x}))$ . Then

$$\hat{\Upsilon}_{t,\mathbf{x},\mathbf{y}} = \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left( \frac{D_i(\mathbf{y})}{h} \right)^\top h^d k_h(D_i(\mathbf{x})) k_h(D_i(\mathbf{y})) (\eta_{i,t,\mathbf{x}} + \xi_{i,t,\mathbf{x}})^2 \mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x})) \right],$$

and we decompose the error into

$$\begin{aligned}
\hat{\Upsilon}_{t,\mathbf{x},\mathbf{y}} - \Upsilon_{t,\mathbf{x},\mathbf{y}} &= \Delta_{1,t,\mathbf{x},\mathbf{y}} + \Delta_{2,t,\mathbf{x},\mathbf{y}} + \Delta_{3,t,\mathbf{x},\mathbf{y}}, \\
\Delta_{1,t,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left( \frac{D_i(\mathbf{y})}{h} \right)^\top h^d k_h(D_i(\mathbf{x})) k_h(D_i(\mathbf{y})) \xi_{i,t,\mathbf{x}}^2 \mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x})) \right], \\
\Delta_{2,t,\mathbf{x},\mathbf{y}} &= 2\mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left( \frac{D_i(\mathbf{y})}{h} \right)^\top h^d k_h(D_i(\mathbf{x})) k_h(D_i(\mathbf{y})) \eta_{i,t,\mathbf{x}} \xi_{i,t,\mathbf{x}} \mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x})) \right], \\
\Delta_{3,t,\mathbf{x},\mathbf{y}} &= \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left( \frac{D_i(\mathbf{y})}{h} \right)^\top h^d k_h(D_i(\mathbf{x})) k_h(D_i(\mathbf{y})) \eta_{i,t,\mathbf{x}}^2 \mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x})) \right] \\
&\quad - \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) \mathbf{r}_p \left( \frac{D_i(\mathbf{y})}{h} \right)^\top h^d k_h(D_i(\mathbf{x})) k_h(D_i(\mathbf{y})) \eta_{i,t,\mathbf{x}}^2 \mathbb{1}_{\mathcal{J}_t}(D_i(\mathbf{x})) \right].
\end{aligned}$$

By Assumption SA-2,  $k_h(D_i(\mathbf{x})) \neq 0$  implies  $\|\mathbf{r}_p(D_i(\mathbf{x})/h)\|_2 \lesssim 1$ . Hence by Lemma SA-3.2 and SA-3.3,

$$\begin{aligned}
& \max_{t \in \{0,1\}} \max_{1 \leq i \leq n} \sup_{\mathbf{x} \in \mathcal{B}} |\xi_{i,t,\mathbf{x}}| \\
&= \max_{t \in \{0,1\}} \max_{1 \leq i \leq n} \sup_{\mathbf{x} \in \mathcal{B}} |\mathbf{r}_p(D_i(\mathbf{x}))^\top (\hat{\gamma}_{t,\mathbf{x}} - \gamma_{t,\mathbf{x}}^*)| \mathbb{1}(k_h(D_i(\mathbf{x})) \geq 0) \\
&= \max_{t \in \{0,1\}} \max_{1 \leq i \leq n} \sup_{\mathbf{x} \in \mathcal{B}} |\mathbf{r}_p(D_i(\mathbf{x}))^\top \mathbf{H}^{-1}(\hat{\Psi}_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} + (\hat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1}) \mathbf{U}_{t,\mathbf{x}})| \mathbb{1}(k_h(D_i(\mathbf{x})) \geq 0) \\
&\leq \max_{t \in \{0,1\}} \sup_{\mathbf{x} \in \mathcal{B}} \|\hat{\Psi}_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}}\|_2 + \max_{t \in \{0,1\}} \sup_{\mathbf{x} \in \mathcal{B}} \|(\hat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1}) \mathbf{U}_{t,\mathbf{x}}\|_2 \\
&\lesssim \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d},
\end{aligned}$$

where

$$\mathbf{U}_{t,\mathbf{x}} = \mathbb{E}_n \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) k_h(D_i(\mathbf{x})) \theta_{t,\mathbf{x}}^*(\mathbf{X}_i) \mathbb{1}_{\mathcal{I}_t}(D_i(\mathbf{x})) \right].$$

Assuming  $\frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \rightarrow \infty$ , similar maximal inequality as in the proof of Lemma SA-3.2 shows

$$\begin{aligned}
\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \|\Delta_{1,t,\mathbf{x},\mathbf{y}}\| &\lesssim_{\mathbb{P}} \max_{t \in \{0,1\}} \max_{1 \leq i \leq n} \sup_{\mathbf{x} \in \mathcal{B}} |\xi_{i,t,\mathbf{x}}|^2 \lesssim \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right)^2, \\
\sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \|\Delta_{2,t,\mathbf{x},\mathbf{y}}\| &\lesssim_{\mathbb{P}} \max_{t \in \{0,1\}} \max_{1 \leq i \leq n} \sup_{\mathbf{x} \in \mathcal{B}} |\xi_{i,t,\mathbf{x}}| \lesssim \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d}.
\end{aligned} \tag{SA-6.3}$$

Consider the  $(\mu, \nu)$  entry of  $\Delta_{3,t,\mathbf{x},\mathbf{y}}$ . Consider the class

$$\mathcal{F} = \left\{ (\mathbf{z}, u) \mapsto \left( \frac{\mathcal{d}(\mathbf{z}, \mathbf{x})}{h} \right)^{\mu+\nu} h^d k_h(\mathcal{d}(\mathbf{z}, \mathbf{x})) k_h(\mathcal{d}(\mathbf{z}, \mathbf{y})) (u - \mathbf{r}_p(\mathcal{d}(\mathbf{z}, \mathbf{x}))^\top \gamma_{t,\mathbf{x}}^*)^2 : \mathbf{x}, \mathbf{y} \in \mathcal{X} \right\}.$$

By Assumption SA-2 and SA-1(v), we have  $\sup_{f \in \mathcal{F}} \mathbb{E}[f(\mathbf{X}_i, Y_i)^2]^{1/2} \lesssim h^{-d/2}$ . Moreover, Assumption SA-2 and Equation (SA-6.2) imply there exists  $C_1, C_2 > 0$  such that  $F(\mathbf{z}, u) = C_1 h^{-d}(u^2 + C_2)$  is an envelope function for  $\mathcal{F}$ , with

$$\mathbb{E} \left[ \max_{1 \leq i \leq n} F(\mathbf{X}_i, Y_i)^2 \right]^{\frac{1}{2}} \lesssim C_1 h^{-d} (\mathbb{E} \left[ \max_{1 \leq i \leq n} Y_i^4 \right]^{\frac{1}{2}} + C_2) \lesssim C_1 h^{-d} (\mathbb{E} \left[ \max_{1 \leq i \leq n} Y_i^{2+v} \right]^{\frac{2}{2+v}} + C_2) \lesssim h^{-d} n^{\frac{2}{2+v}}.$$

Apply Chernozhukov et al. (2014b, Corollary 5.1) similarly as in Lemma SA-3.3 gives

$$\mathbb{E} \left[ \sup_{f \in \mathcal{F}} |\mathbb{E}_n[f(\mathbf{X}_i, Y_i)] - \mathbb{E}[f(\mathbf{X}_i, Y_i)]| \right] \lesssim \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}.$$

Finite dimensionality of  $\Delta_{3,t,\mathbf{x},\mathbf{y}}$  then implies

$$\mathbb{E} \left[ \sup_{\mathbf{x}, \mathbf{y} \in \mathcal{X}} \|\Delta_{3,t,\mathbf{x},\mathbf{y}}\| \right] \lesssim \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}. \tag{SA-6.4}$$

Putting together Equations (SA-6.3), (SA-6.4) and Lemma SA-3.2 gives the result.  $\blacksquare$

### SA-6.6 Proof of Theorem SA-3.1

All analysis in Lemma SA-3.2 and Lemma SA-3.3 can be done when the index set is the singleton  $\{\mathbf{x}\}$  instead of  $\mathcal{B}$ , replacing (Chernozhukov et al., 2014b, Corollary 5.1) by Bernstein inequality, and gives for any  $\mathbf{x} \in \mathcal{B}$ ,

$$\begin{aligned} |\mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}}| &\lesssim_{\mathbb{P}} \sqrt{\frac{1}{nh^d}} + \frac{1}{n^{\frac{1+v}{2+v}} h^d}, \\ |\mathbf{e}_1^\top (\hat{\Psi}_{t,\mathbf{x}}^{-1} - \Psi_{t,\mathbf{x}}^{-1}) \mathbf{O}_{t,\mathbf{x}}| &\lesssim_{\mathbb{P}} \sqrt{\frac{1}{nh^d}} \left( \sqrt{\frac{1}{nh^d}} + \frac{1}{n^{\frac{1+v}{2+v}} h^d} \right). \end{aligned}$$

The decomposition Equation (SA-3.1) then gives the result. ■

### SA-6.7 Proof of Theorem SA-3.2

Define  $\bar{T}_{\text{dis}}(\mathbf{x}) = \Xi_{\mathbf{x},\mathbf{x}}^{-1/2} \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}}$ . Notice that if we define

$$Z_{n,i} = \frac{1}{n} \Xi_{\mathbf{x},\mathbf{x}}^{-1/2} \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) k_h(D_i(\mathbf{x})) (Y_i - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t),$$

then  $\bar{T}_{\text{dis}}(\mathbf{x}) = \sum_{i=1}^n Z_{n,i}$ . Moreover,  $\mathbb{E}[Z_{n,i}] = 0$  and  $\mathbb{V}[Z_{n,i}] = n^{-1}$ . By Berry-Essen Theorem,

$$\begin{aligned} \sup_{u \in \mathbb{R}} |\mathbb{P}(\bar{T}_{\text{dis}}(\mathbf{x}) \leq u) - \Phi(u)| &\lesssim \sum_{i=1}^n \mathbb{E}[|Z_{n,i}|^3] \\ &= \sum_{i=1}^n n^{-3} \Xi_{\mathbf{x},\mathbf{x}}^{-3/2} \mathbb{E} \left[ |\mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) k_h(D_i(\mathbf{x})) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) (Y_i - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x})))|^3 \right] \\ &\lesssim n^{-2} \Xi_{\mathbf{x},\mathbf{x}}^{-3/2} \mathbb{E}[|k_h(D_i(\mathbf{x}))(Y_i - \theta_{t,\mathbf{x}}^*(D_i(\mathbf{x})))|^3] \\ &\lesssim n^{-2} \Xi_{\mathbf{x},\mathbf{x}}^{-3/2} \mathbb{E}[|k_h(D_i(\mathbf{x}))(\mathbb{E}[|Y_i|^3 | \mathbf{X}_i] + |\theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))|^3)|] \\ &\lesssim (nh^d)^{-1/2}, \end{aligned}$$

where in the third line we used  $\sup_{\mathbf{x} \in \mathcal{B}} \|\mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) k_h(D_i(\mathbf{x}))\| \lesssim 1$  holds almost surely in  $\mathbf{X}_i$ , and in the last line we used  $\Xi_{\mathbf{x},\mathbf{x}} \gtrsim (nh^d)^{-1/2}$  from Lemma SA-3.5, Assumption SA-1(v) so that  $\mathbb{E}[|Y_i|^3 | \mathbf{X}_i] \lesssim 1$  and

$$\theta_{t,\mathbf{x}}^*(D_i(\mathbf{x})) = \gamma_t^*(\mathbf{x})^\top \mathbf{r}_p(D_i(\mathbf{x})) = (\Psi_{t,\mathbf{x}} \mathbf{S}_{t,\mathbf{x}})^{-1} \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right),$$

implying  $\max_{1 \leq i \leq n} \sup_{\mathbf{x} \in \mathcal{B}} |\theta_{t,\mathbf{x}}^*(D_i(\mathbf{x}))| \lesssim 1$  for  $t \in \{0, 1\}$ . The counterpart of Theorem SA-3.4 gives

$$|\hat{T}_{\text{dis}}(\mathbf{x}) - \bar{T}_{\text{dis}}(\mathbf{x})| \lesssim_{\mathbb{P}} \frac{1}{\sqrt{nh^d}} + \frac{1}{n^{\frac{v}{2+v}} h^d} + \sqrt{nh^d} \sum_{t \in \{0,1\}} |\mathfrak{B}_{n,t}(\mathbf{x})|.$$

Putting together we have

$$\mathbb{P}(\tau \in \hat{\mathbf{I}}_{\text{dis}}(\mathbf{x}, \alpha)) = \mathbb{P}(|\hat{T}_{\text{dis}}(\mathbf{x})| \leq \mathbf{c}_\alpha) = \mathbb{P}(|\bar{T}_{\text{dis}}(\mathbf{x})| \leq \mathbf{c}_\alpha) + o(1) = 2(1 - \Phi(\mathbf{c}_\alpha)) + o(1) = 1 - \alpha + o(1). ■$$

### SA-6.8 Proof of Theorem SA-3.3

The statement follows from Lemma SA-3.2, Lemma SA-3.3 and the decomposition Equation (SA-3.1).  $\blacksquare$

### SA-6.9 Proof of Theorem SA-3.4

We make the decomposition based on Equation (SA-3.1) and convergence of  $\widehat{\Xi}_{\mathbf{x},\mathbf{x}}$ ,

$$\begin{aligned}\widehat{T}_{\text{dis}}(\mathbf{x}) - \overline{T}_{\text{dis}}(\mathbf{x}) &= \widehat{\Xi}_{\mathbf{x},\mathbf{x}}^{-1/2} \left( \sum_{t \in \{0,1\}} (-1)^{\frac{t+1}{2}} (\widehat{\theta}_{t,\mathbf{x}}(0) - \theta_{t,\mathbf{x}}(0)) \right) - \Xi_{\mathbf{x},\mathbf{x}}^{-1/2} \left( \sum_{t \in \{0,1\}} (-1)^{\frac{t+1}{2}} \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} \right) \\ &= \widehat{\Xi}_{\mathbf{x},\mathbf{x}}^{-1/2} \left( \sum_{t \in \{0,1\}} (-1)^{\frac{t+1}{2}} (\widehat{\theta}_{t,\mathbf{x}}(0) - \theta_{t,\mathbf{x}}(0)) - \sum_{t \in \{0,1\}} (-1)^{\frac{t+1}{2}} \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} \right) \quad (= \Delta_{1,\mathbf{x}}) \\ &\quad + (\widehat{\Xi}_{\mathbf{x},\mathbf{x}}^{-1/2} - \Xi_{\mathbf{x},\mathbf{x}}^{-1/2}) \sum_{t \in \{0,1\}} (-1)^{\frac{t+1}{2}} \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} \quad (= \Delta_{2,\mathbf{x}})\end{aligned}$$

By Lemma SA-3.2 and SA-3.3, and the decomposition Equation (SA-3.1),

$$\begin{aligned}\sup_{\mathbf{x} \in \mathcal{X}} \left| \sum_{t \in \{0,1\}} (-1)^{\frac{t+1}{2}} (\widehat{\theta}_{t,\mathbf{x}}(0) - \theta_{t,\mathbf{x}}(0)) - \sum_{t \in \{0,1\}} (-1)^{\frac{t+1}{2}} \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} \right| \\ \lesssim_{\mathbb{P}} \sqrt{\frac{\log(1/h)}{nh^d}} \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right) + \sup_{\mathbf{x} \in \mathcal{B}} \sum_{t \in \{0,1\}} |\theta_{t,\mathbf{x}}^*(0) - \theta_{t,\mathbf{x}}(0)|.\end{aligned}$$

Together with Lemma SA-3.5,

$$\sup_{\mathbf{x} \in \mathcal{B}} |\Delta_{1,\mathbf{x}}| \lesssim_{\mathbb{P}} \frac{\log(1/h)}{\sqrt{nh^d}} + \frac{(\log(1/h))^{\frac{3}{2}}}{n^{\frac{1+v}{2+v}} h^d} + \sqrt{nh^d} \sup_{\mathbf{x} \in \mathcal{B}} \sum_{t \in \{0,1\}} |\theta_{t,\mathbf{x}}^*(0) - \theta_{t,\mathbf{x}}(0)|. \quad (\text{SA-6.5})$$

By Lemma SA-3.2, Lemma SA-3.3 and Lemma SA-3.5, and assume  $\frac{n^{\frac{v}{2+v}} h^d}{\log(1/h)} \rightarrow \infty$ , then

$$\begin{aligned}\sup_{\mathbf{x} \in \mathcal{X}} \left| \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{O}_{t,\mathbf{x}} \left( \Xi_{\mathbf{x},\mathbf{x}}^{-1/2} - \widehat{\Xi}_{\mathbf{x},\mathbf{x}}^{-1/2} \right) \right| &\lesssim_{\mathbb{P}} \sqrt{nh^d} \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{1+v}{2+v}} h^d} \right) \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right) \\ &= \sqrt{\log(1/h)} \left( 1 + \sqrt{\frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d}} \right) \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right) \\ &\lesssim \sqrt{\log(1/h)} \left( \sqrt{\frac{\log(1/h)}{nh^d}} + \frac{\log(1/h)}{n^{\frac{v}{2+v}} h^d} \right).\end{aligned}$$

Hence

$$\sup_{\mathbf{x} \in \mathcal{B}} |\Delta_{2,\mathbf{x}}| \lesssim_{\mathbb{P}} \frac{\log(1/h)}{\sqrt{nh^d}} + \frac{(\log(1/h))^{\frac{3}{2}}}{n^{\frac{1+v}{2+v}} h^d}. \quad (\text{SA-6.6})$$

Putting together Equations (SA-6.5), (SA-6.6) give the result.  $\blacksquare$

## SA-6.10 Proof of Theorem SA-3.5

Without loss of generality, we can assume  $\mathcal{X} = [0, 1]^d$ , and  $\mathbb{Q}_{\mathcal{F}_t} = \mathbb{P}_X$  is a valid surrogate measure for  $\mathbb{P}_X$  with respect to  $\mathcal{G}$ , and  $\phi_{\mathcal{G}} = \text{Id}$  is a valid normalizing transformation (as in Lemma SA-4.1). This implies the constants  $c_1$  and  $c_2$  from Lemma SA-4.1 are all 1.

By similar arguments as in the proof of Theorem SA-2.7, we get properties of  $\mathcal{G}$  as follows:

$$\mathbf{M}_{\mathcal{G}} \lesssim h^{-d/2}, \quad \mathbf{E}_{\mathcal{G}} \lesssim h^{d/2}, \quad \text{TV}_{\mathcal{G}} \lesssim h^{d/2-1}, \quad \sup_Q N(\mathcal{G}, \|\cdot\|_{Q,2}, \varepsilon(2c+1)^{d+1}\mathbf{M}_{\mathcal{G}}) \leq 2c'\varepsilon^{-d-1} + 2.$$

By definition of  $\theta^*(\cdot)$ , for each  $\mathbf{x} \in \mathcal{B}$ ,  $t \in \{0, 1\}$ ,

$$\theta_{t,\mathbf{x}}^*(\mathcal{d}(\mathbf{u}, \mathbf{x})) = \gamma_t^*(\mathbf{x})^\top \mathbf{r}_p(\mathcal{d}(\mathbf{u}, \mathbf{x})) = (\mathbf{H}^{-1} \Psi_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}})^\top \mathbf{r}_p(\mathcal{d}(\mathbf{u}, \mathbf{x})) = (\Psi_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}})^\top \mathbf{r}_p\left(\frac{\mathcal{d}(\mathbf{u}, \mathbf{x})}{h}\right),$$

recalling

$$\Psi_{t,\mathbf{x}} = \mathbb{E} \left[ \mathbf{r}_p\left(\frac{D_i(\mathbf{x})}{h}\right) \mathbf{r}_p\left(\frac{D_i(\mathbf{x})}{h}\right)^\top k_h(D_i(\mathbf{x})) \mathbb{1}_{\mathcal{I}_t}(D_i(\mathbf{x})) \right], \quad \mathbf{S}_{t,\mathbf{x}} = \mathbb{E} \left[ \mathbf{r}_p\left(\frac{D_i(\mathbf{x})}{h}\right) k_h(D_i(\mathbf{x})) Y_i \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right].$$

We can check that  $\|\Psi_{t,\mathbf{x}}^{-1}\| \lesssim 1$ ,  $\|\mathbf{S}_{t,\mathbf{x}}\| \lesssim 1$  and

$$\mathbf{M}_{\mathcal{H}_t} \lesssim h^{-d/2}, \quad \mathbf{E}_{\mathcal{H}_t} \lesssim h^{d/2}, \quad t \in \{0, 1\}.$$

In what follows, we verify the entropy and total variation properties of  $\mathcal{H}$ . Using product rule we can verify

$$\sup_{\mathbf{u} \in \mathcal{X}} \sup_{\mathbf{x}, \mathbf{x}' \in \mathcal{B}} \frac{|\theta_{t,\mathbf{x}}^*(\mathcal{d}(\mathbf{u}, \mathbf{x})) - \theta_{t,\mathbf{x}}^*(\mathcal{d}(\mathbf{u}, \mathbf{x}'))|}{\|\mathbf{x} - \mathbf{x}'\|} \lesssim h^{-1}.$$

Define  $f_{t,\mathbf{x}}(\cdot) = \frac{h^{-d/2}}{\sqrt{n\Xi_{\mathbf{x},\mathbf{x}}}} \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{r}_p(\cdot) K(\cdot) (\Psi_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}})^\top \mathbf{r}_p(\cdot)$ . Then,

$$\mathfrak{K}_t(\mathbf{u}; \mathbf{x}) \theta_{t,\mathbf{x}}^*(\mathcal{d}(\mathbf{u}, \mathbf{x})) = h^{-d/2} f_{t,\mathbf{x}}\left(\frac{\mathcal{d}(\mathbf{u}, \mathbf{x})}{h}\right), \quad \mathbf{u} \in \mathcal{X}, \mathbf{x} \in \mathcal{B}, t \in \{0, 1\}.$$

Take  $\mathcal{H}_t = \{\mathfrak{K}_t(\cdot; \mathbf{x}) \theta_{t,\mathbf{x}}^*(\mathcal{d}(\cdot, \mathbf{x})) : \mathbf{x} \in \mathcal{B}\}$ ,  $t \in \{0, 1\}$ . For  $t \in \{0, 1\}$ ,  $f_{t,\mathbf{x}}$  satisfies:

$$\begin{aligned} (i) \text{ boundedness} & \quad \sup_{\mathbf{x} \in \mathcal{B}} \sup_{\mathbf{u} \in \mathcal{X}} |f_{t,\mathbf{x}}(\mathbf{u})| \leq c, \\ (ii) \text{ compact support} & \quad \text{supp}(f_{t,\mathbf{x}}(\cdot)) \subseteq [-c, c]^d, \forall \mathbf{x} \in \mathcal{B}, \\ (iii) \text{ Lipschitz continuity} & \quad \sup_{\mathbf{x} \in \mathcal{B}} \sup_{\mathbf{u}, \mathbf{u}' \in \mathcal{X}} \frac{|f_{t,\mathbf{x}}(\mathbf{u}) - f_{t,\mathbf{x}}(\mathbf{u}')|}{\|\mathbf{u} - \mathbf{u}'\|} \leq c \\ & \quad \sup_{\mathbf{u} \in \mathcal{X}} \sup_{\mathbf{x}, \mathbf{x}' \in \mathcal{B}} \frac{|f_{t,\mathbf{x}}(\mathbf{u}) - f_{t,\mathbf{x}'}(\mathbf{u})|}{\|\mathbf{x} - \mathbf{x}'\|} \leq ch^{-1}, \end{aligned}$$

for some constant  $c$  not depending on  $n$ . Then, by an argument similar to Cattaneo et al. (2024, Lemma 7), there exists a constant  $c'$  only depending on  $c$  and  $d$  that for any  $0 \leq \varepsilon \leq 1$ ,

$$\sup_Q N\left(h^{d/2} \mathcal{H}_t, \|\cdot\|_{Q,1}, (2c+1)^{d+1} \varepsilon\right) \leq c' \varepsilon^{-d-1} + 1,$$

where supremum is taken over all finite discrete measures. Taking a constant envelope function  $M_{\mathcal{H}_t} = (2c+1)^{d+1}h^{-d/2}$ , we have for any  $0 < \varepsilon \leq 1$ ,

$$\sup_Q N(\mathcal{H}_t, \|\cdot\|_{Q,1}, \varepsilon M_{\mathcal{H}_t}) \leq \mathbf{c}' \varepsilon^{-d-1} + 1.$$

By Lemma SA-5.2, above implies the uniform covering number for  $\mathcal{H}_t$  satisfies

$$N_{\mathcal{H}_t}(\varepsilon) \leq 4\mathbf{c}'(\varepsilon/2)^{-d-1}, \quad 0 < \varepsilon \leq 1.$$

Since  $\mathcal{H} \subseteq \mathcal{H}_0 + \mathcal{H}_1$ , here  $+$  denotes the Minkowski sum, with  $M_{\mathcal{H}}$  taken to be  $M_{\mathcal{H}_0} + M_{\mathcal{H}_1}$ , a bound on the uniform covering number of  $\mathcal{H}$  can be given by

$$N_{\mathcal{H}}(\varepsilon) \leq 16(\mathbf{c}')^2(\varepsilon/2)^{-2d-2}, \quad 0 < \varepsilon \leq 1.$$

With the assumption that  $\mathcal{L}(E_{t,\mathbf{x}}) \leq Ch^{d-1}$  for  $E_{t,\mathbf{x}} = \{\mathbf{y} \in \mathcal{A}_t : (\mathbf{y} - \mathbf{x})/h \in \text{Supp}(K)\}$  for all  $t \in \{0,1\}$ ,  $\mathbf{x} \in \mathcal{B}$ , and the fact that  $\text{TV}_{\mathcal{H}_t} \lesssim h^{d/2-1}$  for  $t \in \{0,1\}$ , the same argument as in the paragraph **Total Variation** in the proof of Theorem SA-2.7 shows

$$\text{TV}_{\mathcal{H}} \lesssim h^{d/2-1}.$$

Now apply Lemma SA-4.2 with  $\mathcal{G}, \mathcal{H}$  defined in Equation (SA-3.3) and  $\mathcal{R} = \{\text{Id}\}$ , noticing that

$$(\overline{\text{T}}_{\text{dis}} : \mathbf{x} \in \mathcal{B}) = (A_n(g, h, r) : (g, h, r) \in \mathcal{F} \times \mathcal{R}), \quad \mathcal{F} = \{(g_{\mathbf{x}}, h_{\mathbf{x}}) : \mathbf{x} \in \mathcal{B}\} \subseteq \mathcal{G} \times \mathcal{H},$$

the result then follows. ■

### SA-6.11 Proof of Theorem SA-3.6

The result follows from Theorem SA-3.5, Theorem SA-3.4, Lemma SA-3.5 and similar arguments as the proof of Theorem SA-2.9. ■

## SA-7 Proofs of Distance-Based Bias Results

### SA-7.1 Proof of Lemma 2

#### SA-7.1.1 Upper Bound

The proof is essentially the proof for Lemma SA-3.4 with the data generating process ranging over  $\mathcal{P}$ . By Lemma SA-3.1 and Equation (SA-6.1), we have

$$\begin{aligned}
& \sup_{\mathbb{P} \in \mathcal{P}} \sup_{\mathbf{x} \in \mathcal{B}} |\mathfrak{B}_{n,t}(\mathbf{x})| \\
&= \sup_{\mathbb{P} \in \mathcal{P}} \sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbf{S}_{t,\mathbf{x}} - \mu_t(\mathbf{x}) \right| \\
&= \sup_{\mathbb{P} \in \mathcal{P}} \sup_{\mathbf{x} \in \mathcal{B}} \left| \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) k_h(D_i(\mathbf{x})) \mathbf{r}_p(D_i(\mathbf{x}))^\top (\mu_t(\mathbf{X}_i) - \mu_t(\mathbf{x}), 0, \dots, 0) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_t) \right] \right| \\
&\lesssim \sup_{\mathbb{P} \in \mathcal{P}} \sup_{\mathbf{x} \in \mathcal{B}} \sup_{\mathbf{z} \in \mathcal{X}} \left| \mathbf{e}_1^\top \Psi_{t,\mathbf{x}}^{-1} \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right) k_h(D_i(\mathbf{x})) \mathbf{r}_p \left( \frac{D_i(\mathbf{x})}{h} \right)^\top \right] \right| \\
&\quad \cdot \sup_{\mathbb{P} \in \mathcal{P}} \sup_{\mathbf{x} \in \mathcal{B}} \sup_{\mathbf{z} \in \mathcal{X}} |\mu_t(\mathbf{x}) - \mu_t(\mathbf{z})| \mathbb{1}(k_h(\mathcal{A}(\mathbf{z}, \mathbf{x})) > 0) \\
&\lesssim h.
\end{aligned}$$

#### SA-7.1.2 Lower Bound

The lower bound is proved by considering the following data generating process. Suppose  $\mathbf{X}_i \sim \text{Unif}([-2, 2]^2)$ , and  $\mu_0(x_1, x_2) = 0$  and  $\mu_1(x_1, x_2) = x_2$  for all  $(x_1, x_2) \in \mathcal{X} = [-2, 2]^2$ . Suppose  $Y_i(0) \sim \text{N}(\mu_0(\mathbf{X}_i), 1)$  and  $Y_i(1) \sim \text{N}(\mu_1(\mathbf{X}_i), 1)$ . Define the treatment and control region by  $\mathcal{A}_1 = \{(x, y) \in \mathcal{X} : x \geq 0, y \geq 0\}$ ,  $\mathcal{A}_0 = \mathcal{X} \setminus \mathcal{A}_1$ ,  $\mathcal{B} = \{(x, y) \in \mathbb{R} : 0 \leq x \leq 2, y = 0 \text{ or } x = 0, 0 \leq y \leq 2\}$ . Suppose  $Y_i = \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_0)Y_i(0) + \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1)Y_i(1)$ . Suppose we choose  $\mathcal{A}$  to be the Euclidean distance and  $D_i(\mathbf{x}) = \|\mathbf{X}_i - \mathbf{x}\|$ . In this case, although the underlying conditional mean functions  $\mu_t$ ,  $t \in \{0, 1\}$  are smooth, the conditional mean given distance  $\theta_{t,\mathbf{x}}$  may not even be differentiable. In this example,

$$\theta_{1,(s,0)}(r) = \begin{cases} \frac{2}{\pi r}, & \text{if } 0 \leq r \leq s, \\ \frac{r+s}{\pi - \arccos(s/r)}, & \text{if } r > s. \end{cases}$$

Figure SA-1 plots  $r \mapsto \theta_{1,(3/4,0)}(r)$  with the notation  $\mathbf{x}_s = (s, 0)$ .

Under this data generating process, we can show

$$\inf_{0 < h < 1} \sup_{\mathbf{x} \in \mathcal{B}} \frac{|\mathfrak{B}_{n,1}(\mathbf{x}) - \mathfrak{B}_{n,0}(\mathbf{x})|}{h} > 0.$$

The proof proceeds in two steps. First, we show a scaling property of the asymptotic bias under our example, which gives a reduction to fixed- $h$  bias calculation. Second, we prove the lower bound via the reduction from previous step.

**Step 1: A Scaling Property.** Let  $0 < h < 1, 0 < s < 1, 0 < C < 1$ . Define  $h' = Ch$  and  $s' = Cs$ . Here

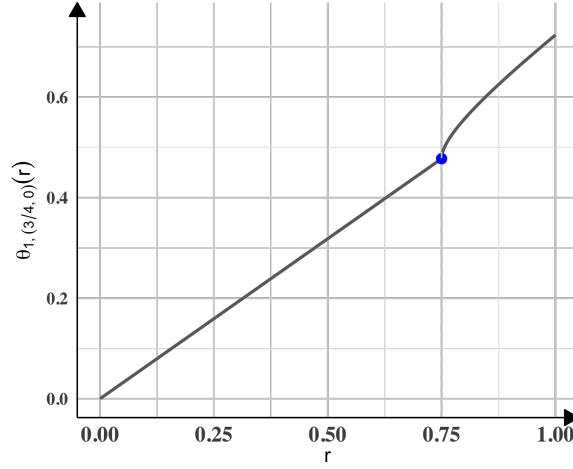


Figure SA-1. Conditional Mean Given Distance with One Kink

$C$  is the scaling factor and denote  $\mathbf{x}_s = (s, 0)$  and  $\mathbf{x}_{s'} = (s', 0)$ . Denote bias for  $\mathbf{x}_{s'}$  under bandwidth  $h'$  to be

$$\begin{aligned} \text{bias}_{n,1}(h', s') &= \mathbf{e}_1^\top \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i((s', 0))}{h'} \right) \mathbf{r}_p \left( \frac{D_i((s', 0))}{h'} \right)^\top k_{h'}(D_i((s', 0))) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \right]^{-1} \\ &\quad \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i((s', 0))}{h'} \right) k_{h'}(D_i((s', 0))) (\mu_1(\mathbf{X}_i - (s', 0))) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \right], \end{aligned} \quad (\text{SA-7.1})$$

where we have used the fact that  $\mu_1$  is linear in our example, hence  $\mu_1(\mathbf{X}_i) - \mu_1((s', 0)) = \mu_1(\mathbf{X}_i - (s', 0))$ . We reserve the notation  $\mathfrak{B}_{n,t}$ ,  $t = 0, 1$ , to the bias when bandwidth is  $h$ , that is,

$$\mathfrak{B}_{n,t}(\mathbf{x}_s) \equiv \text{bias}_{n,t}(h, s), \quad h \in (0, 1), s \in (0, 1), t = 0, 1.$$

Inspecting each element of the last vector, for all  $l \in \mathbb{N}$ ,

$$\begin{aligned} &\mathbb{E} \left[ \left( \frac{\|\mathbf{X}_i - (s', 0)\|}{h'} \right)^l k_{h'}(\|\mathbf{X}_i - (s', 0)\|) (\mu_1(\mathbf{X}_i - (s', 0))) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \right] \\ &= \int_0^2 \int_0^2 \left( \frac{1}{h'} \right)^2 \left( \frac{\|(u' - s', v')\|}{h'} \right)^l k \left( \frac{\|(u' - s', v')\|}{h'} \right) \mu_1((u', v') - (s', 0)) \frac{1}{4} du' dv' \\ &\stackrel{(1)}{=} \int_0^{2/C} \int_0^{2/C} \left( \frac{1}{Ch} \right)^2 \left( \frac{\|(Cu - Cs, Cv)\|}{Ch} \right)^l k \left( \frac{\|(Cu - Cs, Cv)\|}{Ch} \right) \mu_1(C(u - s, v)) \frac{C^2}{4} dudv \\ &= \int_0^{2/C} \int_0^{2/C} \left( \frac{1}{h} \right)^2 \left( \frac{\|(u - s, v)\|}{h} \right)^l k \left( \frac{\|(u - s, v)\|}{h} \right) C \mu_1((u - s, v)) \frac{1}{4} dudv \\ &\stackrel{(2)}{=} \int_0^2 \int_0^2 \left( \frac{1}{h} \right)^2 \left( \frac{\|(u - s, v)\|}{h} \right)^l k \left( \frac{\|(u, v) - (s, 0)\|}{h} \right) C \mu_1((u, v) - (s, 0)) \frac{1}{4} dudv \\ &= C \mathbb{E} \left[ \left( \frac{\|\mathbf{X}_i - (s, 0)\|}{h} \right)^l k_h(\|\mathbf{X}_i - (s, 0)\|) \mu_1(\mathbf{X}_i - (s, 0)) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \right], \end{aligned}$$

where in (1) we have used a change of variable  $(u, v) = \frac{1}{C}(u', v')$ , and (2) holds since  $k \left( \frac{\|\cdot - (s, 0)\|}{h} \right)$  is supported in  $(s, 0) + hB(0, 1)$ , which is contained in  $[0, 2] \times [0, 2] \subseteq [0, 2/C] \times [0, 2/C]$  for all  $0 < h < 1$ ,  $0 < s < 1$ ,

$0 < C < 1$ . This means

$$\begin{aligned} & \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i((s', 0))}{h'} \right) k_{h'}(D_i((s', 0))) (\mu_1(\mathbf{X}_i - (s', 0))) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \right] \\ &= C \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i((s, 0))}{h} \right) k_h(D_i((s, 0))) (\mu_1(\mathbf{X}_i - (s, 0))) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \right]. \end{aligned}$$

Similarly, for all  $l \in \mathbb{N}$  and  $0 < h < 1$ ,  $0 < s < 1$ ,  $0 < C < 1$ ,

$$\mathbb{E} \left[ \left( \frac{D_i((s', 0))}{h'} \right)^l k_{h'}(D_i((s', 0))) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \right] = \mathbb{E} \left[ \left( \frac{D_i((s, 0))}{h} \right)^l k_h(D_i((s, 0))) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \right],$$

implying

$$\begin{aligned} & \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i((s', 0))}{h'} \right) \mathbf{r}_p \left( \frac{D_i((s', 0))}{h'} \right)^\top k_{h'}(D_i((s', 0))) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \right] \\ &= \mathbb{E} \left[ \mathbf{r}_p \left( \frac{D_i((s, 0))}{h} \right) \mathbf{r}_p \left( \frac{D_i((s, 0))}{h} \right)^\top k_h(D_i((s, 0))) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \right]. \end{aligned}$$

It then follows that for all  $0 < h < 1$ ,  $0 < s < 1$ ,  $0 < C < 1$ ,

$$\text{bias}_{n,1}(h', s') = C \text{bias}_{n,1}(h, s).$$

Moreover, for all  $0 < h < 1$ ,  $0 < s < h$ ,

$$\mathfrak{B}_{n,1}(\mathbf{x}_s) = \text{bias}_{n,1}(h, s) = h \text{bias}_{n,1} \left( 1, \frac{s}{h} \right). \quad (\text{SA-7.2})$$

Since  $\mu_0 \equiv 0$ , it is easy to check that

$$\mathfrak{B}_{n,0}(\mathbf{x}_s) = \text{bias}_{n,0}(h, s) \equiv 0, \quad 0 < h < 1, 0 < s < h.$$

**Step 2: Lower Bound on Bias.** Now we want to show  $\sup_{0 \leq s \leq 1} |\text{bias}_{n,1}(1, s) - \text{bias}_{n,0}(1, s)| > 0$ . By Equation (SA-7.1),

$$\begin{aligned} \text{bias}_{n,1}(1, s) - \text{bias}_{n,0}(1, s) &= \mathbf{e}_1^\top \boldsymbol{\Psi}_s^{-1} \mathbf{S}_s - \mu_1(\mathbf{x}_s) - 0 = \mathbf{e}_1^\top \boldsymbol{\Psi}_s^{-1} \mathbf{S}_s, \\ \boldsymbol{\Psi}_s &= \mathbb{E} \left[ \mathbf{r}_p(D_i(\mathbf{x}_s)) \mathbf{r}_p(D_i(\mathbf{x}_s))^\top k(D_i(\mathbf{x}_s)) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1) \right], \\ \mathbf{S}_s &= \mathbb{E} [\mathbf{r}_p(D_i(\mathbf{x}_s)) k(D_i(\mathbf{x}_s)) \mu_1(\mathbf{X}_i) \mathbb{1}(\mathbf{X}_i \in \mathcal{A}_1)]. \end{aligned}$$

Changing to polar coordinates, we have

$$\begin{aligned} \boldsymbol{\Psi}_s &= \int_0^\infty \int_{\Theta_s(r)}^\pi \mathbf{r}_p(r) \mathbf{r}_p(r)^\top K(r) r d\theta dr, \\ \mathbf{S}_s &= \int_0^\infty \int_{\Theta_s(r)}^\pi \mathbf{r}_p(r) K(r) r \sin(\theta) r d\theta dr, \end{aligned}$$

with

$$\Theta_s(r) = \begin{cases} 0, & \text{if } 0 \leq r \leq s, \\ \arccos(s/r), & \text{if } r > s. \end{cases}$$

For notation simplicity, denote

$$\begin{aligned} \mathbf{A}(s) &= \int_0^\infty \int_{\Theta_s(u)}^\pi \mathbf{r}_p(u) \mathbf{r}_p(u)^\top k(u) u d\theta du = \mathbf{A}_1(s) + \mathbf{A}_2(s), \\ \mathbf{B}(s) &= \int_0^\infty \int_{\Theta_s(u)}^\pi \mathbf{r}_p(u) k(u) u \sin(\theta) u d\theta du = \mathbf{B}_1(s) + \mathbf{B}_2(s), \end{aligned}$$

where

$$\begin{aligned} \mathbf{A}_1(s) &= \int_0^s \int_0^\pi \mathbf{r}_p(u) \mathbf{r}_p(u)^\top k(u) u d\theta du = \pi \int_0^s \mathbf{r}_p(u) \mathbf{r}_p(u)^\top k(u) u du, \\ \mathbf{A}_2(s) &= \int_s^\infty \int_{\arccos(s/u)}^\pi \mathbf{r}_p(u) \mathbf{r}_p(u)^\top k(u) u d\theta du = \int_s^\infty (\pi - \arccos(s/u)) \mathbf{r}_p(u) \mathbf{r}_p(u)^\top k(u) u du, \\ \mathbf{B}_1(s) &= \int_0^s \int_0^\pi \mathbf{r}_p(u) k(u) u \sin(\theta) u d\theta du = 2 \int_0^s \mathbf{r}_p(u) k(u) u^2 du, \\ \mathbf{B}_2(s) &= \int_s^\infty \int_{\arccos(s/u)}^\pi \mathbf{r}_p(u) k(u) u \sin(\theta) u d\theta du = \int_s^\infty (1 + \frac{s}{u}) \mathbf{r}_p(u) k(u) u^2 du. \end{aligned}$$

Evaluating the above at zero gives

$$\mathbf{A}(0) = \frac{\pi}{2} \int_0^\infty u \mathbf{r}_p(u) \mathbf{r}_p(u)^\top k(u) du, \quad \mathbf{B}(0) = \int_0^\infty u^2 \mathbf{r}_p(u) k(u) du.$$

Hence

$$\text{bias}_{n,1}(1,0) - \text{bias}_{n,0}(1,0) = \mathbf{e}_1^\top \mathbf{A}(0)^{-1} \mathbf{B}(0) = \mathbf{e}_1^\top \mathbf{A}(0)^{-1} \left[ \frac{2}{\pi} \mathbf{A}(0) \mathbf{e}_2 \right] = 0. \quad (\text{SA-7.3})$$

Taking derivatives with respect to  $s$ , we have

$$\begin{aligned} \dot{\mathbf{A}}_1(s) &= \pi \mathbf{r}_p(s) \mathbf{r}_p(s)^\top k(s) s, \\ \dot{\mathbf{A}}_2(s) &= -\pi \mathbf{r}_p(s) \mathbf{r}_p(s)^\top k(s) s + \int_s^\infty \frac{1}{\sqrt{u^2 - s^2}} u \mathbf{r}_p(u) \mathbf{r}_p(u)^\top k(u) du, \\ \dot{\mathbf{B}}_1(s) &= 2 \mathbf{r}_p(s) k(s) s^2, \\ \dot{\mathbf{B}}_2(s) &= -2 \mathbf{r}_p(s) k(s) s^2 + \int_s^\infty u \mathbf{r}_p(u) k(u) du. \end{aligned}$$

Evaluating the above at zero gives

$$\dot{\mathbf{A}}(0) = \int_0^\infty \mathbf{r}_p(u) \mathbf{r}_p(u)^\top k(u) du, \quad \dot{\mathbf{B}}(0) = \int_0^\infty u \mathbf{r}_p(u) k(u) du.$$

Using matrix calculus, we know

$$\begin{aligned} & \left. \frac{d}{ds} \text{bias}_{n,1}(1, s) - \text{bias}_{n,0}(1, s) \right|_{s=0} \\ &= \left. \frac{d}{ds} \mathbf{e}_1^\top \mathbf{A}(s)^{-1} \mathbf{B}(s) \right|_{s=0} \end{aligned} \quad (\text{SA-7.4})$$

$$\begin{aligned} &= -\mathbf{e}_1^\top \mathbf{A}(0)^{-1} \dot{\mathbf{A}}(0) [\mathbf{A}(0)^{-1} \mathbf{B}(0)] + \mathbf{e}_1^\top \mathbf{A}(0)^{-1} \dot{\mathbf{B}}(0) \\ &= -\mathbf{e}_1^\top \mathbf{A}(0)^{-1} \dot{\mathbf{A}}(0) \left[ \frac{2}{\pi} \mathbf{e}_2 \right] + \mathbf{e}_1^\top \left[ \frac{2}{\pi} \mathbf{e}_1 \right] \end{aligned} \quad (\text{SA-7.5})$$

$$\begin{aligned} &= -\frac{2}{\pi} \mathbf{e}_1^\top \mathbf{A}(0)^{-1} \int_0^\infty \begin{bmatrix} u \\ u^2 \\ \dots \\ u^{p+1} \end{bmatrix} k(u) du + \mathbf{e}_1^\top \left[ \frac{2}{\pi} \mathbf{e}_1 \right] \\ &= -\frac{4}{\pi^2} + \frac{2}{\pi}. \end{aligned} \quad (\text{SA-7.6})$$

Combining Equations (SA-7.3) and (SA-7.4), and the fact that  $\frac{d}{ds} \text{bias}_{n,1}(1, s) - \text{bias}_{n,0}(1, s)$  is continuous in  $s$ , we can show  $\sup_{0 \leq s \leq 1} |\text{bias}_{n,1}(1, s) - \text{bias}_{n,0}(1, s)| > 0$ . Combining with Equation (SA-7.2), we have

$$\begin{aligned} \inf_{0 < h < 1} \sup_{\mathbf{x} \in \mathcal{B}} \frac{|\mathfrak{B}_{n,1}(\mathbf{x}) - \mathfrak{B}_{n,0}(\mathbf{x})|}{h} &\geq \inf_{0 < h < 1} \sup_{0 < s < h} \frac{|\text{bias}_{n,1}(s, h) - \text{bias}_{n,0}(s, h)|}{h} \\ &= \inf_{0 < h < 1} \sup_{0 < s < h} \left| \text{bias}_{n,1} \left( 1, \frac{s}{h} \right) \right| \\ &> 0. \end{aligned}$$

■

## SA-7.2 Proof of Lemma 3

The proof of part (i) follows from part (ii) with  $\mathcal{B} \cap B(\mathbf{x}, \varepsilon)$  as the boundary. To prove part (ii), without loss of generality, we assume that  $\iota = p + 1$ , and want to show  $\sup_{\mathbf{x} \in \mathcal{B}^o} |\mathfrak{B}_{n,t}(\mathbf{x})| \lesssim h^{p+1}$ . This means we have assumed that  $\mathcal{B}$  has a one-to-one curve length parametrization  $\gamma$  that is  $C^{p+3}$  with curve length  $L$ , there exists  $\varepsilon, \delta > 0$  such that for all  $\mathbf{x} \in \gamma([\delta, L - \delta])$  and  $0 < r < \varepsilon$ ,  $S(\mathbf{x}, r)$  intersects  $\mathcal{B}$  with two points,  $s(\mathbf{x}, r)$  and  $t(\mathbf{x}, r)$ . Define  $a(\mathbf{x}, r)$  and  $b(\mathbf{x}, r)$  to be the number in  $[0, 2\pi]$  such that

$$[a(\mathbf{x}, r), b(\mathbf{x}, r)] = \{\theta : \mathbf{x} + r(\cos \theta, \sin \theta) \in \mathcal{A}_1\}.$$

Then, for  $\mathbf{x} \in \mathcal{B}$  and  $0 < r < \varepsilon$ ,  $\theta_{1,\mathbf{x}}(r)$  has the following explicit representation:

$$\theta_{1,\mathbf{x}}(r) = \frac{\int_{a(\mathbf{x},r)}^{b(\mathbf{x},r)} \mu_1(\mathbf{x} + r(\cos \theta, \sin \theta)) f_X(\mathbf{x} + r(\cos \theta, \sin \theta)) d\theta}{\int_{a(\mathbf{x},r)}^{b(\mathbf{x},r)} f_X(\mathbf{x} + r(\cos \theta, \sin \theta)) d\theta}.$$

**Step 1: Curve length v.s. Distance to  $\gamma(0)$**

W.l.o.g., assume  $\gamma(0) = \mathbf{x}$  and  $\gamma'(0) = (1, 0)$ . Let  $T : [0, \infty) \rightarrow [0, \infty)$  to be a continuous increasing function that satisfies

$$\|\gamma \circ T(r)\|^2 = r^2, \quad \forall r \in [0, h].$$

**Initial Case:**  $l = 1, 2, 3$ . We will show that  $T$  is  $C^l$  on  $(0, h)$ . For notational simplicity, define another function  $\phi : [0, \infty) \rightarrow [0, \infty)$  by  $\phi(t) = \|\gamma(t)\|^2$ . Using implicit derivations iteratively,

$$\begin{aligned} \phi \circ T(r) &= r^2, \\ \phi'(T(r))T'(r) &= 2r, \\ \phi''(T(r))(T'(r))^2 + \phi'(T(r))T''(r) &= 2, \\ \phi'''(T(r))(T'(r))^3 + 3\phi''(T(r))T'(r)T''(r) + \phi'(T(r))T'''(r) &= 0. \end{aligned} \tag{1}$$

From the above equalities, we get

$$\begin{aligned} T'(r) &= \frac{2r}{\phi'(T(r))}, \\ T''(r) &= \frac{2 - \phi''(T(r))(T'(r))^2}{\phi'(T(r))}, \\ T'''(r) &= -\frac{\phi'''(T(r))(T'(r))^3 + 3\phi''(T(r))T'(r)T''(r)}{\phi'(T(r))}. \end{aligned}$$

Since we have assumed  $\gamma$  is  $C^{p+3}$  on  $(0, h)$ ,  $\phi$  is also  $C^{p+1}$  on  $(0, h)$ . It follows from the above calculation that  $T$  is  $C^{p+3}$  on  $(0, h)$ . In order to find the limit of derivatives of  $T$  at 0, we need

$$\begin{aligned} \phi(t) &= \gamma_1(t)^2 + \gamma_2(t)^2, & \phi(0) &= 0, \\ \phi'(t) &= 2\gamma_1(t)\gamma_1'(t) + 2\gamma_2(t)\gamma_2'(t), & \phi'(0) &= 0, \\ \phi''(t) &= 2\gamma_1'(t)\gamma_1'(t) + 2\gamma_1(t)\gamma_1''(t) + 2\gamma_2'(t)\gamma_2'(t) + 2\gamma_2(t)\gamma_2''(t), & \phi''(0) &= 2, \\ \phi'''(t) &= 6\gamma_1'(t)\gamma_1''(t) + 2\gamma_1(t)\gamma_1'''(t) + 6\gamma_2'(t)\gamma_2''(t) + 2\gamma_2(t)\gamma_2'''(t). \end{aligned}$$

Using L'Hôpital's rule

$$\begin{aligned} \lim_{r \downarrow 0} T'(r) &= \lim_{r \downarrow 0} \frac{2}{\phi''(T(r))T'(r)} = \frac{2}{2 \lim_{r \downarrow 0} T'(r)} \implies \lim_{r \downarrow 0} T'(r) = 1, \\ \lim_{r \downarrow 0} T''(r) &= \lim_{r \downarrow 0} \frac{-\phi'''(T(r))(T'(r))^3 - \phi''(T(r))2T'(r)T''(r)}{\phi''(T(r))T'(r)} \\ &= \frac{-\phi^{(3)}(0) - 4 \lim_{r \downarrow 0} T''(r)}{2} \\ &= \frac{-\phi^{(3)}(0)}{6} \end{aligned}$$

$$\begin{aligned}
\lim_{r \downarrow 0} T^{(3)}(r) &= -\lim_{r \downarrow 0} \frac{\phi^{(4)}(T(r))(T'(r))^4 + \phi^{(3)}(T(r))3(T'(r))^2 T''(r) + 3\phi^{(3)}(T(r))(T'(r))^2 T''(r)}{\phi''(T(r))T'(r)} \\
&\quad + \lim_{r \downarrow 0} \frac{3\phi''(T(r))T'(r)T^{(3)}(r)}{\phi''(T(r))T'(r)} \\
&= -\frac{\phi^{(4)}(0) - (\phi^{(3)}(0))^2 + 6\lim_{r \downarrow 0} T^{(3)}(r)}{2} \\
&= -\frac{\phi^{(4)}(0) - (\phi^{(3)}(0))^2}{8}.
\end{aligned}$$

**Induction Step:**  $l \geq 4$ . Assume  $\lim_{r \downarrow 0} T^{(i)}(r)$  exists and is finite for  $0 \leq i \leq l-2$  and there exists a function  $q(r)$  such that (i)  $q(r)$  is a polynomial of  $\phi^{(j)}(T(r))$ ,  $1 \leq j \leq l-1$  and  $T^{(k)}(r)$ ,  $1 \leq k \leq l-2$ , (ii)  $\lim_{r \downarrow 0} q(r) = 0$  and (iii)

$$q(r) + \phi'(T(r))T^{(l-1)}(r) = 0. \quad (2)$$

For  $l = 4$ , this assumption can be verified from Equation (1). Using L'hospital's rule,

$$\begin{aligned}
\lim_{r \downarrow 0} T^{(l-1)}(r) &= \lim_{r \downarrow 0} -\frac{q(r)}{\phi'(T(r))} \\
&\stackrel{L'h}{=} \lim_{r \downarrow 0} -\frac{q'(r)}{\phi''(T(r))T'(r)}.
\end{aligned}$$

From the previous paragraph,  $\lim_{r \downarrow 0} \phi''(T(r))T'(r)$  exists and is finite. And  $q'(r)$  is a polynomial of  $\phi^{(j)}(T(r))$ ,  $1 \leq j \leq l$  and  $T^{(k)}(r)$ ,  $1 \leq k \leq l-1$ . Hence  $\lim_{r \downarrow 0} T^{(l-1)}(r)$  can be solved from the following equation and is finite:

$$\lim_{r \downarrow 0} q'(r) + \lim_{r \downarrow 0} \phi''(T(r))T'(r) \cdot \lim_{r \downarrow 0} T^{(l-1)}(r) = 0. \quad (3)$$

Taking derivatives on both sides of Equation (2),

$$q'(r) + \phi''(T(r))T'(r)T^{(l-1)}(r) + \phi'(T(r))T^{(l)}(r) = 0.$$

Take  $q_2(r) = q'(r) + \phi''(T(r))T'(r)T^{(l-1)}(r)$ . Then, (i)  $q_2(r)$  is a polynomial of  $\phi^{(j)}(T(r))$ ,  $1 \leq j \leq l$  and  $T^{(k)}(r)$ ,  $1 \leq k \leq l-1$ , (ii)  $\lim_{r \downarrow 0} q_2(r) = 0$ , and (iii)

$$q_2(r) + \phi'(T(r))T^{(l)}(r) = 0.$$

Continue this argument till  $l = p+3$ ,  $\lim_{r \downarrow 0} T^{(j)}(r)$  exists and is a polynomial of  $\phi^{(0)}(0), \dots, \phi^{(j+1)}(0)$ , which implies that it is bounded by a constant only depending on  $\gamma$ .

**Step 2:**  $(p+1)$ -times continuously differentiable  $S_r$

We use the notation  $\gamma(t) = (\gamma_1(t), \gamma_2(t))$ . Define

$$A(t) = \angle \gamma(t) - \gamma(0), \gamma'(0) = \arcsin \left( \frac{\gamma_2(t)}{\|\gamma(t)\|} \right).$$

Since  $\gamma$  is  $C^{p+3}$ , we can Taylor expand  $\gamma$  at 0 to get

$$\gamma(t) = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 1 \\ 0 \end{pmatrix} t + \begin{pmatrix} u_2 \\ v_2 \end{pmatrix} t^2 + \dots + \begin{pmatrix} u_{p+2} \\ v_{p+2} \end{pmatrix} t^{p+2} + \begin{pmatrix} R_1(t) \\ R_2(t) \end{pmatrix},$$

where we have used the fact that  $\gamma'_2(0) = 0$  and  $\|\gamma'(0)\| = 1$  and

$$R_1(t) = \int_0^t \frac{\gamma_1^{(p+3)}(s)(t-s)^{p+2}}{(p+2)!} ds, \quad R_2(t) = \int_0^t \frac{\gamma_2^{(p+3)}(s)(t-s)^{p+2}}{(p+2)!} ds.$$

Since  $\gamma$  is  $C^{p+3}$ ,  $R_1(t)/t$  and  $R_2(t)/t$  are  $C^{p+3}$  on  $(0, \infty)$ . We *claim* that  $\lim_{t \downarrow 0} \frac{d^v}{dt^v}(R_1(t)/t)$  exists and is uniformly bounded for all  $\mathbf{x} \in \mathcal{B}$ , for all  $0 \leq v \leq p+1$ . Define  $\varphi(t) = R_1(t)/t$ . Then

$$\begin{aligned} \varphi'(t) &= -\frac{R_1(t)}{t^2} + \frac{R_1'(t)}{t}, \\ \varphi''(t) &= \frac{2R_1(t)}{t^3} - \frac{2R_1'(t)}{t^2} + \frac{R_1''(t)}{t}, \\ \varphi^{(3)}(t) &= -\frac{6R_1(t)}{t^4} + \frac{6R_1'(t)}{t^3} - \frac{3R_1^{(2)}(t)}{t^2} + \frac{R_1^{(3)}(t)}{t} \quad \dots \end{aligned}$$

where

$$R_1'(t) = \int_0^t \frac{\gamma_1^{(p+1)}(s)(t-s)^{p-1}}{(p-1)!} ds, \quad R_1''(t) = \int_0^t \frac{\gamma_1^{(p+1)}(s)(t-s)^{p-2}}{(p-2)!} ds, \quad \dots$$

Since  $\gamma_1$  is  $C^{p+3}$ , there exists  $C_1 > 0$  only depending on  $\gamma$  such that for all  $0 \leq v \leq p+3$ ,  $|\frac{d^v}{dt^v} R_1(t)| \leq C_1 t^{p+1-v}$ . Hence

$$\lim_{r \downarrow 0} \varphi^{(j)}(r) = 0, \quad \forall 0 \leq j \leq p+1.$$

Similarly,  $\lim_{r \downarrow 0} \frac{d^v}{dt^v}(R_2(t)/t)$  exists and is uniformly bounded for all  $0 \leq v \leq p+1$ . Then

$$\frac{\gamma_2(t)}{\|\gamma(t)\|} = \frac{v_2 t + \dots + v_{p+2} t^{p+2} + R_2(t)/t}{\sqrt{(1 + u_2 t + \dots + u_{p+2} t^{p+2} + R_1(t)/t)^2 + (v_2 t + \dots + v_{p+2} t^{p+2} + R_2(t)/t)^2}}, \quad t > 0.$$

Notice that  $\gamma_2(t)/\|\gamma(t)\|$  is of the form

$$p(t)(1 + q(t))^\alpha,$$

where  $\alpha < 0$  and  $p(t), q(t)$  are  $C^{p+1}$  on  $(0, \infty)$  with  $\lim_{r \downarrow 0} d^v/dt^v p(t)$  and  $\lim_{r \downarrow 0} d^v/dt^v q(t)$  finite. Since the derivative of  $p(t)(1 + q(t))^\alpha$  is

$$p'(t)(1 + q(t))^\alpha + p(t)\alpha(1 + q(t))^{\alpha-1} q'(t),$$

which is the sum of two terms of the form  $p_2(t)(1 + q_2(t))^\alpha$  with  $p_2$  and  $q_2$  functions that are  $C^p$  with finite limits at 0. Continue this argument, we see that  $\frac{\gamma_2(\cdot)}{\|\gamma(\cdot)\|}$  is  $C^{p+1}$  on  $(0, \infty)$  and  $\lim_{r \downarrow 0} \frac{d^v}{dt^v}(\gamma_2(t)/\|\gamma(t)\|)$  exist and are uniformly bounded for all  $\mathbf{x} \in \mathcal{B}$  and for all  $0 \leq v \leq p+1$ .

Since  $\arcsin$  is  $C^{p+1}$  with bounded (higher order derivatives) on  $[-1/2, 1/2]$ ,  $A$  is  $C^{p+1}$  on  $(0, \delta)$  and for all  $0 \leq v \leq p+1$ ,  $\lim_{r \downarrow 0} A^{(v)}(t)$  exist and are uniformly bounded for all  $\mathbf{x} \in \mathcal{B}$ .

**Step 3:  $(p+1)$ -times continuously differentiable conditional density**

By the previous two steps,  $a(\mathbf{x}, r) = A \circ T(r)$  is  $C^{p+1}$  on  $(0, \infty)$  with  $|\lim_{r \downarrow 0} \frac{d^v}{dr^v} a(\mathbf{x}, r)| < \infty$ . Similarly, we can show that  $b(\mathbf{x}, r)$  is  $C^{p+1}$  in  $r$  with finite limits at  $r = 0$ . By the assumption that  $f_X$  is  $C^{p+1}$  and bounded below by  $\underline{f}$ ,  $\theta_{1,\mathbf{x}}$  is  $C^{p+1}$  with  $\lim_{r \downarrow 0} \frac{d^v}{dr^v} \theta_{1,\mathbf{x}}(r)$  uniformly bounded for all  $\mathbf{x} \in \mathcal{B}$  and for all  $0 \leq v \leq p+1$ .

This completes the proof. ■

### SA-7.3 Proof of Theorem 6

Let  $s > 0$  be a parameter that is chosen later. Consider the following two data generating processes.

**Data Generating Process  $\mathbb{P}_0$ .** Let  $\mathcal{X} = \{r(\cos \theta, \sin \theta) : 0 \leq r \leq 1, 0 \leq \theta \leq \Theta(r)\}$ , where

$$\Theta(r) = \begin{cases} \pi, & 0 \leq r < s, \\ \theta_k, & s + ks^2 \leq r < s + (k+1)s^2, 0 \leq k < K, \\ \theta_K, & s + Ks^2 \leq r < 1, \end{cases}$$

with  $K = \lfloor \frac{1-s}{s^2} \rfloor$  and  $\theta_k$  is the unique zero of

$$\frac{\sin(\theta)}{\theta} = \frac{(k + \frac{1}{2})s^2}{s + (k + \frac{1}{2})s^2}$$

over  $\theta \in [0, \pi]$ , and  $\theta_K$  is the unique zero of

$$\frac{\sin(\theta)}{\theta} = \frac{Ks^2 + 1 - s}{s + Ks^2 + 1}$$

over  $\theta \in [0, \pi]$ . Suppose  $\mathbf{X}_i$  has density  $f_X$  given by

$$f_X(r(\cos \theta, \sin \theta)) = \frac{1}{2\Theta(r)}, \quad 0 \leq r \leq 1, 0 \leq \theta \leq \Theta(r).$$

Suppose

$$\mu_0(x_1, x_2) = \frac{1}{2} + \frac{1}{100}x_1, \quad (x_1, x_2) \in \mathbb{R}^2.$$

Suppose  $Y_i = \mathbb{1}(\eta_i \leq \mu(\mathbf{X}_i))$  where  $(\eta_i : i : 1, \dots, n)$  are i.i.d. random variables independent of  $(\mathbf{X}_i : 1, \dots, n)$ . Let  $\eta_0(r) = \mathbb{E}_{\mathbb{P}_0}[Y_i | \|\mathbf{X}_i - (0, 0)\| = r]$ , for  $r \geq 0$ . In particular,  $\text{bd}(\mathcal{X})$  has length  $\pi + 2$ . Hence,  $\text{bd}(\mathcal{X})$  is a rectifiable curve.

**Data Generating Process  $\mathbb{P}_1$ .** Let  $\mathcal{X} = \{r(\cos \theta, \sin \theta) : 0 \leq r \leq 1, 0 \leq \theta \leq \pi/2\}$ ,  $\mathbf{X}_i$  is uniformly distributed on  $\mathcal{X}$ , and

$$\mu_1(x_1, x_2) = \frac{1}{2} + \frac{1}{100}(x_1 - s), \quad (x_1, x_2) \in \mathbb{R}^2.$$

Suppose  $Y_i = \mathbb{1}(\eta_i \leq \mu(\mathbf{X}_i))$  where  $(\eta_i : 1, \dots, n)$  are i.i.d random variables independent to  $(\mathbf{X}_i : 1, \dots, n)$ . Let  $\eta_1(r) = \mathbb{E}_{\mathbb{P}_1}[Y_i | \|\mathbf{X}_i - (0, 0)\| = r]$ , for  $r \geq 0$ . In particular,  $\text{bd}(\mathcal{X})$  has length  $\pi/2 + 2$ . Hence,  $\text{bd}(\mathcal{X})$  is a rectifiable curve.

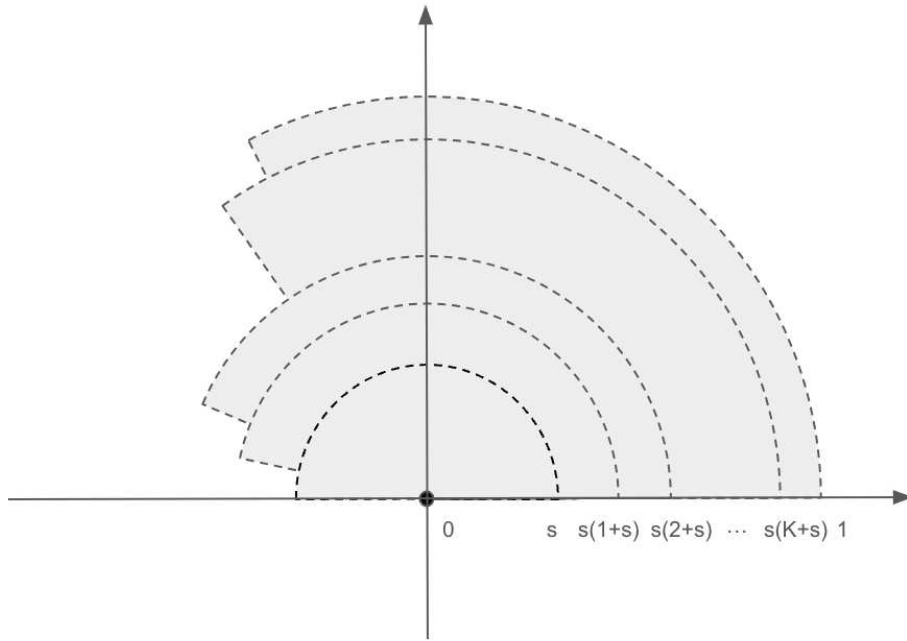


Figure SA-2.  $\mathcal{X}$  from DGP  $\mathbb{P}_0$

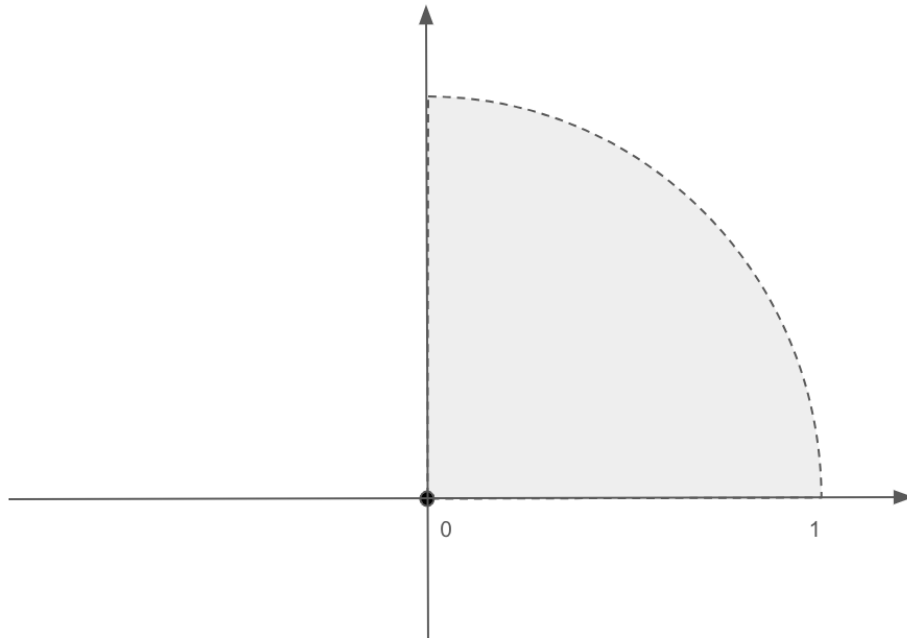


Figure SA-3.  $\mathcal{X}$  from DGP  $\mathbb{P}_1$

**Minimax Lower Bound.** First, we show under the previous two models,  $\mathbb{P}_0(\|\mathbf{X}_i\| \leq r) = \mathbb{P}_1(\|\mathbf{X}_i\| \leq r)$  for all  $r \geq 0$ . Since in  $\mathbb{P}_1$ ,  $\mathbf{X}_i$  is uniform distributed on  $\mathbb{R}$ , we know  $\mathbb{P}_1(\|\mathbf{X}_i\| \leq r) = r^2$ ,  $0 \leq r \leq 1$ .

$$\mathbb{P}_0(\|\mathbf{X}_i\| \leq r) = \int_0^r \int_0^{\Theta(s)} \frac{1}{2\Theta(s)} s d\theta ds = r^2, \quad 0 \leq r \leq 1.$$

Hence, choosing  $(0, 0)$  as the point of evaluation in both  $\mathbb{P}_0$  and  $\mathbb{P}_1$ , we have

$$\begin{aligned} & d_{\text{KL}}(\mathbb{P}_0(\|\mathbf{X}_i - (0, 0)\|, Y_i), \mathbb{P}_1(\|\mathbf{X}_i - (0, 0)\|, Y_i)) \\ &= \int_0^\infty \int_{-\infty}^\infty d\mathbb{P}_0(r, y) \log \frac{d\mathbb{P}_0(r, y)}{d\mathbb{P}_1(r, y)} \\ &= \int_0^\infty \int_{-\infty}^\infty d\mathbb{P}_0(r) d\mathbb{P}_0(y|r) \log \frac{d\mathbb{P}_0(r) d\mathbb{P}_0(y|r)}{d\mathbb{P}_1(r) d\mathbb{P}_1(y|r)} \\ &= \int_0^\infty d\mathbb{P}_0(r) \int_{-\infty}^\infty d\mathbb{P}_0(y|r) \log \frac{d\mathbb{P}_0(y|r)}{d\mathbb{P}_1(y|r)} \\ &= 2 \int_0^1 d_{\text{KL}}(\text{Bern}(\eta_0(r)), \text{Bern}(\eta_1(r))) r dr. \end{aligned}$$

Under  $\mathbb{P}_0$ ,  $\mathbf{X}_i$  is uniformly distributed on  $\{r(\cos \theta, \sin \theta) : 0 \leq \theta \leq \Theta(r)\}$  for each  $0 < r \leq 1$ . Hence

$$\eta_0(r) = \frac{1}{2} + \frac{1}{100} \frac{1}{\Theta(r)} \int_0^{\Theta(r)} r \cos(u) du - \frac{s}{100} = \frac{1}{2} + \frac{1}{100} r \frac{\sin(\Theta(r))}{\Theta(r)}.$$

Thus, for  $0 \leq k < K$ ,

$$\begin{aligned} \eta_0\left(s + \left(k + \frac{1}{2}\right)s^2\right) &= \frac{1}{2} + \frac{1}{100} \left( \left(s + \left(k + \frac{1}{2}\right)s^2\right) \frac{\sin(\Theta_k)}{\Theta_k} \right) \\ &= \frac{1}{2} + \frac{1}{100} \left( \left(s + \left(k + \frac{1}{2}\right)s^2\right) \frac{\left(k + \frac{1}{2}\right)s^2}{s + \left(k + \frac{1}{2}\right)s^2} \right) \\ &= \eta_1\left(s + \left(k + \frac{1}{2}\right)s^2\right). \end{aligned}$$

Since both  $\eta_0$  and  $\eta_1$  are 1-Lipschitz on all intervals  $[s + ks^2, s + (k+1)s^2]$  for all  $0 \leq k < K$ , we know  $|\eta_0(r) - \eta_1(r)| \leq 2s^2$  for all  $r \in [s, 1]$ . Moreover,  $\eta_0(r) = \frac{1}{2}$  for all  $0 \leq r \leq s$  and  $\eta_1(r) = \frac{1}{2} + \frac{1}{100}(r^2 - s)$ . Hence  $|\eta_0(r) - \eta_1(r)| \leq s$  for all  $0 \leq r \leq s$ . Hence,

$$\begin{aligned} \int_0^1 d_{\text{KL}}(\text{Bern}(\eta_0(r)), \text{Bern}(\eta_1(r))) r dr &\leq \int_0^1 d_{\chi^2}(\text{Bern}(\eta_0(r)), \text{Bern}(\eta_1(r))) r dr \\ &= \int_0^1 (\eta_1(r) \left( \frac{\eta_0(r) - \eta_1(r)}{\eta_1(r)} \right)^2 + (1 - \eta_1(r)) \left( \frac{\eta_0(r) - \eta_1(r)}{1 - \eta_1(r)} \right)^2) r dr \\ &\leq \frac{1}{\frac{1}{2} - \frac{3}{100}} \int_0^1 (\eta_0(r) - \eta_1(r))^2 r dr \\ &\leq \frac{1}{\frac{1}{2} - \frac{3}{100}} \int_0^s s^2 r dr + \frac{1}{\frac{1}{2} - \frac{3}{100}} \int_s^1 (2s^2)^2 r dr \\ &\leq \frac{5}{\frac{1}{2} - \frac{3}{100}} s^4. \end{aligned}$$

Moreover,  $|\mu_0(0, 0) - \mu_1(0, 0)| = \frac{1}{100}s$ . Hence, by [Tsybakov \(2008, Theorem 2.2 \(iii\)\)](#), take  $\frac{5}{\frac{1}{2} - \frac{3}{100}} s_*^4 = \frac{\log 2}{n}$ ,

and conclude that

$$\inf_{T_n \in \mathcal{T}} \sup_{\mathbb{P} \in \mathcal{P}} \sup_{\mathbf{x} \in \mathcal{R}(\mathbb{P})} \mathbb{E}_{\mathbb{P}}[|T_n(\mathbf{U}_n(\mathbf{x})) - \mu(\mathbf{x})|] \geq \frac{1}{1600} s_* \gtrsim n^{-\frac{1}{4}}.$$

This concludes the proof. ■

## SA-8 Proofs for Section SA-4

### SA-8.1 Proof of Lemma SA-4.1

We will use a truncation argument. Let  $\tau_n \gtrsim 1$  be the level of truncation. For each  $r \in \mathcal{R}$ , define

$$\tilde{r}(y) = r(y) \mathbb{1}(|y| \leq \tau_n), \quad y \in \mathbb{R},$$

and define the class  $\tilde{\mathcal{R}} = \{\tilde{r} : r \in \mathcal{R}\}$ . For an overview of our argument, suppose  $Z_n^R$  is a mean-zero Gaussian process indexed by  $\mathcal{G} \times \mathcal{R} \cup \mathcal{G} \times \tilde{\mathcal{R}}$ , whose existence will be shown below, then we can decompose by:

$$R_n(g, r) - Z_n^R(g, r) = [R_n(g, \tilde{r}) - Z_n^R(g, \tilde{r})] + [R_n(g, r) - R_n(g, \tilde{r})] + [Z_n^R(g, r) - Z_n^R(g, \tilde{r})].$$

**Part 1: Gaussian strong approximation for the truncated process** —  $\|R_n(g, \tilde{r}) - Z_n^R(g, \tilde{r})\|_{\mathcal{G} \times \mathcal{R}}$

Observe that  $\mathbf{M}_{\tilde{\mathcal{R}}, \mathcal{Y}} \lesssim \tau_n$  and  $\mathbf{pTV}_{\tilde{\mathcal{R}}, \mathcal{Y}} \lesssim \tau_n$ , and  $\tilde{\mathcal{R}}$  is a VC-type class with envelope  $M_{\tilde{\mathcal{R}}, \mathcal{Y}} = M_{\mathcal{R}, \mathcal{Y}} \mathbb{1}(|\cdot| \leq \tau_n)$  over  $\mathcal{Y}$  with constants  $\mathbf{c}_{\mathcal{R}, \mathcal{Y}}$  and  $\mathbf{d}_{\mathcal{R}, \mathcal{Y}}$ . Then, Cattaneo and Yu (2025, Theorem 2) with  $\mathbf{v} = \tau_n$  and  $\alpha = 0$  for the class of functions  $\mathcal{G}$  and  $\tilde{\mathcal{R}}$  implies on a possibly enlarged probability space, there exists a sequence of mean-zero Gaussian processes  $(Z_n^R(g, r) : (g, r) \in \mathcal{G} \times \tilde{\mathcal{R}})$  with almost sure continuous trajectories on  $(\mathcal{G} \times \tilde{\mathcal{R}}, \rho_{\mathbb{P}})$  such that  $\mathbb{E}[R_n(g_1, r_1)R_n(g_2, r_2)] = \mathbb{E}[Z_n^R(g_1, r_1)Z_n^R(g_2, r_2)]$  for all  $(g_1, r_1), (g_2, r_2) \in \mathcal{G} \times \tilde{\mathcal{R}}$ , and

$$\begin{aligned} & \mathbb{E}[\|R_n(g, \tilde{r}) - Z_n^R(g, \tilde{r})\|_{\mathcal{G} \times \mathcal{R}}] \\ & \leq C_1 \tau_n \left( \sqrt{d} \min \left\{ \frac{(\mathbf{c}_1^d \mathbf{M}_{\mathcal{G}}^{d+1} \mathbf{TV}^d \mathbf{E}_{\mathcal{G}})^{\frac{1}{2d+2}}}{n^{1/(2d+2)}}, \frac{(\mathbf{c}_1^{\frac{d}{2}} \mathbf{c}_2^{\frac{d}{2}} \mathbf{M}_{\mathcal{G}} \mathbf{TV}^{\frac{d}{2}} \mathbf{E}_{\mathcal{G}} \mathbf{L}^{\frac{d}{2}})^{\frac{1}{d+2}}}{n^{1/(d+2)}} \right\} ((\mathbf{d} + \mathbf{k}) \log(cn))^{3/2} + \frac{(\mathbf{d} + \mathbf{k}) \log(cn)}{\sqrt{n}} \mathbf{M}_{\mathcal{G}} \right) \\ & = C_1 \tau_n (\sqrt{d} \mathbf{r}_n ((\mathbf{d} + \mathbf{k}) \log(cn))^{\frac{3}{2}} + \frac{(\mathbf{d} + \mathbf{k}) \log(cn)}{\sqrt{n}} \mathbf{M}_{\mathcal{G}}). \end{aligned}$$

**Part 2: Truncation error for the empirical process** —  $\|R_n(g, r) - R_n(g, \tilde{r})\|_{\mathcal{G} \times \mathcal{R}}$  Consider the class of differences due to truncation, that is,  $\Delta \mathcal{R} = \{r - \tilde{r} : r \in \mathcal{R}\}$ . Our assumptions imply  $\mathcal{G} \times \Delta \mathcal{R}$  is VC-type in the sense that for all  $0 < \varepsilon < 1$ ,

$$\sup_{\mathbb{Q}} N(\mathcal{G} \times \Delta \mathcal{R}, \|\cdot\|_{\mathbb{Q}, 2}, \varepsilon) \|\mathbf{M}_{\mathcal{G}}(M_{\mathcal{R}, \mathcal{Y}} - M_{\tilde{\mathcal{R}}, \mathcal{Y}})\|_{\mathbb{Q}, 2} \leq \mathbf{c}_{\mathcal{G}} \mathbf{c}_{\mathcal{R}, \mathcal{Y}} (\varepsilon^2/4)^{-\mathbf{d}_{\mathcal{G}} - \mathbf{d}_{\mathcal{R}, \mathcal{Y}}},$$

where  $\sup$  is over all finite discrete measure on  $\mathbb{R}^{d+1}$ , and  $M_{\tilde{\mathcal{R}}, \mathcal{Y}}(y) = M_{\mathcal{R}, \mathcal{Y}}(y) \mathbb{1}(|y| \leq \tau_n)$ . We can check that  $\mathbf{M}_{\mathcal{G}}(M_{\mathcal{R}, \mathcal{Y}} - M_{\tilde{\mathcal{R}}, \mathcal{Y}})$  is an envelope function for  $\mathcal{G} \times \Delta \mathcal{R}$ , since all functions in  $\Delta \mathcal{R}$  are evaluated to zero

on  $[-\tau_n, \tau_n]$ . Denote  $\mathbf{X} = (\mathbf{x}_i)_{1 \leq i \leq n}$ ,

$$\begin{aligned} \mathbb{E} \left[ \max_{1 \leq i \leq n} \mathbf{M}_{\mathcal{G}}^2 (M_{\mathcal{R}, \mathcal{Y}}(y_i) - M_{\tilde{\mathcal{R}}, \mathcal{Y}}(y_i))^2 \middle| \mathbf{X} \right]^{\frac{1}{2}} &\lesssim \mathbf{M}_{\mathcal{G}} \mathbb{E} \left[ \left( \max_{1 \leq i \leq n} M_{\mathcal{R}, \mathcal{Y}}(y_i) \right)^2 \middle| \mathbf{X} \right]^{\frac{1}{2}} \lesssim \mathbf{M}_{\mathcal{G}} n^{\frac{1}{2+v}}, \\ \sup_{(g, r) \in \mathcal{G} \times \mathcal{R}} \mathbb{E} [g(\mathbf{x}_i)^2 r(y_i)^2 \mathbb{1}(|y_i| \geq \tau_n^{1/\alpha})]^{\frac{1}{2}} &\lesssim \sup_{(g, r) \in \mathcal{G} \times \mathcal{R}} \mathbb{E} \left[ g(\mathbf{x}_i)^2 \mathbb{E}[r(y_i)^{2+v} | \mathbf{x}_i]^{\frac{2}{2+v}} \mathbb{P}(|y_i| \geq \tau_n | \mathbf{x}_i)^{\frac{v}{2+v}} \right] \\ &\lesssim \sqrt{\mathbf{M}_{\mathcal{G}} \mathbf{E}_{\mathcal{G}} \tau_n}. \end{aligned}$$

By Jensen's inequality, we also have

$$\begin{aligned} \mathbb{E} \left[ \max_{1 \leq i \leq n} \mathbf{M}_{\mathcal{G}}^2 (\mathbb{E}[M_{\mathcal{R}, \mathcal{Y}}(y_i) - M_{\tilde{\mathcal{R}}, \mathcal{Y}}(y_i) | \mathbf{x}_i])^2 \middle| \mathbf{X} \right]^{\frac{1}{2}} &\lesssim \mathbf{M}_{\mathcal{G}} n^{\frac{1}{2+v}}, \\ \sup_{(g, r) \in \mathcal{G} \times \mathcal{R}} \mathbb{E} [g(\mathbf{x}_i)^2 \mathbb{E}[r(y_i) - \tilde{r}(y_i) | \mathbf{x}_i]^2]^{\frac{1}{2}} &\lesssim \sqrt{\mathbf{M}_{\mathcal{G}} \mathbf{E}_{\mathcal{G}} \tau_n^{-v}}, \\ \mathbb{E} [\mathbf{M}_{\mathcal{G}}^2 (M_{\mathcal{R}, \mathcal{Y}}(y_i) - M_{\tilde{\mathcal{R}}, \mathcal{Y}}(y_i))^2]^{1/2} &\lesssim \mathbf{M}_{\mathcal{G}} \tau_n^{-v/2}. \end{aligned}$$

Denote  $A = (\mathbf{c}_{\mathcal{G}} \mathbf{c}_{\mathcal{R}})^{\frac{1}{2\mathbf{d}_{\mathcal{G}} + 2\mathbf{d}_{\mathcal{R}}}} / 4$  and  $D = 2\mathbf{d}_{\mathcal{G}} + 2\mathbf{d}_{\mathcal{R}}$ , [Chernozhukov et al. \(2014b\)](#), Corollary 5.1) gives

$$\begin{aligned} \mathbb{E} [\|R_n(g, r) - R_n(g\tilde{r})\|_{\mathcal{G} \times \mathcal{R}}] &\lesssim \mathbb{E} \left[ \sup_{g \in \mathcal{G}} \sup_{h \in \Delta \mathcal{R}} \frac{1}{\sqrt{n}} \sum_{i=1}^n g(\mathbf{x}_i) (h(y_i) - \mathbb{E}[h(y_i) | \mathbf{x}_i]) \right] \\ &\lesssim \sqrt{D \mathbf{M}_{\mathcal{G}} \mathbf{E}_{\mathcal{G}} \tau_n^{-v} \log(A \sqrt{\mathbf{M}_{\mathcal{G}} / \mathbf{E}_{\mathcal{G}}})} + \frac{D \mathbf{M}_{\mathcal{G}} n^{\frac{1}{2+v}}}{\sqrt{n}} \log(A \sqrt{\mathbf{M}_{\mathcal{G}} / \mathbf{E}_{\mathcal{G}}}) \\ &\lesssim \sqrt{D \log(A \sqrt{\mathbf{M}_{\mathcal{G}} / \mathbf{E}_{\mathcal{G}}}) \sqrt{\mathbf{M}_{\mathcal{G}} \mathbf{E}_{\mathcal{G}} \tau_n^{-v/2}}} + \frac{D \log(A \sqrt{\mathbf{M}_{\mathcal{G}} / \mathbf{E}_{\mathcal{G}}}) \mathbf{M}_{\mathcal{G}}}{\sqrt{n^{\frac{v}{2+v}}}}. \end{aligned}$$

**Part 3: Truncation error for the Gaussian process** —  $\|Z_n^R(g, r) - Z_n^R(g, \tilde{r})\|_{\mathcal{G} \times \mathcal{R}}$  Our assumptions imply  $\mathcal{G} \times \tilde{\mathcal{R}} \cup \mathcal{G} \times \mathcal{R}$  is VC-type w.r.p envelope function  $2\mathbf{M}_{\mathcal{G}} \mathbf{M}_{\mathcal{R}, \mathcal{Y}}$  in the sense that for all  $0 < \varepsilon < 1$ ,

$$\sup_{\mathbb{Q}} N(\mathcal{G} \times \mathcal{R} \cup \mathcal{G} \times \tilde{\mathcal{R}}, \|\cdot\|_{\mathbb{Q}, 2}, 2\varepsilon \|\mathbf{M}_{\mathcal{G}} \mathbf{M}_{\mathcal{R}, \mathcal{Y}}\|_{\mathbb{Q}, 2}) \leq \mathbf{c}_{\mathcal{G}} \mathbf{c}_{\mathcal{R}} (\varepsilon^2/4)^{-\mathbf{d}_{\mathcal{G}} - \mathbf{d}_{\mathcal{R}}},$$

where  $\sup$  is over all finite discrete measure on  $\mathbb{R}^{d+1}$ . Hence  $\mathcal{G} \times \tilde{\mathcal{R}} \cup \mathcal{G} \times \mathcal{R}$  is pre-Gaussian, and on some probability space, there exists a mean-zero Gaussian process  $\tilde{Z}_n^R$  indexed by  $\mathcal{F} = \mathcal{G} \times \tilde{\mathcal{R}} \cup \mathcal{G} \times \mathcal{R}$  with the same covariance structure as  $R_n$ , and has almost sure continuous path w.r.p the metric  $\rho$ , given by

$$\rho((g_1, r_1), (g_2, r_2)) = \mathbb{E}[(Z_n^R(g_1, r_1) - Z_n^R(g_2, r_2))^2]^{\frac{1}{2}} = \mathbb{E}[(R_n(g_1, r_1) - R_n(g_2, r_2))^2]^{\frac{1}{2}}, (g_1, r_1), (g_2, r_2) \in \mathcal{F}.$$

Recall the definition of  $\mathcal{G} \times \Delta \mathcal{R}$  in Part 2. Then, we have shown previously that

$$\sigma \equiv \sup_{f \in \mathcal{G} \times \Delta \mathcal{R}} \rho(f, f) \leq \sqrt{\mathbf{M}_{\mathcal{G}} \mathbf{E}_{\mathcal{G}} \tau_n^{-v}},$$

Our assumptions imply for all  $0 < \varepsilon < 1$ ,

$$N(\mathcal{G} \times \mathcal{R} \cup \mathcal{G} \times \tilde{\mathcal{R}}, \rho, \rho(2\varepsilon \mathbf{M}_{\mathcal{G}} M_{\mathcal{R}, \mathcal{Y}}, 2\varepsilon \|\mathbf{M}_{\mathcal{G}} M_{\mathcal{R}, \mathcal{Y}}\|^{1/2})) \leq \mathbf{c}_{\mathcal{G}} \mathbf{c}_{\mathcal{R}} (\varepsilon^2/4)^{-\mathbf{d}_{\mathcal{G}} - \mathbf{d}_{\mathcal{R}}}$$

Denote  $A = (\mathbf{c}_{\mathcal{G}}\mathbf{c}_{\mathcal{R}})^{\frac{1}{2\mathbf{d}_{\mathcal{G}}+2\mathbf{d}_{\mathcal{R}}}}/4$  and  $D = 2\mathbf{d}_{\mathcal{G}} + 2\mathbf{d}_{\mathcal{R}}$ . Then, by [van der Vaart and Wellner \(1996, Corollary 2.2.8\)](#), choose any  $(g_0, r_0) \in \mathcal{G} \times \mathcal{R}$ , we have

$$\begin{aligned} \mathbb{E} \left[ \|\bar{Z}_n^R(g, r) - \bar{Z}_n^R(g, \tilde{r})\|_{\mathcal{G} \times \mathcal{R}} \right] &\lesssim \mathbb{E} [|\bar{Z}_n^R(g_0, r_0) - \bar{Z}_n^R(g_0, \tilde{r}_0)|] + \int_0^\sigma \sqrt{\log \left( \mathbf{c}_{\mathcal{G}}\mathbf{c}_{\mathcal{R}} \left( \frac{\mathbf{M}_{\mathcal{G}}}{\varepsilon} \right)^{\mathbf{d}_{\mathcal{G}}+\mathbf{d}_{\mathcal{R}}} \right)} d\varepsilon \\ &\leq \sqrt{D \log(A \sqrt{\mathbf{M}_{\mathcal{G}}/\mathbf{E}_{\mathcal{G}}}) \sqrt{\mathbf{M}_{\mathcal{G}}\mathbf{E}_{\mathcal{G}}} \tau_n^{-v/2}} \\ &\lesssim \sqrt{(\mathbf{d}_{\mathcal{G}} + \mathbf{d}_{\mathcal{R}, \mathcal{Y}}) \log(\mathbf{c}_{\mathcal{G}}\mathbf{c}_{\mathcal{R}, \mathcal{Y}} \mathbf{k} n) \sqrt{\mathbf{M}_{\mathcal{G}}\mathbf{E}_{\mathcal{G}}} \tau_n^{-v/2}}. \end{aligned}$$

Since  $(\bar{Z}_n^R(g, r) : g \in \mathcal{G}, r \in \mathcal{R})$  has the same distribution as  $(Z_n^R(g, r) : g \in \mathcal{G}, r \in \mathcal{R})$ , we know from Vorob'ev–Berkes–Philipp theorem ([Dudley, 2014, Theorem 1.31](#)) that  $\bar{Z}_n^R$  can be constructed on the same probability space as  $(\mathbf{x}_i, y_i)_{1 \leq i \leq n}$  and  $Z_n^R$ , such that  $\bar{Z}_n^R$  and  $Z_n^R$  coincide on  $\mathcal{G} \times \mathcal{R}$ . By an abuse of notation, call  $\bar{Z}_n^R$  now  $Z_n^R$ , the outputted Gaussian process.

**Part 4: Putting Together** It follows from the definition of  $\tilde{\mathcal{R}}$  and the previous three parts that if we choose  $\tau_n$  such that

$$\mathbf{r}_n \tau_n \asymp \sqrt{\mathbf{M}_{\mathcal{G}}\mathbf{E}_{\mathcal{G}}} \tau_n^{-v/2},$$

then the approximation error can be bounded by

$$\begin{aligned} \mathbb{E}[\|R_n - Z_n^R\|_{\mathcal{G} \times \mathcal{R}}] &\lesssim (\mathbf{d} \log(\mathbf{c}n))^{3/2} \mathbf{r}_n^{\frac{v}{v+2}} (\sqrt{\mathbf{M}_{\mathcal{G}}\mathbf{E}_{\mathcal{G}}})^{\frac{2}{v+2}} + \mathbf{d} \log(\mathbf{c}n) \mathbf{M}_{\mathcal{G}} n^{-\frac{v/2}{2+v}} \\ &\quad + \mathbf{d} \log(\mathbf{c}n) \mathbf{M}_{\mathcal{G}} n^{-1/2} \left( \frac{\sqrt{\mathbf{M}_{\mathcal{G}}\mathbf{E}_{\mathcal{G}}}}{\mathbf{r}_n} \right)^{\frac{2}{v+2}}, \end{aligned}$$

where  $\mathbf{d} = \mathbf{d}_{\mathcal{G}} + \mathbf{d}_{\mathcal{R}, \mathcal{Y}} + \mathbf{k}$ , and  $\mathbf{c} = \mathbf{c}_{\mathcal{G}}\mathbf{c}_{\mathcal{R}, \mathcal{Y}}\mathbf{k}$ . ■

## SA-8.2 Proof of Lemma [SA-4.2](#)

Since  $A_n$  is the addition of two  $M_n$  processes, indexed by  $\mathcal{G} \times \mathcal{R}$  and  $\mathcal{H} \times \mathcal{S}$  respectively, the Gaussian strong approximation error essentially depends on the *worst case scenario* between  $\mathcal{G}$  and  $\mathcal{H}$ , and between  $\mathcal{R}$  and  $\mathcal{S}$ . Hence (1) taking maximums  $\mathbf{E} = \max\{\mathbf{E}_{\mathcal{G}}, \mathbf{E}_{\mathcal{H}}\}$ ,  $\mathbf{M} = \max\{\mathbf{M}_{\mathcal{G}}, \mathbf{M}_{\mathcal{H}}\}$  and  $\mathbf{TV} = \max\{\mathbf{TV}_{\mathcal{G}}, \mathbf{TV}_{\mathcal{H}}\}$ ; (2) noticing that  $A_n$  is still indexed by a VC-type class of functions, we can get the claimed result.

For a more rigor proof, we can not apply [Cattaneo and Yu \(2025, Theorem SA.1\)](#) on  $(M_n(g, r) : g \in \mathcal{G}, r \in \mathcal{R})$  and  $(M_n(h, s) : h \in \mathcal{H}, s \in \mathcal{S})$  directly, since this ignores the dependence structure between the two empirical processes. However, we can still project the functions onto a Haar basis, and control the *strong approximation error for projected process* and the *projection error* as in the proof of [Cattaneo and Yu \(2025, Theorem SA.1\)](#) and show both errors can be controlled via *worst case scenario* between  $\mathcal{G}$  and  $\mathcal{H}$ , and between  $\mathcal{R}$  and  $\mathcal{S}$ .

**Reductions:** Here we present some reductions to our problem. By the same argument as in Section SA-II.3 (Proofs of Theorem 1) in the supplemental appendix of [Cattaneo and Yu \(2025\)](#), we can show there exists  $\mathbf{u}_i, 1 \leq i \leq n$  i.i.d Uniform $([0, 1]^d)$  on a possibly enlarged probability space, such that

$$f(\mathbf{x}_i) = f(\phi_{\mathcal{G} \cup \mathcal{H}}^{-1}(\mathbf{u}_i)), \quad \forall f \in \mathcal{G} \cup \mathcal{H}, \forall 1 \leq i \leq n.$$

With the help of [Cattaneo and Yu \(2025, Lemma SA.10\)](#), we can assume w.l.o.g. that  $\mathbf{x}_i$ 's are i.i.d Uniform( $\mathcal{X}$ ) with  $\mathcal{X} = [0, 1]^d$ , and  $\phi_{\mathcal{G} \cup \mathcal{H}} : [0, 1]^d \rightarrow [0, 1]^d$  is the identity function. Although we assume  $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[|Y_i|^{2+v} | \mathbf{X}_i = \mathbf{x}] < \infty$ , we first present the result under the assumption  $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[\exp(|y_i|) | \mathbf{x}_i = \mathbf{x}] \leq 2$ , which is the same as in [Cattaneo and Yu \(2025, Theorem 2\)](#). Also in correspondence to the notations in [Cattaneo and Yu \(2025, Theorem 2\)](#), we set  $\alpha = 1$  throughout this proof.

**Cell Constructions and Projections:** The constructions here are the same as those in [Cattaneo and Yu \(2025\)](#), and we present them here for completeness. Let  $\mathcal{A}_{M,N}(\mathbb{P}, 1) = \{\mathcal{C}_{j,k} : 0 \leq k < 2^{M+N-j}, 0 \leq j \leq M+N\}$  be an axis-aligned cylindered quasi-dyadic expansion of  $\mathbb{R}^{d+1}$ , with depth  $M$  for the main subspace  $\mathbb{R}^d$  and depth  $N$  for the multiplier subspace  $\mathbb{R}$ , with respect to  $\mathbb{P}$ , the joint distribution of  $(\mathbf{x}_i, y_i)$  taking values in  $\mathbb{R}^d \times \mathbb{R}$ , as in [Cattaneo and Yu \(2025, Definition SA.4\)](#). To see what  $\mathcal{A}_{M,N}(\mathbb{P}, 1)$  is, it can be given by the following iterative partition procedure:

1. *Initialization* ( $q = 0$ ): Take  $\mathcal{C}_{M+N-q,0} = \mathcal{X} \times \mathbb{R}$  where  $\mathcal{X} = [0, 1]^d$ .
2. *Iteration* ( $q = 1, \dots, M$ ): Given  $\mathcal{C}_{K-l,k}$  for  $0 \leq l \leq q-1, 0 \leq k < 2^l$ , take  $s = (q \bmod d) + 1$ , and construct  $\mathcal{C}_{K-q,2k} = \mathcal{C}_{K-q+1,k} \cap \{(\mathbf{x}, y) \in [0, 1]^d \times \mathbb{R} : \mathbf{e}_s^\top \mathbf{x} \leq c_{K-q+1,k}\}$  and  $\mathcal{C}_{K-q,2k+1} = \mathcal{C}_{K-q+1,k} \cap \{(\mathbf{x}, y) \in [0, 1]^d \times \mathbb{R} : \mathbf{e}_s^\top \mathbf{x} > c_{K-q+1,k}\}$  such that  $\mathbb{P}(\mathcal{C}_{K-q,2k})/\mathbb{P}(\mathcal{C}_{K-q+1,k}) \in [\frac{1}{1+\rho}, \frac{\rho}{1+\rho}]$  for all  $0 \leq k < 2^{q-1}$ . Continue until  $(\mathcal{C}_{N,k} : 0 \leq k < 2^M)$  has been constructed. By construction, for each  $0 \leq l < M$ ,  $\mathcal{C}_{N,l} = \mathcal{X}_{0,l} \times \mathcal{Y}_{0,N,0}$ , with  $\mathcal{Y}_{0,N,0} = \mathbb{R}$ .
3. *Iteration* ( $q = M+1, \dots, M+N$ ): Given  $\mathcal{C}_{K-l,k}$  for  $0 \leq l \leq q-1, 0 \leq k < 2^l$ , each  $\mathcal{C}_{M+N-q,k}$  can be written as  $\mathcal{X}_{0,l} \times \mathcal{Y}_{l,M+N-q,m}$  with  $k = 2^{q-M}l + m$ . Construct  $\mathcal{C}_{M+N-q-1,2k} = \mathcal{X}_{0,l} \times \mathcal{Y}_{l,M+N-q-1,2m}$  and  $\mathcal{C}_{M+N-q-1,2k+1} = \mathcal{X}_{0,l} \times \mathcal{Y}_{l,M+N-q-1,2m+1}$ , such that there exists some  $\mathbf{c}_{M+N-q,k} \in \mathbb{R}$  with  $\mathcal{Y}_{l,M+N-q-1,2m} = \mathcal{Y}_{l,M+N-q,m} \cap (-\infty, \mathbf{c}_{M+N-q,k})$  and  $\mathcal{Y}_{l,M+N-q-1,2m+1} = \mathcal{Y}_{l,M+N-q,m} \cap (\mathbf{c}_{M+N-q,k}, \infty)$ ,  $\mathbb{P}(y_i \in \mathcal{Y}_{l,M+N-q-1,2m} | \mathbf{x}_i \in \mathcal{X}_{0,l}) = \mathbb{P}(y_i \in \mathcal{Y}_{l,M+N-q-1,2m+1} | \mathbf{x}_i \in \mathcal{X}_{0,l}) = \frac{1}{2} \mathbb{P}(y_i \in \mathcal{Y}_{l,M+N-q-1,m} | \mathbf{x}_i \in \mathcal{X}_{0,l})$ .

Consider the projection  $\Pi_1(\mathcal{A}_{M,N}(\mathbb{P}, 1))$  given in Equation (SA-7) in [Cattaneo and Yu \(2025\)](#), noticing that  $\mathcal{A}_{M,N}(\mathbb{P}, 1)$  is one special instance of  $\mathcal{C}_{M,N}(\mathbb{P}, \rho)$ . That is, define  $e_{j,k} = \mathbb{1}_{\mathcal{C}_{j,k}}$  and  $\tilde{e}_{j,k} = e_{j-1,2k} - e_{j-1,2k+1}$ ,

$$\Pi_1(\mathcal{C}_{M,N}(\mathbb{P}, \rho))[g, r] = \gamma_{M+N,0}(g, r)e_{M+N,0} + \sum_{1 \leq j \leq M+N} \sum_{0 \leq k < 2^{M+N-j}} \tilde{\gamma}_{j,k}(g, r)\tilde{e}_{j,k}, \quad (\text{SA-8.1})$$

where  $e_{j,k} = \mathbb{1}(\mathcal{C}_{j,k})$  and  $\tilde{e}_{j,k} = \mathbb{1}(\mathcal{C}_{j-1,2k}) - \mathbb{1}(\mathcal{C}_{j-1,2k+1})$ , and

$$\gamma_{j,k}(g, r) = \begin{cases} \mathbb{E}[g(X)r(Y) | X \in \mathcal{X}_{j-N,k}], & \text{if } N \leq j \leq M+N, \\ \mathbb{E}[g(X) | X \in \mathcal{X}_{0,l}] \cdot \mathbb{E}[r(Y) | X \in \mathcal{X}_{0,l}, Y \in \mathcal{Y}_{l,0,m}], & \text{if } j < N, k = 2^{N-j}l + m, \end{cases}$$

and  $\tilde{\gamma}_{j,k}(g, r) = \gamma_{j-1,2k}(g, r) - \gamma_{j-1,2k+1}(g, r)$ . We will use  $\Pi_1$  as a shorthand for  $\Pi_1(\mathcal{C}_{M,N}(\mathbb{P}, \rho))$ .

For simplicity, we denote  $\Pi_1(\mathcal{A}_{M,N}(\mathbb{P}, 1))$  by  $\Pi_1$  instead. Now define the projected empirical process

$$\Pi_1 A_n(g, h, r, s) = \Pi_1 M_n(g, r) + \Pi_1 M_n(h, s), \quad g \in \mathcal{G}, h \in \mathcal{H}, r \in \mathcal{R}, s \in \mathcal{S},$$

where  $\Pi_1 M_n(g, r)$  and  $\Pi_1 M_n(h, s)$  are given in Equation (SA-10) in Cattaneo and Yu (2025), that is,

$$\begin{aligned}\Pi_1 M_n(g, r) &= \frac{1}{\sqrt{n}} \sum_{i=1}^n (\Pi_1[g, r](\mathbf{x}_i, y_i) - \mathbb{E}[\Pi_1[g, r](\mathbf{x}_i, y_i)]), \\ \Pi_1 M_n(h, s) &= \frac{1}{\sqrt{n}} \sum_{i=1}^n (\Pi_1[h, s](\mathbf{x}_i, y_i) - \mathbb{E}[\Pi_1[h, s](\mathbf{x}_i, y_i)]).\end{aligned}$$

**Construction of Gaussian Process** Suppose  $(\tilde{\xi}_{j,k} : 0 \leq k < 2^{M+N-j}, 1 \leq j \leq M+N)$  are i.i.d. standard Gaussian random variables. Take  $F_{(j,k),m}$  to be the cumulative distribution function of  $(S_{j,k} - mp_{j,k})/\sqrt{mp_{j,k}(1-p_{j,k})}$ , where  $p_{j,k} = \mathbb{P}(\mathcal{C}_{j-1,2k})/\mathbb{P}(\mathcal{C}_{j,k})$  and  $S_{j,k}$  is a  $\text{Bin}(m, p_{j,k})$  random variable, and  $G_{(j,k),m}(t) = \sup\{x : F_{(j,k),m}(x) \leq t\}$ . We define  $U_{j,k}, \tilde{U}_{j,k}$ 's via the following iterative scheme:

1. *Initialization:* Take  $U_{M+N,0} = n$ .
2. *Iteration:* Suppose we've defined  $U_{l,k}$  for  $j < l \leq M+N, 0 \leq k < 2^{M+N-l}$ , then solve for  $U_{j,k}$ 's s.t.

$$\begin{aligned}\tilde{U}_{j,k} &= \sqrt{U_{j,k} p_{j,k} (1 - p_{j,k})} G_{(j,k), U_{j,k}} \circ \Phi(\tilde{\xi}_{j,k}), \\ \tilde{U}_{j,k} &= (1 - p_{j,k}) U_{j-1,2k} - p_{j,k} U_{j-1,2k+1} = U_{j-1,2k} - p_{j,k} U_{j,k}, \\ U_{j-1,2k} + U_{j-1,2k+1} &= U_{j,k}, \quad 0 \leq k < 2^{M+N-j}.\end{aligned}$$

Continue till we have defined  $U_{0,k}$  for  $0 \leq k < 2^{M+N}$ .

Then,  $\{U_{j,k} : 0 \leq j \leq K, 0 \leq k < 2^{M+N-j}\}$  have the same joint distribution as  $\{\sum_{i=1}^n e_{j,k}(\mathbf{x}_i, y_i) : 0 \leq j \leq K, 0 \leq k < 2^{M+N-j}\}$ . By Vorob'ev–Berkes–Philipp theorem (Dudley, 2014, Theorem 1.31),  $\{\tilde{\xi}_{j,k} : 0 \leq k < 2^{M+N-j}, 1 \leq j \leq M+N\}$  can be constructed on a possibly enlarged probability space such that the previously constructed  $U_{j,k}$  satisfies  $U_{j,k} = \sum_{i=1}^n e_{j,k}(\mathbf{x}_i)$  almost surely for all  $0 \leq j \leq M+N, 0 \leq k < 2^{M+N-j}$ . We will show  $\tilde{\xi}_{j,k}$ 's can be given as a Brownian bridge indexed by  $\tilde{e}_{j,k}$ 's.

Since all of  $\mathcal{G}, \mathcal{H}, \mathcal{R}$  and  $\mathcal{S}$  are VC-type, we can show  $\mathcal{G} \times \mathcal{H} + \mathcal{R} \times \mathcal{S}$  is also VC-type, here  $+$  is the Minkowski sum. Hence  $\mathcal{F} = \mathcal{G} \times \mathcal{H} + \mathcal{R} \times \mathcal{S} \cup \Pi_1[G \times \mathcal{H} + \mathcal{R} \times \mathcal{S}]$  is pre-Gaussian.

Then, by Skorohod Embedding lemma (Dudley, 2014, Lemma 3.35), on a possibly enlarged probability space, we can construct a Brownian bridge  $(Z_n(f) : f \in \mathcal{F})$  that satisfies

$$\tilde{\xi}_{j,k} = \frac{\mathbb{P}(\mathcal{C}_{j,k})}{\sqrt{\mathbb{P}(\mathcal{C}_{j-1,2k})\mathbb{P}(\mathcal{C}_{j-1,2k+1})}} Z_n(\tilde{e}_{j,k}),$$

for  $0 \leq k < 2^{M+N-j}, 1 \leq j \leq M+N$ . Moreover, call

$$V_{j,k} = \sqrt{n} Z_n(e_{j,k}), \quad \tilde{V}_{j,k} = \sqrt{n} Z_n(\tilde{e}_{j,k}), \quad \tilde{\xi}_{j,k} = \frac{\mathbb{P}(\mathcal{C}_{j,k})}{\sqrt{n\mathbb{P}(\mathcal{C}_{j-1,2k})\mathbb{P}(\mathcal{C}_{j-1,2k+1})}} \tilde{V}_{j,k}.$$

for  $0 \leq k < 2^{K-j}, 1 \leq j \leq K$ . We have for  $g \in \mathcal{G}, h \in \mathcal{H}, r \in \mathcal{R}, s \in \mathcal{S}$ ,

$$\begin{aligned}\sqrt{n}\Pi_1 A_n(g, h, r, s) &= \sum_{j=1}^{M+N} \sum_{0 \leq k < 2^{M+N-j}} (\tilde{\gamma}_{j,k}[g, r] + \tilde{\gamma}_{j,k}[h, s]) \tilde{U}_{j,k}, \\ \sqrt{n}\Pi_1 Z_n(g, h, r, s) &= \sum_{j=1}^{M+N} \sum_{0 \leq k < 2^{M+N-j}} (\tilde{\gamma}_{j,k}[g, r] + \tilde{\gamma}_{j,k}[h, s]) \tilde{V}_{j,k}.\end{aligned}$$

**Decomposition** Fix one  $(g, h, r, s) \in \mathcal{G} \times \mathcal{H} \times \mathcal{R} \times \mathcal{S}$ , we decompose by

$$\begin{aligned} & A_n(g, h, r, s) - Z_n(g, h, r, s) \\ &= \underbrace{\Pi_1 A_n(g, h, r, s) - \Pi_1 Z_n(g, h, r, s)}_{\text{strong approximation (SA) error for projected}} + \underbrace{A_n(g, h, r, s) - \Pi_1 A_n(g, h, r, s) + \Pi_1 Z_n(g, h, r, s) - Z_n(g, h, r, s)}_{\text{projection error}}. \end{aligned}$$

**SA error for Projected Process** The strong approximation error essentially depends on the Hilbertian pseudo norm

$$\sum_{j=1}^{M+N} \sum_{0 \leq k < 2^{M+N-j}} (\tilde{\gamma}_{j,k}[g, r] + \tilde{\gamma}_{j,k}[h, s])^2 \leq 2 \sum_{j=1}^{M+N} \sum_{0 \leq k < 2^{M+N-j}} (\tilde{\gamma}_{j,k}[g, r])^2 + 2 \sum_{j=1}^{M+N} \sum_{0 \leq k < 2^{M+N-j}} (\tilde{\gamma}_{j,k}[h, s])^2.$$

Hence, [Cattaneo and Yu \(2025, Lemma SA.19\)](#) gives with probability at least  $1 - 2e^{-t}$ ,

$$|\Pi_1 A_n(g, h, r, s) - \Pi_1 Z_n(g, h, r, s)| \leq C_1 C_\alpha \sqrt{\frac{N^{2\alpha+1} 2^M \mathbf{EM}}{n}} t + C_1 C_\alpha \sqrt{\frac{(\|\Pi_1[g, r]\|_\infty + \|\Pi_1[h, s]\|_\infty)^2 (M+N)}{n}} t,$$

where  $C_1 > 0$  is a universal constant and  $C_\alpha = 1 + (2\alpha)^{\alpha/2}$ .

**Projection Error** For the projection error, we use the simple observation that

$$|A_n(g, h, r, s) - \Pi_1 A_n(g, h, r, s)| \leq |M_n(g, r) - \Pi_1 M_n(g, r)| + |M_n(h, s) - \Pi_1 M_n(h, s)|,$$

and [Cattaneo and Yu \(2025, Lemma SA.23\)](#) to get for all  $t > N$ ,

$$\begin{aligned} \mathbb{P}\left[|A_n(g, h, r, s) - \Pi_1 A_n(g, h, r, s)| > C_2 \sqrt{C_{2\alpha}} \sqrt{N^2 \mathbf{V} + 2^{-N} \mathbf{M}^2} t^{\alpha+\frac{1}{2}} + C_2 C_\alpha \frac{\mathbf{M}}{\sqrt{n}} t^{\alpha+1}\right] &\leq 4ne^{-t}, \\ \mathbb{P}\left[|Z_n(g, h, r, s) - \Pi_1 Z_n(g, h, r, s)| > C_2 \sqrt{C_{2\alpha}} \sqrt{N^2 \mathbf{V} + C_2 C_\alpha 2^{-N} \mathbf{M}^2} t^{\frac{1}{2}} + C_2 C_\alpha \frac{\mathbf{M}}{\sqrt{n}} t\right] &\leq 4ne^{-t}, \end{aligned}$$

where  $C_\alpha = 1 + (2\alpha)^{\frac{\alpha}{2}}$  and  $C_{2\alpha} = 1 + (4\alpha)^\alpha$  and  $C_2$  is a constant that only depends on the distribution of  $(\mathbf{x}_1, y_1)$ , with

$$\mathbf{V} = \min\{2\mathbf{M}, \sqrt{d} \mathbf{L} 2^{-M/d}\} 2^{-M/d} \mathbf{TV}_{\mathcal{H}}.$$

**Uniform SA Error:** Since all of  $\mathcal{G}$ ,  $\mathcal{H}$ ,  $\mathcal{R}$  and  $\mathcal{S}$  are VC-type class, from a union bound argument and the same control over fluctuation error as in [Cattaneo and Yu \(2025, Lemma SA.18\)](#), denoting  $\mathcal{F} = \mathcal{G} \times \mathcal{H} \times \mathcal{R} \times \mathcal{S}$ , we get for all  $t > 0$  and  $0 < \delta < 1$ ,

$$\mathbb{P}\left[\|A_n - A_n \circ \pi_{\mathcal{F}_\delta}\|_{\mathcal{F}} + \|Z_n - Z_n \circ \pi_{\mathcal{F}_\delta}\|_{\mathcal{F}} > C_1 C_\alpha \mathbf{F}_n(t, \delta)\right] \leq \exp(-t),$$

where  $C_\alpha = 1 + (2\alpha)^{\frac{\alpha}{2}}$  and

$$\mathbf{F}_n(t, \delta) = J(\delta) \mathbf{M} + \frac{(\log n)^{\alpha/2} \mathbf{M} J^2(\delta)}{\delta^2 \sqrt{n}} + \frac{\mathbf{M}}{\sqrt{n}} t + (\log n)^\alpha \frac{\mathbf{M}}{\sqrt{n}} t^\alpha.$$

where

$$\begin{aligned} J(\delta) &= 3\delta \left( \sqrt{\mathbf{d}_{\mathcal{G}} \log\left(\frac{2\mathbf{c}_{\mathcal{G}}}{\delta}\right)} + \sqrt{\mathbf{d}_{\mathcal{H}} \log\left(\frac{2\mathbf{c}_{\mathcal{H}}}{\delta}\right)} + \sqrt{\mathbf{d}_{\mathcal{R}} \log\left(\frac{2\mathbf{c}_{\mathcal{R}}}{\delta}\right)} + \sqrt{\mathbf{d}_{\mathcal{S}} \log\left(\frac{2\mathbf{c}_{\mathcal{S}}}{\delta}\right)} \right) \\ &\lesssim \sqrt{\mathbf{d} \log(\mathbf{c}/\delta)}, \end{aligned}$$

recalling  $\mathbf{c} = \mathbf{c}_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} + \mathbf{c}_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} + \mathbf{c}_{\mathcal{R}, \mathcal{Y}} + \mathbf{c}_{\mathcal{S}, \mathcal{Y}} + \mathbf{k}$ ,  $\mathbf{d} = \mathbf{d}_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} \mathbf{d}_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}} \mathbf{d}_{\mathcal{R}, \mathcal{Y}} \mathbf{d}_{\mathcal{S}, \mathcal{Y}} \mathbf{k}$ . Choosing the optimal  $M^*, N^*$  gives  $\mathbb{P}[\|A_n - Z_n^A\|_{\mathcal{F}} > C_1 \mathbf{v} \mathbf{T}_n(t)] \leq C_2 e^{-t}$  for all  $t > 0$ , where

$$\mathbf{T}_n(t) = \min_{\delta \in (0,1)} \{A_n(t, \delta) + F_n(t, \delta)\},$$

with

$$\begin{aligned} A_n(t, \delta) &= \sqrt{d} \min \left\{ \left( \frac{\mathbf{c}_1^d \mathbf{E} \mathbf{T} \mathbf{V}^d \mathbf{M}^{d+1}}{n} \right)^{\frac{1}{2(d+1)}}, \left( \frac{\mathbf{c}_1^d \mathbf{c}_2^d \mathbf{E}^2 \mathbf{M}^2 \mathbf{T} \mathbf{V}^d \mathbf{L}^d}{n^2} \right)^{\frac{1}{2(d+2)}} \right\} (t + \log(n\mathbf{N}(\delta)N^*))^{\alpha+1} \\ &\quad + \sqrt{\frac{\mathbf{M}^2(M^* + N^*)}{n}} (\log n)^{\alpha} (t + \log(n\mathbf{N}(\delta)N^*))^{\alpha+1}, \\ F_n(t, \delta) &= J(\delta) \mathbf{M} + \frac{(\log n)^{\alpha/2} \mathbf{M} J^2(\delta)}{\delta^2 \sqrt{n}} + \frac{\mathbf{M}}{\sqrt{n}} \sqrt{t} + (\log n)^{\alpha} \frac{\mathbf{M}}{\sqrt{n}} t^{\alpha}, \end{aligned}$$

where

$$\begin{aligned} \mathcal{V}_{\mathcal{R}} &= \{\theta(\cdot, r) : r \in \mathcal{R}\}, \\ \mathbf{N}(\delta) &= \mathbf{N}_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}(\delta/2, \mathbf{M}_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}) \mathbf{N}_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}(\delta/2, \mathbf{M}_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}) \mathbf{N}_{\mathcal{R}, \mathcal{Y}}(\delta/2, M_{\mathcal{R}}) \mathbf{N}_{\mathcal{S}, \mathcal{Y}}(\delta/2, M_{\mathcal{S}, \mathcal{Y}}), \\ J(\delta) &= 2J_{\mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}(\mathcal{G}, \mathbf{M}_{\mathcal{G}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}, \delta/2) + 2J_{\mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}(\mathcal{H}, \mathbf{M}_{\mathcal{H}, \mathcal{Q}_{\mathcal{G} \cup \mathcal{H}}}, \delta/2) + 2J_{\mathcal{Y}}(\mathcal{R}, M_{\mathcal{R}, \mathcal{Y}}, \delta/2) + 2J_{\mathcal{Y}}(\mathcal{S}, M_{\mathcal{S}, \mathcal{Y}}, \delta/2), \\ M^* &= \left\lfloor \log_2 \min \left\{ \left( \frac{\mathbf{c}_1 n \mathbf{T} \mathbf{V}}{\mathbf{E}} \right)^{\frac{d}{d+1}}, \left( \frac{\mathbf{c}_1 \mathbf{c}_2 n \mathbf{L} \mathbf{T} \mathbf{V}}{\mathbf{E} \mathbf{M}} \right)^{\frac{d}{d+2}} \right\} \right\rfloor, \\ N^* &= \left\lceil \log_2 \max \left\{ \left( \frac{n \mathbf{M}^{d+1}}{\mathbf{c}_1^d \mathbf{E} \mathbf{T} \mathbf{V}^d} \right)^{\frac{1}{d+1}}, \left( \frac{n^2 \mathbf{M}^{2d+2}}{\mathbf{c}_1^d \mathbf{c}_2^d \mathbf{T} \mathbf{V}^d \mathbf{L}^d \mathbf{E}^2} \right)^{\frac{1}{d+2}} \right\} \right\rceil. \end{aligned}$$

**Truncation Argument for  $y_i$ 's with Finite Moments** The above result is derived under the assumption that  $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[\exp(|y_i|)|\mathbf{x}_i = \mathbf{x}] < \infty$ . For the result under the condition  $\sup_{\mathbf{x} \in \mathcal{X}} \mathbb{E}[|y_i|^{2+v}|\mathbf{x}_i = \mathbf{x}] < \infty$ , we can use the same truncation argument as in Section SA-8.1 (proof of Lemma SA-4.1) and the VC-type conditions for  $\mathcal{G}, \mathcal{H}, \mathcal{R}, \mathcal{S}$  to get the stated conclusions.  $\blacksquare$

## References

- Cattaneo, M. D., Chandak, R., Jansson, M., and Ma, X. (2024), “Boundary Adaptive Local Polynomial Conditional Density Estimators,” *Bernoulli*, 30, 3193–3223.
- Cattaneo, M. D., and Yu, R. R. (2025), “Strong Approximations for Empirical Processes Indexed by Lipschitz Functions,” *Annals of Statistics*.
- Chernozhukov, V., Chetverikov, D., and Kato, K. (2014a), “Anti-Concentration and Honest, Adaptive Confidence Bands,” *Annals of Statistics*, 42, 1787–1818.
- (2014b), “Gaussian Approximation of Suprema of Empirical Processes,” *Annals of Statistics*, 42, 1564–1597.
- Chernozhuokov, V., Chetverikov, D., Kato, K., and Koike, Y. (2022), “Improved central limit theorem and bootstrap approximations in high dimensions,” *Annals of Statistics*, 50, 2562–2586.
- Dudley, R. M. (2014), *Uniform central limit theorems*, Vol. 142, Cambridge university press.
- Federer, H. (2014), *Geometric measure theory*, Springer.
- Folland, G. (2002), *Advanced Calculus*, Featured Titles for Advanced Calculus Series, Prentice Hall.
- Giné, E., and Nickl, R. (2016), *Mathematical Foundations of Infinite-dimensional Statistical Models*, New York: Cambridge University Press.
- Simon, L. et al. (1984), *Lectures on geometric measure theory*, Centre for Mathematical Analysis, Australian National University Canberra.
- Tsybakov, A. (2008), *Introduction to Nonparametric Estimation*, Springer.
- van der Vaart, A. W., and Wellner, J. A. (1996), *Weak Convergence and Empirical Processes*, Springer.